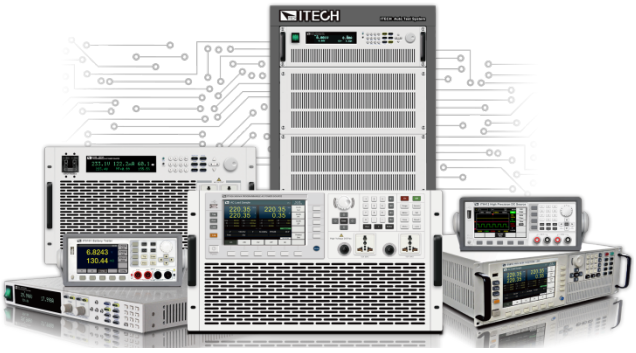
Multi-channel Modular Power System
IT2700 Programming Guide



Model: IT2700
Version: V1.0

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Safety Notices

CAUTION

A CAUTION sign denotes a hazard. It calls attention to an operating procedure or practice that, if not correctly performed or adhered to, could result in damage to the product or loss of important data. Do not proceed beyond a CAUTION sign until the indicated conditions are fully understood and met.

WARNING

A WARNING sign denotes a hazard. It calls attention to an operating procedure or practice that, if not correctly performed or adhered to, could result in personal injury or death. Do not proceed beyond a WARNING sign until the indicated conditions are fully understood and met.

NOTE

A NOTE sign denotes important hint. It calls attention to tips or supplementary information that is essential for users to refer to.

Quality Certification and Assurance

We certify that IT2700 series power supply meets all the published specifications at time of shipment from the factory.

Warranty

ITECH warrants that the product will be free from defects in material and workmanship under normal use for a period of one (1) year from the date of delivery (except those described in the Limitation of Warranty below).

For warranty service or repair, the product must be returned to a service center designated by ITECH.

- The product returned to ITECH for warranty service must be shipped PREPAID. And ITECH will pay for return of the product to customer.
- If the product is returned to ITECH for warranty service from overseas, all the freights, duties and other taxes shall be on the account of customer.

Limitation of Warranty

This Warranty will be rendered invalid if the product is:

- Damaged resulting from customer-wired circuits or customer-supplied parts or accessories;
- Modified or repaired by customer without authorization;
- Damaged resulting from customer-wired circuits or use in an environment not designated by us;
- The product model or serial number is altered, deleted, removed or made illegible by customer;
- Damaged as a result of accidents, including but not limited to lightning, moisture, fire, improper use or negligence.

Safety Symbols

⎓	Direct current	|	ON (power)
∼	Alternating current	○	OFF (power)
⎓ ∼	Both direct and alternating current	⎍	Power-on state
⏚	Chassis (earth ground) symbol.	⎏	Power-off state
⏚	Earth (ground) terminal	⊥	Reference terminal
⚠	Caution	+	Positive terminal
⚠	Warning (refer to this manual for specific Warning or Caution information)	−	Negative terminal

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	A chassis terminal	-	-

Safety Precautions

The following safety precautions must be observed during all phases of operation of this instrument. Failure to comply with these precautions or specific warnings elsewhere in this manual will constitute a default under safety standards of design, manufacture and intended use of the instrument. ITECH assumes no liability for the customer's failure to comply with these precautions.

WARNING

- **Do not use the instrument if it is damaged. Before operation, check the casing to see whether it cracks. Do not operate the instrument in the presence of inflammable gasses, vapors or dusts.**
- **The instrument is provided with a power cord during delivery and should be connected to a socket with a protective earth terminal, a junction box or a three-phase distribution box. Before operation, be sure that the instrument is well grounded.**
- **Check all marks on the instrument before connecting the instrument to power supply.**
- **Use electric wires of appropriate load. All loading wires should be capable of bearing maximum short-circuit of electronic load without overheating. If there are multiple loads, each pair of the load power cord must be carry out the full rated short-circuit output current of the power securely.**
- **Ensure the voltage fluctuation of mains supply is less than 10% of the working voltage range in order to reduce risks of fire and electric shock.**
- **Do not install alternative parts on the instrument or perform any unauthorized modification.**
- **Do not use the instrument if the detachable cover is removed or loosen.**
- **To prevent the possibility of accidental injuries, be sure to use the power adapter supplied by the manufacturer only.**
- **We do not accept responsibility for any direct or indirect financial damage or loss of profit that might occur when using the instrument.**
- **This instrument is used for industrial purposes, do not apply this product to IT power supply system.**
- **Never use the instrument with a life-support system or any other equipment subject to safety requirements.**

WARNING

- SHOCK HAZARD Ground the Instrument. This product is provided with a protective earth terminal. To minimize shock hazard, the instrument must be connected to the AC mains through a grounded power cable, with the ground wire firmly connected to an electrical ground (safety ground) at the power outlet or distribution box. Any interruption of the protective (grounding)

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conductor or disconnection of the protective earth terminal will cause a potential shock hazard that could result in injury or death.

- Before applying power, verify that all safety precautions are taken. All connections must be made with the instrument turned off, and must be performed by qualified personnel who are aware of the hazards involved. Improper actions can cause fatal injury as well as equipment damage.
- SHOCK HAZARD, LETHAL VOLTAGES This product can input the dangerous voltage that can cause personal injury, and the operator must always be protected from electric shock. Ensure that the input electrodes are either insulated or covered using the safety covers provided, so that no accidental contact with lethal voltages can occur.
- Never touch cables or connections immediately after turning off the instrument. Verify that there is no dangerous voltage on the electrodes or sense terminals before touching them.

CAUTION

- **Failure to use the instrument as directed by the manufacturer may render its protective features void.**
- **Always clean the casing with a dry cloth. Do not clean the internals.**
- **Make sure the vent hole is always unblocked.**

Environmental Conditions

The instrument is designed for indoor use and an area with low condensation. The table below shows the general environmental requirements for the instrument.

Environmental Conditions	Requirements
Operating temperature	0°C～40°C
Operating humidity	20%～80%(non-condensation)
Storage temperature	-10°C～70 °C
Altitude	Operating up to 2,000 meters
Installation category	II
Pollution degree	Pollution degree 2

Note

To make accurate measurements, allow the instrument to warm up for 30 min.

Regulatory Markings

CE	The CE mark indicates that the product complies with all the relevant European legal directives. The specific year (if any) affixed refers to the year when the design was approved.
	The instrument complies with the WEEE Directive (2002/96/EC) marking requirement. This affix product label indicates that you must not discard the electrical/electronic product in domestic household waste.

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	This symbol indicates the time period during which no hazardous or toxic substances are expected to leak or deteriorate during normal use. The expected useful life of the product is 10 years. The product can be used safely during the 10-year Environment Friendly Use Period (EFUP). Upon expiration of the EFUP, the product must be immediately recycled.

Waste Electrical and Electronic Equipment (WEEE) Directive



2002/96/EC Waste Electrical and Electronic Equipment (WEEE) Directive

This product complies with the WEEE Directive (2002/96/EC) marking requirement. This affix product label indicates that you must not discard the electrical/electronic product in domestic household waste.

Product Category

With reference to the equipment classifications described in the Annex 1 of the WEEE Directive, this instrument is classified as a “Monitoring and Control Instrument”.

To return this unwanted instrument, contact your nearest ITECH office.

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Compliance Information

Complies with the essential requirements of the following applicable European Directives, and carries the CE marking accordingly:

- Electromagnetic Compatibility (EMC) Directive 2014/30/EU
- Low-Voltage Directive (Safety) 2014/35/EU

Conforms with the following product standards:

EMC Standard

IEC 61326-1:2012/ EN 61326-1:2013 [1][2][3]

Reference Standards

CISPR 11:2009+A1:2010/ EN 55011:2009+A1:2010 (Group 1, Class A)
IEC 61000-4-2:2008/ EN 61000-4-2:2009
IEC 61000-4-3:2006+A1:2007+A2:2010/ EN 61000-4-3:2006+A1:2008+A2:2010
IEC 61000-4-4:2004+A1:2010/ EN 61000-4-4:2004+A1:2010
IEC 61000-4-5:2005/ EN 61000-4-5:2006
IEC 61000-4-6:2008/ EN 61000-4-6:2009
IEC 61000-4-11:2004/ EN 61000-4-11:2004

1. The product is intended for use in non-residential/non-domestic environments. Use of the product in residential/domestic environments may cause electromagnetic interference.
2. Connection of the instrument to a test object may produce radiations beyond the specified limit.
3. Use high-performance shielded interface cable to ensure conformity with the EMC standards listed above.

Safety Standard

IEC 61010-1:2010/ EN 61010-1:2010

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Chapter1 SCPI Introduction

1.1 Overview

SCPI is short for Standard Commands for Programmable Instruments which defines a communication method of bus controller and instrument. It is based on ASCII and supply for testing and measuring instruments. SCPI command is based on hierarchical architecture which also known as tree system. In this system, Relevant Command is returned to a common node or root, so that a subsystem is formed. A part of OUTPut subsystem is listed below:

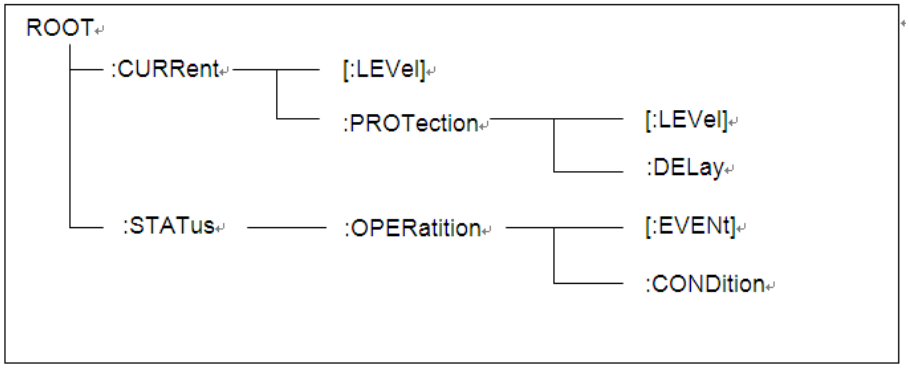
```
OUTPut:
    SYNC {OFF|0|ON|1}
    SYNC:
        MODE {NORMal|CARRier}
        POLarity {NORMal|INVerted}
```

OUTPut is the root keyword, SYNC is the second level keyword, and MODE and POLarity are the third level keywords. A colon (:) is used to separate the command keyword from the next level keyword.

1.2 Command Type of SCPI

SCPI has two types of commands, common and subsystem.

- Common commands generally are not related to specific operation but to controlling overall instrument functions, such as reset, status, and synchronization. All commoncommands consist of a three-letter mnemonic preceded by an asterisk: ***RST *IDN? *SRE 8**.
- Subsystem commands perform specific instrument functions. They are organized into an inverted tree structure with the "root" at the top. The following figure shows a portion of a subsystem command tree, from which you access the commands located along the various paths.



Multiple Commands in a Message

Multiple SCPI commands can be combined and sent as a single message with one message terminator. There are two important considerations when sending several commands within a single message:

- Use a semicolon to separate commands within a message.
- Head paths influence how the instrument interprets commands.

We consider the head path as a string which will be inserted in front of every

command of a message. As for the first command of a message, the head path is a null string; for each subsequent command, the head path is a string which is defined to form the current command until and including the head of the last colon separator. A message with two combined commands:

CURR:LEV 3;PROT:STAT OFF

The Example indicates the effect of semicolon and explains the concept of head path. Since the head path is defined to be "CURR" after "curr: lev 3", the head of the second command, "curr", is deleted and the instrument explains the second command as:

CURR:PROT:STAT OFF

If "curr" is explicitly included in the second command, it is semantically wrong. Since combining it with the head path will become "CURR:CURR:PROT:STAT OFF", resulting in wrong command.

Movement in the Subsystem

In order to combine commands from different subsystems, you need to be able to reset the header path to a null string within a message. You do this by beginning the command with a colon (:), which discards any previous header path. For Example, you could clear the output protection and check the status of the Operation Condition register in one message by using a root specifier as follows:

PROTection:CLEAr;:STATus:OPERation:CONDition?

The following message shows how to combine commands from different subsystems as well as within the same subsystem:

POWer:LEVel 200;PROTection 28; :CURRent:LEVel 3;PROTection:STATe ON

Note the use of the optional header LEVel to maintain the correct path within the voltage and current subsystems, and the use of the root specifier to move between subsystems.

Including Common Commands

You can combine common commands with subsystem commands in the same message. Treat the common command as a message unit by separating it with a semicolon (the message unit separator). Common commands do not affect the header path; you may insert them anywhere in the message.

Case Sensitivity

Common commands and SCPI commands are not case sensitive. You can use upper or lower, for Example:

***RST = *rst**

Long-Form and Short-Form Versions

A SCPI command word can be sent in its long-form or short-form version. However, the short-form version is indicated by upper case characters. Examples:

:SYSTem:ERRor? long-form

:SYST:ERR? short form

:SYSTem:ERR? long-form and short-form combination

Note that each command word must be in long-form or short-form, and not something in between.

For Example, **:SYSTe:Erro?** is illegal and will generate an error. The command will not be executed.

Query

Observe the following precautions with queries:

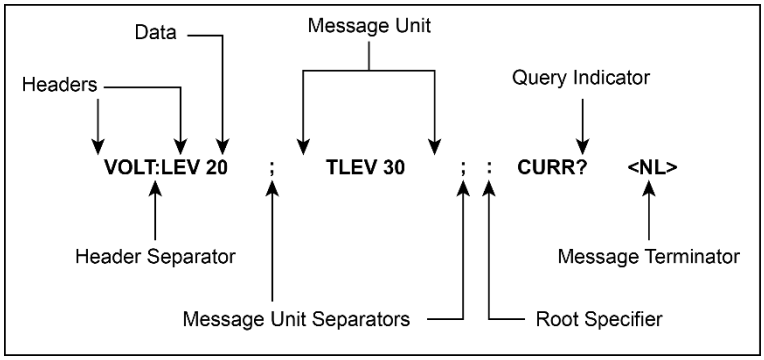
- Set up the proper number of variables for the returned data. For Example, if you are reading back a measurement array, you must dimension the array according to the number of measurements that you have placed in the measurement buffer.
- Read back all the results of a query before sending another command to the instrument. Otherwise a Query Interrupted error will occur and the unreturned data will be lost.

1.3 Message Type of SCPI

There are two types of SCPI messages, program and response.

- Program message: A program message consists of one or more properly formatted SCPI commands sent from the controller to the instrument. The message, which may be sent at any time, requests the instrument to perform some action.
- Response message: A response message consists of data in a specific SCPI format sent from the instrument to the controller. The instrument sends the message only when commanded by a program message called a "query."

The next figure illustrates SCPI message structure:



The message unit

The simplest SCPI command is a single message unit consisting of a command header (or keyword) followed by a message terminator. The message unit may include a parameter after the header. The parameter can be numeric or a string.

ABORt<NL>

VOLTage 20<NL>

Headers

Headers, also referred to as keywords, are instructions recognized by the

instrument. Headers may be either in the long form or the short form. In the long form, the header is completely spelled out, such as VOLTAGE, STATUS and DELAY. In the short form, the header has only the first three or four letters, such as VOLT, STAT and DEL.

Query indicator

Following a header with a question mark turns it into a query (**VOLTage?**, **VOLTage:PROTection?**). If a query contains a parameter, place the query indicator at the end of the last header (**VOLTage:PROTection?MAX**).

Message unit separator

When two or more message units are combined into a compound message, separate the units with a semicolon (**STATus:OPERation?;QUEStionable?**).

Root specifier

When it precedes the first header of a message unit, the colon becomes the root specifier. It tells the command parser that this is the root or the top node of the command tree.

Message terminator

A terminator informs SCPI that it has reached the end of a message. Three permitted message terminators are:

- newline (<NL>), decimal 10 or hexadecimal 0X0A in ASCII.
- end or identify (<END>)
- both of the above (<NL><END>).

In the Examples of this guide, there is an assumed message terminator at the end of each message.

Command execution rules

- Commands execute in the order that they are presented in the program message.
- An invalid command generates an error and, of course, is not executed.
- Valid commands that precede an invalid command in a multiple command program message are executed.
- Valid commands that follow an invalid command in a multiple command program message are ignored.

1.4 Response Data Type

Character strings returned by query statements may take either of the following forms, depending on the length of the returned string:

- **<CRD>**: character response data. Permits the return of character strings.
- **<AARD>**: arbitrary ASCII response data. Permits the return of undelimited 7-bit ASCII. This data type has an implied message terminator.
- **<SRD>**: string response data. Returns string parameters enclosed in double quotes.
- **<Block>**: arbitrary block data.

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Response messages

A response message is the message sent by the instrument to the computer in response to a query command.

Sending a response message

After sending a query command, the response message is placed in the Output Queue. When the instrument is then addressed to talk, the response message is sent from the Output Queue to the computer

Multiple response messages

If you send more than one query command in the same program message, the multiple response messages for all the queries is sent to the computer when the instrument is addressed to talk. The responses are sent in the order that the query commands were sent and are separated by semicolons (;). Items within the same query are separated by commas (,). The following Example shows the response message for a program message that contains four single item query commands:

0; 1; 1; 0

Response message terminator (RMT)

Each response is terminated with an LF (line feed) and EOI (end or identify). The following Example shows how a multiple response message is terminated:

0; 1; 1; 0; <RMT>

Message exchange protocol

Two rules summarize the message exchange protocol:

- **Rule 1**: You must always tell the instrument what to send to the computer.

  The following two steps must always be performed to send information from the instrument other computer:
  1. Send the appropriate query command(s) in a program message.
  2. Address the instrument to talk.
- **Rule 2**: The complete response message must be received by the computer before another program message can be sent to the instrument.

1.5 Command Format

Formats for command display are as follows:

[SOURce[1|2]:]VOLTage:UNIT {VPP|VRMS|DBM}

[SOURce[1|2]:]FREQuency:CENTer
{<frequency>|MINimum|MAXimum|DEFault}

Based on the command syntax, most commands (and certain Parameter) are expressed in both upper and lower cases. Upper case refers to abbreviation of commands. Shorter program line may send commands in abbreviated format. Long-format commands may be sent to ensure better program readability.

For Example, both formats of VOLT and VOLTAGE are acceptable in the above syntax statements. Upper or lower case may be used. Therefore, formats of VOLTAGE, volt and Volt are all acceptable. Other formats (such as VOL and VOLTAG) are invalid and will cause errors.

- Parameter options with given command strings are included in the brace ({ }). The brace is not sent along with command strings.

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- Vertical stripes (|) separate several parameter options with given command strings. For Example, {VPP|VRMS|DBM} indicates that you may assign "APP", "VRMS" or "DBM" in the above commands. Vertical stripes are not sent along with command strings.
- Angle brackets (< >) in the second Example indicates that a value must be assigned to the parameter in the brace. For Example, the parameter in the angle bracket is <frequency> in the above syntax statements. Angle brackets are not sent along with command strings. You must assign a value (such as "FREQ:CENT 1000") to the parameter, unless you select other options displayed in the syntax (such as "FREQ:CENT MIN").
- Some syntax elements (such as nodes and Parameter) are included in square brackets ([ ]). It indicates that these elements can be selected and omitted. Angle brackets are not sent along with command strings. If no value is assigned to the optional Parameter, the instrument will select a default value. In the above Examples, "SOURce[1|2]" indicates that you may refer to source channel 1 by "SOURce" or "SOURce1" or "SOUR1" or "SOUR". In addition, since the whole SOURce node is optional (in the square bracket), you can refer to the channel 1 by omitting the whole SOURce node. It is because the channel 1 is the default channel for SOURce language node. On the other hand, if you want to refer to channel 2, "SOURce2" or "SOUR2" must be used in the program line.

Colon (:)

It is used to separate key words of a command with the key words in next level. As shown below:

APPL:SIN 455E3,1.15,0.0

In this Example, APPLy command assigns a sine wave with frequency of 455 KHz, amplitude of 1.15 V and DC offset of 0.0 V.

Semicolon (;)

It is used to separate several commands in the same subsystem and can also minimize typing. For Example, to send the following command string:

TRIG:SOUR EXT; COUNT 10

has the same effect as sending the following two commands:

TRIG:SOUR EXT
TRIG:COUNT 10

Question mark (?)

You can insert question marks into a command to query current values of most Parameter. For Example, the following commands will trigger to set the count as 10:

TRIG:COUN 10

Then, you may query count value by sending the following command:

TRIG:COUN?

You may also query the allowable minimum or maximum count as follows:

TRIG:COUN?MIN
TRIG:COUN?MAX

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Comma (,)

If a command requires several Parameter, then a comma must be used to separate adjacent Parameter.

Space

You must use blank characters, [TAB] or [Space] to separate Parameter with key words of commands.

Common commands (*)

The IEEE-488.2 standard defines a set of common commands that perform functions such as reset, self-test, and status operations. Common commands always start with a asterisk (*) and occupy 3 character sizes, including one or more Parameter. Key words of a command and the first parameter are separated by a space. Semicolon (;) can separate several commands as follows:

***RST; *CLS; *ESE 32; *OPC?**

Command terminator

Command strings sent to the instrument must end with a <Newline> (<NL>) character. IEEE-488 EOI (End or Identify) information can be used as <NL> character to replace termination command string of <NL> character. It is acceptable to place one <NL> after a <Enter>. Termination of command string always resets current SCPI command path to root level.

NOTE

As for every SCPI message with one query sent to the instrument, the instrument will use a <NL> or newline sign (EOI) to terminate response of return. For Example, if "DISP:TEXT?" is sent, <NL> will be placed after the returned data string to terminate response. If an SCPI message includes several queries separated by semicolon (such as "DISP?;DISP:TEXT?"), <NL> will terminate response returned after response to the last query. In all cases, the program must read <NL> in response before another command is sent to the instrument, otherwise errors will be caused.

1.6 Data Type

SCPI language defines several data types used for program message and response messages.

- Numerical parameter

  Commands requiring numerical parameter support the notations of all common decimal notations, including optional signs, decimal points, scientific notation, etc. Special values of numerical parameter are also acceptable, such as MIN, MAX and DEF. In addition, suffixes for engineering units can also be sent together with numerical parameter (including M, k, m or u). If the command accepts only some specific values, the instrument will automatically round the input parameter to acceptable values. The following commands require numerical parameter of frequency value:
  [SOURce[1|2]:]FREQuency:CENTer {<Frequency>|MINimum|MAXimum}
  – **<NR1>**: represents an integer value, such as 273;
  – **<NR2>**: represents a real number in floating-point format, such as .273;
  – **<NR3>**: represents a real number in scientific notation, such as 2.73E+2;
  – **<Nrf>**: The extensible form includes <NR1>, <NR2> and <NR3>;
  – **<Nrf+>**: The extensible decimal form includes <Nrf>, MIN, MAX and DEF. MIN and MAX are the minimum and maximum finite number. Within the range

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of the parameter definition, DEF is the default of the parameter.

- Discrete parameter

  Discrete parameter are used for settings with limited number of programming values (such as IMMediate, EXTernal or BUS). They can use short and long format like key words of commands. They may be expressed in both upper and lower case. The query response always returns uppercase Parameter in short format. The following commands require discrete parameter in voltage unit:
  [SOURce[1|2]:]VOLTage:UNIT {VPP|VRMS|DBM}

- Boolean parameter

  Boolean parameter refer to true or false binary conditions. In case of false conditions, the instrument will accept "OFF" or "0". In case of true conditions, the instrument will accept "ON" or "1". In query of Boolean settings, the instrument will always return "0" or "1". Boolean parameter are required by the following commands:
  DISPlay {OFF|0|ON|1}

- ASCII string parameter

  String parameter may actually include all ASCII character sets. Character strings must start and end with paired quotation marks; and single quotation marks or double quotation marks are both allowed. Quotation mark separators may also act as one part of a string, they can be typed twice without any character added between them. String parameter is used in the following command:
  DISPlay:TEXT <quoted string>
  For Example, the following commands display message of "WAITING..." (without quotation marks) on the front panel of the instrument.
  DISP:TEXT "WAITING..."
  Single quotation marks may also be used to display the same message.
  DISP:TEXT 'WAITING...'
  – **<SPD>**: string program data. String parameters enclosed in single or double quotes.
  – **<CPD>**: character program data.

1.7 Remote Interface Connection

IT2700 series electronic load are equipped with three communication interfaces as standard: USB, LAN and CAN. Users can choose any one to realize communication with the computer. For detailed introduction of remote interface connection, please refer to the content in the user manual.

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Chapter2 STATus Commands

The status registers are programmed to allow you to determine the operating conditions of the instrument at any time. The instrument contains three groups of status registers; operational, QUEStionable, and standard event. The operational and questionable status groups consist of condition, enable, and event registers and NTR and PTR filters, respectively.

STATus:QUEStionable[:EVENt]? (@<chanlist>)

This command can be used to read the value of the Query Problem Event Register. This is a read-only register, which can be used to store bolted event information through the PTR/NTR filter, and the value of the query event register is cleared to zero after this command is executed.

Command Syntax

STATus:QUEStionable[:EVENt]? (@<chanlist>)

Parameter

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

RST Value

<NR1> This is a decimal value with a binary weighted sum.

Example

STAT:QUES? (@1) // Read the value of the query event register to determine the current instrument status.

Related Command

STATus:QUEStionable:ENABle

STATus:QUEStionable:NTRansition <value>, (@<chanlist>)

STATus:QUEStionable:PTRansition <value>, (@<chanlist>)

STATus:QUEStionable:CONDition? (@<chanlist>)

This command reads the value of the query condition register. This is a read-only register, which directly reflects the suspicious status of the instrument.

The bits in the status are set to 1 or 0 and are not locked. The value of the query condition register will not be cleared after this command is executed.

Query Syntax

STATus:QUEStionable:CONDition? (@<chanlist>)

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Parameter

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

Return Parameters

<NR1> This is a decimal value with a binary weighted sum.

Example

STATus:QUEStionable:CONDition? (@1) //Read the value of the Channel 1 Query Suspicious Status Condition Register.

STATus:QUEStionable:ENABle <value>,(@<chanlist>)

STATus:QUEStionable:ENABle? (@<chanlist>)

The commands set and query the value of the Suspicious Problem Enable Register.Programming parameters determine which bits in the Suspicious Question Register will cause a QUES location 1 in the Status Bit Tuple Register when they are 1.STATus:PRESet clears all bits in the Enable Register.

Command Syntax

STATus:QUEStionable:ENABle <value>,(@<chanlist>)

Parameter

0-65535|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

Query Syntax

STATus:QUEStionable:ENABle? (@<chanlist>)

Return Parameters

<NR1>

Example

STAT:QUES:ENAB 128,(@1) // Set the value of the Channel 1 Suspicious Problem Enable Register to 128, where 128 means bit7 is in place.

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STATus:QUEStionable:NTRansition <value>, (@<chanlist>)

STATus:QUEStionable:NTRansition? (@<chanlist>)

Sets and queries the value of the NTR (Negative Translation) register for suspected problems.These registers act as polarity filters between the Suspicious Condition and Suspicious Event registers.When a bit in the NTR register is set and queried as a 1, a 1 to 0 transition of the corresponding bit in the Suspicious Condition Register causes that bit to be set and queried in the Suspicious Event Register.

STATus:PRESet clears all bits in the NTR register.

Command Syntax

STATus:QUEStionable:NTRansition <value>, (@<chanlist>)

Parameter

0-65535|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

Query Syntax

STATus:QUEStionable:NTRansition? (@<chanlist>)

Return Parameters

<value> This is a decimal value with a binary weighted sum.

Example

STAT:QUES:NTR 16,(@1) //Set the negative hop of the channel 1 issue register to 16.

STAT:QUES:NTR (@1) //Queries the value of the negative jump in the channel 1 issue register.

STATus:QUEStionable:PTRansition <value>, (@<chanlist>)

STATus:QUEStionable:PTRansition? (@<chanlist>)

Sets and queries the value of the PTR (Positive Translation) registers for suspicious issues. These registers act as polarity filters between the Suspicious Condition and Suspicious Event registers.When a bit in the PTR register is set and queried to 1, a 0 to 1 transition of the corresponding bit in the Suspicious Condition Register causes that bit to be set and queried in the Suspicious Event Register.

STATus:PRESet sets all bits in the PTR register.

Command Syntax

STATus:QUEStionable:PTRansition <value>,(@<chanlist>)

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Parameter

0-65535|MINimum|MAXimum

See chapter29 Status Register Definitions for detailed status bit definitions.

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

Query Syntax

STATus:QUEStionable:PTRansition? (@<chanlist>)

Return Parameters

<NR1> This is a decimal value with a binary weighted sum.

Example

STAT:QUES:PTR 128,(@1) // Set the channel 1 suspicious problem register positive jump bit7 to 1, and monitor when the register state of bit7 changes from 0 to 1.

STAT:QUES:PTR? (@1) // Queries the value of the channel 1 Suspicious Problem Register positive jump.

STATus:OPERation[:EVENt]? (@<chanlist>)

This command can be used to read the value of the operation event register. This is a read-only register that stores all events passed by the PTR/NTR filter.After this command is executed, the value of the operation event register is cleared to zero.

The return value is the sum of the binary weighted values of all bits set in the register. For example, if 3 bits (value 8) and 5 bits (value 32) are set (and the corresponding bits are enabled), the query will return +40.

Command Syntax

STATus:OPERation[:EVENt]? (@<chanlist>)

Parameter

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

Return Parameters

<NR1>

Example

STAT:OPER? (@1) // Query Channel 1 Operation Event Register

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Related Command

STATus:OPERation:ENABle

STATus:OPERation:CONDition? (@<chanlist>)

Queries the Condition Register of the Operational Status Group, a read-only register that holds the real-time (non-locked) operational status of the instrument.NOTE: The bits in the condition register reflect the current state; if the state disappears, the corresponding bits are cleared. The return value is the binary-weighted sum of all the bits in the register.

Command Syntax

STATus:OPERation:CONDition? (@<chanlist>)

Parameter

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

Return Parameters

<NR1> This is a decimal value with a binary weighted sum.

Example

STAT:OPER:COND? (@1) // Read Channel 1 Operation Status Condition Register

STATus:OPERation:ENABle <value>,(@<chanlist>)

STATus:OPERation:ENABle? (@<chanlist>)

Sets and queries bits in the Operation Status Group Enable Register.The Enable Register is a mask used to enable specific bits in the Operation Event Register to set and query the OPER (Operation Summary) bit of the Status Byte Register.STATus:PRESet will clear all bits in the Enable Register.

Command Syntax

STATus:OPERation:ENABle <value>, (@<chanlist>)

Parameter

0-65535|MINimum|MAXimum

See chapter29 Status Register Definitions for detailed status bit definitions.

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

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Query Syntax

STATus:OPERation:ENABle?

Return Parameters

<NR1>

Example

STAT:OPER:ENAB 128,(@1) //Set bit7 in the channel 1 operation status group enable register

STAT:OPER:ENAB? (@1) //Query Channel 1 Operation Status Group Enable Register

STATus:OPERation:NTRansition <value>,(@<chanlist>)

STATus:OPERation:NTRansition?

Set and query the value of the NTR (Negative Transition) registers. These registers act as polarity filters between the operation condition and operation event registers.

When a bit in the NTR register is set to 1, a 1 to 0 transition of the corresponding bit in the operation condition register causes that bit to be set in the operation event register.

STATus:PRESet clears all bits in the NTR register.

Command Syntax

STATus:OPERation:NTRansition <value>, (@<chanlist>)

Parameter

0-65535|MINimum|MAXimum

See chapter29 Status Register Definitions for detailed status bit definitions.

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

Query Syntax

STATus:OPERation:NTRansition? (@1) // Query Channel 1 Negative Leap Register

Return Parameters

<NR1>

Example

STATus:OPERation:NTRansition 128,(@1) //Set the value of the channel 1 NTR register we to 128

STAT:OPER:NTR? (@1) //Queries the value of the Channel 1 NTR register

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STATus:OPERation:PTRansition <value>,(@<chanlist>)

STATus:OPERation:PTRansition?

Set and query the value of the PTR (Positive Transition) registers. These registers act as polarity filters between the operation condition and operation event registers.

When a bit in the PTR register is set to 1, a 0 to 1 transition of the corresponding bit in the operation condition register causes the bit to be set in the operation event register.

STATus:PRESet sets all bits in the PTR register.

Command Syntax

STATus:OPERation:PTRansition <value>, (@<chanlist>)

Parameter

0-65535|MINimum|MAXimum

See chapter29 Status Register Definitions for detailed status bit definitions.

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

Query Syntax

STATus:OPERation:PTRansition? (@1)// Queries the value of the Channel 1PTR (Positive Transition) register.

Return Parameters

<NR1>

Example

STAT:OPER:PTR 128,(@1) //Sets the value of the channel 1PTR (Positive Transition) register.

STAT:OPER:PTR? (@1) //Queries the value of the Channel 1PTR (Positive Transition) register.

STATus:PRESet

Reset all Enable, PTR, and NTR registers.

Command Syntax

STATus:PRESet

Example

STAT:PRES

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Chapter3 DISPlay Subsystem

This section is used to control the display of the front panel, the layout of the main interface, and the hold state of the interface.

DISPlay[:WINDow][:STATe] 0|OFF|1|ON

DISPlay[:WINDow}[:STATe]?

The commands can be used to set and query the front panel status, ON indicates that the front panel is normally open, Off indicates that the front panel is closed.

Command Syntax

DISPlay[:WINDow][:STATe] 0|OFF|1|ON

Parameter

0|OFF|1|ON

Query Syntax

DISPlay[:WINDow}[:STATe]?

Return Parameters

0 or 1

Example

DISP OFF //Turn off panel display

DISP? //Query the current panel display status

DISPlay[:WINDow]:VIEW METer1|METer4|METer8

DISPlay[:WINDow]:VIEW?

The commands set and query the display of the output channels of the big screen, including displaying one, four, and eight.

Command Syntax

DISPlay[:WINDow]:VIEW METer1|METer4|METer8

Parameter

METer1|METer4|METer8

Query Syntax

DISPlay[:WINDow]:VIEW?

Return Parameters

MET1,MET4 or MET8

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Example

DISPlay:VIEW METER4 //Set the front panel display to four modules

DISP:VIEW? //Query current panel display module

DISPlay[:WINDow]:VIEW:DEFault METer1|METer4|METer8

DISPlay[:WINDow]:VIEW:DEFault?

The commands set and query the display of default output channels of the big screen, including displaying one, four, and eight.

Command Syntax

DISPlay[:WINDow]:VIEW:DEFault METer1|METer4|METer8

Parameter

METer1|METer4|METer8

Query Syntax

DISPlay[:WINDow]:VIEW:DEFault?

Return Parameters

MET1,MET4 or MET8

Example

DISP:VIEW:DEFault METER4 //Set the default display of the front panel to four modules

DISP:VIEW:DEFault? //Queries the initial display status of the large front panel

DISPlay[:WINDow]:TYPE <METer|SCOPe|DLOG>

DISPlay[:WINDow]:TYPE?

The commands set and query the type of measurement, oscilloscope or logger displayed in the interface. Only for large screen.

Command Syntax

DISPlay[:WINDow]:TYPE <METer|SCOPe|DLOG>

Parameter

METer|SCOPe| DLOG

Query Syntax

DISPlay[:WINDow]:TYPE?

Return Parameters

MET,SCOP or DLOG

Example

DISP:TYPE MET //Set the type of display to Meter

DISP:TYPE? //Query Display Type

DISPlay[:WINDow]:HOLD[:STATe] 0|OFF|1|ON

DISPlay[:WINDow]:HOLD[:STATe]?

The commands set and query the hold state of the front panel display, ON for hold state, OFF to cancel hold state.

Command Syntax

DISPlay[:WINDow]:HOLD[:STATe] 0|OFF|1|ON

Parameter

0|OFF|1|ON

Query Syntax

DISPlay[:WINDow]:HOLD[:STATe]?

Return Parameters

0 or 1

Example

DISP:HOLD ON //Setting the Front Panel to Hold

DISP:HOLD? //Queries the initial display status of the large front panel

DISPlay[:WINDow]:VIEW:SMALl SINGle|MULTi

DISPlay[:WINDow]:VIEW:SMALl?

The commands set and query the display of small-screen output channels, including displaying one or more than one.SINGle is single, MULTi is more than one. (Note: This command is only applicable to IT2703 model)

Command Syntax

DISPlay[:WINDow]:VIEW:SMALl SINGle|MULTi

Parameter

SINGle|MULTi

Query Syntax

DISPlay[:WINDow]:VIEW:SMALl?

Return Parameters

SING or MULT

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Example

DISP:VIEW:SMALl MULT //Setting the small front panel display to multiple

DISP:VIEW:SMALl? //Check the display of the front panel of the small screen

DISPlay[:WINDow]:VIEW:SMALl:DEFault SINGle|MULTi

DISPlay[:WINDow]:VIEW:SMALl:DEFault?

The commands set and query the default display of small screen output channels, including displaying one or more. (Note: This command is only applicable to IT2703 models)

Command Syntax

DISPlay[:WINDow]:VIEW:SMALl:DEFault SINGle|MULTi

Parameter

SINGle|MULTi

Query Syntax

DISPlay[:WINDow]:VIEW:SMAL:DEFaultl?

Return Parameters

SING or MULT

Example

DISP:VIEW:DEFault MULT //Setting the default display of the small screen front panel to multiple channels

DISP:VIEW:DEFault? //Query the default display of the small front panel

Chapter4 FORMat Subsystem

The FORMat command is used to specify the format in which the measurement data is transmitted.

FORMat[:DATA] ASCii|REAL

FORMat[:DATA]?

The commands set and query the format of the returned data

Command Syntax

FORMat[:DATA] ASCii|REAL

Parameter

ASCii|REAL

ASCII Returns the data as ASCII bytes in the corresponding numeric format. Numbers are separated by commas.

REAL returns the data in a finite-length block of data IEEE single-precision floating-point numbers.

RST Value

ASCII

Query Syntax

FORMat[:DATA]?

Return Parameters

ASC or REAL

Example

```
FORMat ASCII          //Setting the data format format to ASCII
FORM?                 //Format of query data
```

FORMat:BORDer NORMal|SWAPped

FORMat:BORDer?

The commands set and query the byte order used in the transfer of data blocks as binary data points

Command Syntax

FORMat:BORDer NORMal|SWAPped

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Parameter

NORMal|SWAPped // Normal:The highest significant byte (MSB) of each data point is the first byte.SWAPped:The least significant byte (LSB) of each data point is the first byte.

RST Value

NORMal

Query Syntax

FORMat:BORDer?

Return Parameters

NORM or SWAP

Example

```
FORM:BORD SWAP      //Setting the SWAPped order
FORM:BORD?          //Order of query bytes
```

Chapter5 SYSTem Subsystem

System commands control system functions that are not directly related to output control, measurement or status functions, in addition to providing settings related to communication, preferences, and so on.

SYSTem:VERSion?

This command is used to return the version of the SCPI parser.

Query Syntax

SYSTem:VERSion?

Return Parameters

<xx.xx.xx>

Example

SYST:VERS?

SYSTem:ERRor[:NEXT]?

This command queries the error information of the instrument response to the command.When the ERROR indicator on the front panel is red, it means that one or more errors have occurred in the instrument's response to the command.Multiple sets of error messages can be stored in the error queue. Send this command once to read an error message from the error queue and clear the error message from the error queue.

- Error messages follow the FIFO (first-in-first-out) principle. The first error that is stored will be the first to be returned.When you have read all the errors in the error queue, the ERROR indicator turns gray. The instrument's buzzer will beep once when an error occurs (in the case of an alarm message buzzer alert switch on).
- If more than the stored number of error messages occur, the last message stored in the queue will be replaced by “-350”, meaning “too many errors”.If the error message in the error message queue is not read, other error messages will not be stored in the error message queue. If there is no error message in the error message queue when the error message is read, “+0” will be returned, meaning “no error”.
- Error messages in the error queue will be cleared if the instrument is turned off or if the *CLS (clear status) command is sent. The *RST command will not clear the error messages in the error queue.

Query Syntax

SYSTem:ERRor?

Return Parameters

<Error code, “Error description” >

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Example

SYST:ERR? //Query an error message about an instrument's response to a command

SYSTem:CHANnel[:COUNt]?

This command is used to read the number of channels in the frame.

Query Syntax

SYSTem:CHANnel[:COUNt]?

Return Parameters

<number of channels>

Example

SYST:CHAN? //Query the number of channels in the frame

SYSTem:CHANnel:MODel? (@<chanlist>)

This command is used to read the product name of the channel.

Query Syntax

SYSTem:CHANnel:MODel? (@<chanlist>)

Return Parameters

<Description of product name>

Example

SYST:CHAN:MOD? (@1) //Query channel 1 module name

SYSTem:CHANnel:OPTion? (@<chanlist>)

This command is used to read the installed options for the channel.

Query Syntax

SYSTem:CHANnel:OPTion? (@<chanlist>)

Return Parameters

<Option Name Description>

Example

SYST:CHAN:OPT? (@1) //Query the description of the installed options for channel 1

SYSTem:CHANnel:DESCription? (@<chanlist>)

This command is used to read the description of the channel.

Query Syntax

SYSTem:CHANnel:DESCription? (@<chanlist>)

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Return Parameters

<return to description of channel module>

Example

SYST:CHAN:DESC? (@1) //Query the description of channel 1

SYSTem:CHANnel:SERial? (@<chanlist>)

This command is used to read the sequence number of the channel.

Query Syntax

SYSTem:CHANnel:SERial? (@<chanlist>)

Return Parameters

<return serial number of channel module>

Example

SYST:CHAN:SER? (@1) //Query the serial number of channel 1

SYSTem:CHANnel:STATus? (@<chanlist>)

This command is used to read the status of the channel.

Query Syntax

SYSTem:CHANnel:STATus? (@<chanlist>)

Return Parameters

0 or 1

0: The channel module does not exist or is not detected.

1: This channel module exists.

Example

SYST:CHAN:STAT? (@1) //Queries whether the module for channel 1 exists

SYSTem:CHANnel:SPEC? (@<chanlist>)

This command is used to read the specifications of the channel.

Query Syntax

SYSTem:CHANnel:SPEC? (@<chanlist>)

Return Parameters

<Module Specification Description>

Example

SYST:CHAN:SPEC? (@1) //Check the specifications of Channel 1 Module

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SYSTem:CHANnel:CATegory? (@<chanlist>)

This command is used to read the model major category of the channel.

Query Syntax

SYSTem:CHANnel:CATegory? (@<chanlist>)

Return Parameters

0,1,2 or 3

0: DC Power supply

1: DC Load

2: AC Power supply

3: AC Load

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example

SYST:CHAN:CAT? (@1) //Check the model type of Channel 1 module

SYSTem:CHANnel:TYPE? (@<chanlist>)

This command is used to read the model subcategory of the channel.

Query Syntax

SYSTem:CHANnel:TYPE? (@<chanlist>)

Return Parameters

0,1,2,3,4 or 5

0: Bipolar Power Supply

1: Unipolar Power Supply

2: DC Load

3: High Precision Source Meter

4: AC Power supply

5: AC Load

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example

SYST:CHAN:TYPE? (@1) //Query Channel 1 Module Model Subcategories

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SYSTem:COMMunicate:RLSTate LOCal|REMote|RWLock

SYSTem:COMMunicate:RLSTate?

The commands set and query the remote or local status.

This command has the same function as SYSTem:REMote/ SYSTem:LOCal/ SYSTem:RWLock.

Command Syntax

SYSTem:COMMunicate:RLSTate LOCal|REMote|RWLock

Parameter

LOCal|REMote|RWLock

LOCal: Instrument operating mode switched to local mode

REMote: The instrument switches to remote control mode.

RWLock: device to the remote state and locks the local Local button.

Query Syntax

SYSTem:COMMunicate:RLSTate?

Return Parameters

LOC,REM or RWL

Example

SYST:COMM:RLST REM //Setting to Remote Status

SYST:COMM:RLST? //Query current remote or local status

SYSTem:REMote

This command is used to set the device to the remote control state.

Command Syntax

SYSTem:REMote

Example

SYST:REM //Switch to remote control status

SYSTem:LOCal

This command is used to set the device to the local control state.

Command Syntax

SYSTem:LOCal

Example

SYST:LOC //Switch to local control

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SYSTem:RWLock

This command is used to set the device to the remote state and lock the local LOCAL key.

Command Syntax

SYSTem:RWLock

Example

SYST:RWL //Switch to remote status and lock the Local key

SYSTem:KEY <0-255>

SYSTem:KEY?

This command is used to simulate a keystroke.

Command Syntax

SYSTem:KEY <0-255>

Parameter:

<0-255>

Query Syntax

SYSTem:KEY?

Queries the keystroke value of the most recent simulation.

Return Parameters

<0-255>

Example

SYST:KEY 1 //Analog button 1

SYST:KEY? //Queries the last simulated keystroke

SYSTem:CLEar

This command clears the error sequence.

Command Syntax

SYSTem:CLEar

Example

SYST:CLE //Clearing Error Sequences

SYSTem:BEEPer[:IMMediate]

This command sets the buzzer to sound once.

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Command Syntax

SYSTem:BEEPer[:IMMediate]

Example

SYST:BEEP //Produces one buzzer sound

SYSTem:BEEPer:CLICk 0|OFF|1|ON

SYSTem:BEEPer:CLICk?

The commands set and query whether the buzzer responds when the key is clicked.

Command Syntax

SYSTem:BEEPer:CLICk 0|OFF|1|ON

Parameter

0|OFF|1|ON

1|ON: The buzzer will respond when the button is clicked

0|OFF: Buzzer does not respond when key is clicked

Query Syntax

SYSTem:BEEPer:CLICk?

Return Parameters

0 or 1

Example

SYST:BEEP:CLIC ON //Sets the buzzer response to ON when a key is clicked.

SYST:BEEP:CLIC? //Queries the status of the buzzer response when a key is clicked.

SYSTem:BEEPer:ALARm 0|OFF|1|ON

SYSTem:BEEPer:ALARm?

This command is used to set or query whether the beep responds in the alarm state.

Command Syntax

SYSTem:BEEPer:ALARm 0|OFF|1|ON

Parameter

0|OFF|1|ON

1|ON: Buzzer will respond in alarm condition

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0|OFF: Buzzer will not respond in alarm condition

Query Syntax

SYSTem:BEEPer:ALARm?

Return Parameters

0 or 1

Example

SYST:BEEP:ALAR ON //Setting the buzzer response to turn on in the alarm state

SYST:BEEP:ALAR? //Queries whether the buzzer responds in the alarm state

SYSTem:COMMunicate:USB:TYPE DEVice|HOST

SYSTem:COMMunicate:USB:TYPE?

Sets or queries the USB interface type.The IT2700 series instrument USB can be set to Host or Device type, which is used to set the USB interface to be used for communication or as a storage device interface.When set to Host, it serves as the storage device interface, and when set to Device, it serves as the communication interface.

Command Syntax

SYSTem:COMMunicate:USB:TYPE DEVice|HOST

Parameter

DEVice|HOST

DEVice: As a storage device interface

HOST: To serve as a communication interface

Query Syntax

SYSTem:COMMunicate:USB:TYPE?

Return Parameters

DEV or HOST

Example

SYST:COMM:USB:TYPE HOST //Set the USB port type to Host

SYST:COMM:USB:TYPE? //Query USB port type

SYSTem:COMMunicate:SELect TMC|VCP

SYSTem:COMMunicate:SELect?

The command selects or queries the communication method in USB

communication mode.The instrument will automatically detect the communication mode, if using USB communication, the user needs to choose TMC mode or VCP mode. Other communication methods are invalid.

Command Syntax

SYSTem:COMMunicate:SELect TMC|VCP

Parameter

TMC|VCP

TMC:USB_TMC protocol communication

VCP Virtual Serial Protocol (VSP) communication

Query Syntax

SYSTem:COMMunicate:SELect?

Return Parameters

TMC or VCP

Example

SYST:COMM:SEL VCP //Select USB communication mode as VCP mode

SYST:COMM:SEL? //Query the current USB communication mode

SYSTem:COMMunicate:LAN:IP[:CONFiguration] <ip>

SYSTem:COMMunicate:LAN:IP[:CONFiguration]?

This command is used to set or query the IP address of the instrument.

Command Syntax

SYSTem:COMMunicate:LAN:IP[:CONFiguration] <ip>

Parameter

<IP address>

Query Syntax

SYSTem:COMMunicate:LAN:IP[:CONFiguration]?

Return Parameters

<IP address>

Example

SYST:COMM:LAN:IP "192.168.0.11" //Set the IP address to 192.168.0.11

SYST:COMM:LAN:IP? //Query current IP address

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SYSTem:COMMunicate:LAN:IP[:CONFiguration]:MODE AUTO|MANual

SYSTem:COMMunicate:LAN:IP[:CONFiguration]:MODE?

This command is used to set or query the IP mode of LAN port.

Command Syntax

SYSTem:COMMunicate:LAN:IP[:CONFiguration]:MODE AUTO|MANual

Parameter

AUTO|MANual

MANual:The user manually sets the IP-related Parameter.

AUTO:The system automatically configures IP-related Parameters.

Query Syntax

SYSTem:COMMunicate:LAN:IP[:CONFiguration]:MODE?

Return Parameters

AUTO or MAN

Example

SYST:COMM:LAN:IP:MODE AUTO //Set the IP mode of the LAN interface to AUTO mode

SYST:COMM:LAN:IP:MODE? //Query IP address setting mode

SYSTem:COMMunicate:LAN:SMASk <smask>

SYSTem:COMMunicate:LAN:SMASk?

This command is used to set or query the subnet mask of LAN.

Command Syntax

SYSTem:COMMunicate:LAN:SMASk <smask>

Parameter

<subnet mask>

Query Syntax

SYSTem:COMMunicate:LAN:SMASk?

Return Parameters

<subnet mask>

Example

SYST:COMM:LAN:SMAS "192.168.27.1" //Setting the Subnet Mask of a LAN

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SYST:COMM:LAN:SMAS? //Query the current subnet mask

SYSTem:COMMunicate:LAN:DGATeway <gateway>

SYSTem:COMMunicate:LAN:DGATeway?

This command is used to set the gateway address of the LAN.

Command Syntax

SYSTem:COMMunicate:LAN:DGATeway <gateway>

Parameter

<Gateway>

Query Syntax

SYSTem:COMMunicate:LAN:DGATeway?

Return Parameters

<Gateway>

Example

SYST:COMM:LAN:DGAT "255.255.255.0" //Set the LAN gateway to 255.255.255.0

SYST:COMM:LAN:DGAT? //> This command is used to query the gateway address of the LAN

SYSTem:COMMunicate:LAN:RAWSocketport <port>

SYSTem:COMMunicate:LAN:RAWSocketport?

This command is used to set or query the socket port number of LAN.

Command Syntax

SYSTem:COMMunicate:LAN:RAWSocketport <port>

Parameter

<port number>

Query Syntax

SYSTem:COMMunicate:LAN:RAWSocketport?

Return Parameters

<port number>

Example

SYST:COMM:LAN:RAWS 30001 //Set the LAN socket port number to 30001

SYST:COMM:LAN:RAWS? //This command queries the port number of

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the LAN

SYSTem:BRIGhtness:LEVel <value>

SYSTem:BRIGhtness:LEVel?

This command is used to set or query the screen brightness of the current instrument, the setting range is 1-10. note that if the model does not have a display panel, this command will report an error.

Command Syntax

SYSTem:BRIGhtness:LEVel <value>

Parameter

1-10

Query Syntax

SYSTem:BRIGhtness:LEVel?

Return Parameters

<Brightness>

Example

SYST:BRIG:LEV 10 //Adjust the instrument background brightness to 10

SYST:BRIG:LEV? //This command queries the current LCD screen brightness.

SYSTem:TOUCh[:STATe] 0|OFF|1|ON

SYSTem:TOUCh[:STATe]?

This command is used to set or query the touch screen status of the current screen. Note that the touch function of 1U model cannot be turned off.

Command Syntax

SYSTem:TOUCh[:STATe] <0|OFF|1|ON>

Parameter

0|OFF|1|ON

0|OFF Indicates that the touch screen is off.

1|ON Indicates that the touch screen is on.

Query Syntax

SYSTem:TOUCh[:STATe]?

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Return Parameters

0 or 1

Example

SYST:TOUC 0 //Turn the instrument's touch screen off

SYST:TOUC? //This command is used to query the touch panel status

SYSTem:LANGuage ENGLish|CHINese

SYSTem:LANGuage?

This command is used to set or query the language display of the current instrument, either Chinese or English.

Command Syntax

SYSTem:LANGuage ENGLish|CHINese

Parameter

ENGLish|CHINese

Query Syntax

SYSTem:LANGuage?

Return Parameters

ENGL or CHIN

Example

SYST:LANG CHIN //Switch the language display of the instrument to Chinese

SYST:LANG? //Query the current language setting

SYSTem:SOFT:KEYBoard[:STATe] 0|OFF|1|ON

SYSTem:SOFT:KEYBoard[:STATe]?

The command sets or queries the UI soft keyboard switch of the current instrument.

Command Syntax

SYSTem:SOFT:KEYBoard[:STATe] 0|OFF|1|ON

Parameter

0|OFF|1|ON

Query Syntax

SYSTem:SOFT:KEYBoard[:STATe]?

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Return Parameters

0 or 1

Example

SYST:SOFT:KEYB 1 //Turn the instrument's on-screen soft keyboard on

SYST:SOFT:KEYB? //This command is used to query the current soft keyboard status

SYSTem:KNOB:IMMediate:EFFective 0|OFF|1|ON

SYSTem:KNOB:IMMediate:EFFective?

Set or query whether the Parameter adjusted by the knob of the machine takes effect immediately.

Command Syntax

SYSTem:KNOB:IMMediate:EFFective 0|OFF|1|ON

Parameter

0|OFF|1|ON

Query Syntax

SYSTem:KNOB:IMMediate:EFFective?

Return Parameters

0 or 1

Example

SYST:KNOB:IMM:EFF 1 //Turn the instrument knob on for immediate effect

SYST:KNOB:IMM:EFF? //Query Knob Immediate Function Status

SYSTem:RUN:TIME?

Queries the current device runtime.

Running time more than one day returns: xxd hh:mm:ss, xxd indicates the number of days, with two spaces between the number of days and the minutes, hours, and seconds that follow it

In a day: hh:mm:ss

Query Syntax

SYSTem:RUN:TIME?

Return Parameters

<xxd hh:mm:ss>

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Example

SYST:RUN:TIME? //Query the current device runtime

SYSTem:GROup:DEFine A|B,(@<chanlist>)

Define multiple channels as a group. Note that only channels with the same model can be edited into a group.

Command Syntax

SYSTem:GROup:DEFine (@<chanlist>)

Parameter

A|B: groups

<chanlist>: channel number

(@1) –Channel 1
(@1,4) – Channels 1 and 4
(@1:4) – Channels 1 through 4, including channels 1,2,3,4

Example

SYST:GRO:DEF A,(@1,3,5) //Set channels 1, 3, and 5 to group A

SYSTem:GROup:CATalog?

Queries information about the currently defined group.

Query Syntax

SYSTem:GROup:CATalog?

Return Parameters

<group definition>

For example, the query returns “1”, “3”, “4,7”, “5”, “6”, “8”, which means that channels 4 and 7 are defined in a group.

Example

SYST:GRO:CAT? //Queries information about groups defined in the current system

SYSTem:GROup:DELete <A|B>,(@<chanlist>)

Deletes the specified channel in the group.

Command Syntax

SYSTem:GROup:DELete <A|B>,(@<chanlist>)

Parameter

A|B: groups

<chanlist>: channel number

(@1) –Channel 1

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(@1,4) –Channels 1 and 4
(@1:4) – Channels 1 through 4, including channels 1,2,3,4

Example

SYST:GRO:DEL B,(@3) //Remove channel 3 from group B

SYSTem:GROup:DELete:ALL

Deletes the defined group.

Command Syntax

SYSTem:GROup:DELete:ALL

Example

SYST:GRO:DEL:ALL //Deletes the defined group.

SYSTem:GROup:APPLy

Submit the created group information and make the group information effective. Note that only channels with the same model can be edited to a group.

Command Syntax

SYSTem:GROup:APPLy

Example

SYST:GRO:APPL //Submit group information

SYSTem:REBoot

Reboot the device to its powered-on state.

Command Syntax

SYSTem:REBoot

Example

SYST:REB //Reboot the device

SYSTem:SECurity:IMMediate

Clear all user-saved data and reboot.

Command Syntax

SYSTem:SECurity:IMMediate

Example

SYST:SEC:IMM //Clear all user-saved data and reboot

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SYSTem:SOURce:SLEW:MODE RATE|TIME

SYSTem:SOURce:SLEW:MODE?

Sets and queries the power supply's ramp mode. The power supply's ramp mode can be selected as either ramp or time. The bipolar power module, DC power module and SMU module support.

Command Syntax

SYSTem:SOURce:SLEW:MODE RATE|TIME

Parameter

RATE|TIME

Query Syntax

SYSTem:SOURce:SLEW:MODE?

Return Parameters

RATE or TIME

Example

SYST:SOUR:SLEW:MODE RATE //Sets the slope mode of the source to the slope value

SYST:SOUR:SLEW:MODE? //Slope pattern of the query source

SYSTem:LOAD:CURRent:SYMBol POSitive|NEGative

SYSTem:LOAD:CURRent:SYMBol?

Set or query the sign of the load current. You can select whether the load current is displayed as positive or negative.

Command Syntax

SYSTem:LOAD:CURRent:SYMBol POSitive|NEGative

Parameter

POSitive|NEGative

Query Syntax

SYSTem:LOAD:CURRent:SYMBol?

Return Parameters

POS or NEG

Example

SYST:LOAD:CURR:SYMB POS //Set the sign of the current to a positive sign

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Chapter6 ABORt Subsystem

The abort command will cancel any triggered operation and return the triggering system to the idle state.The *RST command can also be used to execute the abort command. The transient output system includes arb, sequence, cdarb, list, sine sweep, and load transient.

ABORt[:ALL] (@<chanlist>)

This command is used to stop all operations of the specified channel, including measurement, DLOG, ELOG, and transient system functions.

Command Syntax

ABORt[:ALL]

Parameter:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

Example

ABOR (@1) //Stops channel 1 measurements, DLOG, ELOG, transient system functions.

ABORt:ACQuire (@<chanlist>)

This command is used to stop (meter) the measurement function.

Command Syntax

ABORt:ACQuire

Parameter:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

Example

ABOR:ACQ (@1) //Stop the measurement function of channel 1

ABORt:DLOG

This command is used to stop the data logger.

Command Syntax

ABORt:DLOG

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Example

ABOR:DLOG //Stopping the data logger function

ABORt:ELOG (@<chanlist>)

This command is used to stop the external logger.

Command Syntax

ABORt:ELOG (@<chanlist>)

Parameter:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

Example

ABORt:ELOG (@1) //Disable external logger function for channel 1

ABORt:TRANsient (@<chanlist>)

This command is used to terminate the transient system.

Command Syntax:

ABORt:TRANsient (@<chanlist>)

Parameter:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

Example:

ABORt:TRAN (@1) //Stop the transient system function of channel 1

ABORt:IMPedance (@<chanlist>)

This command is used to terminate the impedance test.

Command Syntax:

ABORt:IMPedance (@<chanlist>)

Parameter:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

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Example:

abort:imp (@1) //Stop the impedance test function of channel 1

ABORt:EIS (@<chanlist>)

This command is used to terminate the EIS.

Command Syntax

ABORt:EIS (@<chanlist>)

Parameter:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

Example:

abort:eis (@1) // Stop the EIS test function of channel 1

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Chapter7 INITiate Subsystem

The start command will initialize the trigger system.This moves the trigger system from the “idle” state to the “waiting for trigger” state, which then enables the instrument to receive the trigger.An event on the selected trigger source can generate a trigger that moves the trigger system from the “waiting for trigger” state to the “running state”Transient output systems include arbitrary wave (arb), sequence, constant dwell arbitrary wave (cdarb), list, sine sweep, and load transient.

INITiate[:IMMediate]:ACQuire (@<chanlist>)

This command is used to initialize (meter) the measurement function.

Command Syntax

INITiate[:IMMediate]:ACQuire (@<chanlist>)

Parameter:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

Example

INIT:ACQ (@1) //Initialize the measurement function of channel 1

INITiate[:IMMediate]:DLOG <"filename">

This command is used to initialize the internal data logger system and specify the name of the export file

Command Syntax

INITiate[:IMMediate]:DLOG <"filename">

Parameter:

<"filename">

Example

INIT:DLOG “Dlogfile” //Initialize the DLOG function with the file name “Dlogfile”.

INITiate[:IMMediate]:ELOG (@<chanlist>)

This command is used to initialize the external datalogger system.

Command Syntax

INITiate[:IMMediate]:ELOG (@<chanlist>)

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Example

INIT:ELOG (@1) //Initialize the ELOG function of channel 1

INITiate[:IMMediate]:TRANsient (@<chanlist>)

This command is used to initialize the transient system.

Command Syntax

INITiate[:IMMediate]:TRANsient (@<chanlist>)

Example

INIT:TRAN (@1) //Initialize transient system functions for channel 1

INITiate:CONTinuous[:ACQuire] 0|OFF|1|ON,(@<chanlist>)

INITiate:CONTinuous[:ACQuire]? (@<chanlist>)

This command is used to set or query the continuous startup state of the acquisition trigger system.

Command Syntax

INITiate:CONTinuous[:ACQuire] 0|OFF|1|ON,(@<chanlist>)

Parameter

0|OFF|1|ON

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax

INITiate:CONTinuous[:ACQuire]? (@<chanlist>)

Return Parameters

0 or 1

Example

INIT:CONT 0,(@1) //Set the continuous startup state of the acquisition trigger system to 0

INIT:CONT? (@1) //Query the continuous startup status of the acquisition trigger system

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INITiate:CONTinuous:TRANsient 0|OFF|1|ON,(@<chanlist>)

INITiate:CONTinuous:TRANsient? (@<chanlist>)

This command is used to set or query the continuous startup state of the transient trigger system.

Command Syntax:

INITiate:CONTinuous:TRANsient 0|OFF|1|ON,(@<chanlist>)

Parameter

0|OFF|1|ON

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax

INITiate:CONTinuous:TRANsient? (@<chanlist>)

Return Parameters

0 or 1

Example

INIT:CONT:TRAN 0,(@1) //Set the continuous start state of the transient trigger system to 0

INIT:CONT:TRAN? (@1) //Querying the continuous startup status of a transient triggered system

INITiate[:IMMediate]:IMPedance (@<chanlist>)

This command is used to initialize the impedance.

Command Syntax:

INITiate[:IMMediate]:IMPedance (@<chanlist>)

Example

INIT:IMP (@1) //initialize the impedance

INITiate:CONTinuous:IMPedance 0|OFF|1|ON,(@<chanlist>)

INITiate:CONTinuous:IMPedance? (@<chanlist>)

This command is used to set and query the continuity of impedance measurement trigger system initialization.

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Command Syntax:

INITiate:CONTinuous:IMPedance 0|OFF|1|ON,(@<chanlist>)

Parameter

0|OFF|1|ON

Query Syntax

INITiate:CONTinuous:IMPedance? (@<chanlist>)

Return Parameters

0 or 1

Example

INIT:CONT:IMP OFF (@1) //Disabled Channel1 Impedance Measurement Trigger System Initialization Continuous 0

INIT:CONT:IMP? (@1) // Query Channel 1 Impedance Measurement Trigger System Continuous Status

INITiate[:IMMediate]:EIS (@<chanlist>)

This command is used to initialize the EIS.

Command Syntax:

INITiate[:IMMediate]:EIS (@<chanlist>)

Example

INIT:EIS (@1) //This command is used to initialize the EIS

Chapter8 OUTPut Subsystem

The OUTPut command is used for input and output related configuration items. the INPut is more commonly used for load related commands.

INPut|OUTPut[:STATe] 0|OFF|1|ON , (@<chanlist>)

INPut|OUTPut[:STATe]? (@<chanlist>)

The commands set and query the output status of the instrument.

Command Syntax:

INPut|OUTPut[:STATe] 0|OFF|1|ON , (@<chanlist>)

Parameter:

0|OFF|1|ON

<chanlist>:Channel number

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channel 1 through Channel 4, including Channel 1234.

RST Value:

0

Query Syntax:

INPut|OUTPut[:STATe]? (@<chanlist>)

Return Parameters:

0 or 1

Example

OUTPut ON , (@1) //Turns on the output of channel 1.

OUTPut? (@1) //Queries the output status of channel 1.

INPut|OUTPut[:STATe]:COUPle:CHANnel ALL|None|[<channel> {,<channel>}]

INPut|OUTPut[:STATe]:COUPle:CHANnel?

The commands set and query the channel for output coupling.

Command Syntax:

INPut|OUTPut[:STATe]:COUPle:CHANnel ALL|None|[<channel> {,<channel>}]

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Parameter:

ALL|None|<1,2,3,4,5,6,7,8>

Query Syntax:

INPut|OUTPut[:STATe]:COUPle:CHANnel?

Return Parameters:

<coupled channels>

Example

OUTP:COUP:CHAN 1,2,4 //Set the channel 1, 2, and 4 output coupling.

OUTP:COUP:CHAN? //Queries the channel to which the output is coupled.

INPut|OUTPut[:STATe]:COUPle:DOFFset <value>

INPut|OUTPut[:STATe]:COUPle:DOFFset? [MINimum|MAXimum]

This instruction is used to set and query the delay offset. (Unit is s)

Command Syntax:

INPut|OUTPut[:STATe]:COUPle:DOFFset <value>

Parameter:

<0.0-1.023>|MINimum|MAXimum

Query Syntax:

INPut|OUTPut[:STATe]:COUPle:DOFFset? [MINimum|MAXimum]

Return Parameters:

<delay offset value>

Example

OUTPut:COUPle:DOFFset 0.02 //Set the offset of the delay to 20ms

OUTPut:COUPle:DOFFset? //Query Delay Offset

INPut|OUTPut[:STATe]:COUPle:DOFFset:MODE <AUTO|MANual>

INPut|OUTPut[:STATe]:COUPle:DOFFset:MODE?

This instruction is used to set and query the mode of the delay offset

Command Syntax:

INPut|OUTPut[:STATe]:COUPle:DOFFset:MODE <AUTO|MANual>

Parameter:

AUTO|MANual

Query Syntax:

INPut|OUTPut[:STATe]:COUPle:DOFFset:MODE?

Return Parameters:

AUTO or MAN

Example

OUTPut:COUPle:DOFF:MODE AUTO //Set delay offset mode to auto

OUTPut:COUPle:DOFF:MODE? //Query Delay Offset Mode

INPut|OUTPut[:STATe]:COUPle:MAX:DOFFset?

This instruction is used to query the maximum delay offset of the machine.

Command Syntax:

INPut|OUTPut[:STATe]:COUPle:MAX:DOFFset?

Return Parameters:

<maximum delay offset value>

Example

OUTPut:COUPle:MAX:DOFFset? //Query Delay Maximum Late Offset

INPut|OUTPut[:STATe]:DELay:FALL <value>,(@<chanlist>)

INPut|OUTPut[:STATe]:DELay:FALL? [MINimum|MAXimum,] (@<chanlist>)

This instruction is used to set and query the delay time when the output is turned off. (Unit is s)

Command Syntax:

INPut|OUTPut[:STATe]:DELay:FALL <value>, (@<chanlist>)

Parameter:

<0-1023>|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

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Query Syntax:

INPut|OUTPut[:STATe]:DELay:FALL? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<delay time value>

Example

OUTPut:DELay:FALL 10, (@1) //Set the delay time when channel 1 output is turned off to 10 seconds

OUTPut:DELay:FALL? (@1) //Query the delay time when the output of channel 1 is turned off

INPut|OUTPut[:STATe]:DELay:RISE <value>,(@<chanlist>)

INPut|OUTPut[:STATe]:DELay:RISE? [MINimum|MAXimum,] (@<chanlist>)

This instruction is used to set and query the delay time when the output is turned on. (Unit is s)

Command Syntax:

INPut|OUTPut[:STATe]:DELay:RISE <value>, (@<chanlist>)

Parameter:

<0-1023>|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

INPut|OUTPut[:STATe]:DELay:RISE? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<delay time value>

Example

OUTPut:DELay:RISE 10, (@1) //Set the delay time when the channel 1 output is turned on to 10 seconds

OUTPut:DELay:RISE? (@1) //Queries the delay time when the output of channel 1 is turned on.

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INPut|OUTPut:INHibit:COUPle:CHANnel ALL|None|[<channel> {,<channel>}]

INPut|OUTPut:INHibit:COUPle:CHANnel?

This instruction is used to set and query the channel for output suppression coupling.

Command Syntax:

INPut|OUTPut:INHibit:COUPle:CHANnel ALL|None|[<channel> {,<channel>}]

Parameter:

ALL|None|[<channel> {,<channel>}]

Query Syntax:

INPut|OUTPut:INHibit:COUPle:CHANnel?

Return Parameters:

coupled channel

Example

OUTPut:INHibit:COUPle:CHANnel ALL //Set all channel output rejection coupling

OUTPut:INHibit:COUPle:CHANnel? //Query the channel of output suppression coupling

INPut|OUTPut:INHibit:MODE <LATChing|LIVE|OFF>

INPut|OUTPut:INHibit:MODE?

This instruction is used to set and query the mode of output suppression.

Command Syntax:

INPut|OUTPut:INHibit:MODE <LATChing|LIVE|OFF>

Parameter:

LATChing|LIVE|OFF

Query Syntax:

INPut|OUTPut:INHibit:MODE?

Return Parameters:

LATC,LIVE or OFF

Example

OUTPut:INHibit:MODE LATChing //Setting Output Inhibit to Lock Mode

OUTPut:INHibit:MODE? //Query Output Inhibit Mode

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INPut|OUTPut:PON:STATe <RST|LAST|LOFF>

INPut|OUTPut:PON:STATe?

This instruction is used to set and query the status of power up. RST means restore factory value, LAST means the status before last power off, LOFF means LAST+OFF.

Command Syntax:

INPut|OUTPut:PON:STATe <RST|LAST|LOFF>

Parameter:

RST|LAST|LOFF

Query Syntax:

INPut|OUTPut:PON:STATe?

Return Parameters:

RST,LAST or LOFF

Example

OUTPut:PON:STATe RST //Setting the power-up state to reset

OUTPut:PON:STATe? //Query power-up status

INPut|OUTPut:PROTection:CLEar (@<chanlist>)

This command is used to clear the protection status.

Command Syntax:

INPut|OUTPut:PROTection:CLEar (@<chanlist>)

Parameter:

<chanlist>:Channel number.

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example

OUTPut:PROT:CLE (@1) //Clear locked channel 1 protection status

INPut|OUTPut:PROTection:COUPle <0|OFF|1|ON>

INPut|OUTPut:PROTection:COUPle?

This command is used to set or query the state of protection coupling.

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Command Syntax:

INPut|OUTPut:PROTection:COUPle <0|OFF|1|ON>

Parameter:

0|OFF|1|ON

Query Syntax:

INPut|OUTPut:PROTection:COUPle?

Return Parameters:

0 or 1

Example

OUTP:PROT:COUPle ON //Open coupling protection

OUTP:PROT:COUPle? //Query coupling protection

INPut|OUTPut:PROTection:COUPle:CHANnel ALL|None|[<channel>{,<channel>}]

INPut|OUTPut:PROTection:COUPle:CHANnel?

The commands set and query the channel for protection coupling.

Command Syntax:

INPut|OUTPut:PROTection:COUPle:CHANnel

ALL|None|[<channel> {,<channel>}]

Parameter:

ALL|None|[<channel> {,<channel>}]

Query Syntax:

INPut|OUTPut:PROTection:COUPle:CHANnel?

Return Parameters:

<coupled channels>

Example

OUTP:PROT:COUP:CHAN 1,2,3 //Setting channel 1,2,3 protection couplingOUTP:PROT:COUP:CHAN? //Query protection coupled channels

INPut|OUTPut:PROTection:WDOG[:STATe] <0|OFF|1|ON>

INPut|OUTPut:PROTection:WDOG[:STATe]?

The commands set and query the enable state of the software watchdog.

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Command Syntax:

INPut|OUTPut:PROTection:WDOG[:STATe] <0|OFF|1|ON>

Parameter:

0|OFF|1|ON

RST Value:

OFF

Query Syntax:

INPut|OUTPut:PROTection:WDOG[:STATe]?

Return Parameters:

0 or 1

Example

OUTP:PROT:WDOG ON //Open the software Watchdog

OUTP:PROT:WDOG? //Query the enable status of the software watchdog

INPut|OUTPut:PROTection:WDOG:DELay <value>

INPut|OUTPut:PROTection:WDOG:DELay? [MINimum|MAXimum]

This command sets the delay time of the software watchdog. (Unit is s)

Command Syntax:

INPut|OUTPut:PROTection:WDOG:DELay <value>

Parameter:

1-3600|MINimum|MAXimum

RST Value:

60

Query Syntax:

INPut|OUTPut:PROTection:WDOG:DELay? [MINimum|MAXimum]

Return Parameters:

<delay time value>

Example

OUTPut:PROTection:WDOG:DELay 10 //Set the watchdog delay to 10s

OUTPut:PROTection:WDOG:DELay? //Query the delay time of the software watchdog

INPut|OUTPut:PROTection:FOLDback[:MODE] <OFF|CC|CV>,(@<chanlist>)

INPut|OUTPut:PROTection:FOLDback[:MODE]? (@<chanlist>)

The commands set and query the mode of foldback protection, which is only supported in power mode.

Command Syntax:

INPut|OUTPut:PROTection:FOLDback[:MODE] <OFF|CC|CV>,(@<chanlist>)

Parameter:

OFF|CC|CV

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

RST Value:

OFF

Query Syntax:

INPut|OUTPut:PROTection:FOLDback[:MODE]? (@<chanlist>)

Return Parameters:

OFF,CC or CV

Example

OUTPut:PROTection:FOLDback CV,(@1) //Set the mode of channel 1 foldback protection to CV

OUTPut:PROTection:FOLDback? (@1) //Inquiry channel 1 discounted return protection model

INPut|OUTPut:PROTection:FOLDback:DELay <value>,(@<chanlist>)

INPut|OUTPut:PROTection:FOLDback:DELay? [MINimum|MAXimum] (@<chanlist>)

The commands set and query the delay time of foldback protection, which is only supported in power mode. (Unit is s)

Command Syntax:

INPut|OUTPut:PROTection:FOLDback:DELay <value>,(@<chanlist>)

Parameter:

0.0-10.0|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

RST Value:

2

Query Syntax:

INPut|OUTPut:PROTection:FOLDback:DELay? [MINimum|MAXimum] (@<chanlist>)

Return Parameters:

<delay time value>

Example

OUTPut:PROTection:FOLDback:DELay 10,(@1) //Set the delay time for channel 1 foldback protection to 10s

OUTPut:PROTection:FOLDback:DELay? (@1) //Query delay time for channel 1 foldback protection

INPut|OUTPut:RELay:LOCK[:STATe] <0|OFF|1|ON>,(@<chanlist>)

INPut|OUTPut:RELay:LOCK[:STATe] ? (@<chanlist>)

This command enables or disables the latched-off state of the output relay.

Command Syntax:

INPut|OUTPut:RELay:LOCK[:STATe] <0|OFF|1|ON>,(@<chanlist>)

Parameter:

0|OFF|1|ON

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

RST Value:

ON

Query Syntax:

INPut|OUTPut:RELay:LOCK[:STATe]? (@<chanlist>)

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Return Parameters:

0 or 1

Example

OUTPut:RELay:LOCK ON,(@1) //Enable latching off state of channel 1 output relay

OUTPut:RELay:LOCK? (@1) //Query the latched-off status of the channel 1 output relay

INPut|OUTPut:REGulation:SPEed <LOW|HIGH1|HIGH2|HIGH3> (@<chanlist>)

INPut|OUTPut:REGulation:SPEed? (@<chanlist>)

The commands set and query the loop regulation speed.

Unipolar source, bipolar source, load mode only supports LOW, HIGH1.

Only SMU module supports high2|high3

Command Syntax:

INPut|OUTPut:REGulation:SPEed <LOW|HIGH1|HIGH2|HIGH3> (@<chanlist>)

Parameter:

LOW|HIGH1|HIGH2|HIGH3

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

RST Value:

HIGH1

Query Syntax:

INPut|OUTPut:REGulation:SPEed? <LOW|HIGH1|HIGH2|HIGH3> (@<chanlist>)

Return Parameters:

LOW,HIGH1,HIGH2 or HIGH3

Example

OUTPut:REGulation:SPEed LOW, (@1) //Setting Channel 1 Loop Adjustment Speed to Low

OUTPut:REGulation:SPEed? (@1) //Query channel 1 loop regulation speed

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INPut|OUTPut:FUNCtion:MODE <FIXed|ARB|BCHarge|BEMulation|TRANsient|SEQuence|CDARB|SSWeep|LIST|BDIScharge>,(@<chanlist>)

INPut|OUTPut:FUNCtion:MODE? (@<chanlist>)

The commands set and query the function mode of input and output. The bipolar power module, DC power module and SMU module support.

Command Syntax:

INPut|OUTPut:FUNCtion:MODE <FIXed|ARB|BCHarge|BEMulation|TRANsient|SEQuence|CDARB|SSWeep|LIST|BDIScharge>,(@<chanlist>)

Parameter:

<FIXed|ARB|BCHarge|BEMulation|TRANsient|SEQuence|CDARB|SSWeep|LIST|BDIScharge>

//Bipolar power module:

FIXed|ARB|BCHarge|BEMulation|TRANsient|SEQuence|CDARB|SSWeep|LIST|BDIScharge

// DC power module: FIXed|ARB|BCHarge|SEQuence|CDARB|SSWeep|LIST

//load module:

FIXed|ARB|BDIScharge|TRANsient|SEQuence|CDARB|SSWeep|LIST

//SMU module:

FIXed|ARB|BCHarge|BEMulation|TRANsient|SEQuence|CDARB|SSWeep|LIST|BDIScharge|EIS

RST Value:

FIXed

Query Syntax:

INPut|OUTPut:FUNCtion:MODE? (@<chanlist>)

Example

OUTPut:FUNCtion:MODE FIXed,(@1) //Set channel 1 operating mode to fixed mode

OUTPut:FUNCtion:MODE? (@1) //Query the operating mode of channel 1

INPut|OUTPut:OFF:VOLTage:RTZero <0|OFF|1|ON>,(@<chanlist>)

INPut|OUTPut:OFF:VOLTage:RTZero? (@<chanlist>)

This command is used to set the state of output and query OFF port voltage zeroing.

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Command Syntax:

INPut|OUTPut:OFF:VOLTage:RTZero <0|OFF|1|ON>,(@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

ON

Query Syntax:

INPut|OUTPut:OFF:VOLTage:RTZero? (@<chanlist>)

Return Parameters:

0 or 1

Example

OUTPut:OFF:VOLTage:RTZero ON,(@1) //Setting the port voltage to zero when the channel 1 output is turned off

OUTPut:OFF:VOLTage:RTZero? (@1) //Query the output shutdown port zero status of channel 1

INPut|OUTPut:SHORt[:STATe] <0|OFF|1|ON>, (@<chanlist>)

INPut|OUTPut:SHORt[:STATe]? (@<chanlist>)

The commands set and query the status of short circuit simulation. (Supported only when load module)

Command Syntax:

INPut|OUTPut:SHORt[:STATe] <0|OFF|1|ON>, (@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

ON

Query Syntax:

INPut|OUTPut:SHORt[:STATe]? (@<chanlist>)

Return Parameters:

0 or 1

Example

OUTPut:SHORt ON, (@1) //Simulating a short circuit at the load input port of channel 1

OUTPut:SHORt? (@1) //Query the short-circuit status of channel 1 load input

port

INPut|OUTPut:TMODe <HIGHz|LOWZ>,(@<chanlist>)

INPut|OUTPut:TMODe? (@<chanlist>)

This command is used to set and query the output mode when the machine is OFF. Supported only by source load modules.

Command Syntax:

INPut|OUTPut:TMODe <HIGHz|LOWZ>,(@<chanlist>)

Parameter:

HIGHz|LOWZ

RST Value:

HIGHz

Query Syntax:

INPut|OUTPut:TMODe? (@<chanlist>)

Return Parameters:

HIGH or LOWZ

Example

```
OUTP:TMOD HIGH,(@1)      // set the off mode of Channel 1
OUTP:TMOD? (@1)          // Query the off mode of Channel 1
```

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Chapter9 Bipolar Power Commands (SOURce Subsystem)

This chapter describes the commands related to the bipolar power module.

9.1 Function Commands

[SOURce:]FUNCtion <CURRent|VOLTage>,(@<chanlist>)

[SOURce:]FUNCtion? (@<chanlist>)

This command is used to set and query the priority of the function.

Command Syntax:

[SOURce:]FUNCtion CURRent|VOLTage,(@<chanlist>)

Parameter:

<mode>: CURRent|VOLTage

Query Syntax:

[SOURce:]FUNCtion? (@<chanlist>)

RST Value:

VOLTage

Return Parameters:

CURR,VOLT

Example:

FUNC CURR,(@1) //Set Channel 1 function priority to current priority

FUNC? (@1) //Queries the Channel 1 function priority.

9.2 Voltage Commands

[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the immediate value of voltage. (Unit is V)

Command Syntax:

[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

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Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

Query Syntax:

[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

RST Value:

Default

Return Parameters:

<immediate value of voltage>

Example:

VOLT 100, (@1) //Set channel 1 voltage value to 100V

VOLT? (@1) //Query the current voltage value of channel 1

[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude] <value>,(@<chanlist>)

[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the value when the voltage is triggered. (Unit is V)

Command Syntax:

[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

None

Query Syntax:

[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Trigger value of voltage>

Example:

VOLT:TRIG 50, (@1) //Setting the voltage trigger value for channel 1 to 50V

VOLT? (@1) //Query the voltage trigger value of channel 1

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[SOURce:]VOLTage:LIMit:HIGH[:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]VOLTage:LIMit:HIGH[:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Sets or queries the upper voltage limit value for current priority mode. (Unit is V)

Command Syntax:

VOLTage:LIMit:HIGH[:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:LIMit:HIGH[:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<upper limit value of voltage>

Example:

VOLT:LIMit:HIGH 50,(@1) //Set the upper voltage limit value for channel 1 current priority mode to 50V

VOLT:LIMit:HIGH? (@1) //Query the upper voltage limit value of channel 1 current priority mode

[SOURce:]VOLTage:LIMit:LOW[:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]VOLTage:LIMit:LOW[:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Sets or queries the voltage lower limit value in current priority mode, this command is only required when the power supply module is bipolar. (Unit is V)

Command Syntax:

[SOURce:]VOLTage:LIMit:LOW[:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>: 0-maximum|MINimum|MAXimum

<chanlist>: Channel number.

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:LIMit:LOW[:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Lower limit value of voltage>

Example:

VOLT:LIM:LOW 10,(@1) //Setting the lower voltage limit value in channel 1 current priority mode

VOLT:LIM:LOW? (@1) //Query the voltage lower limit value in channel 1 current priority mode

[SOURce:]VOLTage:PROTection[:LOCal][:POSitive][:LEVel] <value>,(@<chanlist>)

[SOURce:]VOLTage:PROTection[:LOCal][:POSitive][:LEVel]? [MINimum|MAXimum,](@<chanlist>)

The command sets or queries the positive voltage protection value of overvoltage protection. (Unit is V)

Command Syntax:

[SOURce:]VOLTage:PROTection[:LOCal][:POSitive][:LEVel] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

maximum

Query Syntax:

[SOURce:]VOLTage:PROTection[:LOCal][:POSitive][:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Protection value for positive voltage>

Example

VOLT:PROT 300, (@1) //Setting the positive voltage protection value for channel 1 overvoltage protection

VOLT:PROT? (@1) //Query the positive voltage protection value of

channel 1 overvoltage protection

[SOURce:]VOLTage:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:PROTection:STATe? (@<chanlist>)

The commands set and query the enable state of overvoltage protection. (Unit is V)

Command Syntax:

[SOURce:]VOLTage:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

Parameter:

<0|OFF|1|ON>:0-OFF|1-ON

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:PROTection:STATe? (@<chanlist>)

Return Parameters:

<Protection enable state of voltage>

Example:

VOLT:PROT:STAT ON, (@1) //Setting the enable state of channel 1 overvoltage protection to ON

VOLT:PROT:STAT? (@1) //Queries the enable state of channel 1 overvoltage protection.

[SOURce:]VOLTage:PROTection:DELay[:TIME] <value>,(@<chanlist>)

[SOURce:]VOLTage:PROTection:DELay[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

Setting and querying the delay time for overvoltage protection, (in seconds)

Command Syntax:

[SOURce:]VOLTage:PROTection:DELay[:TIME] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

Query Syntax:

[SOURce:]VOLTage:PROTection:DELay[:TIME]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Delay value for voltage protection>

Example:

VOLT:PROT:DEL 30, (@1) //Set the delay time for channel 1 overvoltage protection to 30s.

VOLT:PROT:DEL? (@1) //Queries the delay time for channel 1 overvoltage protection.

[SOURce:]VOLTage:UNDer:PROTection[:POSitive][:LEVel] <value>,(@<chanlist>)

[SOURce:]VOLTage:UNDer:PROTection[:POSitive] [:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Set and query the positive voltage protection value for undervoltage protection. (Unit is V)

Command Syntax:

[SOURce:]VOLTage:UNDer:PROTection[:POSitive][:LEVel] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

maximum

Query Syntax:

[SOURce:]VOLTage:UNDer:PROTection[:POSitive] [:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Positive voltage protection value for undervoltage>

Example:

VOLT:UND:PROT 10, (@1) //Set the positive voltage protection value for channel 1 undervoltage protection to 10V

VOLT:UND:PROT? (@1) //Queries the positive voltage protection value for channel 1 undervoltage protection.

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[SOURce:]VOLTage:UNDer:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:UNDer:PROTection:STATe? (@<chanlist>)

Sets and queries the enable state of undervoltage protection.

Command Syntax:

[SOURce:]VOLTage:UNDer:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

Parameter:

<0|OFF|1|ON>:0-OFF|1-ON

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:UNDer:PROTection:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

VOLT:UND:PROT:STAT ON,(@1) //Set the enable state of channel 1 undervoltage protection to ON.

VOLT:UND:PROT:STAT? (@1) //Queries the enable state of channel 1 undervoltage protection.

[SOURce:]VOLTage:UNDer:PROTection:DELay[:TIME] <value>,(@<chanlist>)

[SOURce:]VOLTage:UNDer:PROTection:DELay[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

Sets the delay time for undervoltage protection. (in seconds)

Command Syntax:

[SOURce:]VOLTage:UNDer:PROTection:DELay[:TIME] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

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Query Syntax:

[SOURce:]VOLTage:UNDer:PROTection:DELay[:TIME]?

[MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Delay time for undervoltage protection>

Example:

VOLT:UND:PROT:DEL 10, (@1) //Set channel 1 undervoltage protection delay time to 10s

VOLT:UND:PROT:DEL? (@1) //Query channel 1 undervoltage protection delay time

[SOURce:]VOLTage:UNDer:PROTection:WARM[:TIME] <value>,(@<chanlist>)

[SOURce:]VOLTage:UNDer:PROTection:WARM[:TIME]? [MINimum|MAXimum,](@<chanlist>)

Set and query the warm-up time for undervoltage protection. (in seconds)

Command Syntax:

[SOURce:]VOLTage:UNDer:PROTection:WARM[:TIME] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

Query Syntax:

[SOURce:]VOLTage:UNDer:PROTection:WARM[:TIME]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Warm-up time for undervoltage protection>

Example:

VOLT:UND:PROT:WARM 10, (@1) //Set channel 1 undervoltage protection delay time to 10s

VOLT:UND:PROT:WARM? (@1) //Query channel 1 undervoltage protection delay time

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[SOURce:]VOLTage:RESistance[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]VOLTage:RESistance[:LEVel][:IMMediate][:AMPLitude]?(@<chanlist>)

Sets and queries the internal resistance value of the power supply in voltage priority mode. (Unit is Ω)

Valid only when the internal resistance function is enabled.

Command Syntax:

[SOURce:]VOLTage:RESistance[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:MAXimum|MINimum|DEFault

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:RESistance[:LEVel][:IMMediate][:AMPLitude]?(@<chanlist>)

Return Parameters:

<internal resistance value>

Example:

VOLT:RES 0.1,(@1)	//Set the internal resistance of channel 1 to 0.1Ω.
VOLT:RES? (@1)	//Query the internal resistance of channel 1

[SOURce:]VOLTage:RESistance:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:RESistance:STATe? (@<chanlist>)

Sets and queries the status of the internal resistance function of the power supply in voltage priority mode.

Command Syntax:

[SOURce:]VOLTage:RESistance:STATe <0|OFF|1|ON>,(@<chanlist>)

Parameter:

<0|OFF|1|ON>:0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:RESistance:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

VOLT:RES:STAT 0 //Set channel 1 internal resistance function status to 0

VOLT:RES:STAT? (@1) //Query the current internal resistance function status of channel 1

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate] <value>,(@<chanlist>)

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Set and query the voltage rise slope. (Unit is V/ms)

If the [SOURce:]VOLTage:SLEW:COUPle instruction setting state is ON, this instruction sets the voltage upward ramp downward while simultaneously setting the voltage downward ramp downward.

Command Syntax:

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate] <value>,(@<chanlist>)

Parameter:

<value>:minimum-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Rise slope of voltage>

Example:

VOLT:SLEW 10, (@1) //Set channel 1 voltage rise slope to 10V/ms

VOLT:SLEW? (@1) //Query channel 1 voltage rise slope

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[SOURce:]VOLTage:SLEW:NEGative[:IMMediate] <value>,(@<chanlist>)

[SOURce:]VOLTage:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,](@<chanlist>)

Set and query the voltage drop slope. (Unit is V/ms)

If the [SOURce:]VOLTage:SLEW:COUPle instruction setting state is ON, this instruction sets the voltage falling slope while simultaneously setting the voltage rising slope.

Command Syntax:

[SOURce:]VOLTage:SLEW:NEGative[:IMMediate] <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]VOLTage:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Slope of voltage drop>

Example:

VOLT:SLEW:NEG 10,(@1) //Set channel 1 voltage drop slope to 10V/ms

VOLT:SLEW:NEG? (@1) //Query channel 1 voltage drop slope

[SOURce:]VOLTage:SLEW:COUPle <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:SLEW:COUPle? (@<chanlist>)

Set and query the tracking status of the voltage slope.

Command Syntax:

[SOURce:]VOLTage:SLEW:COUPle <0|OFF|1|ON>,(@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

1

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Query Syntax:

[SOURce:]VOLTage:SLEW:COUPle? (@<chanlist>)

Return Parameters:

0 or 1

Example

VOLT:SLEW:COUP ON, (@1) //Enable channel 1 voltage slope tracking state

VOLT:SLEW:COUP? (@1) //Query channel 1 voltage slope tracking status

[SOURce:]VOLTage:SLEW[:POSitive]:MAXimum <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Enable or disable voltage rise slope maximum.

Command Syntax:

[SOURce:]VOLTage:SLEW[:POSitive]:MAXimum 0|OFF|1|ON,(@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Return Parameters:

0 or 1

Example:

VOLT:SLEW:MAX ON, (@1) //Enable Channel 1 Voltage Rise Slope Maximum Function State

VOLT:SLEW:MAX? (@1) //Query Channel 1 Voltage Rise Slope Maximum Function Status

[SOURce:]VOLTage:SLEW:NEGative:MAXimum <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:SLEW:NEGative:MAXimum? (@<chanlist>)

Enable or disable voltage drop slope maximum.

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Command Syntax:

[SOURce:]VOLTage:SLEW:NEGative:MAXimum 0|OFF|1|ON,(@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:SLEW:NEGative:MAXimum? (@<chanlist>)

Return Parameters:

0 or 1

Example:

VOLT:SLEW:NEG:MAX ON,(@1) //Enable channel 1 voltage drop slope maximum state

VOLT:SLEW:NEG:MAX? (@1) //Query channel 1 voltage drop slope maximum value status

9.3 Current Commands

[SOURce:]CURRent[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]CURRent[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Sets the immediate value of the current. (Unit is A)

Command Syntax:

[SOURce:]CURRent[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

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Return Parameters:

<immediate value of current>

Example:

CURR 3,(@1) //3ASet channel 1 current to 3A immediately

CURR? (@1) //Query channel 1 current immediate value

[SOURce:]CURRent[:LEVel]:TRIGgered[:AMPLitude] <value>,(@<chanlist>)

[SOURce:]CURRent[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Sets or queries the trigger value of the current. (Unit is A)

Command Syntax:

[SOURce:]CURRent[:LEVel]:TRIGgered[:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

None

Query Syntax:

[SOURce:]CURRent[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Trigger value of voltage>

Example:

CURR:TRIG 20, (@1) //Set channel 1 current trigger value to 20A

CURR:TRIG ? (@1) //Query channel 1 current trigger value

[SOURce:]CURRent:LIMit[:POSitive][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]CURRent:LIMit[:POSitive][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Sets the current limit value for voltage priority mode. (Unit is A)

With the bipolar power supply function, if the [SOURce:]CURRent:LIMit:COUPle instruction setting status is ON, this instruction sets the positive current limit value while simultaneously setting the negative current limit value.

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Command Syntax:

[SOURce:]CURRent:LIMit[:POSitive][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:LIMit[:POSitive][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Positive limiting value of current>

Example:

CURR:LIM 50,(@1) //Set channel 1 positive current limit value to 50A

CURR:LIM? (@1) //Query channel 1 positive current limit value

[SOURce:]CURRent:LIMit:NEGative[:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]CURRent:LIMit:NEGative[:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Sets the negative current limit value for voltage priority mode. (Unit is A)

If the [SOURce:]CURRent:LIMit:COUPle instruction setting status is ON, this instruction sets the negative current limit value and also sets the positive current limit value synchronously.

Command Syntax:

[SOURce:]CURRent:LIMit:NEGative[:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:minimum - 0|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:LIMit:NEGative[:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

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Return Parameters:

<Negative limit value of current>

Example:

CURR:LIM:NEG -2,(@1) //Set the negative current limit value for channel 1 to -2A.

CURR:LIM:NEG? (@1) //Query channel 1 negative current limit value

[SOURce:]CURRent:LIMit:COUPle <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]CURRent:LIMit:COUPle? (@<chanlist>)

Sets and queries the tracking status of positive and negative current limit values.

Command Syntax:

[SOURce:]CURRent:LIMit:COUPle <0|OFF|1|ON>,(@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

ON

Query Syntax:

[SOURce:]CURRent:LIMit:COUPle? (@<chanlist>)

Return Parameters:

0 or 1

Example:

CURR:LIM:COUP ON,(@1) //Turns the tracking state of channel 1 current positive and negative limits on.

CURR:LIM:COUP? (@1) //Queries the tracking status of channel 1 current positive and negative limit values

[SOURce:]CURRent:PROTection[:LEVel][:POSitive] <level>,(@<chanlist>)

[SOURce:]CURRent:PROTection[:LEVel][:POSitive]? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the positive current protection value of overcurrent protection. (Unit is A)

Command Syntax:

[SOURce:]CURRent:PROTection[:LEVel][:POSitive] <level>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

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<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:PROTection[:LEVel][:POSitive]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Positive current protection value for overcurrent protection>

Example:

CURR:PROT 30, (@1) //Setting the positive current protection value for channel 1 overcurrent protection to 30A

CURR:PROT? (@1) //Query the positive current protection value of channel 1 overcurrent protection

[SOURce:]CURRent:PROTection[:LEVel]:NEGative <level>,(@<chanlist>)

[SOURce:]CURRent:PROTection[:LEVel]:NEGative? [MINimum|MAXimum,](@<chanlist>)

This command sets the negative current protection value for overcurrent protection. (Unit is A)

Command Syntax:

[SOURce:]CURRent:PROTection[:LEVel]:NEGative <level>,(@<chanlist>)

Parameter:

<value>:minimum-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:PROTection[:LEVel]:NEGative? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Negative current protection value for overcurrent protection>

Example:

CURR:PROT:NEG -2,(@1) //Set channel 1 overcurrent protection negative value to -2A

CURR:PROT:NEG? (@1) //Query channel 1 overcurrent protection

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negative value

[SOURce:]CURRent:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]CURRent:PROTection:STATe? (@<chanlist>)

The commands set and query the enable state of overcurrent protection.

Command Syntax:

[SOURce:]CURRent:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

Parameter:

<value>:<0|OFF|1|ON>

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]CURRent:PROTection:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

CURR:PROT:STAT ON, (@1) //Setting the channel 1 current protection function to the ON state

CURR:PROT:STAT? (@1) //Query channel 1 current protection function status

[SOURce:]CURRent:PROTection:DELay[:TIME] <value>,(@<chanlist>)

[SOURce:]CURRent:PROTection:DELay[:TIME]? [MINimum|MAXimum,](@<chanlist>)

This command is used to set or query the delay time of overcurrent protection. (Unit is in seconds)

Command Syntax:

[SOURce:]CURRent:PROTection:DELay[:TIME] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

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Query Syntax:

[SOURce:]CURRent:PROTection:DELay[:TIME]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Delay time for overcurrent protection>

Example:

CURR:PROT:DEL 10,(@1) //Set the delay time for channel 1 overcurrent protection to 10 seconds

CURR:PROT:DEL? (@1) //Query delay time for channel 1 overcurrent protection

[SOURce:]CURRent:UNDer:PROTection[:LEVel][:POSitive] <level>,(@<chanlist>)

[SOURce:]CURRent:UNDer:PROTection[:LEVel][:POSitive] [MINimum|MAXimum,](@<chanlist>)

Set and query the positive current protection value for undercurrent protection. (Unit is A)

Command Syntax:

[SOURce:]CURRent:UNDer:PROTection[:LEVel][:POSitive] <level>,(@<chanlist>)

Parameter:

<value>:minimum-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

maximum

Query Syntax:

[SOURce:]CURRent:UNDer:PROTection[:LEVel][:POSitive] [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Positive current protection value for undercurrent protection>

Example:

VOLT:UND:PROT 10, (@1) //Set channel 1 undercurrent protection positive current value to 10A

VOLT:UND:PROT? (@1) //Query channel 1 undercurrent protection positive current value

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[SOURce:]CURRent:UNDer:PROTection[:LEVel]:NEGative <level>,(@<chanlist>)

[SOURce:]CURRent:UNDer:PROTection[:LEVel]:NEGative? [MINimum|MAXimum,](@<chanlist>)

Set and query the negative current protection value for undercurrent protection. (Unit is A)

Command Syntax:

[SOURce:]CURRent:UNDer:PROTection[:LEVel]:NEGative <level>,(@<chanlist>)

Parameter:

<value>:minimum-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:UNDer:PROTection[:LEVel]:NEGative? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Negative current protection value for undercurrent protection>

Example:

CURR:UND:PROT:NEG 10, (@1) //Set channel 1 undercurrent protection negative current value to 10A

CURR:UND:PROT:NEG? (@1) //Query channel 1 undercurrent protection negative current value

[SOURce:]CURRent:UNDer:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]CURRent:UNDer:PROTection:STATe? (@<chanlist>)

Sets and queries the enable state of undercurrent protection.

Command Syntax:

[SOURce:]CURRent:UNDer:PROTection:STATe 0|OFF|1|ON,(@<chanlist>)

Parameter:

<value>:<0|OFF|1|ON>

<chanlist>:Channel number.

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RST Value:

0

Query Syntax:

[SOURce:]CURRent:UNDer:PROTection:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

CURR:UND:PROT:STATe ON,(@1) //Set the enable state of channel 1 undercurrent protection to ON

CURR:UND:PROT:STATe? (@1) //Query the enable status of channel 1 undercurrent protection

[SOURce:]CURRent:UNDer:PROTection:DELay[:TIME] <value>,(@<chanlist>)

[SOURce:]CURRent:UNDer:PROTection:DELay[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

Sets or queries the delay time for undercurrent protection. (Unit is in seconds)

Command Syntax:

[SOURce:]CURRent:UNDer:PROTection:DELay[:TIME] <level>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

Query Syntax:

[SOURce:]CURRent:UNDer:PROTection:DELay[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Delay time for undercurrent protection>

Example:

CURR:UND:PROT:DEL 10,(@1) //Set the delay time for channel 1 undercurrent protection to 10 seconds

CURR:UND:PROT:DEL? (@1) //Query the delay time for channel 1 undercurrent protection

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[SOURce:]CURRent:UNDer:PROTection:WARM[:TIME] <value>,(@<chanlist>)

[SOURce:]CURRent:UNDer:PROTection:WARM[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

Sets the warm-up time for undercurrent protection. (in seconds)

Command Syntax:

[SOURce:]CURRent:UNDer:PROTection:WARM[:TIME] <level>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

Query Syntax:

[SOURce:]CURRent:UNDer:PROTection:WARM[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Warm-up time for undercurrent protection>

Example:

CURR:UND:PROT:WARM 10,(@1) //Set the warm-up time for channel 1 undercurrent protection to 10 seconds

CURR:UND:PROT:WARM? (@1) //Query the warm-up time for channel 1 undercurrent protection

[SOURce:]CURRent:SLEW[:POSitive][:IMMediate] <value>,(@<chanlist>)

[SOURce:]CURRent:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,](@<chanlist>)

Set and query the current rise slope. (Unit is A/ms)

If the [SOURce:]CURRent:SLEW:COUPle instruction setting state is ON, this instruction sets the current up and down slopes simultaneously with the voltage down slope.

Command Syntax:

[SOURce:]CURRent:SLEW[:POSitive][:IMMediate] <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

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RST Value:

Default

Query Syntax:

[SOURce:]CURRent:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Ramp rate of current rise

Example:

CURR:SLEW 10, (@1) //Set channel 1 current rise slope to 10 A/ms

CURR:SLEW? (@1) //Query channel 1 current rise slope

[SOURce:]CURRent:SLEW:NEGative[:IMMediate] <value>,(@<chanlist>)

[SOURce:]CURRent:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Setting and querying the current drop slope. (Unit is A/ms)

If the [SOURce:]CURRent:SLEW:COUPle instruction setting state is ON, this instruction sets the current down and up slope simultaneously with the voltage up slope.

Command Syntax:

[SOURce:]CURRent:SLEW:NEGative[:IMMediate] <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Slope of current drop>

Example:

CURR:SLEW:NEG 10, (@1) //Set channel 1 current down slope to 10 A/ms

CURR:SLEW:NEG? (@1) //Query channel 1 current descent slope

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[SOURce:]CURRent:SLEW:COUPle 0|OFF|1|ON, (@<chanlist>)

[SOURce:]CURRent:SLEW:COUPle? (@<chanlist>)

Set and query the tracking status of the current slope.

Command Syntax:

[SOURce:]CURRent:SLEW:COUPle 0|OFF|1|ON, (@<chanlist>)

Parameter:

<value>:0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]CURRent:SLEW:COUPle? (@<chanlist>)

Return Parameters:

0 or 1

Example:

CURR:SLEW:COUP ON, (@1) //Enable tracking state of channel 1 current slope

CURR:SLEW:COUP? (@1) //Queries the tracking status of channel 1 current slope

[SOURce:]CURRent:SLEW[:POSitive]:MAXimum 0|OFF|1|ON, (@<chanlist>)

[SOURce:]CURRent:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Enable or disable current ramp maximum.

Command Syntax:

[SOURce:]CURRent:SLEW[:POSitive]:MAXimum 0|OFF|1|ON, (@<chanlist>)

Parameter:

<value>:0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]CURRent:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Return Parameters:

0 or 1

Example:

CURR:SLEW:MAX ON, (@1) //Enable channel 1 current rise slope max.

CURR:SLEW:MAX? (@1) //Query channel 1 current rise slope maximum value status

[SOURce:]CURRent:SLEW:NEGative:MAXimum 0|OFF|1|ON,(@<chanlist>)

[SOURce:]CURRent:SLEW:NEGative:MAXimum? (@<chanlist>)

Enable or disable current ramp down maximum.

Command Syntax:

[SOURce:]CURRent:SLEW:NEGative:MAXimum 0|OFF|1|ON, (@<chanlist>)

Parameter:

MAXimum|MINimum|DEFault

RST Value:

0

Query Syntax:

[SOURce:]CURRent:SLEW:NEGative:MAXimum? (@<chanlist>)

Return Parameters:

<Maximum value of current descent slope>

Example:

CURR:SLEW:NEG:MAX ON, (@1) //Enable channel 1 current down slope maximum value

CURR:SLEW:NEG:MAX? (@1) //Query channel 1 current descent slope maximum value status

9.4 Power Commands

[SOURce:]POWer:LIMit[:POSitive][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]POWer:LIMit[:POSitive][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Set and query the power positive limit value. (in W)

If the [SOURce:]POWer:LIMit:COUPle instruction setting status is ON, this instruction sets the positive power limit value and also sets the negative power

limit value synchronously.

Command Syntax:

[SOURce:]POWer:LIMit[:POSitive][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

maximum

Query Syntax:

[SOURce:]POWer:LIMit[:POSitive][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Power positive limit value>

Example:

POW:LIM 75,(@1) //Set channel 1 power upper limit value to 75W

POW:LIM? (@1) //Query channel 1 power upper limit value

[SOURce:]POWer:LIMit:NEGative[:IMMediate][:AMPLitude]<value>,(@<chanlist>)

[SOURce:]POWer:LIMit:NEGative[:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Set and query the negative limit value of power. (in W)

If the [SOURce:]POWer:LIMit:COUPle instruction setting status is ON, this instruction sets the negative power limit value and also sets the positive power limit value synchronously.

Command Syntax:

[SOURce:]POWer:LIMit:NEGative[:IMMediate][:AMPLitude]<value>,(@<chanlist>)

Parameter:

<value>:minimum - maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

minimum

Query Syntax:

[SOURce:]POWer:LIMit:NEGative[:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Negative limit value for power>

Example:

POW:LIM:NEG 20,(@1) //Set channel 1 power negative limit value to 20W

POW:LIM:NEG? (@1) //Query channel 1 power negative limit value

[SOURce:]POWer:LIMit:COUPle 0|OFF|1|ON,(@<chanlist>)

[SOURce:]POWer:LIMit:COUPle? (@<chanlist>)

This command is used to set or query the state of positive and negative power limit value coupling function.

Command Syntax:

[SOURce:]POWer:LIMit:COUPle 0|OFF|1|ON,(@<chanlist>)

Parameter:

0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

ON

Query Syntax:

[SOURce:]POWer:LIMit:COUPle? (@<chanlist>)

Return Parameters:

0 or 1

Example:

POW:LIM:COUP ON,(@1) //Enable channel 1 positive and negative power limit value coupling

POW:LIM:COUP? (@1) //Query channel 1 positive and negative power limit value coupling status

[SOURce:]POWer:PROTection:DELay[:TIME] <value>,(@<chanlist>)

[SOURce:]POWer:PROTection:DELay[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the delay time of overpower protection. (Unit is second)

Command Syntax:

[SOURce:]POWer:PROTection:DELay[:TIME] <value>,(@<chanlist>)

Parameter:

<TIME>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

Query Syntax:

[SOURce:]POWer:PROTection:DELay[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Delay time for overpower protection>

Example:

POW:PROT:DEL 10,(@1) //Set channel 1 overpower protection delay time to 10s

POW:PROT:DEL? (@1) //Query channel 1 overpower protection delay time

[SOURce:]POWer:PROTection:STATe 0|OFF|1|ON,(@<chanlist>)

[SOURce:]POWer:PROTection:STATe? (@<chanlist>)

This command is used to enable or disable the state of overpower protection.

Command Syntax:

[SOURce:]POWer:PROTection:STATe 0|OFF|1|ON,(@<chanlist>)

Parameter:

0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

OFF

Query Syntax:

[SOURce:]POWer:PROTection:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

POWer:PROTection:STATe ON,(@1) //Set channel 1 overpower protection

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status to ON

POWer:PROTection:STATe? (@1) //Query channel 1 over power protection status

[SOURce:]POWer:PROTection[:LEVel][:POSitive] <value>,(@<chanlist>)

[SOURce:]POWer:PROTection[:LEVel][:POSitive]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the positive protection value for overpower protection. (in W)

Command Syntax:

[SOURce:]POWer:PROTection[:LEVel][:POSitive] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

maximum

Query Syntax:

[SOURce:]POWer:PROTection[:LEVel][:POSitive]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<positive protection value for overpower protection>

Example:

POWer:PROTection 100, (@1) //Set channel 1 over power positive protection value to 100W

POWer:PROTection? (@1) //Query channel 1 over power positive protection value

[SOURce:]POWer:PROTection[:LEVel]:NEGative <value>,(@<chanlist>)

[SOURce:]POWer:PORTection[:LEVel]:NEGative? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the negative protection value for overpower protection. (in W)

Command Syntax:

[SOURce:]POWer:PROTection[:LEVel]:NEGative <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

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RST Value:

-maximum

Query Syntax:

[SOURce:]POWer:PORTection[:LEVel]:NEGative? [MINimum|MAXimum,](@<chanlist>)*RST-MAX

Return Parameters

<Negative protection value for power protection>

Example:

POW:PORT:NEG 10, (@1) //Set channel 1 overpower protection negative value to 10W

POW:PORT:NEG? (@1) //Query channel 1 over power protection negative value

9.5 Resistance Commands

[SOURce:]SINK:RESistance[:LEVel][:IMMediate][:AMPLitude] <value>, (@<chanlist>)

[SOURce:]SINK:RESistance[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Sets and queries the level of the SINK resistor in CC mode.

Command Syntax:

[SOURce:]SINK:RESistance[:LEVel][:IMMediate][:AMPLitude] <value>, (@<chanlist>)

Parameter:

minimum-maximum|MAXimum|MINimum|DEFault

RST Value:

Default

Query Syntax:

[SOURce:]SINK:RESistance[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters

<SINK resistance value>

Example

SINK:RESistance 50, (@1) //Sets the level of the SINK resistor to 50Ω

SINK:RESistance? (@1) //Query the level of the SINK resistor

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[SOURce:]SINK:RESistance:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]SINK:RESistance:STATe? (@<chanlist>)

Sets and queries the status of the SINK resistor in CC priority mode.

Command Syntax:

[SOURce:]SINK:RESistance:STATe <0|OFF|1|ON>,(@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

0

Query Syntax:

[SOURce:]SINK:RESistance:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

```
SINK:RESistance:STATe ON, (@1)      //Enable SINK resistor state
SINK:RESistance:STATe? (@1)         //Query the SINK resistor status
```

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Chapter10 DC Power Commands (SOURce Subsystem)

This chapter describes the unipolar power module commands.

10.1 Function Commands

[SOURce:]FUNCtion <CURRent|VOLTage>,(@<chanlist>)

[SOURce:]FUNCtion? (@<chanlist>)

This command is used to set and query the priority of the function.

Command Syntax:

[SOURce:]FUNCtion CURRent|VOLTage,(@<chanlist>)

Parameter:

<mode>: CURRent|VOLTage

Query Syntax:

[SOURce:]FUNCtion? (@<chanlist>)

RST Value:

VOLTage

Return Parameters:

CURR or VOLT

Example:

FUNC CURR,(@1) //Set Channel 1 function priority to current priority

FUNC? (@1) //Queries the Channel 1 function priority.

10.2 Voltage Commands

[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the immediate value of voltage. (in V)

Command Syntax:

[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

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Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

Query Syntax:

[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

RST Value:

Default

Return Parameters:

<immediate value of voltage>

Example:

VOLT 100, (@1) //Set channel 1 voltage value to 100V

VOLT? (@1) //Query the current voltage value of channel 1

[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude] <value>,(@<chanlist>)

[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the value when the voltage is triggered. (Unit is V)

Command Syntax:

[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

None

Query Syntax:

[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Trigger value of voltage>

Example:

VOLT:TRIG 50, (@1) //Setting the voltage trigger value for channel 1 to 50V

VOLT? (@1) //Query the voltage trigger value of channel 1

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[SOURce:]VOLTage:LIMit:HIGH[:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]VOLTage:LIMit:HIGH[:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Sets or queries the upper voltage limit value for current priority mode. (Unit is V)

Command Syntax:

VOLTage:LIMit:HIGH[:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:LIMit:HIGH[:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<upper limit value of voltage>

Example:

VOLT:LIMit:HIGH 50,(@1) //Set the upper voltage limit value for channel 1 current priority mode to 50V

VOLT:LIMit:HIGH? (@1) //Query the upper voltage limit value of channel 1 current priority mode

[SOURce:]VOLTage:LIMit:LOW[:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]VOLTage:LIMit:LOW[:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Sets or queries the voltage lower limit value in current priority mode, this command is only required when the power supply module is bipolar. (Unit is V)

Command Syntax:

[SOURce:]VOLTage:LIMit:LOW[:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

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RST Value:

0

Query Syntax:

[SOURce:]VOLTage:LIMit:LOW[:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Lower limit value of voltage>

Example:

VOLT:LIM:LOW 10,(@1) //Setting the lower voltage limit value in channel 1 current priority mode

VOLT:LIM:LOW? (@1) //Query the voltage lower limit value in channel 1 current priority mode

[SOURce:]VOLTage:PROTection[:LOCal][:POSitive][:LEVel] <value>,(@<chanlist>)

[SOURce:]VOLTage:PROTection[:LOCal][:POSitive][:LEVel]? [MINimum|MAXimum,](@<chanlist>)

The command sets or queries the positive voltage protection value of overvoltage protection. (Unit is V)

Command Syntax:

[SOURce:]VOLTage:PROTection[:LOCal][:POSitive][:LEVel] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

maximum

Query Syntax:

[SOURce:]VOLTage:PROTection[:LOCal][:POSitive][:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Protection value for positive voltage>

Example

VOLT:PROT 300, (@1) //Setting the positive voltage protection value for channel 1 overvoltage protection

VOLT:PROT? (@1) //Query the positive voltage protection value of

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channel 1 overvoltage protection

[SOURce:]VOLTage:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:PROTection:STATe? (@<chanlist>)

The commands set and query the enable state of overvoltage protection. (Unit is V)

Command Syntax:

[SOURce:]VOLTage:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

Parameter:

<0|OFF|1|ON>:0-OFF|1-ON

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:PROTection:STATe? (@<chanlist>)

Return Parameters:

<Protection enable state of voltage>

Example:

VOLT:PROT:STAT ON, (@1) //Enable channel 1 overvoltage protection state

VOLT:PROT:STAT? (@1) //Queries the enable state of channel 1 overvoltage protection.

[SOURce:]VOLTage:PROTection:DELay[:TIME] <value>,(@<chanlist>)

[SOURce:]VOLTage:PROTection:DELay[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

Setting and querying the delay time for overvoltage protection, (in seconds)

Command Syntax:

[SOURce:]VOLTage:PROTection:DELay[:TIME] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

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Query Syntax:

[SOURce:]VOLTage:PROTection:DELay[:TIME]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Delay value for voltage protection>

Example:

VOLT:PROT:DEL 30, (@1) //Set the delay time for channel 1 overvoltage protection to 30s.

VOLT:PROT:DEL? (@1) //Query the delay time for channel 1 overvoltage protection.

[SOURce:]VOLTage:UNDer:PROTection[:POSitive][:LEVel] <value>,(@<chanlist>)

[SOURce:]VOLTage:UNDer:PROTection[:POSitive][:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Set and query the positive voltage protection value for undervoltage protection. (Unit is V)

Command Syntax:

[SOURce:]VOLTage:UNDer:PROTection[:POSitive][:LEVel] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

maximum

Query Syntax:

[SOURce:]VOLTage:UNDer:PROTection[:POSitive][:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Positive voltage protection value for undervoltage>

Example:

VOLT:UND:PROT 10, (@1) //Set the positive voltage protection value for channel 1 undervoltage protection to 10V

VOLT:UND:PROT? (@1) //Queries the positive voltage protection value for channel 1 undervoltage protection.

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[SOURce:]VOLTage:UNDer:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:UNDer:PROTection:STATe? (@<chanlist>)

Sets and queries the enable state of undervoltage protection.

Command Syntax:

[SOURce:]VOLTage:UNDer:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

Parameter:

<0|OFF|1|ON>:0-OFF|1-ON

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:UNDer:PROTection:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

VOLT:UND:PROT:STAT ON,(@1) //Enable channel 1 undervoltage protection.

VOLT:UND:PROT:STAT? (@1) //Queries the enable state of channel 1 undervoltage protection.

[SOURce:]VOLTage:UNDer:PROTection:DELay[:TIME] <value>,(@<chanlist>)

[SOURce:]VOLTage:UNDer:PROTection:DELay[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

Sets the delay time for undervoltage protection. (in seconds)

Command Syntax:

[SOURce:]VOLTage:UNDer:PROTection:DELay[:TIME] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

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Query Syntax:

[SOURce:]VOLTage:UNDer:PROTection:DELay[:TIME]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Delay time for undervoltage protection>

Example:

VOLT:UND:PROT:DEL 10, (@1) //Set channel 1 undervoltage protection delay time to 10s

VOLT:UND:PROT:DEL? (@1) //Query channel 1 undervoltage protection delay time

[SOURce:]VOLTage:UNDer:PROTection:WARM[:TIME] <value>,(@<chanlist>)

[SOURce:]VOLTage:UNDer:PROTection:WARM[:TIME]? [MINimum|MAXimum,](@<chanlist>)

Set and query the warm-up time for undervoltage protection. (in seconds)

Command Syntax:

[SOURce:]VOLTage:UNDer:PROTection:WARM[:TIME] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

Query Syntax:

[SOURce:]VOLTage:UNDer:PROTection:WARM[:TIME]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Warm-up time for undervoltage protection>

Example:

VOLT:UND:PROT:WARM 10, (@1) //Set the undervoltage protection warm time to 10s

VOLT:UND:PROT:WARM? (@1) //Query undervoltage protection warm time

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[SOURce:]VOLTage:RESistance[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]VOLTage:RESistance[:LEVel][:IMMediate][:AMPLitude]?(@<chanlist>)

Sets and queries the internal resistance value of the power supply in voltage priority mode. (Unit is Ω)

Valid only when the internal resistance function is enabled.

Command Syntax:

[SOURce:]VOLTage:RESistance[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:MAXimum|MINimum|DEFault

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:RESistance[:LEVel][:IMMediate][:AMPLitude]?(@<chanlist>)

Return Parameters:

<internal resistance value>

Example:

VOLT:RES 0.1,(@1)	//Set the internal resistance of channel 1 to 0.1Ω.
VOLT:RES? (@1)	//Query the internal resistance of channel 1

[SOURce:]VOLTage:RESistance:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:RESistance:STATe? (@<chanlist>)

Sets and queries the status of the internal resistance function of the power supply in voltage priority mode.

Command Syntax:

[SOURce:]VOLTage:RESistance:STATe <0|OFF|1|ON>,(@<chanlist>)

Parameter:

<0|OFF|1|ON>:0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:RESistance:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

VOLT:RES:STAT 0,(@1) //Set channel 1 internal resistance function status to off

VOLT:RES:STAT? (@1) //Query the current internal resistance function status of channel 1

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate] <value>,(@<chanlist>)

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Set and query the voltage rise slope. (Unit is V/ms)

If the [SOURce:]VOLTage:SLEW:COUPle instruction setting state is ON, this instruction sets the voltage upward ramp downward while simultaneously setting the voltage downward ramp downward.

Command Syntax:

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate] <value>,(@<chanlist>)

Parameter:

<value>:minimum-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Rise slope of voltage>

Example:

VOLT:SLEW 10, (@1) //Set channel 1 voltage rise slope to 10V/ms

VOLT:SLEW? (@1) //Query channel 1 voltage rise slope

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[SOURce:]VOLTage:SLEW:NEGative[:IMMediate] <value>,(@<chanlist>)

[SOURce:]VOLTage:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,](@<chanlist>)

Set and query the voltage drop slope. (Unit is V/ms)

If the [SOURce:]VOLTage:SLEW:COUPle instruction setting state is ON, this instruction sets the voltage falling slope while simultaneously setting the voltage rising slope.

Command Syntax:

[SOURce:]VOLTage:SLEW:NEGative[:IMMediate] <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]VOLTage:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Slope of voltage drop>

Example:

VOLT:SLEW:NEG 10,(@1) //Set channel 1 voltage drop slope to 10 V/ms

VOLT:SLEW:NEG? (@1) //Query channel 1 voltage drop slope

[SOURce:]VOLTage:SLEW:COUPle <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:SLEW:COUPle? (@<chanlist>)

Set and query the tracking status of the voltage slope.

Command Syntax:

[SOURce:]VOLTage:SLEW:COUPle <0|OFF|1|ON>,(@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

1

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Query Syntax:

[SOURce:]VOLTage:SLEW:COUPle? (@<chanlist>)

Return Parameters:

0 or 1

Example:

VOLT:SLEW:COUP ON, (@1) //Enable channel 1 voltage slope tracking state

VOLT:SLEW:COUP? (@1) //Query channel 1 voltage slope tracking status

[SOURce:]VOLTage:SLEW[:POSitive]:MAXimum <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Enable or disable voltage rise slope maximum.

Command Syntax:

[SOURce:]VOLTage:SLEW[:POSitive]:MAXimum <0|OFF|1|ON>,(@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Return Parameters:

0 or 1

Example:

VOLT:SLEW:MAX ON, (@1) //Enable Channel 1 Voltage Rise Slope Maximum Function State

VOLT:SLEW:MAX? (@1) //Query Channel 1 Voltage Rise Slope Maximum Function Status

[SOURce:]VOLTage:SLEW:NEGative:MAXimum <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:SLEW:NEGative:MAXimum? (@<chanlist>)

Enable or disable voltage drop slope maximum.

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Command Syntax:

[SOURce:]VOLTage:SLEW:NEGative:MAXimum <0|OFF|1|ON>,(@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:SLEW:NEGative:MAXimum? (@<chanlist>)

Return Parameters:

0 or 1

Example:

VOLT:SLEW:NEG:MAX ON,(@1) //Enable channel 1 voltage drop slope maximum state

VOLT:SLEW:NEG:MAX? (@1) //Query channel 1 voltage drop slope maximum value status

10.3 Current Commands

[SOURce:]CURRent[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]CURRent[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Sets the immediate value of the current. (in A)

Command Syntax:

[SOURce:]CURRent[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

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Return Parameters:

<immediate value of current>

Example:

CURR 3,(@1) //Set channel 1 current to 3A immediately

CURR? (@1) //Query channel 1 current immediate value

[SOURce:]CURRent[:LEVel]:TRIGgered[:AMPLitude] <value>,(@<chanlist>)

[SOURce:]CURRent[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Sets or queries the trigger value of the current. (Unit is A)

Command Syntax:

[SOURce:]CURRent[:LEVel]:TRIGgered[:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

None

Query Syntax:

[SOURce:]CURRent[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Trigger value of voltage>

Example:

CURR:TRIG 20, (@1) //Set channel 1 current trigger value to 20A

CURR:TRIG ? (@1) //Query channel 1 current trigger value

[SOURce:]CURRent:LIMit[:POSitive][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]CURRent:LIMit[:POSitive][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Sets the current limit value for voltage priority mode. (Unit is A)

Command Syntax:

[SOURce:]CURRent:LIMit[:POSitive][:IMMediate][:AMPLitude]

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<value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:LIMit[:POSitive][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Positive limiting value of current>

Example:

CURR:LIM 50,(@1) //Set channel 1 positive current limit value to 50A

CURR:LIM? (@1) //Query channel 1 positive current limit value

[SOURce:]CURRent:PROTection[:LEVel][:POSitive] <level>,(@<chanlist>)

[SOURce:]CURRent:PROTection[:LEVel][:POSitive]? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the positive current protection value of overcurrent protection. (Unit is A)

Command Syntax:

[SOURce:]CURRent:PROTection[:LEVel][:POSitive] <level>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:PROTection[:LEVel][:POSitive]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Positive current protection value for overcurrent protection>

Example:

CURR:PROT 30, (@1) //Setting the positive current protection value for

channel 1 overcurrent protection to 30A

CURR:PROT? (@1) //Query the positive current protection value of channel 1 overcurrent protection

[SOURce:]CURRent:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]CURRent:PROTection:STATe? (@<chanlist>)

The commands set and query the enable state of overcurrent protection.

Command Syntax:

[SOURce:]CURRent:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

Parameter:

<value>:<0|OFF|1|ON>

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]CURRent:PROTection:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

CURR:PROT:STAT ON, (@1) //Setting the channel 1 current protection function to the ON state

CURR:PROT:STAT? (@1) //Query channel 1 current protection function status

[SOURce:]CURRent:PROTection:DELay[:TIME] <value>,(@<chanlist>)

[SOURce:]CURRent:PROTection:DELay[:TIME]? [MINimum|MAXimum,](@<chanlist>)

This command is used to set or query the delay time of overcurrent protection. (Unit is in seconds)

Command Syntax:

[SOURce:]CURRent:PROTection:DELay[:TIME] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

Query Syntax:

[SOURce:]CURRent:PROTection:DELay[:TIME]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Delay time for overcurrent protection>

Example:

CURR:PROT:DEL 10,(@1) //Set the delay time for channel 1 overcurrent protection to 10 seconds

CURR:PROT:DEL? (@1) //Query delay time for channel 1 overcurrent protection

[SOURce:]CURRent:UNDer:PROTection[:LEVel][:POSitive] <level>,(@<chanlist>)

[SOURce:]CURRent:UNDer:PROTection[:LEVel][:POSitive] [MINimum|MAXimum,](@<chanlist>)

Set and query the positive current protection value for undercurrent protection. (Unit is A)

Command Syntax:

[SOURce:]CURRent:UNDer:PROTection[:LEVel][:POSitive] <level>,(@<chanlist>)

Parameter:

<value>:minimum-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

maximum

Query Syntax:

[SOURce:]CURRent:UNDer:PROTection[:LEVel][:POSitive] [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Positive current protection value for undercurrent protection>

Example:

VOLT:UND:PROT 10, (@1) //Set channel 1 undercurrent protection positive current value to 10A

VOLT:UND:PROT? (@1) //Query channel 1 undercurrent protection positive current value

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[SOURce:]CURRent:UNDer:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]CURRent:UNDer:PROTection:STATe? (@<chanlist>)

Sets and queries the enable state of undercurrent protection.

Command Syntax:

[SOURce:]CURRent:UNDer:PROTection:STATe 0|OFF|1|ON,(@<chanlist>)

Parameter:

<value>:<0|OFF|1|ON>

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]CURRent:UNDer:PROTection:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

CURR:UND:PROT:STATe ON,(@1) //ONSet the enable state of channel 1 undercurrent protection to ON

CURR:UND:PROT:STATe? (@1) //Query the enable status of channel 1 undercurrent protection

[SOURce:]CURRent:UNDer:PROTection:DELay[:TIME] <value>,(@<chanlist>)

[SOURce:]CURRent:UNDer:PROTection:DELay[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

Sets or queries the delay time for undercurrent protection. (Unit is in seconds)

Command Syntax:

[SOURce:]CURRent:UNDer:PROTection:DELay[:TIME] <level>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

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Query Syntax:

[SOURce:]CURRent:UNDer:PROTection:DELay[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<<Delay time for undercurrent protection>

Example:

CURR:UND:PROT:DEL 10,(@1) //Set the delay time for channel 1 undercurrent protection to 10 seconds

CURR:UND:PROT:DEL? (@1) //Query delay time for channel 1 undercurrent protection

[SOURce:]CURRent:UNDer:PROTection:WARM[:TIME] <value>,(@<chanlist>)

[SOURce:]CURRent:UNDer:PROTection:WARM[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

Sets the warm-up time for undercurrent protection. (in seconds)

Command Syntax:

[SOURce:]CURRent:UNDer:PROTection:WARM[:TIME] <level>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

Query Syntax:

[SOURce:]CURRent:UNDer:PROTection:WARM[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Warm-up time for undercurrent protection>

Example:

CURR:UND:PROT:WARM 10,(@1) //Set the warm-up time for channel 1 undercurrent protection to 10 seconds

CURR:UND:PROT:WARM? (@1) //Query the warm-up time for channel 1 undercurrent protection

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[SOURce:]CURRent:SLEW[:POSitive][:IMMediate] <value>,(@<chanlist>)

[SOURce:]CURRent:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,](@<chanlist>)

Set and query the current rise slope. (Unit is A/ms)

If the [SOURce:]CURRent:SLEW:COUPle instruction setting state is ON, this instruction sets the current up and down slopes simultaneously with the voltage down slope.

Command Syntax:

[SOURce:]CURRent:SLEW[:POSitive][:IMMediate] <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Ramp rate of current rise>

Example:

CURR:SLEW 10, (@1)	//Set channel 1 current rise slope to 10 A/ms
CURR:SLEW? (@1)	//Query channel 1 current rise slope

[SOURce:]CURRent:SLEW:NEGative[:IMMediate] <value>,(@<chanlist>)

[SOURce:]CURRent:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Setting and querying the current drop slope. (Unit is A/ms)

If the [SOURce:]CURRent:SLEW:COUPle instruction setting state is ON, this instruction sets the current down and up slope simultaneously with the voltage up slope.

Command Syntax:

[SOURce:]CURRent:SLEW:NEGative[:IMMediate] <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

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RST Value:

Default

Query Syntax:

[SOURce:]CURRent:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Slope of current drop>

Example::

CURR:SLEW:NEG 10, (@1) //Set channel 1 current down slope to 10 A/ms

CURR:SLEW:NEG? (@1) //Query channel 1 current descent slope

[SOURce:]CURRent:SLEW:COUPle 0|OFF|1|ON, (@<chanlist>)

[SOURce:]CURRent:SLEW:COUPle? (@<chanlist>)

Set and query the tracking status of the current slope.

Command Syntax:

[SOURce:]CURRent:SLEW:COUPle 0|OFF|1|ON, (@<chanlist>)

Parameter:

<value>:0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]CURRent:SLEW:COUPle? (@<chanlist>)

Return Parameters:

0 or 1

Example

CURR:SLEW:COUP ON, (@1) //Set the tracking state of channel 1 current slope to ON

CURR:SLEW:COUP? (@1) //Queries the tracking status of channel 1 current slope

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[SOURce:]CURRent:SLEW[:POSitive]:MAXimum 0|OFF|1|ON, (@<chanlist>)

[SOURce:]CURRent:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Enable or disable current ramp maximum.

Command Syntax:

[SOURce:]CURRent:SLEW[:POSitive]:MAXimum 0|OFF|1|ON, (@<chanlist>)

Parameter:

<value>:0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]CURRent:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Return Parameters:

0 or 1

Example

CURR:SLEW:MAX ON, (@1) //Enable channel 1 current rise slope max.

CURR:SLEW:MAX? (@1) //Query channel 1 current rise slope maximum value status

[SOURce:]CURRent:SLEW:NEGative:MAXimum 0|OFF|1|ON,(@<chanlist>)

[SOURce:]CURRent:SLEW:NEGative:MAXimum? (@<chanlist>)

Enable or disable current ramp down maximum.

Command Syntax:

[SOURce:]CURRent:SLEW:NEGative:MAXimum 0|OFF|1|ON, (@<chanlist>)

Parameter:

MAXimum|MINimum|DEFault

RST Value:

0

Query Syntax:

[SOURce:]CURRent:SLEW:NEGative:MAXimum? (@<chanlist>)

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Return Parameters:

<Maximum value of current descent slope>

Example:

CURR:SLEW:NEG:MAX ON, (@1) //Enable channel 1 current down slope maximum value

CURR:SLEW:NEG:MAX? (@1) //Query channel 1 current descent slope maximum value status

10.4 Power Commands

[SOURce:]POWer:LIMit[:POSitive][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]POWer:LIMit[:POSitive][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Set and query the power positive limit value. (in W)

If the [SOURce:]POWer:LIMit:COUPle instruction setting status is ON, this instruction sets the positive power limit value and also sets the negative power limit value synchronously.

Command Syntax:

[SOURce:]POWer:LIMit[:POSitive][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

maximum

Query Syntax:

[SOURce:]POWer:LIMit[:POSitive][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Power positive limit value>

Example:

POW:LIM 75,(@1) //Set channel 1 power upper limit value to 75W

POW:LIM? (@1) //Query channel 1 power upper limit value

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[SOURce:]POWer:PROTection:DELay[:TIME] <value>,(@<chanlist>)

[SOURce:]POWer:PROTection:DELay[:TIME]? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the delay time of overpower protection. (Unit is second)

Command Syntax:

[SOURce:]POWer:PROTection:DELay[:TIME] <value>,(@<chanlist>)

Parameter:

<TIME>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

Query Syntax:

[SOURce:]POWer:PROTection:DELay[:TIME]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Delay time for overpower protection>

Example:

POW:PROT:DEL 10,(@1) //Set channel 1 overpower protection delay time to 10s

POW:PROT:DEL? (@1) //Query channel 1 overpower protection delay time

[SOURce:]POWer:PROTection:STATe 0|OFF|1|ON,(@<chanlist>)

[SOURce:]POWer:PROTection:STATe? (@<chanlist>)

This command is used to enable or disable the state of overpower protection.

Command Syntax:

[SOURce:]POWer:PROTection:STATe 0|OFF|1|ON,(@<chanlist>)

Parameter:

0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

OFF

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Query Syntax:

[SOURce:]POWer:PROTection:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

POWer:PROTection:STATe ON,(@1) //Set channel 1 overpower protection status to ON

POWer:PROTection:STATe? (@1) //Query channel 1 over power protection status

[SOURce:]POWer:PROTection[:LEVel][:POSitive] <value>,(@<chanlist>)

[SOURce:]POWer:PROTection[:LEVel][:POSitive]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the positive protection value for overpower protection. (in W)

Command Syntax:

[SOURce:]POWer:PROTection[:LEVel][:POSitive] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

maximum

Query Syntax:

[SOURce:]POWer:PROTection[:LEVel][:POSitive]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<positive protection value for overpower protection>

Example:

POWer:PROTection 100, (@1) //Set channel 1 over power positive protection value to 100W

POWer:PROTection? (@1) //Query Channel 1 Over Power Positive Protection

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Chapter11 Load Commands (SOURce Subsystem)

11.1 Function Commands

[SOURce:]FUNCtion CURRent|VOLTage|POWer|RESistance|CC+CV|CR+CV|CP+CV|CC+CR|AUTO|BSIM, (@<chanlist>)

[SOURce:]FUNCtion? (@<chanlist>)

This command is used to set and query the priority of the function.

Command Syntax:

[SOURce:]FUNCtion CURRent|VOLTage|POWer|RESistance|CC+CV|CR+CV|CP+CV|CC+CR|AUTO|BSIM, (@<chanlist>)

Parameter:

CURRent|VOLTage|POWer|RESistance|CC+CV|CR+CV|CP+CV|CC+CR|AUTO|BSIM

Query Syntax:

[SOURce:]FUNCtion? (@<chanlist>)

Initial value:

CURRent

Return Parameters:

CURR,VOLT,POW,RES, CC+CV,CR+CV,CP+CV,CC+CR,AUTO or BSIM

Example:

FUNC CURR,(@1) //Set Channel 1 function priority to current priority

FUNC? (@1) //Queries the Channel 1 function priority.

11.2 Voltage Commands

[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the immediate value of voltage. (Unit is V)

Command Syntax:

[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

Query Syntax:

[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

RST Value:

Default

Return Parameters:

<immediate value of voltage>

Example:

VOLT 100, (@1) //Set channel 1 voltage value to 100V

VOLT? (@1) //Query the current voltage value of channel 1

[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude] <value>,(@<chanlist>)

[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the value when the voltage is triggered. (Unit is V)

Command Syntax:

[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

None

Query Syntax:

[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Trigger value of voltage>

Example:

VOLT:TRIG 50, (@1) //Setting the voltage trigger value for channel 1 to

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50V

VOLT? (@1) //Query the voltage trigger value of channel 1

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate] <value>,(@<chanlist>)

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Set and query the voltage rise slope. (Unit is V/ms)

If the [SOURce:]VOLTage:SLEW:COUPle Commands setting state is ON, this Commands sets the voltage upward ramp downward while simultaneously setting the voltage downward ramp downward.

Command Syntax:

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate] <value>,(@<chanlist>)

Parameter:

<value>:minimum-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Rise slope of voltage>

Example:

VOLT:SLEW 10, (@1) //Set channel 1 voltage rise slope to 10V/ms

VOLT:SLEW? (@1) //Query channel 1 voltage rise slope

[SOURce:]VOLTage:SLEW:NEGative[:IMMediate] <value>,(@<chanlist>)

[SOURce:]VOLTage:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,](@<chanlist>)

Set and query the voltage drop slope. (Unit is V/ms)

If the [SOURce:]VOLTage:SLEW:COUPle Commands setting state is ON, this Commands sets the voltage down slope while simultaneously setting the voltage up slope.

Command Syntax:

[SOURce:]VOLTage:SLEW:NEGative[:IMMediate] <value>,(@<chanlist>)

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Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]VOLTage:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Slope of voltage drop>

Example:

VOLT:SLEW:NEG 10,(@1) //Set channel 1 voltage drop slope to 10 V/ms

VOLT:SLEW:NEG? (@1) //Query channel 1 voltage drop slope

[SOURce:]VOLTage:SLEW:COUPle 0|OFF|1|ON,(@<chanlist>)

[SOURce:]VOLTage:SLEW:COUPle? (@<chanlist>)

Set and query the tracking status of the voltage slope.

Command Syntax:

[SOURce:]VOLTage:SLEW:COUPle 0|OFF|1|ON,(@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

1

Query Syntax:

[SOURce:]VOLTage:SLEW:COUPle? (@<chanlist>)

Return Parameters:

0 or 1

Example:

VOLT:SLEW:COUP ON, (@1) //Enable channel 1 voltage slope tracking state

VOLT:SLEW:COUP? (@1) //Query channel 1 voltage slope tracking status

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[SOURce:]VOLTage:SLEW[:POSitive]:MAXimum <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Enable or disable voltage rise slope maximum.

Command Syntax:

[SOURce:]VOLTage:SLEW[:POSitive]:MAXimum <0|OFF|1|ON>,(@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Return Parameters:

0 or 1

Example:

VOLT:SLEW:MAX ON, (@1) //Enable Channel 1 Voltage Rise Slope Maximum Function State

VOLT:SLEW:MAX? (@1) //Query Channel 1 Voltage Rise Slope Maximum Function Status

[SOURce:]VOLTage:SLEW:NEGative:MAXimum <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:SLEW:NEGative:MAXimum? (@<chanlist>)

Enable or disable voltage drop slope maximum.

Command Syntax:

[SOURce:]VOLTage:SLEW:NEGative:MAXimum <0|OFF|1|ON>,(@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:SLEW:NEGative:MAXimum? (@<chanlist>)

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Return Parameters:

0 or 1

Example:

VOLT:SLEW:NEG:MAX ON,(@1) //Enable channel 1 voltage drop slope maximum state

VOLT:SLEW:NEG:MAX? (@1) //Query channel 1 voltage drop slope maximum value status

[SOURce:]VOLTage:INHibit:VON[:LEVel] <value>, (@<chanlist>)

[SOURce:]VOLTage:INHibit:VON[:LEVel]? (@<chanlist>)

Setting and querying the load mode, the level at which the load is started to be carried. (in V)

Command Syntax:

[SOURce:]VOLTage:INHibit:VON[:LEVel] <value>, (@<chanlist>)

Parameter:

minimum-maximum|MINimum|MAXimum

RST Value

0.1

Query Syntax:

[SOURce:]VOLTage:INHibit:VON[:LEVel]? (@<chanlist>)

Return Parameters:

<Voltage value of transient>

Example:

VOLT:INH:VON 15,(@1) //Set the level at which channel 1 begins to carry load to 15V

VOLT:INH:VON? (@1) //Query the level at which channel 1 starts to carry load

[SOURce:]VOLTage:INHibit:VOFF[:LEVel] <value>, (@<chanlist>)

[SOURce:]VOLTage:INHibit:VOFF[:LEVel]? (@<chanlist>)

Sets and queries the load mode, the level at which unloading begins. (in V)

Command Syntax:

[SOURce:]VOLTage:INHibit:VOFF[:LEVel] <value>, (@<chanlist>)

Parameter:

minimum-maximum|MINimum|MAXimum

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RST Value

0

Query Syntax:

[SOURce:]VOLTage:INHibit:VOFF[:LEVel]? (@<chanlist>)

Return Parameters

<Voltage value of transient>

Example:

VOLT:INH:VOFF 15,(@1) //Setting the voltage level for channel 1 unloading 15V

VOLT:INH:VOFF? (@1) //Query the unloaded voltage level of channel 1

[SOURce:]VOLTage:INHibit:VON:MODE <LATChing|LIVE|OFF>,(@<chanlist>)

[SOURce:]VOLTage:INHibit:VON:MODE? (@chanlist)

Sets and queries the mode of undervoltage suppression.

Command Syntax:

[SOURce:]VOLTage:INHibit:VON:MODE <LATChing|LIVE|OFF>,(@<chanlist>)

Parameter:

LATChing|LIVE|OFF

RST Value

LATChing

Query Syntax:

[SOURce:]VOLTage:INHibit:VON:MODE? (@chanlist)

Return Parameters

LATC|LIVE|OFF

Example:

VOLTage:INH:VON:MODE OFF,(@1) //Set the mode of channel 1 undervoltage suppression to OFF

VOLTage:INH:VON:MODE? (@1) //Query the mode of channel 1 undervoltage suppression

[SOURce:]VOLTage:TLEVel [MINimum|MAXimum,] (@<chanlist>)

[SOURce:]VOLTage:TLEVel? [MINimum|MAXimum,] (@<chanlist>)

Sets and queries the voltage level during transients. (in V)

Command Syntax:

[SOURce:]VOLTage:TLEVel [MINimum|MAXimum,] (@<chanlist>)

Parameter:

minimum-maximum|MINimum|MAXimum

RST Value

None

Query Syntax:

[SOURce:]VOLTage:TLEVel? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters

<Voltage value of transient>

Example:

VOLTage:TLEVel 15,(@1) //Setting the voltage level at channel 1 transient 15V

VOLTage:TLEVel? (@1) //Query the voltage level during transient of channel 1

11.3 Current Commands

[SOURce:]CURRent[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]CURRent[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Sets the immediate value of the current. (in A)

Command Syntax:

[SOURce:]CURRent[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

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Return Parameters:

<immediate value of current>

Example

CURR 3,(@1) //Set channel 1 current to 3A immediately

CURR? (@1) //Query channel 1 current immediate value

[SOURce:]CURRent[:LEVel]:TRIGgered[:AMPLitude] <value>,(@<chanlist>)

[SOURce:]CURRent[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Sets or queries the trigger value of the current. (Unit is A)

Command Syntax:

[SOURce:]CURRent[:LEVel]:TRIGgered[:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

None

Query Syntax:

[SOURce:]CURRent[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Trigger value of voltage>

Example:

CURR:TRIG 20, (@1) //Set channel 1 current trigger value to 20A

CURR:TRIG ? (@1) //Query channel 1 current trigger value

[SOURce:]CURRent:LIMit[:POSitive][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]CURRent:LIMit[:POSitive][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Sets the current limit value for voltage priority mode. (Unit is A)

Command Syntax:

[SOURce:]CURRent:LIMit[:POSitive][:IMMediate][:AMPLitude]

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<value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:LIMit[:POSitive][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Positive limiting value of current>

Example:

CURR:LIM 50,(@1) //Set channel 1 positive current limit value to 50A

CURR:LIM? (@1) //Query channel 1 positive current limit value

[SOURce:]CURRent:PROTection[:LEVel][:POSitive] <level>,(@<chanlist>)

[SOURce:]CURRent:PROTection[:LEVel][:POSitive]? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the positive current protection value of overcurrent protection. (Unit is A)

Command Syntax:

[SOURce:]CURRent:PROTection[:LEVel][:POSitive] <level>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:PROTection[:LEVel][:POSitive]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Positive current protection value for overcurrent protection>

Example:

CURR:PROT 30, (@1) //Setting the positive current protection value for

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channel 1 overcurrent protection to 30A

CURR:PROT? (@1) //Query the positive current protection value of channel 1 overcurrent protection

[SOURce:]CURRent:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]CURRent:PROTection:STATe? (@<chanlist>)

The commands set and query the enable state of overcurrent protection.

Command Syntax:

[SOURce:]CURRent:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

Parameter:

<value>:<0|OFF|1|ON>

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]CURRent:PROTection:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

CURR:PROT:STAT ON, (@1) //Setting the channel 1 current protection function to the ON state

CURR:PROT:STA? (@1) //Query channel 1 current protection function status

[SOURce:]CURRent:PROTection:DELay[:TIME] <value>,(@<chanlist>)

[SOURce:]CURRent:PROTection:DELay[:TIME]? [MINimum|MAXimum,](@<chanlist>)

This command is used to set or query the delay time of overcurrent protection. (Unit is in seconds)

Command Syntax:

[SOURce:]CURRent:PROTection:DELay[:TIME] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

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RST Value:

2

Query Syntax:

[SOURce:]CURRent:PROTection:DELay[:TIME]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Delay time for overcurrent protection>

Example:

CURR:PROT:DEL 10,(@1) //Set the delay time for channel 1 overcurrent protection to 10 seconds

CURR:PROT:DEL? (@1) //Query delay time for channel 1 overcurrent protection

[SOURce:]CURRent:SLEW[:POSitive][:IMMediate] <value>,(@<chanlist>)

[SOURce:]CURRent:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,](@<chanlist>)

Set and query the current rise slope. (Unit is A/ms)

If the [SOURce:]CURRent:SLEW:COUPle Commands setting state is ON, this Commands sets the current up and down slopes simultaneously with the voltage down slope.

Command Syntax:

[SOURce:]CURRent:SLEW[:POSitive][:IMMediate] <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Ramp rate of current rise

Example:

CURR:SLEW 10, (@1) //Set channel 1 current rise slope to 10 A/ms

CURR:SLEW? (@1) //Query channel 1 current rise slope

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[SOURce:]CURRent:SLEW:NEGative[:IMMediate] <value>,(@<chanlist>)

[SOURce:]CURRent:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Setting and querying the current drop slope. (Unit is A/ms)

If the [SOURce:]CURRent:SLEW:COUPle Commands setting state is ON, the Commands set the current downward and upward slopes simultaneously with the voltage upward slope.

Command Syntax:

[SOURce:]CURRent:SLEW:NEGative[:IMMediate] <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Slope of current drop>

Example:

CURR:SLEW:NEG 10, (@1) //Set channel 1 current down slope to 10 A/ms

CURR:SLEW:NEG? (@1) //Query channel 1 current descent slope

[SOURce:]CURRent:SLEW:COUPle <0|OFF|1|ON>, (@<chanlist>)

[SOURce:]CURRent:SLEW:COUPle? (@<chanlist>)

Set and query the tracking status of the current slope.

Command Syntax:

[SOURce:]CURRent:SLEW:COUPle 0|OFF|1|ON, (@<chanlist>)

Parameter:

<value>:0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]CURRent:SLEW:COUPle? (@<chanlist>)

Return Parameters:

0 or 1

Example:

CURR:SLEW:COUP ON, (@1) //Set the tracking state of channel 1 current slope to ON

CURR:SLEW:COUP? (@1) //Queries the tracking status of channel 1 current slope

[SOURce:]CURRent:SLEW[:POSitive]:MAXimum <0|OFF|1|ON>, (@<chanlist>)

[SOURce:]CURRent:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Enable or disable current ramp maximum.

Command Syntax:

[SOURce:]CURRent:SLEW[:POSitive]:MAXimum 0|OFF|1|ON, (@<chanlist>)

Parameter:

<value>:0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]CURRent:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Return Parameters:

0|1

Example:

CURR:SLEW:MAX ON, (@1) //Enable channel 1 current rise slope max.

CURR:SLEW:MAX? (@1) //Query channel 1 current rise slope maximum value status

[SOURce:]CURRent:SLEW:NEGative:MAXimum <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]CURRent:SLEW:NEGative:MAXimum? (@<chanlist>)

Enable or disable current ramp down maximum.

Command Syntax:

[SOURce:]CURRent:SLEW:NEGative:MAXimum <0|OFF|1|ON>, (@<chanlist>)

Parameter:

MAXimum|MINimum|DEFault

RST Value:

0

Query Syntax:

[SOURce:]CURRent:SLEW:NEGative:MAXimum? (@<chanlist>)

Return Parameters:

<Maximum value of current down slope>

Example:

CURR:SLEW:NEG:MAX ON, (@1) //Enable channel 1 current down slope maximum value

CURR:SLEW:NEG:MAX? (@1) //Query channel 1 current down slope maximum value status

[SOURce:]CURRent:TLEVel <value> (@<chanlist>)

[SOURce:]CURRent:TLEVel? [MINimum|MAXimum,] (@<chanlist>)

Sets and queries the current value during transients. (Unit is A)

Command Syntax:

[SOURce:]CURRent:TLEVel <value> (@<chanlist>)

Parameter:

minimum-maximum|MINimum|MAXimum

RST Value:

None

Query Syntax:

[SOURce:]CURRent:TLEVel? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Current value at transient>

Example:

CURRent:TLEVel 15,(@1) //Set current level at channel 1 transient to 15A

CURRent:TLEVel? (@1) //Query the current level during transient of channel 1

11.4 Power Commands

[SOURce:]POWer[:LEVel][:IMMediate][:AMPLitude] <value>, (@<chanlist>)

[SOURce:]POWer[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the immediate value of electric power. (Unit is W)

Command Syntax:

[SOURce:]POWer[:LEVel][:IMMediate][:AMPLitude] <value>, (@<chanlist>)

Parameter:

0-maximum|MINimum|MAXimum

RST Value:

0

Query Syntax:

[SOURce:]POWer[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Power Immediate>

Example:

POWer 100, (@1) //Set the power immediate value for channel 1 to 100W

POWer? (@1) //Query the power immediate value of channel 1

[SOURce:]POWer[:LEVel]:TRIGgered[:AMPLitude] <value>, (@<chanlist>)

[SOURce:]POWer[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the value of electrical power when triggered. (Unit is W)

Command Syntax:

[SOURce:]POWer[:LEVel]:TRIGgered[:AMPLitude] <value>, (@<chanlist>)

Parameter:

MAXimum|MINimum|DEFault

RST Value:

<default value>

Query Syntax:

[SOURce:]POWer[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Power Trigger Value>

Example:

POWer:TRIGgered 50, (@1) //Set channel 1 power trigger value to 50W

POWer:TRIGgered? (@1) //Query channel 1 power trigger value

[SOURce:]POWer:PROTection:DELay[:TIME] <value>,(@<chanlist>)

[SOURce:]POWer:PROTection:DELay[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the delay time of overpower protection. (Unit is second)

Command Syntax:

[SOURce:]POWer:PROTection:DELay[:TIME] <value>,(@<chanlist>)

Parameter:

<TIME>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

Query Syntax:

[SOURce:]POWer:PROTection:DELay[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Delay time for overpower protection>

Example:

POW:PROT:DEL 10,(@1) //Set channel 1 overpower protection delay time to 10s

POW:PROT:DEL? (@1) //Query over power protection delay time

[SOURce:]POWer:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]POWer:PROTection:STATe? (@<chanlist>)

This command is used to enable or disable the state of overpower protection.

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Command Syntax:

[SOURce:]POWer:PROTection:STATe 0|OFF|1|ON,(@<chanlist>)

Parameter:

0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

OFF

Query Syntax:

[SOURce:]POWer:PROTection:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

POWer:PROTection:STATe ON,(@1) //Set channel 1 overpower protection status to ON

POWer:PROTection:STATe? (@1) //Query channel 1 over power protection status

[SOURce:]POWer:PROTection[:LEVel][:POSitive] <value>,(@<chanlist>)

[SOURce:]POWer:PROTection[:LEVel][:POSitive]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the positive protection value for overpower protection. (in W)

Command Syntax:

[SOURce:]POWer:PROTection[:LEVel][:POSitive] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

maximum

Query Syntax:

[SOURce:]POWer:PROTection[:LEVel][:POSitive]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<positive protection value for overpower protection>

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Example:

POWer:PROTection 100, (@1) //Set channel 1 over power positive protection value to 100W

POWer:PROTection? (@1) //Query channel 1 over power positive protection value

[SOURce:]POWer:SLEW[:POSitive][:IMMediate] <value>, (@<chanlist>)

[SOURce:]POWer:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Sets the rising slope of the power. (in W/ms)

If the state of the [SOURce:]POWer:SLEW:COUPle Commands setting is ON, the Commands set the upward slope while simultaneously setting the downward slope.

Command Syntax:

[SOURce:]POWer:SLEW[:POSitive][:IMMediate] <value>, (@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

0.0

Query Syntax:

[SOURce:]POWer:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

MAXimum

Example:

POWer:SLEW 2 (@1) //Set channel 1 power ramp setting to 2W/ms

POWer:SLEW? (@1) //Query Channel 1 Power Rise Slope

[SOURce:]POWer:SLEW:NEGative[:IMMediate] <value>, (@<chanlist>)

[SOURce:]POWer:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Set and query the power down slope. (Unit is W/ms)

If the state of the [SOURce:]POWer:SLEW:COUPle Commands setting is ON, the Commands set the descending slope while simultaneously setting the ascending slope.

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Command Syntax:

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate] <value>, (@<chanlist>)

Parameter

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value: <default value>

Query Syntax:

[SOURce:]VOLTage:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<power downward slope>

Example:

POW:SLEW:NEG 10, (@1) //Channel 1 power down slope set to 10W/ms

POW:SLEW:NEG? (@1) //Query channel 1 power down slope

[SOURce:]POWer:SLEW:COUPle <0|OFF|1|ON>, (@<chanlist>)

[SOURce:]POWer:SLEW:COUPle? (@<chanlist>)

The commands set and query the power slope coupling state.

Command Syntax:

[SOURce:]POWer:SLEW:COUPle <0|OFF|1|ON>N, (@<chanlist>)

Parameter:

0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

ON

Query Syntax:

[SOURce:]POWer:SLEW:COUPle? (@<chanlist>)

Return Parameters:

0 or 1

Example:

POW:SLEW:COUP ON, (@1) //Enable power slope coupling state

POW:SLEW:COUP? (@1) //Query power slope coupling state

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[SOURce:]POWer:SLEW[:POSitive]:MAXimum <0|OFF|1|ON>, (@<chanlist>)

[SOURce:]POWer:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Set and query the status of the maximum slope of the power rise slope.

Command Syntax:

[SOURce:]POWer:SLEW[:POSitive]:MAXimum <0|OFF|1|ON>N, (@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

OFF

Query Syntax:

[SOURce:]POWer:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Return Parameters:

0 or 1

Example:

POWer:SLEW:MAXimum ON, (@1) //Enable Channel 1 Power Ramp Up Slope Max

POW:SLEW:MAX? (@1) //Query the enable state of channel 1 with the largest power rise slope

[SOURce:]POWer:SLEW:NEGative:MAXimum <0|OFF|1|ON>, (@<chanlist>)

[SOURce:]POWer:SLEW:NEGative:MAXimum? (@<chanlist>)

Set and query the status of the maximum slope of the power down slope.

Command Syntax:

[SOURce:]POWer:SLEW:NEGative:MAXimum <0|OFF|1|ON>N, (@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

OFF

Query Syntax:

[SOURce:]POWer:SLEW:NEGative:MAXimum? (@<chanlist>)

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Return Parameters:

0 or 1

Example:

POWer:SLEW:NEGative:MAXimum ON, (@1) //Enable channel 1 power down slope maximum

POWer:SLEW:NEGative:MAXimum? (@1) //Enable channel 1 power down slope maximum

[SOURce:]POWer:TLEVel <value> (@<chanlist>)

[SOURce:]POWer:TLEVel? [MINimum|MAXimum,] (@<chanlist>)

Set and query the power value at transient. (in W)

Command Syntax:

[SOURce:]POWer:TLEVel <value> (@<chanlist>)

Parameter:

minimum-maximum|MINimum|MAXimum

RST Value:

0.0

Query Syntax:

[SOURce:]POWer:TLEVel? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Transient power value>

Example:

POWer:TLEVel 15,(@1) //Setting the power level of channel 1 at transient 15W

POWer:TLEVel? (@1) //Query the power level during transient of channel 1

11.5 Resistance Commands

[SOURce:]RESistance[:LEVel][:IMMediate][:AMPLitude] <value>, (@<chanlist>)

[SOURce:]RESistance[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the immediate value of resistance. (Unit is Ω)

Command Syntax:

[SOURce:]RESistance[:LEVel][:IMMediate][:AMPLitude] <value>, (@<chanlist>)

Parameter:

0-maximum|MINimum|MAXimum

Query Syntax:

[SOURce:]RESistance[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

RST Value:

MAXimum

Return Parameters:

<immediate value of resistance>

Example:

RESistance 50, (@1) //Set the immediate value of the resistor for channel 1 to 50Ω

RESistance? (@1) //Queries the immediate value of the resistance of channel 1

[SOURce:]RESistance[:LEVel]:TRIGgered[:AMPLitude] <value>, (@<chanlist>)

[SOURce:]RESistance[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the value of resistance when triggered. (Unit is Ω)

Command Syntax:

[SOURce:]RESistance[:LEVel]:TRIGgered[:AMPLitude] <value>, (@<chanlist>)

Parameter:

minimum-maximum|MINimum|MAXimum

RST Value:

<MAXimum>

Query Syntax:

[SOURce:]RESistance[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Trigger value of the resistor>

Example:

RESistance:TRIGgered 50, (@1) //Setting the resistor trigger value for channel 1 to 50Ω

RESistance:TRIGgered? (@1) //Query the resistor trigger value of channel 1

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[SOURce:]RESistance:SLEW[:POSitive][:IMMediate] <value>, (@<chanlist>)

[SOURce:]RESistance:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Sets and queries the rising slope of the resistor. (Unit is S/ms)

If the [SOURce:]RESistance:SLEW:COUPle instruction setting state is ON, this instruction sets the upward slope while simultaneously setting the downward slope.

Command Syntax:

[SOURce:]RESistance:SLEW[:POSitive][:IMMediate] <value>, (@<chanlist>)

Parameter:

minimum-maximum|MINimum|MAXimum

RST Value:

default

Query Syntax:

[SOURce:]RESistance:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Rising slope of resistance>

Example:

RESistance:SLEW 1, (@1)	//Set the resistor rise slope to 1S/ms
RESistance:SLEW? (@1)	//Query Resistance Rise Slope

[SOURce:]RESistance:SLEW:NEGative[:IMMediate] <value>, (@<chanlist>)

[SOURce:]RESistance:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Sets and queries the falling slope of the resistor. (Unit is S/ms)

If the [SOURce:]RESistance:SLEW:COUPle instruction setting state is ON, this instruction sets the descending slope while simultaneously setting the ascending slope.

Command Syntax:

[SOURce:]RESistance:SLEW:NEGative[:IMMediate] <value>, (@<chanlist>)

Parameter:

minimum-maximum|MINimum|MAXimum

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RST Value:

default

Query Syntax:

[SOURce:]RESistance:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Descent slope of resistance>

Example:

RESistance:SLEW:NEGative 10, (@1) //Set channel 1 resistance drop slope to 10S/ms

RESistance:SLEW:NEGative? (@1) //Query channel 1 resistance descent slope

[SOURce:]RESistance:SLEW:COUPle <0|OFF|1|ON>, (@<chanlist>)

[SOURce:]RESistance:SLEW:COUPle? (@<chanlist>)

The commands set and query the status of resistance slope tracking.

Command Syntax:

[SOURce:]RESistance:SLEW:COUPle <0|OFF|1|ON>, (@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

0

Query Syntax:

[SOURce:]RESistance:SLEW:COUPle? (@<chanlist>)

Return Parameters:

0 or 1

Example:

RESistance:SLEW:COUPle ON, (@1) //Enable channel 1 resistance slope tracking state

RESistance:SLEW:COUPle? (@1) //Query channel 1 resistance slope tracking status

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[SOURce:]RESistance:SLEW[:POSitive]:MAXimum <0|OFF|1|ON>, (@<chanlist>)

[SOURce:]RESistance:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Sets and queries the status of the resistor rising slope maximum slope.

Command Syntax:

[SOURce:]RESistance:SLEW[:POSitive]:MAXimum <0|OFF|1|ON>, (@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

0

Query Syntax:

[SOURce:]RESistance:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Return Parameters:

0 or 1

Example:

RESistance:SLEW:MAXimum ON, (@1) //Enable Resistor Rise Slope Maximum State

RESistance:SLEW:MAXimum? (@1) //Queries the state of the maximum value of the resistance rise slope

[SOURce:]RESistance:SLEW:NEGative:MAXimum <0|OFF|1|ON>, (@<chanlist>)

[SOURce:]RESistance:SLEW:NEGative:MAXimum? (@<chanlist>)

Sets and queries the status of the maximum slope of the resistor descent slope.

Command Syntax:

[SOURce:]RESistance:SLEW:NEGative:MAXimum <0|OFF|1|ON>, (@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

0

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Query Syntax:

[SOURce:]RESistance:SLEW:NEGative:MAXimum? (@<chanlist>)

Return Parameters:

0 or 1

Example:

RES:SLEW:NEG:MAX ON, (@1) //Enable resistor falling slope maximum state

RES:SLEW:NEG:MAX? (@1) //Queries the state of the maximum value of the resistance descent slope

[SOURce:]RESistance:TLEVel <value> (@<chanlist>)

[SOURce:]RESistance:TLEVel? [MINimum|MAXimum,] (@<chanlist>)

Sets and queries the resistance value during transients. (Unit is W)

Command Syntax:

[SOURce:]RESistance:TLEVel <value> (@<chanlist>)

Parameter:

minimum-maximum|MINimum|MAXimum

RST Value:

None

Query Syntax:

[SOURce:]RESistance:TLEVel? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Transient resistance value>

Example:

RES:TLEVel 15,(@1) //Setting the resistance level of channel 1 at transient 15Ω

RES:TLEVel? (@1) //Query the resistor level during transient of channel 1

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Chapter12 SMU Commands (SOURce Subsystem)

This chapter describes the commands related to the SMU module.

12.1 Function Commands

[SOURce:]EMULation <type>, (@<chanlist>)

[SOURce:]EMULation? (@<chanlist>)

This command is used to set and query the simulation mode of the SMU.

Command Syntax:

[SOURce:]EMULation <type>, (@<chanlist>)

Parameter:

PS4Q|PS2Q|PS1Q|CCLoad|CVLoad|CRLoad|VMETer|AMETer|BATTery|CHARger

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]EMULation? (@<chanlist>)

RST Value:

PS4Q

Return Parameters:

PS4Q|PS2Q|PS1Q|CCL|CVL|CRL|VMET|AMET|BATT|CHAR

Example:

```
EMUL CVL,(@1)       // Set the simulation type of CH1 to voltage load.
EMULation? (@1)          //query the simulation mode of ch1
```

[SOURce:]FUNCtion <CURRent|VOLTage>,(@<chanlist>)

[SOURce:]FUNCtion? (@<chanlist>)

This command is used to set and query the priority of the function. This command is active when the type value of the EMULATION command is PS4Q|PS2Q|PS1Q.

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Command Syntax:

[SOURce:]FUNCtion CURRent|VOLTage,(@<chanlist>)

Parameter:

<mode>: CURRent|VOLTage

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]FUNCtion? (@<chanlist>)

RST Value:

VOLTage

Return Parameters:

CURR,VOLT

Example:

FUNC CURR,(@1) //Set Channel 1 function priority to current priority

FUNC? (@1) //Queries the Channel 1 function priority.

12.2 Voltage Commands

[SOURce:]VOLTage:RANGe <value>, (@<chanlist>)

[SOURce:]VOLTage:RANGe? [MINimum|MAXimum,] (@<chanlist>)

This command is used to set and query the voltage priority range.

Command Syntax::

[SOURce:]VOLTage:RANGe <value>, (@<chanlist>)

Arguments:

<value>: minimum-maximum|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax::

[SOURce:]VOLTage:RANGe? [MINimum|MAXimum,] (@<chanlist>)

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复位值:

MAXimum

Returns:

<voltage range>

Example:

VOLT:RANG 30.0,(@1) //Set the voltage range for Channel 1 to 30V.

VOLT:RANG? (@1) //Query the voltage range for Channel 1

[SOURce:]VOLTage:AC[:LEVel][:IMMediate][:AMPLitude] <value|MINimum|MAXimum>,(@<channellist>)

[SOURce:]VOLTage:AC[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<channellist>)

This command is used to set and query the immediate or trigger values for AC voltage.

Command Syntax::

[SOURce:]VOLTage:AC[:LEVel][:IMMediate][:AMPLitude] <value|MINimum|MAXimum>,(@<channellist>)

Arguments:

<value>:minimum-maximum|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax::

[SOURce:]VOLTage:AC[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<channellist>)

复位值:

0

Returns:

<ac voltage level>

Example:

VOLT:AC 5,(@1) //Set the AC voltage instantaneous value of Channel 1 to 5V.

VOLT:AC? (@1) //Query Channel 1 AC Voltage Instantaneous Value

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[SOURce:]VOLTage:AC[:LEVel]:TRIGgered[:AMPLitude] <value|MINimum|MAXimum>,(@<channellist>)

[SOURce:]VOLTage:AC[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<channellist>)

This command is used to set and query the immediate or trigger values for AC voltage.

Command Syntax::

[SOURce:]VOLTage:AC[:LEVel]:TRIGgered[:AMPLitude] <value|MINimum|MAXimum>,(@<channellist>)

Arguments:

<value>:minimum-maximum|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax::

[SOURce:]VOLTage:AC[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<channellist>)

Returns:

<ac voltage trig>

Example:

VOLT:AC:TRIG 5,(@1) // Set the AC voltage trigger value for Channel 1 to 5V.

VOLT:AC:TRIG? (@1) //Query Channel 1 AC Voltage Trigger Value

[SOURce:]VOLTage:LIMit:MODE <NORMal|BATTery>,(@<chanlist>)

[SOURce:]VOLTage:LIMit:MODE? (@<chanlist>)

This command is used to set and query the voltage limit mode.

Command Syntax::

[SOURce:]VOLTage:LIMit:MODE <NORMal|BATTery>,(@<chanlist>)

Arguments:

NORMal|BATTery

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

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Query Syntax::

[SOURce:]VOLTage:LIMit:MODE? (@<chanlist>)

复位值:

NORMal

Returns:

NORM|BATT

Example:

VOLTage:LIMit:MODE BATTery,(@2) //Set Channel 2 voltage limit mode to Battery

VOLTage:LIMit:MODE? (@2) //Query Channel 2 Voltage Limit Mode

[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the immediate value of voltage. (Unit is V)

Command Syntax:

[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number

Query Syntax:

[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

RST Value:

Default

Return Parameters:

<immediate value of voltage>

Example:

VOLT 100, (@1) //Set channel 1 voltage value to 100V

VOLT? (@1) //Query the current voltage value of channel 1

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[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude] <value>,(@<chanlist>)

[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the value when the voltage is triggered. (Unit is V)

Command Syntax:

[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

None

Query Syntax:

[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Trigger value of voltage>

Example:

VOLT:TRIG 50, (@1) //Setting the voltage trigger value for channel 1 to 50V

VOLT? (@1) //Query the voltage trigger value of channel 1

[SOURce:]VOLTage:LIMit[:POSitive][:IMMediate][:AMPLitude] <value> , (@<chanlist>)

[SOURce:]VOLTage:LIMit[:POSitive][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

This command is used to set and query the positive voltage limit value for the CC priority mode voltage.

Command Syntax::

[SOURce:]VOLTage:LIMit[:POSitive][:IMMediate][:AMPLitude] <value> , (@<chanlist>)

Arguments:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

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(@1:4) – Channels 1 through 4, including channel 1234.

复位值:

MAXimum

Query Syntax::

[SOURce:]VOLTage:LIMit[:POSitive][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Returns:

<maximum voltage>

Example:

VOLT:LIM 2.0,(@1) //Set the positive voltage limit for Channel 1 to 2.0V.

VOLT:LIM? (@1) //Query the positive voltage limit value for Channel 1

[SOURce:]VOLTage:LIMit:NEGative[:IMMediate][:AMPLitude] <value>, (@<chanlist>)

[SOURce:]VOLTage:LIMit:NEGative[:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

This command is used to set and query the negative voltage limit value for the CC priority mode voltage.

Command Syntax::

[SOURce:]VOLTage:LIMit:NEGative[:IMMediate][:AMPLitude] <value>, (@<chanlist>)

Arguments:

<value>:maximum*(-1)-0|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

复位值:

MAXimum*(-1)

Query Syntax::

[SOURce:]VOLTage:LIMit:NEGative[:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Returns:

<voltage>

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Example:

VOLT:LIM:NEG -2.0,(@1) //Set the negative voltage limit for Channel 1 to 2.0V.

VOLT:LIM:NEG? (@1) //Negative voltage limit value for channel 1

[SOURce:]VOLTage:LIMit:COUPle 0|OFF|1|ON, (@<chanlist>)

[SOURce:]VOLTage:LIMit:COUPle? (@<chanlist>)

This command is used to set and query the tracking status of voltage limit values, with negative voltage tracking positive voltage.

Command Syntax::

[SOURce:]VOLTage:LIMit:COUPle 0|OFF|1|ON, (@<chanlist>)

Arguments:

0|OFF|1|ON

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

复位值:

ON

Query Syntax::

[SOURce:]VOLTage:LIMit:COUPle? (@<chanlist>)

Returns:

0 or 1

Example

VOLT:LIM:COUP 1,(@1) // Enable Channel 1 Voltage Positive/Negative Limit Coupling

VOLT:LIM:COUP? (@1) // Query Channel 1 Voltage Positive/Negative Limit Values Coupling

[SOURce:]VOLTage:PROTection[:LOCal][:POSitive][:LEVel] <value>,(@<chanlist>)

[SOURce:]VOLTage:PROTection[:LOCal][:POSitive][:LEVel]? [MINimum|MAXimum,](@<chanlist>)

The command sets or queries the positive voltage protection value of overvoltage protection. (Unit is V)

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Command Syntax:

[SOURce:]VOLTage:PROTection[:LOCal][:POSitive][:LEVel] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

maximum

Query Syntax:

[SOURce:]VOLTage:PROTection[:LOCal][:POSitive][:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Protection value for positive voltage>

Example

VOLT:PROT 300, (@1) //Setting the positive voltage protection value for channel 1 overvoltage protection

VOLT:PROT? (@1) //Query the positive voltage protection value of channel 1 overvoltage protection

[SOURce:]VOLTage:PROTection[:LOCal]:NEGative[:LEVel] <value>, (@<chanlist>)

[SOURce:]VOLTage:PROTection[:LOCal]:NEGative[:LEVel]? [MINimum|MAXimum,] (@<chanlist>)

Set and query the negative voltage protection value for overvoltage protection. (Unit: V)

Command Syntax::

[SOURce:]VOLTage:PROTection[:LOCal]:NEGative[:LEVel] <value>, (@<chanlist>)

Arguments:

<value>:maximum*(-1)-0|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

复位值:

MAXimum*(-1)

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Query Syntax::

[SOURce:]VOLTage:UNDer:PROTection[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Returns:

<过电压的正电压保护值>

Example:

VOLT:PROT:NEG 30.0,(@1) //Set the overvoltage negative protection value for Channel 1 to -30.0V.

VOLT:PROT:NEG? (@1) //Query Channel 1 Overvoltage Negative Protection Value

[SOURce:]VOLTage:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:PROTection:STATe? (@<chanlist>)

The commands set and query the enable state of overvoltage protection. (Unit is V)

Command Syntax:

[SOURce:]VOLTage:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

Parameter:

<0|OFF|1|ON>:0-OFF|1-ON

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:PROTection:STATe? (@<chanlist>)

Return Parameters:

<Protection enable state of voltage>

Example:

VOLT:PROT:STAT ON, (@1) //Setting the enable state of channel 1 overvoltage protection to ON

VOLT:PROT:STAT? (@1) //Queries the enable state of channel 1 overvoltage protection.

[SOURce:]VOLTage:PROTection:DELay[:TIME] <value>,(@<chanlist>)

[SOURce:]VOLTage:PROTection:DELay[:TIME]? [MINimum|MAXimum,](@<chanlist>)

Setting and querying the delay time for overvoltage protection, (in seconds)

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Command Syntax:

[SOURce:]VOLTage:PROTection:DELay[:TIME] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

Query Syntax:

[SOURce:]VOLTage:PROTection:DELay[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Delay value for voltage protection>

Example:

VOLT:PROT:DEL 30, (@1) //Set the delay time for channel 1 overvoltage protection to 30s.

VOLT:PROT:DEL? (@1) //Queries the delay time for channel 1 overvoltage protection.

[SOURce:]VOLTage:UNDer:PROTection[:POSitive][:LEVel] <value>,(@<chanlist>)

[SOURce:]VOLTage:UNDer:PROTection[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Set and query the positive voltage protection value for undervoltage protection. (Unit is V)

Command Syntax:

[SOURce:]VOLTage:UNDer:PROTection[:POSitive][:LEVel] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

maximum

Query Syntax:

[SOURce:]VOLTage:UNDer:PROTection[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

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Return Parameters:

<Positive voltage protection value for undervoltage>

Example:

VOLT:UND:PROT 10, (@1) //Set the positive voltage protection value for channel 1 undervoltage protection to 10V

VOLT:UND:PROT? (@1) //Queries the positive voltage protection value for channel 1 undervoltage protection.

[SOURce:]VOLTage:UNDer:PROTection:NEGative[:LEVel] <value>.(@<chanlist>)

[SOURce:]VOLTage:UNDer:PROTection:NEGative[:LEVel]? [MINimum|MAXimum].(@<chanlist>)

Set and query the negative voltage protection value for undervoltage protection. (Unit: V)

Command Syntax::

[SOURce:]VOLTage:UNDer:PROTection:NEGative[:LEVel] <value>.(@<chanlist>)

Arguments:

<value>:maximum*(-1)-0|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

复位值:

MAXimum*(-1)

Query Syntax::

[SOURce:]VOLTage:UNDer:PROTection:NEGative[:LEVel]? [MINimum|MAXimum].(@<chanlist>)

Returns:

< negative voltage>

Example:

VOLT:UND:PROT:NEG -5.0,(@1) // Set the undervoltage negative protection value for Channel 1 to -5.0V.

VOLT:UND:PROT:NEG? (@1) // Query Channel 1 Undervoltage Negative Protection Value

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[SOURce:]VOLTage:UNDer:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:UNDer:PROTection:STATe? (@<chanlist>)

Sets and queries the enable state of undervoltage protection.

Command Syntax:

[SOURce:]VOLTage:UNDer:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

Parameter:

<0|OFF|1|ON>:0-OFF|1-ON

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:UNDer:PROTection:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

VOLT:UND:PROT:STAT ON,(@1) //Set the enable state of channel 1 undervoltage protection to ON.

VOLT:UND:PROT:STAT? (@1) //Queries the enable state of channel 1 undervoltage protection.

[SOURce:]VOLTage:UNDer:PROTection:DELay[:TIME] <value>,(@<chanlist>)

[SOURce:]VOLTage:UNDer:PROTection:DELay[:TIME]? [MINimum|MAXimum,] (@<chanlist>)

Sets the delay time for undervoltage protection. (in seconds)

Command Syntax:

[SOURce:]VOLTage:UNDer:PROTection:DELay[:TIME] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

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Query Syntax:

[SOURce:]VOLTage:UNDer:PROTection:DELay[:TIME]?
[MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Delay time for undervoltage protection>

Example:

VOLT:UND:PROT:DEL 10, (@1) //Set channel 1 undervoltage protection delay time to 10s

VOLT:UND:PROT:DEL? (@1) //Query channel 1 undervoltage protection delay time

[SOURce:]VOLTage:UNDer:PROTection:WARM[:TIME] <value>,(@<chanlist>)

[SOURce:]VOLTage:UNDer:PROTection:WARM[:TIME]? [MINimum|MAXimum,](@<chanlist>)

Set and query the warm-up time for undervoltage protection. (in seconds)

Command Syntax:

[SOURce:]VOLTage:UNDer:PROTection:WARM[:TIME] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

Query Syntax:

[SOURce:]VOLTage:UNDer:PROTection:WARM[:TIME]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Warm-up time for undervoltage protection>

Example:

VOLT:UND:PROT:WARM 10, (@1) //Set channel 1 undervoltage protection delay time to 10s

VOLT:UND:PROT:WARM? (@1) //Query channel 1 undervoltage protection delay time

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[SOURce:]VOLTage:RESistance[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]VOLTage:RESistance[:LEVel][:IMMediate][:AMPLitude]?(@<chanlist>)

Sets and queries the internal resistance value of the power supply in voltage priority mode. (Unit is Ω)

Valid only when the internal resistance function is enabled.

Command Syntax:

[SOURce:]VOLTage:RESistance[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:MAXimum|MINimum|DEFault

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:RESistance[:LEVel][:IMMediate][:AMPLitude]?(@<chanlist>)

Return Parameters:

<internal resistance value>

Example:

VOLT:RES 0.1,(@1)	//Set the internal resistance of channel 1 to 0.1Ω.
VOLT:RES? (@1)	//Query the internal resistance of channel 1

[SOURce:]VOLTage:RESistance:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]VOLTage:RESistance:STATe? (@<chanlist>)

Sets and queries the status of the internal resistance function of the power supply in voltage priority mode.

Command Syntax:

[SOURce:]VOLTage:RESistance:STATe <0|OFF|1|ON>,(@<chanlist>)

Parameter:

<0|OFF|1|ON>:0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:RESistance:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

VOLT:RES:STAT 0 //Set channel 1 internal resistance function status to 0

VOLT:RES:STAT? (@1) //Query the current internal resistance function status of channel 1

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate] <value>,(@<chanlist>)

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Set and query the voltage rise slope. (Unit is V/ms)

If the [SOURce:]VOLTage:SLEW:COUPle instruction setting state is ON, this instruction sets the voltage upward ramp downward while simultaneously setting the voltage downward ramp downward.

Command Syntax:

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate] <value>,(@<chanlist>)

Parameter:

<value>:minimum-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]VOLTage:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Rise slope of voltage>

Example:

VOLT:SLEW 10, (@1) //Set channel 1 voltage rise slope to 10V/ms

VOLT:SLEW? (@1) //Query channel 1 voltage rise slope

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[SOURce:]VOLTage:SLEW:NEGative[:IMMediate] <value>,(@<chanlist>)

[SOURce:]VOLTage:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,](@<chanlist>)

Set and query the voltage drop slope. (Unit is V/ms)

If the [SOURce:]VOLTage:SLEW:COUPle instruction setting state is ON, this instruction sets the voltage falling slope while simultaneously setting the voltage rising slope.

Command Syntax:

[SOURce:]VOLTage:SLEW:NEGative[:IMMediate] <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]VOLTage:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Slope of voltage drop>

Example:

VOLT:SLEW:NEG 10,(@1) //Set channel 1 voltage drop slope to 10 V/ms

VOLT:SLEW:NEG? (@1) //Query channel 1 voltage drop slope

[SOURce:]VOLTage:SLEW:COUPle 0|OFF|1|ON,(@<chanlist>)

[SOURce:]VOLTage:SLEW:COUPle? (@<chanlist>)

Set and query the tracking status of the voltage slope.

Command Syntax:

[SOURce:]VOLTage:SLEW:COUPle 0|OFF|1|ON,(@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

1

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Query Syntax:

[SOURce:]VOLTage:SLEW:COUPle? (@<chanlist>)

Return Parameters:

0 or 1

Example

VOLT:SLEW:COUP ON, (@1) //Enable channel 1 voltage slope tracking state

VOLT:SLEW:COUP? (@1) //Query channel 1 voltage slope tracking status

[SOURce:]VOLTage:SLEW[:POSitive]:MAXimum 0|OFF|1|ON,(@<chanlist>)

[SOURce:]VOLTage:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Enable or disable voltage rise slope maximum.

Command Syntax:

[SOURce:]VOLTage:SLEW[:POSitive]:MAXimum 0|OFF|1|ON,(@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Return Parameters:

0 or 1

Example:

VOLT:SLEW:MAX ON, (@1) //Enable Channel 1 Voltage Rise Slope Maximum Function State

VOLT:SLEW:MAX? (@1) //Query Channel 1 Voltage Rise Slope Maximum Function Status

[SOURce:]VOLTage:SLEW:NEGative:MAXimum 0|OFF|1|ON,(@<chanlist>)

[SOURce:]VOLTage:SLEW:NEGative:MAXimum? (@<chanlist>)

Enable or disable voltage drop slope maximum.

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Command Syntax:

[SOURce:]VOLTage:SLEW:NEGative:MAXimum 0|OFF|1|ON,(@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

0

Query Syntax:

[SOURce:]VOLTage:SLEW:NEGative:MAXimum? (@<chanlist>)

Return Parameters:

0 or 1

Example:

VOLT:SLEW:NEG:MAX ON,(@1) //Enable channel 1 voltage drop slope maximum state

VOLT:SLEW:NEG:MAX? (@1) //Query channel 1 voltage drop slope maximum value status

12.3 Current Commands

[SOURce:]CURRent:RANGe <value>, (@<chanlist>)

[SOURce:]CURRent:RANGe? [MINimum|MAXimum,] (@<chanlist>)

This command is used to set the current measurement range.

Command Syntax::

[SOURce:]CURRent:RANGe <value>, (@<chanlist>)

Arguments:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

RST Value:

MAXimum

Query Syntax::

[SOURce:]CURRent:RANGe? [MINimum|MAXimum,] (@<chanlist>)

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Returns:

<Default>

Example:

CURR;RANG 3.0 (@1) // Set the current range for Channel 1 to 3.0A.

CURR:RANG? (@1) // Current Range for Inquiry Notice 1

[SOURce:]CURRent[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]CURRent[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Sets the immediate value of the current. (Unit is A)

Command Syntax:

[SOURce:]CURRent[:LEVel][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<immediate value of current>

Example:

CURR 3,(@1) //3ASet channel 1 current to 3A immediately

CURR? (@1) //Query channel 1 current immediate value

[SOURce:]CURRent[:LEVel]:TRIGgered[:AMPLitude] <value>,(@<chanlist>)

[SOURce:]CURRent[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Sets or queries the trigger value of the current. (Unit is A)

Command Syntax:

[SOURce:]CURRent[:LEVel]:TRIGgered[:AMPLitude] <value>,(@<chanlist>)

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Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

None

Query Syntax:

[SOURce:]CURRent[:LEVel]:TRIGgered[:AMPLitude]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Trigger value of voltage>

Example:

CURR:TRIG 20, (@1)	//Set channel 1 current trigger value to 20A
CURR:TRIG ? (@1)	//Query channel 1 current trigger value

[SOURce:]CURRent:AC[:LEVel][:IMMediate][:AMPLitude] <value|MINimum|MAXimum>,(@<channellist>)

[SOURce:]CURRent:AC[:LEVel][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<channellist>)

Set and query the instantaneous value of alternating current. (Unit: A)

Command Syntax::

[SOURce:]CURRent:AC[:LEVel][:IMMediate][:AMPLitude]
<value|MINimum|MAXimum>,(@<channellist>)

Arguments:

<value>:minimum-maximum|MINimum|MAXimum (unit: A)

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

RST Value:

0

Query Syntax::

[SOURce:]CURRent:AC[:LEVel][:IMMediate][:AMPLitude]?
[MINimum|MAXimum,](@<channellist>)

Returns:

<AC current level>

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Example:

CURR:AC 2,(@1) // Set the AC current instantaneous value for Channel 1 to 2A.

CURR:AC? (@1) // Query Channel 1 AC Current Instantaneous Value

[SOURce:]CURRent:AC[:LEVel]:TRIGgered[:AMPLitude] <value|MINimum|MAXimum>,(@<channellist>)

[SOURce:]CURRent:AC[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<channellist>)

Set and query the trigger value for alternating current. (Unit: A)

Command Syntax::

[SOURce:]CURRent:AC[:LEVel]:TRIGgered[:AMPLitude] <value|MINimum|MAXimum>,(@<channellist>)

Arguments:

<value>:minimum-maximum|MINimum|MAXimum (单位为 A)

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

RST Value:

0

Query Syntax::

[SOURce:]CURRent:AC[:LEVel]:TRIGgered[:AMPLitude]? [MINimum|MAXimum,](@<channellist>)

Returns:

<AC current triggred level>

Example:

CURR:AC:TRIG 2,(@1) // Set the AC current trigger value for Channel 1 to 2V.

CURR:AC:TRIG? (@1) // Query Channel 1 AC Current Trigger Value

[SOURce:]CURRent:LIMit[:POSitive][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]CURRent:LIMit[:POSitive][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Sets the current limit value for voltage priority mode. (Unit is A)

With the bipolar power supply function, if the [SOURce:]CURRent:LIMit:COUPle

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instruction setting status is ON, this instruction sets the positive current limit value while simultaneously setting the negative current limit value.

Command Syntax:

[SOURce:]CURRent:LIMit[:POSitive][:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:LIMit[:POSitive][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Positive limiting value of current>

Example:

CURR:LIM 50,(@1) //Set channel 1 positive current limit value to 50A

CURR:LIM? (@1) //Query channel 1 positive current limit value

[SOURce:]CURRent:LIMit:NEGative[:IMMediate][:AMPLitude] <value>,(@<chanlist>)

[SOURce:]CURRent:LIMit:NEGative[:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Sets the negative current limit value for voltage priority mode. (Unit is A)

If the [SOURce:]CURRent:LIMit:COUPle instruction setting status is ON, this instruction sets the negative current limit value and also sets the positive current limit value synchronously.

Command Syntax:

[SOURce:]CURRent:LIMit:NEGative[:IMMediate][:AMPLitude] <value>,(@<chanlist>)

Parameter:

<value>:minimum - 0|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

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Query Syntax:

[SOURce:]CURRent:LIMit:NEGative[:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Negative limit value of current>

Example:

CURR:LIM:NEG -2,(@1) //Set the negative current limit value for channel 1 to -2A.

CURR:LIM:NEG? (@1) //Query channel 1 negative current limit value

[SOURce:]CURRent:LIMit:COUPle 0|OFF|1|ON,(@<chanlist>)

[SOURce:]CURRent:LIMit:COUPle? (@<chanlist>)

Sets and queries the tracking status of positive and negative current limit values.

Command Syntax:

[SOURce:]CURRent:LIMit:COUPle 0|OFF|1|ON,(@<chanlist>)

Parameter:

0|OFF|1|ON

RST Value:

ON

Query Syntax:

[SOURce:]CURRent:LIMit:COUPle? (@<chanlist>)

Return Parameters:

0 or 1

Example:

CURR:LIM:COUP ON,(@1) //Turns the tracking state of channel 1 current positive and negative limits on.

CURR:LIM:COUP? (@1) //Queries the tracking status of channel 1 current positive and negative limit values

[SOURce:]CURRent:PROTection[:LEVel][:POSitive] <level>,(@<chanlist>)

[SOURce:]CURRent:PROTection[:LEVel][:POSitive]? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the positive current protection value of overcurrent protection. (Unit is A)

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Command Syntax:

[SOURce:]CURRent:PROTection[:LEVel][:POSitive] <level>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:PROTection[:LEVel][:POSitive]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Positive current protection value for overcurrent protection>

Example:

CURR:PROT 30, (@1) //Setting the positive current protection value for channel 1 overcurrent protection to 30A

CURR:PROT? (@1) //Query the positive current protection value of channel 1 overcurrent protection

[SOURce:]CURRent:PROTection[:LEVel]:NEGative <level>,(@<chanlist>)

[SOURce:]CURRent:PROTection[:LEVel]:NEGative? [MINimum|MAXimum,](@<chanlist>)

This command sets the negative current protection value for overcurrent protection. (Unit is A)

Command Syntax:

[SOURce:]CURRent:PROTection[:LEVel]:NEGative <level>,(@<chanlist>)

Parameter:

<value>:minimum-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:PROTection[:LEVel]:NEGative? [MINimum|MAXimum,](@<chanlist>)

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Return Parameters:

<Negative current protection value for overcurrent protection>

Example:

CURR:PROT:NEG -2,(@1) //Set channel 1 overcurrent protection negative value to -2A

CURR:PROT:NEG? (@1) //Query channel 1 overcurrent protection negative value

[SOURce:]CURRent:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]CURRent:PROTection:STATe? (@<chanlist>)

The commands set and query the enable state of overcurrent protection.

Command Syntax:

[SOURce:]CURRent:PROTection:STATe <0|OFF|1|ON>,(@<chanlist>)

Parameter:

<value>:<0|OFF|1|ON>

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]CURRent:PROTection:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

CURR:PROT:STAT ON, (@1) //Setting the channel 1 current protection function to the ON state

CURR:PROT:STA? (@1) //Query channel 1 current protection function status

[SOURce:]CURRent:PROTection:DELay[:TIME] <value>,(@<chanlist>)

[SOURce:]CURRent:PROTection:DELay[:TIME]? [MINimum|MAXimum,](@<chanlist>)

This command is used to set or query the delay time of overcurrent protection. (Unit is in seconds)

Command Syntax:

[SOURce:]CURRent:PROTection:DELay[:TIME] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

Query Syntax:

[SOURce:]CURRent:PROTection:DELay[:TIME]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Delay time for overcurrent protection>

Example:

CURR:PROT:DEL 10,(@1) //Set the delay time for channel 1 overcurrent protection to 10 seconds

CURR:PROT:DEL? (@1) //Query delay time for channel 1 overcurrent protection

[SOURce:]CURRent:SLEW[:POSitive][:IMMediate] <value>,(@<chanlist>)

[SOURce:]CURRent:SLEW[:POSitive][:IMMediate]? [MINimum|MAXimum,](@<chanlist>)

Set and query the current rise slope. (Unit is A/ms)

If the [SOURce:]CURRent:SLEW:COUPle instruction setting state is ON, this instruction sets the current up and down slopes simultaneously with the voltage down slope.

Command Syntax:

[SOURce:]CURRent:SLEW[:POSitive][:IMMediate] <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:SLEW[:POSitive][:IMMediate]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Ramp rate of current rise

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Example:

CURR:SLEW 10, (@1) //Set channel 1 current rise slope to 10 A/ms

CURR:SLEW? (@1) //Query channel 1 current rise slope

[SOURce:]CURRent:SLEW:NEGative[:IMMediate] <value>,(@<chanlist>)

[SOURce:]CURRent:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Setting and querying the current drop slope. (Unit is A/ms)

If the [SOURce:]CURRent:SLEW:COUPle instruction setting state is ON, this instruction sets the current down and up slope simultaneously with the voltage up slope.

Command Syntax:

[SOURce:]CURRent:SLEW:NEGative[:IMMediate] <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

Default

Query Syntax:

[SOURce:]CURRent:SLEW:NEGative[:IMMediate]? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Slope of current drop>

Example:

CURR:SLEW:NEG 10, (@1) //Set channel 1 current down slope to 10 A/ms

CURR:SLEW:NEG? (@1) //Query channel 1 current descent slope

[SOURce:]CURRent:SLEW:COUPle 0|OFF|1|ON, (@<chanlist>)

[SOURce:]CURRent:SLEW:COUPle? (@<chanlist>)

Set and query the tracking status of the current slope.

Command Syntax:

[SOURce:]CURRent:SLEW:COUPle 0|OFF|1|ON, (@<chanlist>)

Parameter:

<value>:0|OFF|1|ON

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<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]CURRent:SLEW:COUPle? (@<chanlist>)

Return Parameters:

0 or 1

Example:

CURR:SLEW:COUP ON, (@1) //Enable tracking state of channel 1 current slope

CURR:SLEW:COUP? (@1) //Queries the tracking status of channel 1 current slope

[SOURce:]CURRent:SLEW[:POSitive]:MAXimum 0|OFF|1|ON, (@<chanlist>)

[SOURce:]CURRent:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Enable or disable current ramp maximum.

Command Syntax:

[SOURce:]CURRent:SLEW[:POSitive]:MAXimum 0|OFF|1|ON, (@<chanlist>)

Parameter:

<value>:0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

0

Query Syntax:

[SOURce:]CURRent:SLEW[:POSitive]:MAXimum? (@<chanlist>)

Return Parameters:

0 or 1

Example:

CURR:SLEW:MAX ON, (@1) //Enable channel 1 current rise slope max.

CURR:SLEW:MAX? (@1) //Query channel 1 current rise slope maximum value status

[SOURce:]CURRent:SLEW:NEGative:MAXimum 0|OFF|1|ON,(@<chanlist>)

[SOURce:]CURRent:SLEW:NEGative:MAXimum? (@<chanlist>)

Enable or disable current ramp down maximum.

Command Syntax:

[SOURce:]CURRent:SLEW:NEGative:MAXimum 0|OFF|1|ON, (@<chanlist>)

Parameter:

MAXimum|MINimum|DEFault

RST Value:

0

Query Syntax:

[SOURce:]CURRent:SLEW:NEGative:MAXimum? (@<chanlist>)

Return Parameters:

<Maximum value of current descent slope>

Example:

CURR:SLEW:NEG:MAX ON, (@1) //Enable channel 1 current down slope maximum value

CURR:SLEW:NEG:MAX? (@1) //Query channel 1 current descent slope maximum value status

12.4 Power Commands

[SOURce:]POWer:LIMit[:POSitive][:IMMediate][:AMPLitude] <value>,(<@chanlist>)

[SOURce:]POWer:LIMit[:POSitive][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](<@chanlist>)

Set and query the power positive limit value. (in W)

If the [SOURce:]POWer:LIMit:COUPle instruction setting status is ON, this instruction sets the positive power limit value and also sets the negative power limit value synchronously.

Command Syntax:

[SOURce:]POWer:LIMit[:POSitive][:IMMediate][:AMPLitude] <value>,(<@chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

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<chanlist>:Channel number.

RST Value:

maximum

Query Syntax:

[SOURce:]POWer:LIMit[:POSitive][:IMMediate][:AMPLitude]? [MINimum|MAXimum,](<@chanlist>)

Return Parameters:

<Power positive limit value>

Example:

POW:LIM 75,(@1) //Set channel 1 power upper limit value to 75W

POW:LIM? (@1) //Query channel 1 power upper limit value

[SOURce:]POWer:LIMit:NEGative[:IMMediate][:AMPLitude]<value>,(<@chanlist>)

[SOURce:]POWer:LIMit:NEGative[:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (<@chanlist>)

Set and query the negative limit value of power. (in W)

If the [SOURce:]POWer:LIMit:COUPle instruction setting status is ON, this instruction sets the negative power limit value and also sets the positive power limit value synchronously.

Command Syntax:

[SOURce:]POWer:LIMit:NEGative[:IMMediate][:AMPLitude]<value>,(<@chanlist>)

Parameter:

<value>:minimum - maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

minimum

Query Syntax:

[SOURce:]POWer:LIMit:NEGative[:IMMediate][:AMPLitude]? [MINimum|MAXimum,] (<@chanlist>)

Return Parameters:

<Negative limit value for power>

Example:

POW:LIM:NEG 20,(@1) //Set channel 1 power negative limit value to 20W

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POW:LIM:NEG? (@1) //Query channel 1 power negative limit value

[SOURce:]POWer:LIMit:COUPle 0|OFF|1|ON,(@<chanlist>)

[SOURce:]POWer:LIMit:COUPle? (@<chanlist>)

This command is used to set or query the state of positive and negative power limit value coupling function.

Command Syntax:

[SOURce:]POWer:LIMit:COUPle 0|OFF|1|ON,(@<chanlist>)

Parameter:

0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

ON

Query Syntax:

[SOURce:]POWer:LIMit:COUPle? (@<chanlist>)

Return Parameters:

0 or 1

Example:

POW:LIM:COUP ON,(@1) //Enable channel 1 positive and negative power limit value coupling

POW:LIM:COUP? (@1) //Query channel 1 positive and negative power limit value coupling status

[SOURce:]POWer:PROTection:DELay[:TIME] <value>,(@<chanlist>)

[SOURce:]POWer:PROTection:DELay[:TIME]? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the delay time of overpower protection. (Unit is second)

Command Syntax:

[SOURce:]POWer:PROTection:DELay[:TIME] <value>,(@<chanlist>)

Parameter:

<TIME>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

2

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Query Syntax:

[SOURce:]POWer:PROTection:DELay[:TIME]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Delay time for overpower protection>

Example:

POW:PROT:DEL 10,(@1) //Set channel 1 overpower protection delay time to 10s

POW:PROT:DEL? (@1) //Query channel 1 overpower protection delay time

[SOURce:]POWer:PROTection:STATe 0|OFF|1|ON,(@<chanlist>)

[SOURce:]POWer:PROTection:STATe? (@<chanlist>)

This command is used to enable or disable the state of overpower protection.

Command Syntax:

[SOURce:]POWer:PROTection:STATe 0|OFF|1|ON,(@<chanlist>)

Parameter:

0|OFF|1|ON

<chanlist>:Channel number.

RST Value:

OFF

Query Syntax:

[SOURce:]POWer:PROTection:STATe? (@<chanlist>)

Return Parameters:

0 or 1

Example:

POWer:PROTection:STATe ON,(@1) //Set channel 1 overpower protection status to ON

POWer:PROTection:STATe? (@1) //Query channel 1 over power protection status

[SOURce:]POWer:PROTection[:LEVel][:POSitive] <value>,(@<chanlist>)

[SOURce:]POWer:PROTection[:LEVel][:POSitive]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the positive protection value for overpower protection. (in W)

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Command Syntax:

[SOURce:]POWer:PROTection[:LEVel][:POSitive] <value>,(@<chanlist>)

Parameter:

<value>:0-maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

maximum

Query Syntax:

[SOURce:]POWer:PROTection[:LEVel][:POSitive]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<positive protection value for overpower protection>

Example:

POWer:PROTection 100, (@1) //Set channel 1 over power positive protection value to 100W

POWer:PROTection? (@1) //Query channel 1 over power positive protection value

[SOURce:]POWer:PROTection[:LEVel]:NEGative <value>,(@<chanlist>)

[SOURce:]POWer:PORTection[:LEVel]:NEGative? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the negative protection value for overpower protection. (in W)

Command Syntax:

[SOURce:]POWer:PROTection[:LEVel]:NEGative <value>,(@<chanlist>)

Parameter:

<value>:minimum -maximum|MINimum|MAXimum

<chanlist>:Channel number.

RST Value:

-maximum

Query Syntax:

[SOURce:]POWer:PORTection[:LEVel]:NEGative?
[MINimum|MAXimum,](@<chanlist>)*RST-MAX

Return Parameters

<Negative protection value for power protection>

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Example:

POW:PORT:NEG 10, (@1) //Set channel 1 overpower protection negative value to 10W

POW:PORT:NEG? (@1) //Query channel 1 over power protection negative value

12.5 Frequency Commands

[SOURce:]FREQuency[:IMMediate] <value|MINimum|MAXimum>,(@<channellist>)

[SOURce:]FREQuency[:IMMediate]? [MINimum|MAXimum,](@<channellist>)

This command is used to set and query the frequency value.

Command Syntax::

[SOURce:]FREQuency[:IMMediate] <value|MINimum|MAXimum>,(@<channellist>)

Arguments:

<value>:minimum-MAXimum|MINimum|MAXimum (单位为 Hz)

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

RST Value:

DEFault

Query Syntax::

[SOURce:]FREQuency[:IMMediate]? [MINimum|MAXimum,](@<channellist>)

Returns:

<frequency level>

Example:

FREQ 200,(@1) // Set the frequency of Channel 1 to 200Hz.

FREQ? (@1) //Query the frequency of Channel 1

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[SOURce:]FREQuency:TRIGgered <value|MINimum|MAXimum>,(@<channellist>)

[SOURce:]FREQuency:TRIGgered? [MINimum|MAXimum,](@<channellist>)

This command is used to set and query the frequency trigger value.

Command Syntax::

[SOURce:]FREQuency:TRIGgered <value|MINimum|MAXimum>,(@<channellist>)

Arguments:

<value>:minimum-MAXimum|MINimum|MAXimum (Unit: Hz)

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax::

[SOURce:]FREQuency:TRIGgered? [MINimum|MAXimum,](@<channellist>)

Returns:

<frequency triggred level>

Example:

FREQ:TRIG 300,(@1) //Set the trigger frequency for Channel 1 to 300Hz.

FREQ:TRIG? (@1) //Query Channel 1 Trigger Frequency

[SOURce:]AC:STATe <ON|OFF|1|0>,(@<channellist>)

[SOURce:]AC:STATe? (@<channellist>)

This command is used to set and query the AC enable status.

Command Syntax:

[SOURce:]AC:STATe <ON|OFF|1|0>,(@<channellist>)

Arguments:

ON|OFF|1|0

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

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RST Value:

0

Query Syntax:

[SOURce:]AC:STATe? (@<channellist>)

Returns:

0|1

Example:

AC:STAT 1,(@1) // Enable Channel 1 AC Status

AC:STAT? (@1) //Query Channel 1 AC Status

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Chapter13 FETCh & MEASure Subsystem

FETCh[:SCALar]:CURRent[:DC]? (@<chanlist>)

MEASure[:SCALar]:CURRent[:DC]? (@<chanlist>)

This command is used to read the average current value.

Command Syntax:

FETCh[:SCALar]:CURRent? (@<chanlist>)

MEASure[:SCALar]:CURRent[:AC]? (@<chanlist>)

Arguments:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<Current>

Example:

FETC:CURR? (@1) // Read the average current value of Channel 1

MEAS:CURR? (@1) // Read the average current value of Channel 1

FETCh[:SCALar]:CURRent:ACDC? (@<chanlist>)

MEASure[:SCALar]:CURRent:ACDC? (@<chanlist>)

This command is used to read the current effective value.

Command Syntax:

FETCh[:SCALar]:CURRent:ACDC? (@<chanlist>)

MEASure[:SCALar]:CURRent:ACDC? (@<chanlist>)

Arguments:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<Current>

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Example:

FETC:CURR:ACDC? (@1) // Read the RMS value of the current in Channel 1

MEAS:CURR:ACDC? (@1) // Read the RMS value of the current in Channel 1

FETCh[:SCALar]:CURRent:MAXimum? (@<chanlist>)

MEASure[:SCALar]:CURRent:MAXimum? (@<chanlist>)

This command is used to read the maximum current value.

Command Syntax:

FETCh[:SCALar]:CURRent:MAXimum? (@<chanlist>)

MEASure[:SCALar]:CURRent:MAXimum? (@<chanlist>)

Arguments:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<Current>

Example:

FETCh:CURRent:MAX? (@1) //Maximum current value for channel 1

MEASure:CURRent:MAX? (@1) //Maximum current value for channel 1

FETCh[:SCALar]:CURRent:MINimum? (@<chanlist>)

MEASure[:SCALar]:CURRent:MINimum? (@<chanlist>)

This command is used to read the minimum current value.

Command Syntax:

FETCh[:SCALar]:CURRent:MINimum? (@<chanlist>)

MEASure[:SCALar]:CURRent:MINimum? (@<chanlist>)

Arguments:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<Current>

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Example:

FETCh:CURR:MIN? (@1) // read the minimum current value.

MEAS:CURRent:MIN? (@1) // read the minimum current value.

FETCh[:SCALar]:POWer[:DC]? (@<chanlist>)

MEASure[:SCALar]:POWer[:DC]? (@<chanlist>)

This command is used to read the power average value.

Command Syntax:

FETCh[:SCALar]:POWer[:DC]? (@<chanlist>)

MEASure[:SCALar]:POWer[:DC]? (@<chanlist>)

Arguments:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<Power>

Example:

FETCh:POWer? (@1) // read the power average value

MEASure:POWer? (@1) // read the power average value

FETCh[:SCALar]:POWer:MAXimum? (@<chanlist>)

MEASure[:SCALar]:POWer:MAXimum? (@<chanlist>)

This command is used to read the maximum power value.

Command Syntax:

FETCh[:SCALar]:POWer:MAXimum? (@<chanlist>)

MEASure[:SCALar]:POWer:MAXimum? (@<chanlist>)

Arguments:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<Power>

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Example:

FETCh:POWer:MAX? (@1) // Read the maximum power value for Channel 1

MEASure:POWer:MAX? (@1) // Read the maximum power value for Channel 1

FETCh[:SCALar]:POWer:MINimum? (@<chanlist>)

MEASure[:SCALar]:POWer:MINimum? (@<chanlist>)

This command is used to read the minimum power value.

Command Syntax:

FETCh[:SCALar]:POWer:MINimum? (@<chanlist>)

MEASure[:SCALar]:POWer:MINimum? (@<chanlist>)

Arguments:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<Power>

Example:

FETCh:POWer:MIN? (@1)

MEAS:POWer:MIN? (@1) //Read the minimum power value.

FETCh[:SCALar]:VOLTage[:DC]? (@<chanlist>)

MEASure[:SCALar]:VOLTage[:DC]? (@<chanlist>)

This command is used to read the average voltage value.

Command Syntax:

FETCh[:SCALar]:VOLTage[:DC]? (@<chanlist>)

MEASure[:SCALar]:VOLTage[:DC]? (@<chanlist>)

Arguments:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<Voltage>

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Example:

FETC:VOLT? (@1)

MEAS:VOLT? (@1) //Read the average voltage value.

FETCh[:SCALar]:VOLTage:LOCal[:DC]? (@<chanlist>)

This command is used to read the average local voltage during the sense state.

Command Syntax:

FETCh[:SCALar]:VOLTage:LOCal[:DC]? (@<chanlist>)

Arguments:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

FETC:VOLT:LOCal[:DC]? (@1) //Read the average local voltage during the sense state.

FETCh[:SCALar]:VOLTage:ACDC? (@<chanlist>)

MEASure[:SCALar]:VOLTage:ACDC? (@<chanlist>)

This command is used to read the effective value of the voltage.

Command Syntax:

FETCh[:SCALar]:VOLTage:ACDC? (@<chanlist>)

MEASure[:SCALar]:VOLTage:ACDC? (@<chanlist>)

Arguments:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<Voltage>

Example:

FETC:VOLT:ACDC? (@1)

MEAS:VOLT:ACDC? (@1) //Read the effective value of the voltage

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FETCh[:SCALar]:VOLTage:MINimum? (@<chanlist>)

MEASure[:SCALar]:VOLTage:MINimum? (@<chanlist>)

This command is used to read the minimum voltage value.

Command Syntax:

FETCh[:SCALar]:VOLTage:MINimum? (@<chanlist>)

MEASure[:SCALar]:VOLTage:MINimum? (@<chanlist>)

Arguments:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<Voltage>

Example:

FETC:VOLT:MIN? (@1)

MEAS:VOLT:MIN? (@1) //Read the minimum voltage value.

FETCh[:SCALar]:VOLTage:MAXimum? (@<chanlist>)

MEASure[:SCALar]:VOLTage:MAXimum? (@<chanlist>)

This command is used to read the maximum voltage value.

Command Syntax:

FETCh[:SCALar]:VOLTage:MAXimum? (@<chanlist>)

MEASure[:SCALar]:VOLTage:MAXimum? (@<chanlist>)

Arguments:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<Voltage>

Example:

FETC:VOLTage:MAX? (@1)

MEAS:VOLTage:MAX? (@1) //Read the maximum voltage value

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FETCh[:SCALar]:AHOur? (@<chanlist>)

MEASure[:SCALar]:AHOur? (@<chanlist>)

This command is used to read the ampere-hour data corresponding to the channel.

Command Syntax:

FETCh[:SCALar]:AHOur? (@<chanlist>)

MEASure[:SCALar]:AHOur? (@<chanlist>)

Arguments:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<NRf>

Example:

FETCh:AHOur? (@1)

MEAS:AHOur? (@1) // read the AH data

FETCh[:SCALar]:WHOur? (@<chanlist>)

MEASure[:SCALar]:WHOur? (@<chanlist>)

This command is used to read the watt-hour data for the channel.

Command Syntax:

FETCh[:SCALar]:WHOur? (@<chanlist>)

MEASure[:SCALar]:WHOur? (@<chanlist>)

Return

<NRf>

Example:

FETC:WHOur? (@1)

MEAS:WHOur? (@1) //Read the WH data for the CH1

FETCh[:SCALar]:TIME? (@<chanlist>)

MEASure[:SCALar]:TIME? (@<chanlist>)

This command is used to obtain the load time for the corresponding channel.

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Command Syntax:

FETCh[:SCALar]:TIME? (@<chanlist>)

MEASure[:SCALar]:TIME? (@<chanlist>)

Arguments:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<NRf>

Example:

FETC:TIME? (@1)

MEAS:TIME? (@1) //Read the load time for CH1

FETCh[:SCALar][:ALL]? (@<chanlist>)

MEASure[:SCALar][:ALL]? (@<chanlist>)

Read all measurement data at once.

(CURR_DC,VOLT_DC,POW_DC,CURR_ACDC,VOLT_ACDC,CURR_MAX,VOLT_MAX,POW_MAX,CURR_MIN,VOLT_MIN,POW_MIN,AH,WH,TIME)

Command Syntax:

FETCh[:SCALar][:ALL]? (@<chanlist>)

MEASure[:SCALar][:ALL]? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return

<NRf>

Example:

FTEC? (@1)

MEAS? (@1) // Read all measurement data at once for CH1

FETCh:ARRay:CURRent[:DC]? (@<chanlist>)

MEASure:ARRay:CURRent[:DC]? (@<chanlist>)

Returns the cumulative measured current data block value, where the measurement depends on the number of sampling points in the meter.

Command Syntax:

FETCh:ARRay:CURRent[:DC]? (@<chanlist>)

MEASure:ARRay:CURRent[:DC]? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return

<Current>

Example:

FETCh:ARRay:CURRent? (@1)

MEASure:ARRay:CURRent? (@1) //Measure current data block for CH1

FETCh:ARRay:VOLTage[:DC]? (@<chanlist>)

MEASure:ARRay:VOLTage[:DC]? (@<chanlist>)

Returns the cumulative measured voltage data block value, where the measurement depends on the number of sampling points in the meter.

Command Syntax:

FETCh:ARRay:VOLTage[:DC]? (@<chanlist>)

MEASure:ARRay:VOLTage[:DC]? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<NRf>

Example:

FETCh:ARRay:VOLTage? (@1)

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MEASure:ARRay:VOLTage? (@1) // Measure the voltage data block value for Channel 1

FETCh:ARRay:POWer[:DC]? (@<chanlist>)

MEASure:ARRay:POWer[:DC]? (@<chanlist>)

Returns the cumulative measured power data block value, where the measurement depends on the number of sampling points in the meter.

Command Syntax:

FETCh:ARRay:POWer[:DC]? (@<chanlist>)

MEASure:ARRay:POWer[:DC]? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<NRf>

Example:

FETCh:ARRay:POWer? (@1)

MEASure:ARRay:POWer? (@1) //read the cumulative measured power data block

FETCh[:SCALar]:ACMeter:EACTotal[:CURRent]?

Used to read the power feedback from this instrument.

Command Syntax::

FETCh[:SCALar]:ACMeter:EACTotal[:CURRent]?

Example:

FETC:ACM:EACT? // read the power feedback

FETCh[:SCALar]:ACMeter:EACTotal:HISTory?

This command is used to read the instrument's historical power feedback.

Command Syntax::

FETCh[:SCALar]:ACMeter:EACTotal:HISTory?

Example:

FETCh:ACM:EACT:HIST? //Read the instrument's historical power feedback.

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FETCh[:SCALar]:ACMeter:FREQuency?

Used to read the input line frequency of this instrument.

Command Syntax::

FETCh[:SCALar]:ACMeter:FREQuency?

Example:

FETCh:ACM:FREQ? //Read the input line frequency of this instrument.

FETCh:SCOPe:ALL? <ITEM>

Display all measurements for the specified waveform between return markers

(order: m1,m2,delta,min,avg,max,ptp,m1_t,m2_t,delta_t,min_t,avg_t,max_t,pp_t)

Command Syntax:

FETCh:SCOPe:ALL? <ITEM>

Arguments

<U1|I1|U2|I2|U3|I3|U4|I4|U5|I5|U6|I6|U7|I7|U8|I8>

Return:

<NRf>

Example:

FETC:SCOP:ALL? U1 //Read all measurements of CH1

FETCh:SCOPe:CURRent[:DC]? (@<chanlist>)

Return the average value of the oscilloscope's current measurement between the markers.

Command Syntax:

FETCh:SCOPe:CURRent[:DC]? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<NRf>

Example:

FETC:SCOP:CURR? (@1) //Obtain the average value of the current in channel 1 between the markers.

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FETCh:SCOPe:VOLTage[:DC]? (@<chanlist>)

Return the average voltage value of the oscilloscope between the markers.

Command Syntax:

FETCh:SCOPe:VOLTage[:DC]? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<NRf>

Example:

FETC:SCOP:VOLT? (@1) //Return the average voltage value of the oscilloscope between the markers.

FETCh:SCOPe:CURRent:MAXimum? (@<chanlist>)

Return the maximum value of the current oscillating between the markers.

Command Syntax:

FETCh:SCOPe:CURRent:MAXimum? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<NRf>

Example:

FETC:SCOP:CURR:MAX? (@1) // Return the maximum value of the current oscillating between the markers.

FETCh:SCOPe:CURRent:MINimum? (@<chanlist>)

Return the minimum value of the oscilloscope's current measurement between the markers.

Command Syntax:

FETCh:SCOPe:CURRent:MINimum? (@<chanlist>)

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Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<NRf>

Example:

FETC:SCOP:CURR:MIN? (@1) // Return the minimum value of the oscilloscope's current measurement between the markers.

FETCh:SCOPe:VOLTage:MAXimum? (@<chanlist>)

Return the maximum voltage value observed between the return marks.

Command Syntax:

FETCh:SCOPe:VOLTage:MAXimum? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<NRf>

Example:

FETC:SCOP:VOLT:MAX? (@1) // Obtain the maximum value of the voltage across channel 1 between the marks.

FETCh:SCOPe:VOLTage:MINimum? (@<chanlist>)

Return the minimum voltage value observed between the return marks.

Command Syntax:

FETCh:SCOPe:VOLTage:MINimum? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

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Return:

<NRf>

Example:

FETC:SCOP:VOLT:MIN? (@1) //Return the minimum voltage value observed between the return marks.

FETCh:SCOPe:CURRent:PTPeak? (@<chanlist>)

Display the peak-to-peak current value between return marks on the oscilloscope.

Command Syntax:

FETCh:SCOPe:CURRent:PTPeak? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<NRf>

Example:

FETC:SCOP:CURR:PTP? (@1) //Read the Ipeak between return marks

FETCh:SCOPe:VOLTage:PTPeak? (@<chanlist>)

Display the peak-to-peak voltage between the return marks.

Command Syntax:

FETCh:SCOPe:VOLTage:PTPeak? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<NRf>

Example:

FETC:SCOP:VOLT:PTP? (@1) // Obtain the peak-to-peak voltage of channel 1 between the marks

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FETCh:DLOG:ALL? (@<chanlist>)

Return all values measured by the internal recorder in the following sequence: (AHO,WHO,CURR_DC,VOLT_DC,CURR_MAX,VOLT_MAX,CURR_MIN,VOLT_MIN,CURR_PTP,VOLT_PTP)

Command Syntax:

FETCh:DLOG:ALL? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

FETC:DLOG:ALL? (@1) //Return all values measured by the internal recorder

FETCh:DLOG:AHOur? (@<chanlist>)

Return the AH data from the internal recorder.

Command Syntax:

FETCh:DLOG:AHOur? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

FETC:DLOG:AHO? (@1) // Read the AH data from the internal recorder

FETCh:DLOG:WHOur? (@<chanlist>)

Read the internal recorder's watt-hour data.

Command Syntax:

FETCh:DLOG:WHOur? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

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Example:

FETC:DLOG:WHO? (@1) //Read the watt-hour information measured by the internal recorder of Channel 1

FETCh:DLOG:CURRent[:DC]? (@<chanlist>)

Return the average current value from the internal recorder.

Command Syntax:

FETCh:DLOG:CURRent[:DC]? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

FETC:DLOG:CURR? (@1) //Read the average current value from the internal recorder.

FETCh:DLOG:VOLTage[:DC]? (@<chanlist>)

Read the average voltage value from the internal recorder.

Command Syntax:

FETCh:DLOG:VOLTage[:DC]? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

FETC:DLOG:VOLT? (@1) //Read the average voltage value from the internal recorder

FETCh:DLOG:CURRent:MAXimum? (@<chanlist>)

Returns the maximum current value recorded internally.

Command Syntax:

FETCh:DLOG:CURRent:MAXimum? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

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(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

FETC:DLOG:CURR:MAX? (@1) //Returns the maximum current value recorded internally.

FETCh:DLOG:CURRent:MINimum? (@<chanlist>)

Returns the minimum current value recorded internally.

Command Syntax:

FETCh:DLOG:CURRent:MINimum? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

FETC:DLOG:CURR:MIN? (@1) // Returns the minimum current value recorded internally.

FETCh:DLOG:VOLTage:MAXimum? (@<chanlist>)

Return the maximum voltage value recorded internally.

Command Syntax:

FETCh:DLOG:VOLTage:MAXimum? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

FETC:DLOG:VOLT:MAX? (@1) // Return the maximum voltage value recorded internally.

FETCh:DLOG:VOLTage:MINimum? (@<chanlist>)

Returns the minimum voltage recorded internally.

Command Syntax:

FETCh:DLOG:VOLTage:MINimum? (@<chanlist>)

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Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

FETC:DLOG:VOLT:MIN? (@1) // Returns the minimum voltage recorded internally.

FETCh:DLOG:CURRent:PTPeak? (@<chanlist>)

Return the peak-to-peak value of the measured current between time stamps M1 and M2 in the data logger.

Command Syntax:

FETCh:DLOG:CURRent:PTPeak? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<NRf>

Example:

FETC:DLOG:CURR:PTP? (@1) //Return the peak-to-peak current value to the internal recorder.

FETCh:DLOG:VOLTage:PTPeak? (@<chanlist>)

Return the peak-to-peak value of the measured voltage between time stamps M1 and M2 in the data logger.

Command Syntax:

FETCh:DLOG:VOLTage:PTPeak? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

<NRf>

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Example:

FETC:DLOG:VOLT:PTP? (@1) // Return the peak-to-peak voltage of the internal recorder

FETCh:ELOG? <maxrecords>, (@<chanlist>)

Return data from the nearest external data logger

Command Syntax::

FETCh:ELOG? <maxrecords>, (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

FETC:ELOG? (@1) //Return data from the nearest external data logger

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Chapter14 SCOPe Subsystems

SCOPe:AUTO

Oscilloscope function automatically

Command Syntax:

SCOPe:AUTO

Example:

SCOP:AUTO //Starting automatic setup

SCOPe:RUN

Oscilloscope function operation

Command Syntax:

SCOPe:RUN

Example:

SCOP:RUN //Start running the oscilloscope

SCOPe:SINGle

Oscilloscope function single run

Command Syntax:

SCOPe:SINGle

Example:

SCOP:SING //Single Capture Oscilloscope

SCOPe:STOP

Oscilloscope function stops

Command Syntax:

SCOPe:STOP

Example:

SCOP:STOP //Stop Oscilloscope

SCOPe:FORCe:TRIGger

Oscilloscope forced trigger

Command Syntax:

SCOPe:FORCe:TRIGger

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Example:

SCOP:FORCe:TRIGger //Force Trigger Oscilloscope

SCOPe:CLEar

Oscilloscope clear screen

Command Syntax:

SCOPe:CLEar

Example:

SCOP:CLEar //Clear Oscilloscope Display

SCOPe:TIMebase:SCALe <value|MINimum|MAXimum>

SCOPe:TIMebase:SCALe? [MINimum|MAXimum]

The commands set and query the time base of the oscilloscope

Command Syntax:

SCOPe:TIMebase:SCALe <value|MINimum|MAXimum>

Parameters:

<value|MINimum|MAXimum>

Query Syntax:

SCOPe:TIMebase:SCALe? [MINimum|MAXimum]

Return Parameters:

<time base>

Example:

SCOP:TIM:SCAL 200.0ms //Setting the oscilloscope's time base to 200.0ms

SCOP:TIM:SCAL? //Query the time base of the oscilloscope

SCOPe:TIMebase:DELay <value|MINimum|MAXimum>

SCOPe:TIMebase:DELay? [MINimum|MAXimum]

The commands set and query the delay time of the oscilloscope

Command Syntax:

SCOPe:TIMebase:DELay <value|MINimum|MAXimum>

Parameters:

<value|MINimum|MAXimum>

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Query Syntax:

SCOPe:TIMebase:DELay? [MINimum|MAXimum]

Return Parameters:

<delay time value>

Example:

SCOP:TIM:DEL 100.0ms //Set the delay time of the oscilloscope to 100.0ms

SCOP:TIM:DEL? //Query the delay time of the oscilloscope

SCOPe: TRIGger:LEVel <value|MINimum|MAXimum>

SCOPe:TRIGger:LEVel? [MINimum|MAXimum]

The commands set and query the trigger level of the oscilloscope.

Command Syntax:

SCOPe:TRIGger:LEVel <value|MINimum|MAXimum>

Parameters:

<U1|I1|U2|I2|U3|I3|U4|I4|U5|I5|U6|I6|U7|I7|U8|I8>

Query Syntax:

SCOPe:TRIGger:LEVel? [MINimum|MAXimum]

Return Parameters:

<Trigger level value>

Example:

SCOP:TRIG:LEV 3.0 //Set the trigger level of the oscilloscope to 3.0V

SCOP:TRIG:LEV? //Query the trigger level of the oscilloscope

SCOPe:TRIGger:SOURce <source>

SCOPe:TRIGger:SOURce?

The commands set and query the trigger source of the oscilloscope

Command Syntax:

SCOPe:TRIGger:SOURce <source>

Parameters:

U1|I1|U2|I2|U3|I3|U4|I4|U5|I5|U6|I6|U7|I7|U8|I8|PIN1|PIN2|PIN3|PIN4|PIN5|PIN6|PIN7

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Query Syntax:

SCOPe:TRIGger:SOURce?

Return Parameters:

U1,I1,U2,I2,U3,I3,U4,I4,U5,I5,U6,I6,U7,I7,U8,I8,PIN1,PIN2,PIN3,PIN4,PIN5,PIN6 or PIN7

Example:

SCOP:TRIG:SOUR PIN7 //Set the trigger source of the oscilloscope to trigger digital port 7

SCOP:TRIG:SOUR? //Query the trigger source of the oscilloscope

SCOPe:TRIGger:MODE <AUTO|NORMal>

SCOPe:TRIGger:MODE?

The commands set and query the trigger mode of the oscilloscope

Command Syntax:

SCOPe:TRIGger:MODE <AUTO|NORMal>

Parameters:

AUTO|NORMal

Query Syntax:

SCOPe:TRIGger:MODE?

Return Parameters:

AUTO or NORM

Example:

SCOP:TRIG:MODE AUTO //Set the trigger mode of the oscilloscope to Auto

SCOP:TRIG:MODE? //Query the trigger mode of the oscilloscope

SCOPe:LINE:SELection <1-6>,<0|OFF|1|ON>[,ITEM]

SCOPe:LINE:SELection? <1-6>

The commands set and query the selection of each line

Command Syntax:

SCOPe:LINE:SELection <1-6>,<0|OFF|1|ON>[,ITEM]

Parameters:

<1-6>:Lines 1~6

<0|OFF|1|ON>:Open and Close

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<U1|I1|U2|I2|U3|I3|U4|I4|U5|I5|U6|I6|U7|I7|U8|I8>:signal source

Query Syntax:

SCOPe:LINE:SELection?

Return Parameters:

0 or 1,<U1,I1,U2,I2,U3,I3,U4,I4,U5,I5,U6,I6,U7,I7,U8,I8>

Example:

SCOP:LINE:SEL 1,ON,U1 //Select line 1 for voltage, display turns on

SCOP:LINE:SEL? 2 //Queries the status of line 2 and displays the waveform

SCOPe:STATus?

This command queries the status of the oscilloscope

Query Syntax:

SCOPe:STATus?

Return Parameters:

"Stop",“Ready”,”Roll”,“Auto”,“Trig'd”,”Arm”,” Acq complete”

//"Stop":The oscilloscope has stopped capturing data. In this state, the oscilloscope will not update the waveform on the screen, and the user can view and analyze the data that has been captured.

//"Ready ":The oscilloscope is ready to capture new data, but has not yet started. Usually indicates that the oscilloscope is waiting for a trigger event.

//"Roll":This mode is typically used for low frequency signals. The oscilloscope updates the screen continuously, similar to the scrolling of an old-fashioned paper tape recorder. The waveform scrolls from one side of the screen to the other, displaying the input signal in real time.

//"Auto":Auto Trigger Mode. If the oscilloscope does not detect a valid trigger signal within a preset period of time, it will automatically display the waveform. This ensures that the waveform can be viewed continuously even when there is no trigger event.

//"Trig'd":The oscilloscope has detected an event that meets the trigger condition and has started to capture data. This status indicates that the trigger condition has been met and waveform data is being recorded.

//"Arm":The oscilloscope is waiting for a trigger condition to be met to begin data capture. This state is usually used for more complex trigger configurations, such as sequence triggering OR delay triggering.

//"Acq complete":The oscilloscope has completed capturing the data. In this state, all scheduled waveform data has been recorded and the user can begin analysis.

Example:

SCOP:STAT? //Query the status of the oscilloscope

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SCOPe:RSTate?

This command is used to query the operation status of the oscilloscope

Query Syntax:

SCOPe:RSTate?

Return Parameters:

“RUN”,“STOP”,”SINGle”

Example:

SCOP:RST? //Query the operation status of the oscilloscope

SCOPe:WAVeform:DATA?

The command queries the data of the oscilloscope. The return format depends on the setting of FORMat:BORDer and FORMat[:DATA] commands.If the data format is set to ASCII, the value returned will be comma-separated character data.If the data format is set to REAL, the data is returned in the format #<data length bits><8-bit data byte length><transmitted data>, and the transmitted data specifies the byte order in the FORMat:BORDer configuration item.

Query Syntax:

SCOPe:WAVeform:DATA?

Return Parameters:

<Data1>,<Data2>,...

Example:

SCOP:WAV:DATA? //Query oscilloscope processed data

SCOPe:RECord:LENGth <0.6|6|60|600>

SCOPe:RECord:LENGth?

The commands set and query the record length of the oscilloscope in kpts.

Command Syntax:

SCOPe:RECord:LENGth <0.6|6|60|600>

Parameters:

0.6|6|60|600

Query Syntax:

SCOPe:RECord:LENGth?

Return Parameters:

0.6,6,60or600

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Example:

SCOP:REC:LENG 60 //Set the oscilloscope record length to 60kpts

SCOPe:RECord:LENGth? //Querying Oscilloscope Record Length

SCOPe:SAMPle:MODE <NORMal|PEAK>

SCOPe:SAMPle:MODE?

The commands set and query the sampling mode of the oscilloscope

Command Syntax:

SCOPe:SAMPle:MODE <NORMal|PEAK>

Parameters:

NORMal|PEAK

Query Syntax:

SCOPe:SAMPle:MODE?

Return Parameters:

NORMorPEAK

Example:

SCOP:SAMP:MODE NORM //Set the sampling mode of the oscilloscope to NORM

SCOPe:SAMPle:MODE? //Query the sampling mode of the oscilloscope

SCOPe:DATA:TAG?

This command is used to query the sampling data identification of the oscilloscope

Query Syntax:

SCOPe:DATA:TAG?

Return Parameters:

<display start time point value>,<display end time point value> ,<data start time point value> ,<data end time point value> ,<data trigger reference point value>

Example:

SCOP:DATA:TAG? //Querying the oscilloscope's data identification

SCOPe:MARKer1:POINt <value|MINimum|MAXimum>

SCOPe:MARKer1:POINt? [MINimum|MAXimum]

The commands set and query the position of data marker point 1

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Command Syntax:

SCOPe:MARKer1:POINt <value|MINimum|MAXimum>

Parameters:

<0-599|MINimum|MAXimum>

Query Syntax:

SCOPe:MARKer1:POINt? [MINimum|MAXimum]

Return Parameters:

<marker position value>

Example:

SCOP:MARK1:POIN 20.0 //Set the position of measurement marker point 1 to 20

SCOP:MARK1:POIN? //Query the position of measurement marker point 1

SCOPe:MARKer2:POINt <value|MINimum|MAXimum>

SCOPe:MARKer2:POINt? [MINimum|MAXimum]

The commands set and query the position of data marker point 2

Command Syntax:

SCOPe:MARKer2:POINt <value|MINimum|MAXimum>

Parameters:

<0-599|MINimum|MAXimum>

Query Syntax:

SCOPe:MARKer2:POINt? [MINimum|MAXimum]

Return Parameters:

<marker position value>

Example:

SCOP:MARK2:POIN 20.0 //Set the position of the measurement marking point 2 to 20

SCOP:MARK2:POIN? //Query the position of measurement marker point 2

SCOPe:VERTical:COUPle <ITEM>,<DC|AC>

SCOPe:VERTical:COUPle? <ITEM>

The commands set and query the coupling of the oscillometric item

Command Syntax:

SCOPe:VERTical:COUPle <ITEM>,<DC|AC>

Parameters:

<U1|I1|U2|I2|U3|I3|U4|I4|U5|I5|U6|I6|U7|I7|U8|I8>,<DC|AC>

Query Syntax:

SCOPe:VERTical:COUPle? <ITEM>

Return Parameters:

DC or AC

Example:

SCOP:VERT:COUP U1,AC //Set the coupling method of the oscilloscope channel 1 voltage to AC

SCOP:VERT:COUP? //Query the coupling method of the oscillator channel 1 voltage

SCOPe:VERTical:INVert <ITEM>,<0|OFF|1|ON>

SCOPe:VERTical:INVert? <ITEM>

The commands set and query the reverse display state of the oscillometric item

Command Syntax:

SCOPe:VERTical:INVert <ITEM>,<0|OFF|1|ON>

Parameters:

<U1|I1|U2|I2|U3|I3|U4|I4|U5|I5|U6|I6|U7|I7|U8|I8>,<0|OFF|1|ON>

Query Syntax:

SCOPe:VERTical:INVert? <ITEM>

Return Parameters:

0 or 1

Example:

SCOP:VERT:INV I1,ON //Set the current reverse display state of oscilloscope channel 1 to ON.

SCOP:VERT:INV? I1 //Query the current reverse display status of channel 1

SCOPe:VERTical:POSition <1-6>,<MINimum|MAXimum>

SCOPe:VERTical:POSition? <1-6>,[,MINimum|MAXimum]

The commands set and query the position of the oscillometric item

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Command Syntax:

SCOPe:VERTical:POSition <1-6>,<MINimum|MAXimum>

Parameters:

<1-6>,<level|MINimum|MAXimum>

Query Syntax:

SCOPe:VERTical:POSition? <1-6>,[,MINimum|MAXimum]

Return Parameters:

<position value>

Example:

SCOP:VERT:POS 1,1.0 //Set the position of line 1 data to 1div

SCOP:VERT:POS? 1 //Query the position of line 1 data

SCOPe:VERTical:SCALe <ITEM>,<level|MINimum|MAXimum>

SCOPe:VERTical:SCALe? <ITEM>[,MINimum|MAXimum]

The commands set and query the vertical gear of the oscilloscope

Command Syntax:

SCOPe:VERTical:SCALe <ITEM>,<level|MINimum|MAXimum>

Parameters:

<ITEM>,<level|MINimum|MAXimum>

Query Syntax:

SCOPe:VERTical:SCALe? <ITEM>[,MINimum|MAXimum]

Return Parameters:

<scale value>

Example:

SCOP:VERT:SCAL,U1,10.0 //Set the vertical gear of channel 1 voltage to 10.0V

SCOP:VERT:SCAL? U1 //Query vertical gear for channel 1.

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SCOPe:VERTical:HPF:FREQuency <ITEM>,<freq|MINimum|MAXimum>

SCOPe:VERTical:HPF:FREQuency? <ITEM>[,MINimum|MAXimum]

The commands set and query the high-pass cutoff frequency of the oscillometric term AC coupling

Command Syntax:

SCOPe:VERTical:HPF:FREQuency <ITEM>,<freq|MINimum|MAXimum>

Parameters:

<U1|I1|U2|I2|U3|I3|U4|I4|U5|I5|U6|I6|U7|I7|U8|I8> [,MINimum|MAXimum]

Query Syntax:

SCOPe:VERTical:HPF:FREQuency? <ITEM>[,MINimum|MAXimum]

Return Parameters:

<High-pass cutoff frequency value>

Example:

SCOP:VER:HPF:FREQ U1,1.0 //Set the high pass cutoff frequency for channel 1 voltage to 1.0hz

SCOP:VERT:HPF:FREQ? U1 //Query the high pass cutoff frequency for channel 1 voltage.

SCOPe:VERTical:SCALe:CATalog? <ITEM>

The command queries the scale range of the oscilloscope measurement range.

Query Syntax:

SCOPe:VERTical:SCALe:CATalog? <ITEM>

Parameters:

<U1|I1|U2|I2|U3|I3|U4|I4|U5|I5|U6|I6|U7|I7|U8|I8>

Return Parameters:

<div1>,<div2>,<div3>… //The scale is a small to large scale of selections, the selections are arranged in the relationship 1,2,5

Example:

SCOP:VERT:SCAL:CAT? U1 // Query Oscilloscope U1 Scale Range

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SCOPe:TRIGger:LEVel:HYSTeresis <ITEM>,<value|MINimum|MAXimum>

SCOPe:TRIGger:LEVel:HYSTeresis? <ITEM>,[MINimum|MAXimum]

This command is used to set and query the trigger threshold of the oscilloscope.

Command Syntax:

SCOPe:TRIGger:LEVel:HYSTeresis <ITEM>,<value|MINimum|MAXimum>

Arguments

<U1|I1|U2|I2|U3|I3|U4|I4|U5|I5|U6|I6|U7|I7|U8|I8>

<value|MINimum|MAXimum>

查询命令

SCOPe:TRIGger:LEVel:HYSTeresis? <ITEM>,[MINimum|MAXimum]

Returns

<trig level hyst>

Example

SCOP:TRIG:LEV:HYST U1,3.0 //Set the voltage trigger threshold for Oscilloscope Channel 1 to 3.0V.

SCOP:TRIG:LEV:HYST? U1 // Query the trigger threshold for the voltage on Oscilloscope Channel 1

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Chapter15 SENSe Subsystem

This chapter describes the Sense subsystem related commands.

SENSe[:REMote][:STATe] 0|OFF|1|ON,(@<channlist>)

SENSe[:REMote][:STATe]? (@<channlist>)

Set and query the status of remote sensing.

Command Syntax:

SENSe[:REMote][:STATe] 0|OFF|1|ON, (@<channlist>)

Parameters:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return Parameters:

0 or 1

Query Syntax:

SENSe[:REMote][:STATe]? (@<channlist>)

Example:

SENS ON,(@1) //Enable remote sensing for Channel 1

SENS? (@1) //Query the remote sensing function of Channel 1

SENSe:VOLTage[:DC]:RANGe[:UPPer] <value>, (@<chanlist>)

SENSe:VOLTage[:DC]:RANGe[:UPPer]? [MINimum|MAXimum,] (@<chanlist>)

Set and query the voltage range.

Command Syntax:

SENSe:VOLTage[:DC]:RANGe[:UPPer] <value>, (@<chanlist>)

Arguments

<value>: minmimum to maximum|MINimum|MAXimum (Unit:V)

<chanlist>:Channel number.

(@1) – Channel 1

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(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

RST value:

MAXimum

Return:

<voltage range>

Query Syntax

SENSe:VOLTage[:DC]:RANGe[:UPPer]? [MINimum|MAXimum,] (@<chanlist>)

Example:

SENS:VOLT:RANG 30.0,(@1) // Set the voltage range for Channel 1 to 30.0V.

SENS:VOLT:RANG? (@1) // Query the voltage range for Channel 1

SENSe:VOLTage[:DC]:RANGe:AUTO 0|OFF|1|ON, (@<chanlist>)

SENSe:VOLTage[:DC]:RANGe:AUTO? (@<chanlist>)

Enable or disable automatic voltage range.

Command Syntax:

SENSe:VOLTage[:DC]:RANGe[:UPPer] <value>, (@<chanlist>)

Arguments

0|OFF|1|ON

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

RST value:

ON

Returns:

0 or 1

Query Syntax

SENSe:VOLTage[:DC]:RANGe:AUTO? (@<chanlist>)

Example:

SENS:VOLT:RANG:AUTO 0,(@1) // Enable automatic voltage range.

SENS:VOLT:RANG:AUTO? (@1) //Query automatic voltage range status

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SENSe:SWEep:OFFSet:POINts <value>, (@<chanlist>)

SENSe:SWEep:OFFSet:POINts? [MINimum|MAXimum,] (@<chanlist>)

This command defines the offset of the data scan when triggering a measurement. The programmed values range from - (SENS:SWE:POIN setting -1) to 2,000,000,000 (2E9).Positive values indicate the delay time after the trigger occurs and before the sample is taken. Negative values indicate data samples taken before the trigger. The programmable maximum negative value is one less than the maximum allowed value for SENSe:SWEep:POINts.

Command Syntax:

SENSe:SWEep:OFFSet:POINts <value>, (@<chanlist>)

Parameters:

<Value>:[MINimum|MAXimum]

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return Parameters:

<Offset Points>

Query Syntax:

SENSe:SWEep:OFFSet:POINts? [MINimum|MAXimum,] (@<chanlist>)

Example:

SENS:SWE:OFFS:POIN -1024,(@1) //Set the measurement scan offset for Channel 1 to -1024

SENS:SWE:OFFS:POIN? (@1) //Queries the measured scanning offset of Channel 1

SENSe:SWEep:POINts <value>, (@<chanlist>)

SENSe:SWEep:POINts? [MINimum|MAXimum,](@<chanlist>)

This command defines the number of points in the measurement.The number of samples (points) that can be specified depends on the number of measurement Parameters: selected. Up to 8 Parameters: (voltage and current X 4 outputs) can be measured.

Command Syntax:

SENSe:SWEep:POINts <value>, (@<chanlist>)

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Parameters:

<Value>:1 to 16384|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return Parameters:

<scanning point>

Query Syntax:

SENSe:SWEep:POINts? [MINimum|MAXimum,] (@<chanlist>)

Example:

SENS:SWE:POIN 1024,(@1) //Set the number of measurement scan points for Channel 1 to 1024.

SENS:SWE:POIN? (@1) //Queries the number of measurement scan points for Channel 1.

SENSe:SWEep:TINTerval <value>, (@<chanlist>)

SENSe:SWEep:TINTerval? [MINimum|MAXimum,] (@<chanlist>)

This command defines the sampling interval time in seconds.Programmed values range from 5 microseconds 2100.Note that the shortest time interval (fastest speed) that can be specified depends on the number of measurement Parameters: and the model for which the measurement is performed.

Command Syntax:

SENSe:SWEep:TINTerval <value>, (@<chanlist>)

Parameter:

<Value>:0.000005-2100.0|MINimum|MAXimum

<chanlist>:Channel number

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return Parameters:

<sampling interval>

Query Syntax:

SENSe:SWEep:TINTerval? [MINimum|MAXimum,] (@<chanlist>)

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Example:

SENS:SWE:TINT 0.002,(@1) //0.002s.Set the measurement scanning interval of Channel 1 to 0.002s.

SENS:SWE:TINT? (@1) //Queries the measurement scanning interval of Channel 1.

SENSe:NPLCycles <arg|MINimum|MAXimum>,(@<chanlist>)

SENSe:NPLCycles? [MINimum|MAXimum,](@<chanlist>)

This command queries the number of measured IF cycles.

Command Syntax:

SENSe:NPLCycles <arg|MINimum|MAXimum>,(@<chanlist>)

Parameter:

<Value>:

1.0-172032000|MINimum|MAXimum (50HZ)

1.2-2064384000|MINimum|MAXimum (60HZ)

<chanlist>:Channel number

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

RST Value:

10 (50HZ)

12 (60HZ)

Query Syntax:

SENSe:NPLCycles? [MINimum|MAXimum,](@<chanlist>)

Retur:

<nplc>

Example:

SENS:NPLC 10,(@1) //Set the industrial frequency period of Channel 1 to 10.

SENS:NPLC? (@1) //Query the frequency period of Channel 1

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SENSe:APERture <time|MINimum|MAXimum>,(@<chanlist>)

SENSe:APERture? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the measured aperture time.

Command Syntax:

SENSe:APERture <time|MINimum|MAXimum>,(@<chanlist>)

Parameter:

<Value>:0.02-34406400|MINimum|MAXimum

<chanlist>:Channel number

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return:

[MINimum|MAXimum]

Query Syntax:

SENSe:APERture? [MINimum|MAXimum,] (@<chanlist>)

Example:

SENS:APER 0.1,(@1) //Set the measurement aperture time for Channel 1 to 0.1s.

SENS:APER? (@1) //Querying the measurement aperture time of Channel 1

SENSe:AHOur:CLEar (@<chanlist>)

This instruction is used to clear the ampersand data.

Clearing the amp-hour data will clear the corresponding watt-hour data at the same time.

Command Syntax:

SENSe:AHOur:CLEar (@<chanlist>)

Parameter:

<chanlist>:Channel number

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

SENS:AHO:CLE (@1) //Clearing the an hourly statistics for Channel 1

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SENSe:WHOur:CLEar (@<chanlist>)

This instruction is used to clear the watt-hour data.

Clearing the watt-hour data will clear the corresponding amp-hour data at the same time.

Command Syntax:

SENSe:AHOur:CLEar (@<chanlist>)

Parameter:

<chanlist>:Channel number

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

SENS:WHO:CLE (@1) //Clear the watt-hour statistics for Channel 1

SENSe:AHOur:CLEar:AUTO <0|OFF|1|ON>,(@<chanlist>)

SENSe:AHOur:CLEar:AUTO? (@<chanlist>)

This instruction is used to set up the automatic clearing of the amperage data.

The watt-hour data will also be automatically cleared.

Command Syntax:

SENSe:AHOur:CLEar (@<chanlist>)

Parameter:

<0|OFF|1|ON>

<chanlist>:Channel number

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax::

SENSe:AHOur:CLEar:AUTO? (@<chanlist>)

Example:

SENS:AHO:CLE:AUTO 1,(@1) //Enable the auto-clear state of Channel 1 security time data

SENS:AHO:CLE:AUTO? (@1) //Queries the auto-clear status of Channel 1 ATS data.

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SENSe:WHOur:CLEar:AUTO <0|OFF|1|ON>,(@<chanlist>)

SENSe:WHOur:CLEar:AUTO? (@<chanlist>)

This instruction is used to set up automatic clearing of watt-hour data.

An hour data will also be cleared automatically.

Command Syntax:

SENSe:WHOur:CLEar:AUTO <0|OFF|1|ON>,(@<chanlist>)

Parameter:

<0|OFF|1|ON>

<chanlist>:Channel number

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax::

SENSe:WHOur:CLEar:AUTO? (@<chanlist>)

Example:

SENS:WHO:CLE:AUTO ON,(@1) //Enable Channel 1 watt-hour statistics auto-clear state

SENS:WHO:CLE:AUTO? (@1) //Queries the auto-clear status of Channel 1 watt-hour statistics

SENSe:DLOG:FUNCtion:CURRent 0|OFF|1|ON,(@<chanlist>)

SENSe:DLOG:FUNCtion:CURRent? (@<chanlist>)

This command is used to set and query the current recording status of the data logger.

Command Syntax:

SENSe:DLOG:FUNCtion:CURRent 0|OFF|1|ON,(@<chanlist>)

Parameters:

0|OFF|1|ON

<chanlist>:Channel number

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

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Query Syntax:

SENSe:DLOG:FUNCtion:CURRent? (@<chanlist>)

Return Parameters:

0 or 1

Example:

SENS:DLOG:FUNC:CURR ON,(@1) //Turn on data logger Channel 1 current logging

SENS:DLOG:FUNC:CURR? (@1) //Query data logger Channel 1 current logging status

SENSe:DLOG:FUNCtion:VOLTage 0|OFF|1|ON,(@<chanlist>)

SENSe:DLOG:FUNCtion:VOLTage? (@<chanlist>)

This command is used to set and query the voltage recording status of the data logger.

Command Syntax:

SENSe:DLOG:FUNCtion:VOLTage 0|OFF|1|ON,(@<chanlist>)

Parameters:

0|OFF|1|ON

<chanlist>:Channel number

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

SENSe:DLOG:FUNCtion:VOLTage? (@<chanlist>)

Return Parameters:

0 or 1

Example:

SENS:DLOG:FUNC:CURR ON,(@1) //Turn on the data logger Channel 1 voltage logging

SENS:DLOG:FUNC:CURR? (@1) //Query data logger Channel 1 voltage logging status

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SENSe:DLOG:FUNCtion:MINMax 0|OFF|1|ON

SENSe:DLOG:FUNCtion:MINMax?

This command is used to set and query the maximum and minimum value recording function of the data logger.

Command Syntax:

SENSe:DLOG:FUNCtion:MINMax 0|OFF|1|ON

Parameters:

0|OFF|1|ON

Query Syntax:

SENSe:DLOG:FUNCtion:MINMax?

Return Parameters:

0 or 1

Example:

SENS:DLOG:FUNC:MINM ON //Open Data Logger Maximum Minimum Record

SENS:DLOG:FUNC:MINM? //Query Data Logger Maximum and Minimum Value Recording Status

SENSe:DLOG:MARKer1:POINt <value>

SENSe:DLOG:MARKer1:POINt?

This command is used to set and query the position of the data logger time marker M1. (Unit is s)

Command Syntax:

SENSe:DLOG:MARKer1:POINt <value>

Parameters:

-359999999-359999999|MINimum|MAXimum

Query Syntax:

SENSe:DLOG:MARKer1:POINt?

Return Parameters:

<position of marker 1>

Example:

SENS:DLOG:MARKer1:POIN 20 //Set the position of the data logger M1 at 20s

SENS:DLOG:MARKer1:POIN? //Query the location of data logger M1

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SENSe:DLOG:MARKer2:POINt <value>

SENSe:DLOG:MARKer2:POINt?

This command is used to set and query the position of the data logger time marker M2. (Unit is s)

Command Syntax:

SENSe:DLOG:MARKer2:POINt <value>

Parameters:

-359999999-359999999|MINimum|MAXimum

Query Syntax:

SENSe:DLOG:MARKer2:POINt?

Return Parameters:

<position of marker 2>

Example:

SENS:DLOG:MARKer2:POIN 30 //Set the position of data logger M2 at 30s

SENS:DLOG:MARKer2:POIN? //Query the location of data logger M2

SENSe:DLOG:OFFSet <value>

SENSe:DLOG:OFFSet?

This command is used to set and query the trigger offset time percentage of the data logger. (Unit is %)

Command Syntax:

SENSe:DLOG:OFFSet <value>

Parameters:

0.00-100.00|MINimum|MAXimum

Query Syntax:

SENSe:DLOG:OFFSet?

Return Parameters:

<trigger offset>

Example:

SENS:DLOG:OFFS 20 //Setting the data logger's trigger offset bit by 20%

SENS:DLOG:OFFS? //Query Data Logger Trigger Offset

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SENSe:DLOG:PERiod <value>

SENSe:DLOG:PERiod?

This command is used to set and query the integration time of the data logger. (Unit is s)

Command Syntax:

SENSe:DLOG:PERiod <value>

Parameters:

0.00002-60|MINimum|MAXimum

Query Syntax:

SENSe:DLOG:PERiod? [MINimum|MAXimum]

Return Parameters:

<Points time>

Example:

SENS:DLOG:PER 0.02 //Set the integration time of the data logger to 0.02s

SENS:DLOG:PER? //Query the integration time of the data logger

SENSe:DLOG:TIME <value>

SENSe:DLOG:TIME?

This command is used to set and query the time of the data logger. (Unit is s)

Command Syntax:

SENSe:DLOG:TIME <value>

Parameters:

0-359999999|MINimum|MAXimum

Query Syntax:

SENSe:DLOG:TIME? [MINimum|MAXimum]

Return Parameters:

<record time>

Example:

SENS:DLOG:TIME 50 //Set the data logger time to 50s

SENS:DLOG:TIME? //Setting the data logger time

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SENSe:DLOG:TINTerval <value>

SENSe:DLOG:TINTerval?

This command is used to set and query the time interval of the data logger. (Unit is s)

Command Syntax:

SENSe:DLOG:TINTerval <value>

Parameters:

0.00002-60|MINimum|MAXimum

Query Syntax:

SENSe:DLOG:TINTerval? [MINimum|MAXimum]

Return Parameters:

<time interval>

Example:

SENS:DLOG:TINT 0.02 //Set the time interval of the data logger to 0.02s

SENS:DLOG:TINT? //Interval for querying the data logger

SENSe:ELOG:FUNCtion:CURRent 0|OFF|1|ON,(@<chanlist>)

SENSe:ELOG:FUNCtion:CURRent? (@<chanlist>)

This command is used to set and query the current logging status of the external logger.

Command Syntax:

SENSe:ELOG:FUNCtion:CURRent 0|OFF|1|ON, (@<chanlist>)

Parameters:

0|OFF|1|ON

<chanlist>:Channel number

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

SENSe:ELOG:FUNCtion:CURRent? (@<chanlist>)

Return Parameters:

0 or 1

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Example:

SENS:ELOG:FUNC:CURR ON,(@1) //Setting to turn on Channel 1 external logger current logging

SENS:ELOG:FUNC:CURR? (@1) //Queries whether Channel 1 external recorder current logging is turned on or not

SENSe:ELOG:FUNCtion:VOLTage 0|OFF|1|ON,(@<chanlist>)

SENSe:ELOG:FUNCtion:VOLTage? (@<chanlist>)

This command is used to set and query the voltage recording status of the external recorder.

Command Syntax:

SENSe:ELOG:FUNCtion:VOLTage 0|OFF|1|ON, (@<chanlist>)

Parameters:

0|OFF|1|ON

<chanlist>:Channel number

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

SENSe:ELOG:FUNCtion:VOLTage? (@<chanlist>)

Return Parameters:

0 or 1

Example:

SENS:ELOG:FUNC:VOLT ON,(@1) //Setting to turn on Channel 1 external logger voltage recording

SENS:ELOG:FUNC:VOLT? (@1) //Queries whether Channel 1 external recorder voltage recording is turned on or not

SENSe:ELOG:FUNCtion:CURRent:MINMax 0|OFF|1|ON,(@<chanlist>)

SENSe:ELOG:FUNCtion:CURRent:MINMax? (@<chanlist>)

This command is used to set and query the current maximum and minimum recording function of the external recorder.

Command Syntax:

SENSe:ELOG:FUNCtion:CURRent:MINMax 0|OFF|1|ON, (@<chanlist>)

Parameters:

0|OFF|1|ON

<chanlist>:Channel number

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

SENSe:ELOG:FUNCtion:CURRent:MINMax? (@<chanlist>)

Return Parameters:

0 or 1

Example:

SENS:ELOG:CURR:FUNC:MINM ON,(@1) //Setting to turn on Channel 1 external recorder current max. min. recording

SENS:ELOG:CURR:FUNC:MINM? (@1) //Queries whether Channel 1 external recorder current max. min. record is turned on or not

SENSe:ELOG:FUNCtion:VOLTage:MINMax 0|OFF|1|ON,(@<chanlist>)

SENSe:ELOG:FUNCtion:VOLTage:MINMax? (@<chanlist>)

This command is used to set and query the voltage maximum and minimum recording function of the external recorder.

Command Syntax:

SENSe:ELOG:FUNCtion:VOLTage:MINMax 0|OFF|1|ON, (@<chanlist>)

Parameters:

0|OFF|1|ON

<chanlist>:Channel number

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

SENSe:ELOG:FUNCtion:VOLTage:MINMax? (@<chanlist>)

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Return Parameters:

0 or 1

Example:

SENS:ELOG:VOLT:FUNC:MINM ON,(@1) //Setting to turn on Channel 1 external recorder voltage max. min. recording

SENS:ELOG:VOLT:FUNC:MINM? (@1) //Queries whether Channel 1 External Recorder Voltage Maximum Value Minimum Value Record is turned on.

SENSe:ELOG:PERiod <value>,(@<chanlist>)

SENSe:ELOG:PERiod? (@<chanlist>)

This command is used to set and query the integration time of the external recorder. (Unit s)

Command Syntax:

SENSe:ELOG:PERiod <value>, (@<chanlist>)

Parameters:

0.0001-1|MINimum|MAXimum

<chanlist>:Channel number

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

SENSe:ELOG:PERiod? [MINimum|MAXimum,] (@<chanlist>)

Return Parameters:

<Points time>

Example:

SENS:ELOG:PER 0.005,(@1) //Set the Channel 1 external recorder division time to 0.005s.

SENS:ELOG:PER? (@1) //Query Channel 1 external logger integration time

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Chapter16 List Subsystems

[SOURce:]LIST:FUNCtion:TYPE <CURRent|VOLTage|POWer|RESistance>,(@<chanlist>)

[SOURce:]LIST:FUNCtion:TYPE? (@<chanlist>)

This command is used to set the working mode of list file, when power mode, only voltage mode OR current mode can be selected, CC/CV/CP/CR mode is supported in load mode.

Command Syntax:

[SOURce:]LIST:FUNCtion:TYPE <CURRent|VOLTage|POWer|RESistance>,(@<chanlist>)

Parameters:

CURRent|VOLTage|POWer|RESistance

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:FUNCtion:TYPE? (@<chanlist>)

Return Parameters:

CURR, VOLT, POWorRES

Example:

LIST:FUNCtion:TYPE VOLTage,(@1) // Set the list operating mode of Channel 1 to voltage

LIST:FUNCtion:TYPE? (@1) // Querying the list operating mode of Channel 1

[SOURce:]LIST:COUNt <value>, (@<chanlist>)

[SOURce:]LIST:COUNt? [MINimum|MAXimum,](@<chanlist>)

This command is used to set and query the number of repetitions of the LIST. 0 means None limited loop.

Command Syntax:

[SOURce:]LIST:COUNt <value>, (@<chanlist>)

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Parameters:

0-65535|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:COUNt? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<repeat count>

Example:

LIST:COUN 10,(@1) // Set the number of times the list in Channel 1 is repeated to 10 times

LIST:COUN? (@1) // Query the number of times the list in Channel 1 is repeated.

[SOURce:]LIST:CURRent:STEP[:LEVel] <value>,<step>,(@<chanlist>)

[SOURce:]LIST:CURRent:STEP[:LEVel]? [MINimum|MAXimum,]<step>,(@<chanlist>)

The commands set and query the current value of the specified step of LIST. (Unit is A)

Command Syntax:

[SOURce:]LIST:CURRent:STEP[:LEVel] <value>,{<step>,(@<chanlist>)

Parameter:

MINimum-MAXimum|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:CURRent:STEP[:LEVel]? [MINimum|MAXimum,]<step>,(@<chanlist>)

Return Parameters:

<current value>

Example:

LIST:CURRent 1.5,1,(@1) // Set Channel 1 first step list current value to 1.5A

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LIST:CURRent? 1,(@1) // Query Channel 1 first step list current value

[SOURce:]LIST:CURRent:SLEW <value>,<step>, (@<chanlist>)

[SOURce:]LIST:CURRent:SLEW? [MINimum|MAXimum,]<step>, (@<chanlist>)

The commands set and query the current slope of the specified step of LIST. (Unit is A/ms)

Command Syntax:

[SOURce:]LIST:CURRent:SLEW <value>,<step>, (@<chanlist>)

Parameter:

MINimum-MAXimum|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:CURRent:SLEW? [MINimum|MAXimum,]<step>(@<chanlist>)

Return Parameters:

<current slope>

Example:

LIST:CURRent:SLEW 2,1,(@1) // Set Channel 1's list first step current slope to 2

LIST:CURRent:SLEW? 1,(@1) // Query Channel 1's list first step current slope

[SOURce:]LIST:CURRent:POINts <value>,(@<chanlist>)

[SOURce:]LIST:CURRent:POINts? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the number of current list points.

Command Syntax:

[SOURce:]LIST:CURRent:POINts <value>,(@<chanlist>)

Parameters:

0-2000|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

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(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:CURRent:POINts? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<current points>

Example:

LIST:CURRent:POINts 1,(@1) //Sets the number of points in the Channel 1 list of currents to 1.

LIST:CURRent:POINts? (@1) // Queries the number of points of the current in the Channel 1 list.

[SOURce:]LIST:VOLTage:STEP[:LEVel] <value>,<step>, (@<chanlist>)

[SOURce:]LIST:VOLTage:STEP[:LEVel]? [MINimum|MAXimum,]<step>, (@<chanlist>)

The commands set and query the voltage value of the specified step of LIST. (Unit is V)

Command Syntax:

[SOURce:]LIST:VOLTage:STEP[:LEVel] <value>,<step>, (@<chanlist>)

Parameter:

MINimum-MAXimum|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:VOLTage:STEP[:LEVel]? [MINimum|MAXimum,]<step>,(@<chanlist>)

Return Parameters:

<voltage value>

Example:

LIST:VOLTage 10,1,(@1) // Set Channel 1's list first step voltage value to 10

LIST:VOLTage? 1,(@1) // Query the voltage value of the first step of the list of Channel 1

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[SOURce:]LIST:VOLTage:SLEW <value>,<step>, (@<chanlist>)

[SOURce:]LIST:VOLTage:SLEW? [MINimum|MAXimum,]<step>, (@<chanlist>)

The commands set and query the voltage slope of the specified step of LIST. (Unit is V/ms)

Command Syntax:

[SOURce:]LIST:VOLTage:SLEW <value>,<step>, (@<chanlist>)

Parameter:

MINimum-MAXimum|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:VOLTage:SLEW? [MINimum|MAXimum,]<step>,(@<chanlist>)

Return Parameters:

<voltage slope>

Example:

LIST:VOLTage:SLEW 1,2,(@1) //Set the voltage slope of the second step of the list for CH1 to 1

LIST:VOLTage:SLEW? 1,(@1) //Query CH1's list first step voltage slope

[SOURce:]LIST:VOLTage:POINts <value>,(@<chanlist>)

[SOURce:]LIST:VOLTage:POINts? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the number of voltage list points.

Command Syntax:

[SOURce:]LIST:VOLTage:POINts <value>,(@<chanlist>)

Parameters:

1-2000|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

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(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:VOLTage:POINts? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<voltage points>

Example:

LIST:VOLTage:POINts 1,(@1) //Setting List Voltage Points

LIST:VOLTage:POINts? (@1) //Query List Voltage Points

[SOURce:]LIST:POWer:STEP[:LEVel] <value>,<step>, (@<chanlist>)

[SOURce:]LIST:POWer:STEP[:LEVel]? [MINimum|MAXimum,]<step>, (@<chanlist>)

This command is used to set and query the power value of the specified step of LIST. Only the load supports this function, and only the load module supports this command. (Unit is W)

Command Syntax:

[SOURce:]LIST:POWer:STEP[:LEVel] <value>,<step>, (@<chanlist>)

Parameter:

MINimum-MAXimum|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:POWer:STEP[:LEVel]? [MINimum|MAXimum,]<step>,(@<chanlist>)

Return Parameters:

<power value>

Example:

LIST:POWer 100,1,(@1) //Set the power value of the first step of Channel 1's list to 100W.

LIST:POWer? 1,(@1) //Query the power value of the list of Channel 1

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[SOURce:]LIST:POWer:SLEW <value>,<step>, (@<chanlist>)

[SOURce:]LIST:POWer:SLEW? [MINimum|MAXimum,]<step>, (@<chanlist>)

This command is used to set and query the power slope of the specified step of LIST. Only the load supports this function, and only the load module supports this command. (Unit is W/ms)

Command Syntax:

[SOURce:]LIST:POWer:SLEW <value>,<step>, (@<chanlist>)

Parameter:

MINimum-MAXimum|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:POWer:SLEW? [MINimum|MAXimum,]<step>,(@<chanlist>)

Return Parameters:

<power slope>

Example:

LIST:POWer:SLEW 100,1,(@1) //100Set the power slope of the first step of Channel 1's list to 100

LIST:POWer:SLEW? (@1) //Query the power slope of the first step of the list for Channel 1

[SOURce:]LIST:POWer:POINts <value>,(@<chanlist>)

[SOURce:]LIST:POWer:POINts? [MINimum|MAXimum,](@<chanlist>)

The command sets and queries the number of power list points. Only the load supports this function, and only the load module supports this command.

Command Syntax:

[SOURce:]LIST:POWer:POINts <value>,(@<chanlist>)

Parameters:

1-2000|MINimum|MAXimum

<chanlist>:Channel number.

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(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:POWer:POINts? (@<chanlist>)

Return Parameters:

<Power Points>

Example:

LIST:POWer:POINts 1,(@1) //Set the number of list power points for Channel 1 to 1

LIST:POWer:POINts? (@1) //Query the number of list power points for Channel 1.

[SOURce:]LIST:RESistance:STEP[:LEVel] <value>,<step>, (@<chanlist>)

[SOURce:]LIST:RESistance:STEP[:LEVel]? [MINimum|MAXimum,]<step>, (@<chanlist>)

The commands set and query the resistance value of the specified step of LIST. Only the load supports this function, and only the load module supports this command. (Unit is Ω)

Command Syntax:

[SOURce:]LIST:RESistance:STEP[:LEVel] <value>,<step>, (@<chanlist>)

Parameter:

MINimum-MAXimum|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:RESistance:STEP[:LEVel]? [MINimum|MAXimum,]<step>,(@<chanlist>)

Return Parameters:

<resistance value>

Example:

LIST:RESistance 10,1,(@1) //Set Channel 1's list first step resistor value to 10

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LIST:RESistance? (@1) //Query the resistance value of the first step of the channel list

[SOURce:]LIST:RESistance:SLEW <value>,<step>, (@<chanlist>)

[SOURce:]LIST:RESistance:SLEW? [MINimum|MAXimum,]<step>, (@<chanlist>)

The commands set and query the resistance value of the specified step of LIST. Only load supports this function, and only load module supports this command. (Unit is S/ms)

Command Syntax:

[SOURce:]LIST:RESistance:SLEW <value>,<step>, (@<chanlist>)

Parameter:

MINimum-MAXimum|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:RESistance:SLEW? [MINimum|MAXimum,]<step>,(@<chanlist>)

Return Parameters:

<Resistance slope>

Example:

LIST:RESistance:SLEW 100,1,(@1) //Set Channel 1's list resistor slope to 100

LIST:RESistance:SLEW? 1,(@1) //Query the slope of the list resistance of Channel 1

[SOURce:]LIST:RESistance:POINts <value>,(@<chanlist>)

[SOURce:]LIST:RESistance:POINts? (@<chanlist>)

The commands set and query the number of resistor list points. Only the load supports this function, and only the load module supports this command.

Command Syntax:

[SOURce:]LIST:RESistance:POINts <value>,(@<chanlist>)

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Parameters:

1-2000|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:RESistance:POINts? (@<chanlist>)

Return Parameters:

<resistance points>

Example:

LIST:RESistance:POINts 10,(@1) //Set the number of list resistor points for Channel 1 to 10

LIST:RESistance:POINts? (@1) //Queries the number of list resistor points for Channel 1.

[SOURce:]LIST:TERMinate:MODE <NORMal|LAST|OFF>,(@<chanlist>)

[SOURce:]LIST:TERMinate:MODE? (@<chanlist>)

This command is used to set and query the end of list run, there are three end methods here:

NORMal:The output at the end of the list run is a fixed-mode output value

LAST:Holds the last step of output at the end of the list run

OFF:Close the output at the end of the list run and remain in list mode

Command Syntax:

[SOURce:]LIST:TERMinate:MODE <NORMal|LAST|OFF>,(@<chanlist>)

Parameters:

NORMal|LAST|OFF

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:TERMinate:MODE? (@<chanlist>)

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Return Parameters:

NORM,LASTorOFF

Example:

LIST:TERM NORM,(@1) //Set the end-of-list mode of Channel 1 to Normal.

LIST:TERM? (@1) //Query Channel 1's list ending method

[SOURce:]LIST:DWELl <value>,<step>, (@<chanlist>)

[SOURce:]LIST:DWELl? [MINimum|MAXimum,]<step>, (@<chanlist>)

This command is used to set and query the residence time of LIST for the specified number of steps. The unit is s.

Command Syntax:

[SOURce:]LIST:DWELl <value>,<step>, (@ss<chanlist>)

Parameters:

<value>:listSingle-step dwell time. Range 0.000005-10000.00

<step>:Number of steps for a single step, range 1-2000

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:DWELl? [MINimum|MAXimum,],<step>,(@<chanlist>)

Return Parameters:

<time>

Example:

LIST:DWELl 2,1,(@1) // Set the first step of Channel 1's list to reside for 2s

LIST:DWELl? 1, (@1) // Query Channel 1's list first step residence time

[SOURce:]LIST:TOUTput:BOSTep:POINts? (@<chanlist>)

This command queries the number of list points.

Command Syntax:

[SOURce:]LIST:TOUTput:BOSTep:POINts? (@<chanlist>)

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Parameters:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return Parameters:

<voltage points>

Example:

LIST:TOUTput:BOSTep:POINts? (@1) //Query the number of list points in Channel 1.

[SOURce:]LIST:TOUTput:EOSTep:POINts? (@<chanlist>)

This command is used to query the number of query list points.

Command Syntax:

[SOURce:]LIST:TOUTput:EOSTep:POINts? (@<chanlist>)

Parameters:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return Parameters:

<voltage points>

Example:

LIST:TOUTput:EOSTep:POINts? (@1) /Query the list points of Channel 1

[SOURce:]LIST:STEP <ONCE|AUTO> (@<chanlist>)

[SOURce:]LIST:STEP? <chanlist>

The commands set and query whether the current step of the LIST needs to be triggered to run to the next step.

Command Syntax:

[SOURce:]LIST:STEP <ONCE|AUTO> (@<chanlist>)

Parameters:

ONCE|AUTO

<chanlist>:Channel number.

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(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:STEP? (@<chanlist>)

Return Parameters:

ONCEorAUTO

Example:

LIST:STEP AUTO,(@1) //Set Channel 1list trigger run mode to Auto

LIST:STEP? (@1) //Query Channel 1list to trigger the run mode

[SOURce:]LIST:TOUTput:BOSTep[:DATA] <0|OFF|1|ON>,<step>,(@<chanlist>)

[SOURce:]LIST:TOUTput:BOSTep[:DATA]? <step>,(@<chanlist>)

The commands set and query whether the trigger signal is generated before the single step of LIST.

Command Syntax:

[SOURce:]LIST:TOUTput:BOSTep[:DATA] <0|OFF|1|ON>,<step>,(@<chanlist>)

Parameters:

0|OFF|1|ON

<step>:Number of steps for a single step, range 1-2000

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:TOUTput:BOSTep[:DATA]? <step>,(@<chanlist>)

Return Parameters:

0 or 1

Example:

LIST:TOUTput:BOSTep ON,1,(@1) //Enable pre-step trigger output for the first step of the Channel 1 list

LIST:TOUTput:BOSTep? 1,(@1) //Query the pre-step trigger output of the first

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step of the Channel 1 list

[SOURce:]LIST:TOUTput:EOSTep[:DATA] <0|OFF|1|ON>,<step>,(@<chanlist>)

The commands set and query whether the trigger signal is generated after a single step of LIST.

Command Syntax:

[SOURce:]LIST:TOUTput:EOSTep[:DATA] <0|OFF|1|ON>,<step>,(@<chanlist>)

Parameters:

0|OFF|1|ON

<step>:Number of steps for a single step, range 1-2000

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]LIST:TOUTput:EOSTep[:DATA]? <step>,(@<chanlist>)

Return Parameters:

0 or 1

Example:

LIST:TOUTput:EOSTep ON,1,(@1) //Trigger output mode on after setting list step 1

LIST:TOUTput:EOSTep? 1,(@1) //Trigger the output switch after the first step of the query list.

[SOURce:]LIST:COUNt:RUN? (@<chanlist>)

This command queries the number of repetitions of the current LIST run.

Command Syntax:

[SOURce:]LIST:COUNt:RUN? (@<chanlist>)

Parameters:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return Parameters:

<number of times it has been run>

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Example:

LIST:COUN:RUN? (@1) //Query the number of repetitions that list 1 has run.

[SOURce:]LIST:STEP:RUN? (@<chanlist>)

Queries the number of steps of the current LIST run.

Command Syntax:

[SOURce:]LIST:STEP:RUN? (@<chanlist>)

Parameters:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Return Parameters:

<steps>

Example:

LIST:STEP:RUN? (@1) //Queries the number of steps currently running on Channel 1

[SOURce]LIST:FILE:OPEN <string>,<LOCal|USB>,(@<chanlist>)

This command is used to open a list file in the specified storage path.

Command Syntax:

[SOURce]LIST:FILE:OPEN <string>,<LOCal|USB>,(@<chanlist>)

Parameters:

<string>:String file name

<local_type>:LOCcal: local file, USB: USB flash drive file

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

LIST:FILE:OPEN “test.csv”,USB,(@1) // Open the list file named test.csv that Channel 1 has stored on the USB.

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[SOURce]LIST:FILE:SAVE <string>,<LOCal|USB>,(@<chanlist>)

This command is used to save the list file to the specified storage path.

<string>:String file name

<local_type>LOCal: local file, USB: USB flash drive file

Command Syntax:

[SOURce]LIST:FILE:SAVE <string>,<LOCal|USB>,(@<chanlist>)

Example:

LIST:FILE:SAVE “test.csv”,USB,(@1) // Save the LIST file from Channel 1 to the USB with the file name “test.csv”.

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Chapter17 ARB Subsystem

SOURce:]ARB:COUNt <value>, (@<chanlist>)

[SOURce:]ARB:COUNt? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the number of repetitions for any wave. When the repeat count is 0, it means None limit repeat.

Command Syntax:

[SOURce:]ARB:COUNt <value>,(@<chanlist>)

Parameters:

0-65535|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:COUNt? [|MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<repeat count>

Example:

ARB:COUN 10,(@1) //Set the number of running repetitions of the ARB waveform for Channel 1 to 10

ARB:COUN? (@1) //Query the number of ARB waveform run repetitions for Channel 1

[SOURce:]ARB:COUNt:RUN? (@<chanlist>)

Queries the number of times the current ARB has been run.

Command Syntax:

[SOURce:]ARB:COUNt:RUN?

Return Parameters:

<number of times it has been run>

Example:

ARB:COUN:RUN? (@1) //Query Channel 1 for the current number of ARB runs

[SOURce:]ARB:CURRent:CONVert (@<chanlist>)

Converts the specified arbitrary waveform to a point in a sequence of current custom waveforms.

Command Syntax:

[SOURce:]ARB:CURRent:CONVert (@<chanlist>)

Parameters:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

ARB:CURR:CONV (@1) //Convert the currently selected waveform of Channel 1 into a customized waveform.

[SOURce:]ARB:VOLTage:CONVert (@<chanlist>)

Converts the points of a specified arbitrary waveform into a sequence of voltage custom waveforms.

Command Syntax:

[SOURce:]ARB:VOLTage:CONVert (@<chanlist>)

Parameters:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

ARB:VOLT:CONV (@1) //Convert the currently selected voltage waveform of Channel 1 into a customized waveform.

[SOURce:]ARB:POWer:CONVert (@<chanlist>)

Converts a specified arbitrary waveform to a point in a sequence of power-customized waveforms. Supported in load mode only.

Command Syntax:

[SOURce:]ARB:POWer:CONVert (@<chanlist>)

Parameters:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

ARB:POW:CONV (@1) //Convert the currently selected power waveform of Channel 1 into a customized waveform.

[SOURce:]ARB:RESistance:CONVert (@<chanlist>)

Converts the points of a specified arbitrary waveform into a sequence of resistor custom waveforms. Supported in load mode only.

Command Syntax:

[SOURce:]ARB:RESistance:CONVert (@<chanlist>)

Parameters:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

ARB:RES:CONV (@1) //Convert the currently selected resistor waveform of CH1 to a custom waveform

[SOURce:]ARB:FUNCtion:SHAPe <function>, (@<chanlist>)

[SOURce:]ARB:FUNCtion:SHAPe? (@<chanlist>)

Set and query the type of arbitrary waves.

Command Syntax:

[SOURce:]ARB:FUNCtion:SHAPe <function>, (@<chanlist>)

Parameters:

<STEP|RAMP|STAircase|SINusoid|PULSe|TRAPezoid|EXPonential|UDEFined>
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:FUNCtion:SHAPe? (@<chanlist>)

Return Parameters:

STEP,RAMP,STA,SIN,PULS,TRAP,EXP orUDEF

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Example:

ARB:FUNC:SHAP STEP,(@1) //Set the currently selected waveform of Channel 1 to STEP

ARB:FUNC:SHAP? (@1) //Queries the type of waveform selected for Channel 1.

[SOURce:]ARB:FUNCtion:TYPE <type>, (@<chanlist>)

[SOURce:]ARB:FUNCtion:TYPE? (@<chanlist>)

Set and query the priority mode of any wave.

Command Syntax:

[SOURce:]ARB:FUNCtion:TYPE <type>, (@<chanlist>)

Parameters:

CURRent|VOLTage|POWer|RESistance
POWer and RESistance modes are only available in load mode
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:FUNCtion:TYPE? (@<chanlist>)

Return Parameters:

CURR,VOLT,POW orRES

Example:

ARB:FUNCtion:TYPE VOLTage,(@1) //Set the arb operating mode of Channel 1 to voltage priority

ARB:FUNCtion:TYPE? (@1) //Query the arb operating mode of Channel 1

[SOURce:]ARB:TERMinate:MODE <mode>,(@<chanlist>)

[SOURce:]ARB:TERMinate:MODE? (@<chanlist>)

Sets and queries how the ARB ends.

Command Syntax:

ARB:TERMinate:MODE (@<chanlist>)

Parameters:

NORMal|LAST|OFF
<chanlist>:Channel number.

(@1) – Channel 1

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(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:FUNCtion:TYPE? (@<chanlist>)

Return Parameters:

NORM,LASTorOFF

Example:

ARB:TERMinate:MODE LAST,(@1) //Set the arb termination method of Channel 1 to Last

ARB:TERMinate:MODE? (@1) //Query the arb ending method of Channel 1

[SOURce:]ARB:CURRent:EXPonential:END[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:EXPonential:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the end value of the current index waveform.

Command Syntax:

[SOURce:]ARB:CURRent:EXPonential:END[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:EXPonential:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<current value>

Example:

ARB:CURR:EXP:END 1,(@1) //Set the ending current of the exponential waveform of Channel 1 to 1A.

ARB:CURR:EXP:END? (@1) //Queries the ending current of the exponential waveform of Channel 1.

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[SOURce:]ARB:VOLTage:EXPonential:END[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:EXPonential:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the end value of the voltage index waveform.

Command Syntax:

[SOURce:]ARB:VOLTage:EXPonential:END[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:EXPonential:END[:LEVel]?
[|MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:VOLT:EXP:END 10,(@1) //Set the ending voltage of the exponential waveform of Channel 1 to 10V.

ARB:VOLT:EXP:END? (@1) //Query the end voltage of the exponential waveform of Channel 1.

[SOURce:]ARB:POWer:EXPonential:END[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:POWer:EXPonential:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the end value of the power index waveform. Supported in load module only.

Command Syntax:

[SOURce:]ARB:POWer:EXPonential:END[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

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(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:EXPonential:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<power value>

Example:

ARB:POW:EXP:END 100,(@1) //Set the ending power of the exponential waveform of Channel 1 to 100W.

ARB:POW:EXP:END? (@1) //Query the end power of the exponential waveform of Channel 1

[SOURce:]ARB:RESistance:EXPonential:END[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:EXPonential:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the end value of the resistance index waveform. Supported in load module only.

Command Syntax:

[SOURce:]ARB:RESistance:EXPonential:END[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:EXPonential:END[:LEVel]? [|MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:RES:EXP:END 1000,(@1) //Set the ending resistance of the exponential waveform of Channel 1 to 1000Ω.

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ARB:RES:EXP:END? (@1) //Query the exponential waveform ending resistance of Channel 1.

[SOURce:]ARB:CURRent:EXPonential:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Specifies the initial value of the current for an arbitrary wave exponential waveform.

Command Syntax:

[SOURce:]ARB:CURRent:EXPonential:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:EXPonential:STARt[:LEVel]? [|MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<current value>

Example:

ARB:CURR:EXP:STAR 1,(@1) //Set the starting current of the exponential waveform of Channel 1 to 1A.

ARB:CURR:EXP:STAR? (@1) //Queries the starting current of the exponential waveform of Channel 1.

[SOURce:]ARB:VOLTage:EXPonential:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Specifies the initial value of the voltage for the arbitrary wave index waveform.

Command Syntax:

[SOURce:]ARB:VOLTage:EXPonential:END[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum -MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:EXPonential:END[:LEVel]?
[|MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:VOLT:EXP:STAR 10,(@1) //Set the starting voltage of the exponential waveform of Channel 1 to 10V.

ARB:VOLT:EXP:STAR? (@1) //Queries the start voltage of the exponential waveform of Channel 1.

[SOURce:]ARB:POWer:EXPonential:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:POWer:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Specifies the initial value of power for arbitrary wave exponential waveforms. Supported only in load module.

Command Syntax:

[SOURce:]ARB:POWer:EXPonential:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:EXPonential:STARt[:LEVel]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<power value>

Example:

ARB:POW:EXP:STAR 100,(@1) //Set the starting power of the exponential waveform of Channel 1 to 100W.

ARB:POW:EXP:STAR? (@1) //Queries the starting power of the exponential waveform of Channel 1.

[SOURce:]ARB:RESistance:EXPonential:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Specifies the initial value of the resistor for arbitrary wave index waveforms. Supported in load module only.

Command Syntax:

[SOURce:]ARB:RESistance:EXPonential:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:RES:EXP:STAR 1000,(@1) //Set the starting resistance of the exponential waveform of Channel 1 to 1000Ω.

ARB:RES:EXP:STAR? (@1) // Queries the starting resistance of the exponential waveform of Channel 1.

[SOURce:]ARB:CURRent:EXPonential:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Specifies the initial value of the current for an arbitrary wave exponential waveform.

Command Syntax:

[SOURce:]ARB:CURRent:EXPonential:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:EXPonential:STARt[:LEVel]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<current value>

Example:

ARB:CURR:EXP:STAR 1,(@1) //Set the starting current of the exponential waveform of Channel 1 to 1A.

ARB:CURR:EXP:STAR? (@1) //Sets the starting current of the exponential waveform of Channel 1.

[SOURce:]ARB:VOLTage:EXPonential:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Specifies the initial value of the voltage for the arbitrary wave index waveform.

Command Syntax:

[SOURce:]ARB:VOLTage:EXPonential:END[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:EXPonential:END[:LEVel]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:VOLT:EXP:STAR 10,(@1) //Set the starting voltage of the exponential waveform of Channel 1 to 10V.

ARB:VOLT:EXP:STAR? (@1) //Queries the start voltage of the exponential waveform of Channel 1.

[SOURce:]ARB:POWer:EXPonential:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:POWer:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Specifies the initial value of power for arbitrary wave exponential waveforms. Supported only in load mode.

Command Syntax:

[SOURce:]ARB:POWer:EXPonential:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<power value>

Example:

ARB:POW:EXP:STAR 100,(@1) //Set the starting power of the exponential waveform of Channel 1 to 100W.

ARB:POW:EXP:STAR? (@1) //Queries the starting power of the exponential waveform of Channel 1.

[SOURce:]ARB:RESistance:EXPonential:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Specifies the initial value of the resistor for arbitrary wave index waveforms. Supported in load mode only.

Command Syntax:

[SOURce:]ARB:RESistance:EXPonential:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:RES:EXP:STAR 1000,(@1) //Set the starting resistance of the exponential waveform of Channel 1 to 1000Ω.

ARB:RES:EXP:STAR? (@1) //Setting the starting power of the exponential waveform of Channel 1

[SOURce:]ARB:CURRent:EXPonential:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:EXPonential:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Specifies the delay time after an arbitrary exponential waveform is triggered.

Command Syntax:

[SOURce:]ARB:CURRent:EXPonential:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.0|MINimum|MAXimum (in s)
<chanlist>:Channel number.

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(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:EXPonential:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<current value>

Example:

ARB:CURR:EXP:STAR:TIME 1,(@1) //Set the current index waveform current start time of Channel 1 to 1s

ARB:CURR:EXP:STAR:TIME? (@1) //Queries the current index waveform current start time of Channel 1.

[SOURce:]ARB:VOLTage:EXPonential:STARt:TIME<value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:EXPonential:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Specifies the delay time after an arbitrary exponential waveform is triggered.

Command Syntax:

[SOURce:]ARB:VOLTage:EXPonential:STARt:TIME<value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:EXPonential:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<start time>

Example:

ARB:VOLT:EXP:STAR:TIME 1,(@1) //Set the voltage index waveform voltage start time of Channel 1 to 1s

ARB:VOLT:EXP:STAR:TIME? (@1) //Queries the voltage index waveform

voltage start time of Channel 1.

[SOURce:]ARB:POWer:EXPonential:STARt:TIME<value>, (@<chanlist>)

[SOURce:]ARB:POWer:EXPonential:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Specifies the delay time after an arbitrary waveform exponential waveform is triggered. Supported only in load mode.

Command Syntax:

[SOURce:]ARB:POWer:EXPonential:STARt:TIME<value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:EXPonential:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<start time>

Example:

ARB:POW:EXP:STAR:TIME 1,(@1) //Set the power index waveform power onset time for Channel 1 to 1s

ARB:POW:EXP:STAR:TIME? (@1) //Queries the power index waveform power onset time for Channel 1

[SOURce:]ARB:RESistance:EXPonential:STARt:TIME<value>, (@<chanlist>)

[SOURce:]ARB:RESistance:EXPonential:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Specifies the delay time after an arbitrary waveform exponential waveform is triggered. Supported only in load mode.

Command Syntax:

[SOURce:]ARB:RESistance:EXPonential:STARt:TIME<value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:EXPonential:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<delay time>

Example:

ARB:RES:EXP:STAR:TIME 1,(@1) //Set the resistor index waveform resistance start time of Channel 1 to 1s

ARB:RES:EXP:STAR:TIME? (@1) //Queries Channel 1's Resistance Index Waveform Resistance Start Time

[SOURce:]ARB:CURRent:EXPonential:TCONstant <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:EXPonential:TCONstant? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time constant of the exponential waveform.

Command Syntax:

[SOURce:]ARB:CURRent:EXPonential:TCONstant <value>, (@<chanlist>)

Parameters:

0.000005-Time to end of index|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:EXPonential:TCONstant? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time constant>

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Example:

ARB:CURR:EXP:TCON 1,(@1) //Set the current time constant of the current exponential waveform of Channel 1 to 1s

ARB:CURR:EXP:TCON? (@1) //Queries the current time constant of the current index waveform of Channel 1.

[SOURce:]ARB:VOLTage:EXPonential:TCONstant <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:EXPonential:TCONstant? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time constant of the exponential waveform.

Command Syntax:

[SOURce:]ARB:VOLTage:EXPonential:TCONstant <value>, (@<chanlist>)

Parameters:

0.000005-10000.0|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:EXPonential:TCONstant? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time constant>

Example:

ARB:VOLT:EXP:TCON 1,(@1) //Set the voltage time constant of the voltage exponential waveform of Channel 1 to 1s

ARB:VOLT:EXP:TCON? (@1) //Queries the voltage time constant of the voltage index waveform of Channel 1.

[SOURce:]ARB:POWer:EXPonential:TCONstant <value>, (@<chanlist>)

[SOURce:]ARB:POWer:EXPonential:TCONstant? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time constant of the exponential waveform. Supported only in load module.

Command Syntax:

[SOURce:]ARB:POWer:EXPonential:TCONstant <value>, (@<chanlist>)

Parameters:

0.000005-10000.0|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:EXPonential:TCONstant?
[|MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time constant>

Example:

ARB:POW:EXP:TCON 1,(@1) //Set the power time constant of the power index waveform of Channel 1 to 1s

ARB:POW:EXP:TCON? (@1) //Query the power time constant of the power index waveform of Channel 1.

[SOURce:]ARB:RESistance:EXPonential:TCONstant <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:EXPonential:TCONstant? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time constant of the exponential waveform. Supported only in load module.

Command Syntax:

[SOURce:]ARB:RESistance:EXPonential:TCONstant <value>, (@<chanlist>)

Parameters:

0.000005-10000.0|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:EXPonential:TCONstant?
[MINimum|MAXimum,](@<chanlist>)

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Return Parameters:

<time constant>

Example:

ARB:RES:EXP:TCON 1,(@1) //Set the resistance time constant of the resistance index waveform of Channel 1 to 1s
ARB:RES:EXP:TCON? (@1) //Queries the resistance time constant of the resistance index waveform of Channel 1.

[SOURce:]ARB:CURRent:EXPonential:TIME <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:EXPonential:TIME? [MINimum|MAXimum,](@<chanlist>)

Set and query the time from the start of the exponential change to the end of the amplitude.

Command Syntax:

[SOURce:]ARB:CURRent:EXPonential:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:EXPonential:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:CURR:EXP:TIME 1,(@1) //Set the time from the beginning of the exponential change to the end of the current exponential waveform of Channel 1 to 1s.

ARB:CURR:EXP:TIME? (@1) //Queries the time from the beginning of the exponential change to the end of the current exponential waveform of Channel 1.

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[SOURce:]ARB:VOLTage:EXPonential:TIME <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:EXPonential:TIME? [MINimum|MAXimum,](@<chanlist>)

Set and query the time from the start of the exponential change to the end of the amplitude.

Command Syntax:

[SOURce:]ARB:VOLTage:EXPonential:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:EXPonential:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:VOLT:EXP:TIME 1,(@1) //Set the time from the beginning of the exponential change to the end of the voltage exponential waveform of Channel 1 to 1s.

ARB:VOLT:EXP:TIME? (@1) //Queries the time from the beginning of the exponential change to the end of the voltage exponential waveform of Channel 1.

[SOURce:]ARB:POWer:EXPonential:TIME <value>, (@<chanlist>)

[SOURce:]ARB:POWer:EXPonential:TIME? [MINimum|MAXimum,](@<chanlist>)

Setting and querying the time for the amplitude to change from the on start index to the end, Supported only in load module.

Command Syntax:

[SOURce:]ARB:POWer:EXPonential:TIME <value>, (@<chanlist>)

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Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:EXPonential:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:POW:EXP:TIME 1,(@1) //Set the time from the beginning of the exponential change to the end of the power exponential waveform of Channel 1 to 1s.

ARB:POW:EXP:TIME? (@1) //Queries the time from the beginning of the exponential change to the end of the power exponential waveform of Channel 1.

[SOURce:]ARB:RESistance:EXPonential:TIME <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:EXPonential:TIME? [MINimum|MAXimum,](@<chanlist>)

Set and query the time from the start of the exponential change to the end of the amplitude, Supported only in load module.

Command Syntax:

[SOURce:]ARB:RESistance:EXPonential:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:EXPonential:TIME?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:RES:EXP:TIME 1,(@1) //Set the time from the beginning of the exponential change to the end of the resistance exponential waveform of Channel 1 to 1s.

ARB:RES:EXP:TIME? (@1) //Queries the time from the beginning of the exponential change to the end of the resistance exponential waveform of Channel 1.

[SOURce:]ARB:CURRent:PULSe:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:PULSe:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the output hold ends after the pulse is completed.

Command Syntax:

[SOURce:]ARB:CURRent:PULSe:END:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:PULSe:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<end time>

Example:

ARB:CURR:PULS:END:TIME 1,(@1) //Set the current end time of the pulse waveform of Channel 1 to 1s.

ARB:CURR:PULS:END:TIME? (@1) //Queries the current end time of the pulse waveform of Channel 1.

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[SOURce:]ARB:VOLTage:PULSe:END:TIME<value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:PULSe:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Setting and querying the end of output hold after pulse completion

Command Syntax:

[SOURce:]ARB:VOLTage:PULSe:END:TIME<value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:PULSe:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:VOLT:PULS:END:TIME 1,(@1) //Set the voltage end time of the pulse waveform of Channel 1 to 1s.

ARB:VOLT:PULS:END:TIME? (@1) //Queries the voltage end time of the pulse waveform of Channel 1.

[SOURce:]ARB:POWer:PULSe:END:TIME<value>, (@<chanlist>)

[SOURce:]ARB:POWer:PULSe:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Setting and querying the end of output hold after pulse completion, Supported only in load module.

Command Syntax:

[SOURce:]ARB:POWer:PULSe:END:TIME<value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:PULSe:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:POW:PULS:END:TIME 1,(@1) //Setting the power end time of the pulse waveform of Channel 1 to 1s
ARB:POW:PULS:END:TIME? (@1) //Queries the power end time of the pulse waveform of Channel 1.

[SOURce:]ARB:RESistance:PULSe:END:TIME<value>, (@<chanlist>)

[SOURce:]ARB:RESistance:PULSe:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Setting and querying the end of output hold after pulse completion, Supported only in load module.

Command Syntax:

[SOURce:]ARB:RESistance:PULSe:END:TIME<value>, (@<chanlist>)

Parameters:

0.000005-10000.00s|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:PULSe:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:RES:PULS:END:TIME 1,(@1) //Set the resistive end time of the pulse waveform of Channel 1 to 1s.

ARB:RES:PULS:END:TIME? (@1) //Queries the resistance end time of the

pulse waveform of Channel 1.

[SOURce:]ARB:CURRent:PULSe:FREQuency<value>, (@<chanlist>)

[SOURce:]ARB:CURRent:PULSe:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the current frequency of the pulse.

Command Syntax:

[SOURce:]ARB:CURRent:PULSe:FREQuency<value>, (@<chanlist>)

Parameters:

0.00005-100000.0|MINimum|MAXimum (in Hz)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:PULSe:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<frequency>

Example:

ARB:CURR:PULS:FREQ 1,(@1) //Set the current frequency of the pulse waveform of Channel 1 to 1Hz.

ARB:CURR:PULS:FREQ? (@1) //Queries the current frequency of the pulse waveform of Channel 1.

[SOURce:]ARB:VOLTage:PULSe:FREQuency<value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:PULSe:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the voltage frequency of the pulse.

Command Syntax:

[SOURce:]ARB:VOLTage:PULSe:FREQuency<value>, (@<chanlist>)

Parameters:

0.00005-100000.0|MINimum|MAXimum (in Hz)

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<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:PULSe:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<frequency>

Example:

ARB:VOLT:PULS:FREQ 1,(@1) //Set the voltage frequency of the pulse waveform of Channel 1 to 1Hz.

ARB:VOLT:PULS:FREQ? (@1) //Queries the voltage frequency of the pulse waveform of Channel 1.

[SOURce:]ARB:POWer:PULSe:FREQuency<value>, (@<chanlist>)

[SOURce:]ARB:POWer:PULSe:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the power frequency of the power pulse, Supported only in load module.

Command Syntax:

[SOURce:]ARB:POWer:PULSe:FREQuency<value>, (@<chanlist>)

Parameters:

0.00005-100000.0|MINimum|MAXimum (in Hz)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:PULSe:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<frequency>

Example:

ARB:POW:PULS:FREQ 1,(@1) //Set the power frequency of the pulse

waveform of Channel 1 to 1Hz.

ARB:POW:PULS:FREQ? (@1) //Queries the power frequency of the pulse waveform of Channel 1.

[SOURce:]ARB:RESistance:PULSe:FREQuency<value>, (@<chanlist>)

[SOURce:]ARB:RESistance:PULSe:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the resistance frequency of the power pulse, Supported only in load module.

Command Syntax:

[SOURce:]ARB:RESistance:PULSe:FREQuency<value>, (@<chanlist>)

Parameters:

0.00005-100000.0|MINimum|MAXimum (in Hz)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:PULSe:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<frequency>

Example:

ARB:RES:PULS:FREQ 1,(@1) //Set the resistor frequency of the pulse waveform of Channel 1 to 1Hz.

ARB:RES:PULS:FREQ? (@1) //Queries the resistance frequency of the pulse waveform of Channel 1.

[SOURce:]ARB:CURRent:PULSe:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:PULSe:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Set and query the current level before the pulse.

Command Syntax:

[SOURce:]ARB:CURRent:PULSe:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

0.01-500MINimum|MAXimum (in Hz)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:PULSe:STARt[:LEVel]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<current value>

Example:

ARB:CURR:PULS:STAR 1,(@1) //Set the starting current of the pulse waveform of Channel 1 to 1A.

ARB:CURR:PULS:STAR? (@1) //Queries the starting current of the pulse waveform of Channel 1.

[SOURce:]ARB:VOLTage:PULSe:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:PULSe:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Set and query the voltage level before the pulse.

Command Syntax:

[SOURce:]ARB:VOLTage:PULSe:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:PULSe:STARt[:LEVel]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:VOLT:PULS:STAR 1,(@1) //Set the start voltage of the pulse waveform of Channel 1 to 1V.

ARB:VOLT:PULS:STAR? (@1) //Queries the start voltage of the pulse waveform of Channel 1.

[SOURce:]ARB:POWer:PULSe:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:POWer:PULSe:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Set and query the power level before the pulse, Supported only in load module.

Command Syntax:

[SOURce:]ARB:POWer:PULSe:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:PULSe:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<power value>

Example:

ARB:POW:PULS:STAR 1,(@1) //Set the starting power of the pulse waveform of Channel 1 to 1W.

ARB:POW:PULS:STAR? (@1) //Sets the starting power of the pulse waveform of Channel 1.

[SOURce:]ARB:RESistance:PULSe:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:PULSe:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Set and query the resistor level before the pulse, Supported only in load module.

Command Syntax:

[SOURce:]ARB:RESistance:PULSe:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:PULSe:STARt[:LEVel]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:RES:PULS:STAR 1,(@1) //Set the starting resistance of the pulse waveform of Channel 1 to 1Ω.

ARB:RES:PULS:STAR? (@1) //Queries the starting resistance of the pulse waveform of Channel 1.

[SOURce:]ARB:CURRent:PULSe:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:PULSe:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Set and query the delay time of the pulse.

Command Syntax:

[SOURce:]ARB:CURRent:PULSe:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.00||MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:PULSe:STARt:TIME?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:CURR:PULS:STAR:TIME 1,(@1) //Set the current start time of the pulse waveform of Channel 1 to 1s.

ARB:CURR:PULS:STAR:TIME? (1) //Queries the current start time of the pulse waveform of Channel 1.

[SOURce:]ARB:VOLTage:PULSe:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:PULSe:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Set and query the delay time of the pulse.

Command Syntax:

[SOURce:]ARB:VOLTage:PULSe:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:PULSe:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:VOLT:PULS:STAR:TIME 1,(@1) //Set the voltage start time of the pulse waveform of Channel 1 to 1s.

ARB:VOLT:PULS:STAR:TIME? (@1) //Queries the voltage start time of the pulse waveform of Channel 1.

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[SOURce:]ARB:POWer:PULSe:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:POWer:PULSe:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Set and query the delay time of the pulse, supported only in load module.

Command Syntax:

[SOURce:]ARB:POWer:PULSe:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:PULSe:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:POW:PULS:STAR:TIME 1,(@1) //Set the power onset time of the pulse waveform of Channel 1 to 1s.

ARB:POW:PULS:STAR:TIME? (@1) //Queries the power onset time of the pulse waveform of Channel 1.

[SOURce:]ARB:RESistance:PULSe:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:PULSe:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Set and query the delay time of the pulse, supported only in load module.

Command Syntax:

[SOURce:]ARB:RESistance:PULSe:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:PULSe:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:RES:PULS:STAR:TIME 1,(@1) //Set the resistor start time of the pulse waveform of Channel 1 to 1s.

ARB:RES:PULS:STAR:TIME? (@1) //Queries the resistance start time of the pulse waveform of Channel 1.

[SOURce:]ARB:CURRent:PULSe:TOP[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:PULSe:TOP[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the current amplitude of the pulse.

Command Syntax:

[SOURce:]ARB:CURRent:PULSe:TOP[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:PULSe:TOP[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<current value>

Example:

ARB:CURR:PULS:TOP 1,(@1) //Set the peak current of the pulse waveform of Channel 1 to 1A.

ARB:CURR:PULS:TOP? (@1) //Queries the peak current of the pulse

waveform of Channel 1.

[SOURce:]ARB:VOLTage:PULSe:TOP[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:PULSe:TOP[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the voltage amplitude of the pulse.

Command Syntax:

[SOURce:]ARB:VOLTage:PULSe:TOP[:LEVel] <value>, (@<chanlist>)

Parameters:

Minimum-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:PULSe:TOP[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:VOLT:PULS:TOP 1,(@1) //Set the peak voltage of the pulse waveform of Channel 1 to 1V.

ARB:VOLT:PULS:TOP? (@1) //Queries the peak voltage of the pulse waveform of Channel 1.

[SOURce:]ARB:POWer:PULSe:TOP[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:POWer:PULSe:TOP[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Set and query the power amplitude of the pulse, supported only in load module.

Command Syntax:

[SOURce:]ARB:POWer:PULSe:TOP[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in W)

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<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:PULSe:TOP[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<power value>

Example:

ARB:POW:PULS:TOP 1,(@1) //Set the peak power of the pulse waveform of Channel 1 to 1W.

ARB:POW:PULS:TOP? (@1) //Queries the peak power of the pulse waveform of Channel 1.

[SOURce:]ARB:RESistance:PULSe:TOP[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:PULSe:TOP[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Set and query the power amplitude of the pulse, supported only in load module.

Command Syntax:

[SOURce:]ARB:RESistance:PULSe:TOP[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:PULSe:TOP[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Resistance>

Example:

ARB:RES:PULS:TOP 1,(@1) //Set the peak resistance of the pulse waveform of Channel 1 to 1Ω.

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ARB:RES:PULS:TOP? (@1) //Queries the peak resistance of the pulse waveform of Channel 1.

[SOURce:]ARB:CURRent:PULSe:TOP:TIME <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:PULSe:TOP:TIME? [MINimum|MAXimum,](@<chanlist>)

Set and query the pulse width.

Command Syntax:

[SOURce:]ARB:CURRent:PULSe:TOP:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:PULSe:TOP:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:CURR:PULS:TOP:TIME 1,(@1) //Set the current peak time of the pulse waveform of Channel 1 to 1s.

ARB:CURR:PULS:TOP:TIME? (@1) //Queries the current peak time of the pulse waveform of Channel 1.

[SOURce:]ARB:VOLTage:PULSe:TOP:TIME <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:PULSe:TOP:TIME? [MINimum|MAXimum,](@<chanlist>)

Set and query the pulse width.

Command Syntax:

[SOURce:]ARB:VOLTage:PULSe:TOP:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:PULSe:TOP:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:VOLT:PULS:TOP:TIME 1,(@1) //Set the voltage peak time of the pulse waveform of Channel 1 to 1s.

ARB:VOLT:PULS:TOP:TIME? (@1) //Queries the voltage peak time of the pulse waveform of Channel 1.

[SOURce:]ARB:POWer:PULSe:TOP:TIME<value>, (@<chanlist>)

[SOURce:]ARB:POWer:PULSe:TOP:TIME? [MINimum|MAXimum,](@<chanlist>)

Set and query the pulse width, supported only in load module.

Command Syntax:

[SOURce:]ARB:POWer:PULSe:TOP:TIME<value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:PULSe:TOP:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<power value>

Example:

ARB:POW:PULS:TOP:TIME 1,(@1) //Set the power peak time of the pulse waveform of Channel 1 to 1s.

ARB:POW:PULS:TOP:TIME? (@1) //Queries the power peak time of the pulse waveform of Channel 1.

[SOURce:]ARB:RESistance:PULSe:TOP:TIME<value>, (@<chanlist>)

[SOURce:]ARB:RESistance:PULSe:TOP:TIME? [MINimum|MAXimum,](@<chanlist>)

Set and query the pulse width, supported only in load module.

Command Syntax:

[SOURce:]ARB:RESistance:PULSe:TOP:TIME<value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:PULSe:TOP:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:RES:PULS:TOP:TIME 1,(@1) //Set the power peak time of the pulse waveform of Channel 1 to 1s.

ARB:RES:PULS:TOP:TIME? (@1) //Queries the power peak time of the pulse waveform of Channel 1.

[SOURce:]ARB:CURRent:RAMP:END[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:RAMP:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the level after the ramp wave.

Command Syntax:

[SOURce:]ARB:CURRent:RAMP:END[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:RAMP:END[:LEVel]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<current value>

Example:

ARB:CURR:RAMP:END 1,(@1) //Set the end current of the ramp waveform of Channel 1 to 1A.

ARB:CURR:RAMP:END? (@1) //Query the end current of the ramp waveform of Channel 1 as 1A

[SOURce:]ARB:VOLTage:RAMP:END[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:RAMP:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the level after the ramp wave.

Command Syntax:

[SOURce:]ARB:VOLTage:RAMP:END[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum -MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:RAMP:END[:LEVel]?
[MINimum|MAXimum,](@<chanlist>)

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Return Parameters:

<voltage value>

Example:

ARB:VOLT:RAMP:END 1,(@1) //Set the end voltage of the ramp waveform of Channel 1 to 1V.

ARB:VOLT:RAMP:END? (@1) //Queries the end voltage of the ramp waveform of Channel 1.

[SOURce:]ARB:POWer:RAMP:END[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:POWer:RAMP:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the level after the ramp wave, supported only in load module.

Command Syntax:

[SOURce:]ARB:POWer:RAMP:END[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:RAMP:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<power value>

Example:

ARB:POW:RAMP:END 1,(@1) //Setting the end power of the ramp waveform of Channel 1 to 1W

ARB:POW:RAMP:END? (@1) //Query the end power of the ramp waveform of Channel 1.

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[SOURce:]ARB:RESistance:RAMP:END[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:RAMP:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the level after the ramp wave, supported only in load module.

Command Syntax:

[SOURce:]ARB:RESistance:RAMP:END[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:RAMP:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:RES:RAMP:END 1,(@1) //Set the end resistor of the ramp waveform of Channel 1 to 1Ω.

ARB:RES:RAMP:END? (@1) //Queries the end resistance of the ramp waveform of Channel 1.

[SOURce:]ARB:CURRent:RAMP:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:RAMP:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the output hold ends after the ramp wave completes.

Command Syntax:

[SOURce:]ARB:CURRent:RAMP:END[:LEVel] <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:RAMP:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:CURR:RAMP:END:TIME 1,(@1) //Setting the current end time of the ramp waveform of Channel 1 to 1s

ARB:CURR:RAMP:END:TIME? (@1) //Queries the current end time of the ramp waveform of Channel 1.

[SOURce:]ARB:VOLTage:RAMP:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:RAMP:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the output hold ends after the ramp wave completes.

Command Syntax:

[SOURce:]ARB:VOLTage:RAMP:END[:LEVel] <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:RAMP:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<NRf+>

Example:

ARB:VOLT:RAMP:END:TIME 1,(@1) //Set the voltage end time of the ramp waveform of Channel 1 to 1s

ARB:VOLT:RAMP:END:TIME? (@1) //Queries the voltage end time of the ramp waveform of Channel 1.

[SOURce:]ARB:POWer:RAMP:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:POWer:RAMP:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the output hold ends after the ramp wave completes, supported only in load module.

Command Syntax:

[SOURce:]ARB:POWer:RAMP:END[:LEVel] <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:RAMP:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:POW:RAMP:END:TIME 1,(@1) //Set the power end time of the ramp waveform of Channel 1 to 1s

ARB:POW:RAMP:END:TIME? (@1) //Queries the power end time of the ramp waveform of Channel 1.

[SOURce:]ARB:RESistance:RAMP:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:RAMP:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the output hold ends after the ramp wave completes, supported only in load module.

Command Syntax:

[SOURce:]ARB:RESistance:RAMP:END[:LEVel] <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:RAMP:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:RES:RAMP:END:TIME 1,(@1) //Setting the current end time of the ramp waveform of Channel 1 to 1s

ARB:RES:RAMP:END:TIME? (@1) //Queries the current end time of the ramp waveform of Channel 1.

[SOURce:]ARB:CURRent:RAMP:RTIMe <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:RAMP:RTIMe? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the rise time of the ramp waveform.

Command Syntax:

[SOURce:]ARB:CURRent:RAMP:RTIMe <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:RAMP:RTIMe? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

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Example:

ARB:CURR:RAMP:RTIM 1,(@1) //Set the current rise time of the ramp waveform of Channel 1 to 1s

ARB:CURR:RAMP:RTIM? (@1) //Queries the current rise time of the ramp waveform of Channel 1.

[SOURce:]ARB:VOLTage:RAMP:RTIMe <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:RAMP:RTIMe? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the rise time of the ramp waveform.

Command Syntax:

[SOURce:]ARB:VOLTage:RAMP:RTIMe <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:RAMP:RTIMe? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:VOLT:RAMP:RTIM 1,(@1) //Set the voltage rise time of the ramp waveform of Channel 1 to 1s

ARB:VOLT:RAMP:RTIM? (@1) //Queries the voltage rise time of the ramp waveform of Channel 1.

[SOURce:]ARB:POWer:RAMP:RTIMe <value>, (@<chanlist>)

[SOURce:]ARB:POWer:RAMP:RTIMe? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the rise time of the ramp waveform, supported only in load module.

Command Syntax:

[SOURce:]ARB:POWer:RAMP:RTIMe <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:RAMP:RTIMe? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:POW:RAMP:RTIM 1,(@1) //Set the power rise time of the ramp waveform of Channel 1 to 1s

ARB:POW:RAMP:RTIM? (@1) //Query the power rise time of the ramp waveform of Channel 1.

[SOURce:]ARB:RESistance:RAMP:RTIMe <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:RAMP:RTIMe? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the rise time of the ramp waveform, supported only in load module.

Command Syntax:

[SOURce:]ARB:RESistance:RAMP:RTIMe <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:RAMP:RTIMe? [MINimum|MAXimum,](@<chanlist>)

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Return Parameters:

<time>

Example:

ARB:RES:RAMP:RTIM 1,(@1) //Set the resistor rise time of the ramp waveform of Channel 1 to 1s

ARB:RES:RAMP:RTIM? (@1) //Queries the resistance rise time of the ramp waveform of Channel 1.

[SOURce:]ARB:CURRent:RAMP:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:RAMP:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the start level of the ramp waveform.

Command Syntax:

[SOURce:]ARB:CURRent:RAMP:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:RAMP:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<current value>

Example:

ARB:CURR:RAMP:STAR 1,(@1) //Set the starting current of Channel 1's ramp waveform to 1A.

ARB:CURR:RAMP:STAR? (@1) //Queries the starting current of the ramp waveform of Channel 1.

[SOURce:]ARB:VOLTage:RAMP:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:RAMP:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the start level of the ramp waveform.

Command Syntax:

[SOURce:]ARB:VOLTage:RAMP:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:RAMP:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:VOLT:RAMP:STAR 1,(@1) //Set the start voltage of the ramp waveform of Channel 1 to 1V.

ARB:VOLT:RAMP:STAR? (@1) //Queries the start voltage of the ramp waveform of Channel 1.

[SOURce:]ARB:POWer:RAMP:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:POWer:RAMP:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the start level of the ramp waveform. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:RAMP:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:RAMP:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<power value>

Example:

ARB:POW:RAMP:STAR 1,(@1) //Set the starting power of Channel 1's ramp waveform to 1W.

ARB:POW:RAMP:STAR? (@1) //Queries the starting power of the ramp waveform of Channel 1.

[SOURce:]ARB:RESistance:RAMP:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:RAMP:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the start level of the ramp waveform. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:RAMP:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:RAMP:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:RES:RAMP:STAR 1,(@1) //Set the starting resistance of the ramp waveform of Channel 1 to 1Ω.

ARB:RES:RAMP:STAR? (@1) //Queries the starting resistance of the ramp

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waveform of Channel 1.

[SOURce:]ARB:CURRent:RAMP:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:RAMP:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the delay time of the ramp waveform.

Command Syntax:

[SOURce:]ARB:CURRent:RAMP:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:RAMP:STARt:TIME?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:CURR:RAMP:STAR:TIME 1,(@1) //Set the current start time of the ramp waveform of Channel 1 to 1s

ARB:CURR:RAMP:STAR:TIME? (@1) //Queries the current start time of the ramp waveform of Channel 1.

[SOURce:]ARB:VOLTage:RAMP:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:RAMP:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the delay time of the ramp waveform.

Command Syntax:

[SOURce:]ARB:VOLTage:RAMP:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:RAMP:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:VOLT:RAMP:STAR:TIME 1,(@1) //Set the voltage start time of the ramp waveform of Channel 1 to 1s.

ARB:VOLT:RAMP:STAR:TIME? (@1) //Queries the voltage start time of the ramp waveform of Channel 1.

[SOURce:]ARB:POWer:RAMP:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:POWer:RAMP:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the delay time of the ramp waveform. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:RAMP:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:RAMP:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:POW:RAMP:STAR:TIME 1,(@1) //Set the power onset time of the ramp waveform of Channel 1 to 1s.

ARB:POW:RAMP:STAR:TIME? (@1) //Queries the power onset time of the ramp waveform of Channel 1.

[SOURce:]ARB:RESistance:RAMP:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:RAMP:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the delay time of the ramp waveform. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:RAMP:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:RAMP:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:RES:RAMP:STAR:TIME 1,(@1) //Set the resistor start time of the ramp waveform of Channel 1 to 1s.

ARB:RES:RAMP:STAR:TIME? (@1) //Queries the resistor start time of the ramp waveform of Channel 1.

[SOURce:]ARB:CURRent:SINusoid:AMPLitude<value>, (@<chanlist>)

[SOURce:]ARB:CURRent:SINusoid:AMPLitude? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the sine wave amplitude of the ARB current.

Command Syntax:

[SOURce:]ARB:CURRent:SINusoid:AMPLitude<value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:SINusoid:AMPLitude?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<current value>

Example:

ARB:CURR:SIN:AMPL 1,(@1) //Set the current amplitude of the sinusoidal waveform of Channel 1 to 1A.

ARB:CURR:SIN:AMPL? (@1) //Query the current amplitude of sinusoidal waveform of Channel 1.

[SOURce:]ARB:VOLTage:SINusoid:AMPLitude<value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:SINusoid:AMPLitude? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the sine wave amplitude of the ARB voltage.

Command Syntax:

[SOURce:]ARB:VOLTage:SINusoid:AMPLitude<value>, (@<chanlist>)

Parameters:

Minimum-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:SINusoid:AMPLitude?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:VOLT:SIN:AMPL 1,(@1) //Set the voltage amplitude of the sinusoidal waveform of Channel 1 to 1V

ARB:VOLT:SIN:AMPL? (@1) //Query the voltage amplitude of the sinusoidal waveform of Channel 1.

[SOURce:]ARB:POWer:SINusoid:AMPLitude<value>, (@<chanlist>)

[SOURce:]ARB:POWer:SINusoid:AMPLitude? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the sine wave amplitude of the ARB power. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:SINusoid:AMPLitude<value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:SINusoid:AMPLitude? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<power value>

Example:

ARB:POW:SIN:AMPL 1,(@1) //Set the amplitude power of the sinusoidal waveform of Channel 1 to 1A

ARB:POW:SIN:AMPL? (@1) //Query the amplitude power of the sine waveform of Channel 1.

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[SOURce:]ARB:RESistance:SINusoid:AMPLitude<value>, (@<chanlist>)

[SOURce:]ARB:RESistance:SINusoid:AMPLitude? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the sine wave amplitude of the ARB resistor. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:SINusoid:AMPLitude<value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:SINusoid:AMPLitude? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:RES:SIN:AMPL 1,(@1) //Set the amplitude resistance of the sine waveform of Channel 1 to 1Ω.

ARB:RES:SIN:AMPL? (@1) //Queries the amplitude resistance of the sine waveform of Channel 1.

[SOURce:]ARB:CURRent:SINusoid:FREQuency<value>, (@<chanlist>)

[SOURce:]ARB:CURRent:SINusoid:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the frequency of the ARB sine wave.

Command Syntax:

[SOURce:]ARB:CURRent:SINusoid:FREQuency<value>, (@<chanlist>)

Parameters:

0.00001-10000.0||MINimum|MAXimum (in Hz)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:SINusoid:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<frequency>

Example:

ARB:CURR:SIN:FREQ 1,(@1) //Set the current frequency of the sinusoidal waveform of Channel 1 to 1Hz.

ARB:CURR:SIN:FREQ? (@1) //Queries the current frequency of the sine waveform of Channel 1.

[SOURce:]ARB:VOLTage:SINusoid:FREQuency<value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:SINusoid:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the frequency of the ARB sine wave.

Command Syntax:

[SOURce:]ARB:VOLTage:SINusoid:FREQuency<value>, (@<chanlist>)

Parameters:

0.00001-10000.0|MINimum|MAXimum (in Hz)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:SINusoid:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<frequency>

Example:

ARB:VOLT:SIN:FREQ 1,(@1) //Set the voltage frequency of the sinusoidal waveform of Channel 1 to 1Hz.

ARB:VOLT:SIN:FREQ? (@1) //Queries the voltage frequency of the sinusoidal

waveform of Channel 1.

[SOURce:]ARB:POWer:SINusoid:FREQuency<value>, (@<chanlist>)

[SOURce:]ARB:POWer:SINusoid:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the frequency of the ARB sine wave. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:SINusoid:FREQuency<value>, (@<chanlist>)

Parameters:

0.00001-10000.0|MINimum|MAXimum (in Hz)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:SINusoid:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<frequency>

Example:

ARB:POW:SIN:FREQ 1,(@1) //Set the power frequency of the sine waveform of Channel 1 to 1Hz.

ARB:POW:SIN:FREQ? (@1) //Queries the power frequency of the sine waveform of Channel 1.

[SOURce:]ARB:RESistance:SINusoid:FREQuency<value>, (@<chanlist>)

[SOURce:]ARB:RESistance:SINusoid:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the frequency of the ARB sine wave. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:SINusoid:FREQuency<value>, (@<chanlist>)

Parameters:

0.00001-10000.0|MINimum|MAXimum (in Hz)

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:SINusoid:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<frequency>

Example:

ARB:RES:SIN:FREQ 1,(@1) //Set the current frequency of the sinusoidal waveform of Channel 1 to 1Hz.

ARB:RES:SIN:FREQ? (@1) //Queries the current frequency of the sine waveform of Channel 1.

[SOURce:]ARB:CURRent:SINusoid:OFFSet<value>, (@<chanlist>)

[SOURce:]ARB:CURRent:SINusoid:OFFSet? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the bias of the ARB sine wave.

Command Syntax:

[SOURce:]ARB:CURRent:SINusoid:OFFSet<value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:SINusoid:OFFSet? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<current value>

Example:

ARB:CURR:SIN:OFFS 1,(@1) //Setting the bias current of the sine wave of Channel 1 to 1A

ARB:CURR:SIN:OFFS? (@1) //Query the bias current of the sine wave of Channel 1.

[SOURce:]ARB:VOLTage:SINusoid:OFFSet<value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:SINusoid:OFFSet? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the bias of the ARB sine wave.

Command Syntax:

[SOURce:]ARB:VOLTage:SINusoid:OFFSet<value>, (@<chanlist>)

Parameters:

Minimum-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:SINusoid:OFFSet? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<power value>

Example:

ARB:VOLT:SIN:OFFS 1,(@1) //Set the bias voltage of the sine wave of Channel 1 to 1V.

ARB:VOLT:SIN:OFFS? (@1) //Query the bias voltage of the sine wave of Channel 1.

[SOURce:]ARB:POWer:SINusoid:OFFSet<value>, (@<chanlist>)

[SOURce:]ARB:POWer:SINusoid:OFFSet? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the bias of the ARB sine wave.

Command Syntax:

[SOURce:]ARB:POWer:SINusoid:OFFSet<value>, (@<chanlist>)

Parameters:

Minimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:SINusoid:OFFSet? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<power value>

Example:

ARB:POW:SIN:OFFS 1,(@1) //Set the bias power of the sine wave of Channel 1 to 1A

ARB:POW:SIN:OFFS? (@1) //Query the bias power of the sine wave of Channel 1.

[SOURce:]ARB:RESistance:SINusoid:OFFSet<value>, (@<chanlist>)

[SOURce:]ARB:RESistance:SINusoid:OFFSet? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the bias of the ARB sine wave.

Command Syntax:

[SOURce:]ARB:RESistance:SINusoid:OFFSet<value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:SINusoid:OFFSet?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:RES:SIN:OFFS 1,(@1) //Set the bias resistor for the sine wave of Channel 1 to 1Ω.

ARB:RES:SIN:OFFS? (@1) //Query the bias resistance of the sine wave of Channel 1.

[SOURce:]ARB:CURRent:STAircase:END[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:STAircase:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the end level of the step wave.

Command Syntax:

[SOURce:]ARB:CURRent:STAircase:END[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:STAircase:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<current value>

Example:

ARB:CURR:STA:END 1,(@1) //Setting the end current of the step waveform of Channel 1 to 1A

ARB:CURR:STA:END? (@1) //Queries the end current of the step waveform of Channel 1.

[SOURce:]ARB:VOLTage:STAircase:END[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:STAircase:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the end level of the step wave.

Command Syntax:

[SOURce:]ARB:VOLTage:STAircase:END[:LEVel] <value>, (@<chanlist>)

Parameters:

Minimum-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:STAircase:END[:LEVel]?
[MINimum|MAXimum,](@<chanlist>)

Example:

ARB:VOLT:STA:END 1,(@1) //Set the ending voltage of the step waveform of Channel 1 to 1V

ARB:VOLT:STA:END? (@1) //Queries the end voltage of the step waveform of Channel 1.

[SOURce:]ARB:POWer:STAircase:END[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:POWer:STAircase:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the end level of the step wave. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:STAircase:END[:LEVel] <value>, (@<chanlist>)

Parameters:

Minimum-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:STAircase:END[:LEVel]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<power value>

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Example:

ARB:POW:STA:END 1,(@1) //Setting the end power of the step waveform of Channel 1 to 1W

ARB:POW:STA:END? (@1) //Queries the end power of the step waveform of Channel 1.

[SOURce:]ARB:RESistance:STAircase:END[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:STAircase:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the end level of the step wave. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:STAircase:END[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:STAircase:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:RES:STA:END 1,(@1) //Set the ending resistor of the step waveform of Channel 1 to 1Ω.

ARB:RES:STA:END? (@1) //Queries the ending resistance of the step waveform of Channel 1.

[SOURce:]ARB:CURRent:STAircase:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:STAircase:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the step waveform hold ends.

Command Syntax:

[SOURce:]ARB:CURRent:STAircase:END:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:STAircase:END:TIME?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:CURR:STAircase:END:TIME 1,(@1) //Set the current end time of the step waveform of Channel 1 to 1s

ARB:CURR:STAircase:END:TIME? (@1) //Queries the current end time of the step waveform of Channel 1.

[SOURce:]ARB:VOLTage:STAircase:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:STAircase:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the step waveform hold ends.

Command Syntax:

[SOURce:]ARB:VOLTage:STAircase:END:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:STAircase:END:TIME?
[MINimum|MAXimum,](@<chanlist>)

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Return Parameters:

<voltage value>

Example:

ARB:VOLT:STAircase:END:TIME 1,(@1) //Set the current end time of the step waveform of Channel 1 to 1s

ARB:VOLT:STAircase:END:TIME? (@1) //Queries the current end time of the step waveform of Channel 1.

[SOURce:]ARB:POWer:STAircase:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:POWer:STAircase:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the step waveform hold ends. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:STAircase:END:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:STAircase:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:POW:STAircase:END:TIME 1,(@1) //Set the current end time of the step waveform of Channel 1 to 1s

ARB:POW:STAircase:END:TIME? (@1) //Queries the current end time of the step waveform of Channel 1.

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[SOURce:]ARB:RESistance:STAircase:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:STAircase:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the step waveform hold ends. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:STAircase:END:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:STAircase:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:RES:STAircase:END:TIME 1,(@1) //Set the resistor end time of the step waveform of Channel 1 to 1s

ARB:RES:STAircase:END:TIME? (@1) //Queries the resistance end time of the step waveform of Channel 1.

[SOURce:]ARB:CURRent:STAircase:NSTeps <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:STAircase:NSTeps? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the total number of ladder steps.

Command Syntax:

[SOURce:]ARB:CURRent:STAircase:NSTeps <value>, (@<chanlist>)

Parameters:

0-500|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:STAircase:NSTeps? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<steps>

Example:

ARB:CURR:STA:NST 1,(@1) //Set the current step of the stepped waveform of Channel 1 to 1

ARB:CURR:STA:NST? (@1) //Queries the number of current steps in the step waveform of Channel 1.

[SOURce:]ARB:VOLTage:STAircase:NSTeps <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:STAircase:NSTeps? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the total number of ladder steps.

Command Syntax:

[SOURce:]ARB:VOLTage:STAircase:NSTeps <value>, (@<chanlist>)

Parameters:

0-500|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:STAircase:NSTeps? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<steps>

Example:

ARB:VOLT:STA:NST 1,(@1) //Set the voltage step of the stepped waveform of Channel 1 to 1

ARB:VOLT:STA:NST? (@1) //Query the number of voltage steps of the

stepped waveform of Channel 1

[SOURce:]ARB:POWer:STAircase:NSTeps <value>, (@<chanlist>)

[SOURce:]ARB:POWer:STAircase:NSTeps? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the total number of ladder steps. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:STAircase:NSTeps <value>, (@<chanlist>)

Parameters:

0-500|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:STAircase:NSTeps? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Power>

Example:

ARB:POW:STA:NST 1,(@1) //Set the power step of the stepped waveform of Channel 1 to 1

ARB:POW:STA:NST? (@1) //Query the number of power steps of the stepped waveform of Channel 1

[SOURce:]ARB:RESistance:STAircase:NSTeps <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:STAircase:NSTeps? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the total number of ladder steps. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:STAircase:NSTeps <value>, (@<chanlist>)

Parameters:

0-500|MINimum|MAXimum
<chanlist>:Channel number.

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(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:STAircase:NSTeps? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<steps>

Example:

ARB:RES:STA:NST 1,(@1) //Set the number of resistor steps for the step waveform of Channel 1 to 1

ARB:RES:STA:NST? (@1) //Queries the number of resistor steps of the step waveform of Channel 1.

[SOURce:]ARB:CURRent:STAircase:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:STAircase:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Setting and querying before setting and querying the ladder.

Command Syntax:

[SOURce:]ARB:CURRent:STAircase:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum(in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:STAircase:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<current value>

Example:

ARB:CURR:STA:STAR 1,(@1) //Set the starting current of the step waveform of Channel 1 to 1A

ARB:CURR:STA:STAR? (@1) //Queries the starting current of the step

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waveform of Channel 1.

[SOURce:]ARB:VOLTage:STAircase:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:STAircase:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Setting and querying before setting and querying the ladder.

Command Syntax:

[SOURce:]ARB:VOLTage:STAircase:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

Minimum-MAXimum|MINimum|MAXimum(in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:STAircase:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:VOLT:STA:STAR 1,(@1) //Set the start voltage of the step waveform of Channel 1 to 1V

ARB:VOLT:STA:STAR? (@1) //Queries the start voltage of the step waveform of Channel 1.

[SOURce:]ARB:POWer:STAircase:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:POWer:STAircase:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Setting and querying before setting and querying the ladder. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:STAircase:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum(in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:STAircase:STARt[:LEVel]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<power value>

Example:

ARB:POW:STA:STAR 1,(@1) //Set the starting power of the step waveform of Channel 1 to 1W

ARB:POW:STA:STAR? (@1) //Queries the starting power of the step waveform of Channel 1.

[SOURce:]ARB:RESistance:STAircase:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:STAircase:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Setting and querying before setting and querying the ladder. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:STAircase:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum(in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:STAircase:STARt[:LEVel]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<resistance value>

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Example:

ARB:RES:STA:STAR 1,(@1) //Set the starting resistance of the step waveform of Channel 1 to 1Ω.

ARB:RES:STA:STAR? (@1) //Queries the starting resistance of the step waveform of Channel 1.

[SOURce:]ARB:CURRent:STAircase:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:STAircase:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the delay time after the trigger signal is received and before the step waveform starts to be output.

Command Syntax:

[SOURce:]ARB:CURRent:STAircase:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:STAircase:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:CURR:STA:STAR:TIME 1,(@1) //Set the current start time of the step waveform of Channel 1 to 1s

ARB:CURR:STA:STAR:TIME? (@1) //Queries the current start time of the step waveform of Channel 1.

[SOURce:]ARB:VOLTage:STAircase:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:STAircase:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the delay time after the trigger signal is received and before the step waveform starts to be output.

Command Syntax:

[SOURce:]ARB:VOLTage:STAircase:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:STAircase:STARt:TIME?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:VOLT:STA:STAR:TIME 1,(@1) //Set the voltage start time of the step waveform of Channel 1 to 1s

ARB:VOLT:STA:STAR:TIME? (@1) //Queries the voltage start time of the step waveform of Channel 1.

[SOURce:]ARB:POWer:STAircase:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:POWer:STAircase:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the delay time after the trigger signal is received and before the step waveform starts to be output. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:STAircase:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:STAircase:STARt:TIME?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:POW:STA:STAR:TIME 1,(@1) //Set the power onset time of the step waveform of Channel 1 to 1s

ARB:POW:STA:STAR:TIME? (@1) //Query the power start time of the step waveform of Channel 1

[SOURce:]ARB:RESistance:STAircase:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:STAircase:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the delay time after the trigger signal is received and before the step waveform starts to be output. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:STAircase:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:STAircase:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:RES:STA:STAR:TIME 1,(@1) //Set the resistor start time of the step waveform of Channel 1 to 1s

ARB:RES:STA:STAR:TIME? (@1) //Queries the resistance start time of the step waveform of Channel 1.

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[SOURce:]ARB:CURRent:STAircase:TIME <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:STAircase:TIME? [MINimum|MAXimum,](@<chanlist>)

Setting and querying the time required to complete all ladder steps

Command Syntax:

[SOURce:]ARB:CURRent:STAircase:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:STAircase:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:CURR:STA:TIME 1,(@1) //Set the current end time of the step waveform of Channel 1 to 1s

ARB:CURR:STA:TIME? (@1) //Queries the current end time of the step waveform of Channel 1.

[SOURce:]ARB:VOLTage:STAircase:TIME <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:STAircase:TIME? [MINimum|MAXimum,](@<chanlist>)

Setting and querying the time required to complete all ladder steps

Command Syntax:

[SOURce:]ARB:VOLTage:STAircase:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:STAircase:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:VOLT:STA:TIME 1,(@1) //Set the voltage end time of the step waveform of Channel 1 to 1s

ARB:VOLT:STA:TIME? (@1) //Queries the voltage end time of the step waveform of Channel 1.

[SOURce:]ARB:POWer:STAircase:TIME <value>, (@<chanlist>)

[SOURce:]ARB:POWer:STAircase:TIME? [MINimum|MAXimum,](@<chanlist>)

Setting and querying the time required to complete all ladder steps, only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:STAircase:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:STAircase:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:POW:STA:TIME 1,(@1) //Set the power end time for the step waveform of Channel 1 to 1s

ARB:POW:STA:TIME? (@1) //Query the power end time of the step waveform of Channel 1

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[SOURce:]ARB:RESistance:STAircase:TIME <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:STAircase:TIME? [MINimum|MAXimum,](@<chanlist>)

Set and query the time it takes to complete all ladder steps. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:STAircase:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:STAircase:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:RES:STA:TIME 1,(@1) //Set the resistor end time of the step waveform of Channel 1 to 1s

ARB:RES:STA:TIME? (@1) //Queries the resistance end time of the step waveform of Channel 1.

[SOURce:]ARB:CURRent:STEP:END[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:STEP:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Setting and querying after a step leap.

Command Syntax:

[SOURce:]ARB:CURRent:STEP:END[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:STEP:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<current>

Example:

ARB:CURR:STEP:END 1,(@1) //Setting and querying after a step leap.

ARB:CURR:STEP:END? (@1) //Queries the end current of the step waveform of Channel 1.

[SOURce:]ARB:VOLTage:STEP:END[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:STEP:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Setting and querying after a step leap.

Command Syntax:

[SOURce:]ARB:VOLTage:STEP:END[:LEVel] <value>, (@<chanlist>)

Parameters:

Minimum-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:STEP:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:VOLT:STEP:END 1,(@1) //Set the end voltage of the step waveform of Channel 1 to 1V.

ARB:VOLT:STEP:END? (@1) //Queries the end voltage of the step waveform of Channel 1.

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[SOURce:]ARB:POWer:STEP:END[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:POWer:STEP:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Setting and querying after a step leap. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:STEP:END[:LEVel] <value>, (@<chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:STEP:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<power value>

Example:

ARB:POW:STEP:END 1,(@1) //Setting the end power of the step waveform of Channel 1 to 1W

ARB:POW:STEP:END? (@1) //Queries the end power of the step waveform of Channel 1.

[SOURce:]ARB:RESistance:STEP:END[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:STEP:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Setting and querying after a step leap. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:STEP:END[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:STEP:END[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Resistance>

Example:

ARB:RES:STEP:END 1,(@1) //Set the ending resistance of the step waveform of Channel 1 to 1Ω.

ARB:RES:STEP:END? (@1) //Queries the end resistance of the step waveform of Channel 1.

[SOURce:]ARB:CURRent:STEP:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:STEP:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Setup and query before the step leap.

Command Syntax:

[SOURce:]ARB:CURRent:STEP:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:STEP:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<current value>

Example:

ARB:CURR:STEP:STAR 1,(@1) //Set the starting current of the step waveform of Channel 1 to 1A.

ARB:CURR:STEP:STAR? (@1) //Setting the start current of the step waveform of Channel 1

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[SOURce:]ARB:VOLTage:STEP:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:STEP:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Setup and query before the step leap.

Command Syntax:

[SOURce:]ARB:VOLTage:STEP:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

Minimum-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:STEP:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:VOLT:STEP:STAR 1,(@1) //Set the start voltage of the step waveform of Channel 1 to 1V.

ARB:VOLT:STEP:STAR? (@1) //Queries the start voltage of the step waveform of Channel 1.

[SOURce:]ARB:POWer:STEP:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:POWer:STEP:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Setup and query before the step leap. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:STEP:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:STEP:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<Power>

Example:

ARB:POW:STEP:STAR 1,(@1) //Set the starting power of the step waveform of Channel 1 to 1W.

ARB:POW:STEP:STAR? (@1) //Queries the starting power of the step waveform of Channel 1.

[SOURce:]ARB:RESistance:STEP:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:STEP:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Setup and query before the step leap. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:STEP:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:STEP:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:RES:STEP:STAR 1,(@1) //Set the starting resistance of the step waveform of Channel 1 to 1A.

ARB:RES:STEP:STAR? (@1) //Queries the starting resistance of the step

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waveform of Channel 1.

[SOURce:]ARB:CURRent:STEP:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:STEP:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the delay time after a trigger signal is received and before a step occurs.

Command Syntax:

[SOURce:]ARB:CURRent:STEP:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:STEP:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:CURR:STEP:STAR:TIME 1,(@1) //Set the current start time of the step waveform of Channel 1 to 1s

ARB:CURR:STEP:STAR:TIME? (@1) //Queries the current start time of the step waveform of Channel 1.

[SOURce:]ARB:VOLTage:STEP:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:STEP:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the delay time after the trigger signal is received and before the step occurs.

Command Syntax:

[SOURce:]ARB:VOLTage:STEP:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:STEP:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:VOLT:STEP:STAR:TIME 1,(@1) //Set the voltage start time of the step waveform of Channel 1 to 1s

ARB:VOLT:STEP:STAR:TIME? (@1) //Queries the voltage start time of the step waveform of Channel 1.

[SOURce:]ARB:POWer:STEP:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:POWer:STEP:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the delay time after the trigger signal is received and before the step occurs. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:STEP:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:STEP:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:POW:STEP:STAR:TIME 1,(@1) //Set the power onset time of the step waveform of Channel 1 to 1s

ARB:POW:STEP:STAR:TIME? (@1) //Queries the power onset time of the step waveform of Channel 1.

[SOURce:]ARB:RESistance:STEP:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:STEP:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the delay time after the trigger signal is received and before the step occurs. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:STEP:STARt:TIME <value>, (@<chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:STEP:STARt:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:RES:STEP:STAR:TIME 1,(@1) //Set the resistor start time of the step waveform of Channel 1 to 1s

ARB:RES:STEP:STAR:TIME? (@1) //Queries the resistance start time of the step waveform of Channel 1.

[SOURce:]ARB:CURRent:STEP:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:STEP:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the delay time after the trigger signal is received until after the step occurs.

Command Syntax:

[SOURce:]ARB:CURRent:STEP:END:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:STEP:END:TIME?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:CURR:STEP:END:TIME 1,(@1) //Setting the current end time of the step waveform of Channel 1 to 1s

ARB:CURR:STEP:END:TIME? (@1) //Queries the current end time of the step waveform of Channel 1.

[SOURce:]ARB:VOLTage:STEP:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:STEP:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the delay time after the trigger signal is received until after the step occurs.

Command Syntax:

[SOURce:]ARB:VOLTage:STEP:END:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:STEP:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:VOLT:STEP:END:TIME 1,(@1) //Set the voltage end time of the step waveform of Channel 1 to 1s

ARB:VOLT:STEP:END:TIME? (@1) //Queries the voltage end time of the step waveform of Channel 1.

[SOURce:]ARB:POWer:STEP:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:POWer:STEP:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the delay time after the trigger signal is received until after the step occurs. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:STEP:END:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:STEP:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:POW:STEP:END:TIME 1,(@1) //Setting the current end time of the step waveform of Channel 1 to 1s

ARB:POW:STEP:END:TIME? (@1) //Queries the current end time of the step waveform of Channel 1.

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[SOURce:]ARB:RESistance:STEP:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:STEP:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the delay time after the trigger signal is received until after the step occurs. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:STEP:END:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:STEP:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:RES:STEP:END:TIME 1,(@1) //Set the resistor end time of the step waveform of Channel 1 to 1s

ARB:RES:STEP:END:TIME? (@1) //Queries the resistance end time of the step waveform of Channel 1.

[SOURce:]ARB:CURRent:TRAPezoid:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:TRAPezoid:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the output hold ends after the ladder wave completes.

Command Syntax:

[SOURce:]ARB:CURRent:TRAPezoid:END:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:TRAPezoid:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:CURR:TRAP:END:TIME 1,(@1) //Set the current end time of the trapezoidal waveform of Channel 1 to 1s

ARB:CURR:TRAP:END:TIME? (@1) //Queries the current end time of the trapezoidal waveform of Channel 1.

[SOURce:]ARB:VOLTage:TRAPezoid:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:TRAPezoid:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the output hold ends after the ladder wave completes.

Command Syntax:

[SOURce:]ARB:VOLTage:TRAPezoid:END:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:TRAPezoid:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:VOLT:TRAP:END:TIME 1,(@1) //Set the voltage end time of the trapezoidal

waveform of Channel 1 to 1s

ARB:VOLT:TRAP:END:TIME? (@1) //Queries the voltage end time of the trapezoidal waveform of Channel 1.

[SOURce:]ARB:POWer:TRAPezoid:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:POWer:TRAPezoid:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the output hold ends after the ladder wave completes. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:TRAPezoid:END:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:TRAPezoid:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:POW:TRAP:END:TIME 1,(@1) //Set the power end time of the trapezoidal waveform of Channel 1 to 1s

ARB:POW:TRAP:END:TIME? (@1) //Query the power end time of the trapezoidal waveform of Channel 1

[SOURce:]ARB:RESistance:TRAPezoid:END:TIME <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:TRAPezoid:END:TIME? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the output hold ends after the ladder wave completes. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:TRAPezoid:END:TIME <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:TRAPezoid:END:TIME?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:RES:TRAP:END:TIME 1,(@1) //Set the resistor end time of the trapezoidal waveform of Channel 1 to 1s

ARB:RES:TRAP:END:TIME? (@1) //Queries the resistance end time of the trapezoidal waveform of Channel 1.

[SOURce:]ARB:CURRent:TRAPezoid:FTIMe <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:TRAPezoid:FTIMe? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the trapezoidal wave descends along the ramp.

Command Syntax:

[SOURce:]ARB:CURRent:TRAPezoid:FTIMe <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:TRAPezoid:FTIMe?
[MINimum|MAXimum,](@<chanlist>)

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Return Parameters:

<time>

Example:

ARB:CURR:TRAP:FTIM 1,(@1) //Set the current fall time of the trapezoidal waveform of Channel 1 to 1s

ARB:CURR:TRAP:FTIM? (@1) //Query the current fall time of the trapezoidal waveform of Channel 1

[SOURce:]ARB:VOLTage:TRAPezoid:FTIMe <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:TRAPezoid:FTIMe? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the trapezoidal wave descends along the ramp.

Command Syntax:

[SOURce:]ARB:VOLTage:TRAPezoid:FTIMe <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:TRAPezoid:FTIMe? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:VOLT:TRAP:FTIM 1,(@1) //Set the voltage drop time of the trapezoidal waveform of Channel 1 to 1s

ARB:VOLT:TRAP:FTIM? (@1) //Query the voltage drop time of the trapezoidal waveform of Channel 1.

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[SOURce:]ARB:POWer:TRAPezoid:FTIMe <value>, (@<chanlist>)

[SOURce:]ARB:POWer:TRAPezoid:FTIMe? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the trapezoidal wave descends along the ramp. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:TRAPezoid:FTIMe <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:TRAPezoid:FTIMe? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:POW:TRAP:FTIM 1,(@1) //Set the power down time of the trapezoidal waveform of Channel 1 to 1s

ARB:POW:TRAP:FTIM? (@1) //Query the power down time of the trapezoidal waveform of Channel 1

[SOURce:]ARB:RESistance:TRAPezoid:FTIMe <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:TRAPezoid:FTIMe? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the trapezoidal wave descends along the ramp. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:TRAPezoid:FTIMe <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:TRAPezoid:FTIMe? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:RES:TRAP:FTIM 1,(@1) //Set the resistance fall time of the trapezoidal waveform of Channel 1 to 1s

ARB:RES:TRAP:FTIM? (@1) //Query the resistance fall time of the trapezoidal waveform of Channel 1.

[SOURce:]ARB:CURRent:TRAPezoid:RTIMe <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:TRAPezoid:RTIMe? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the trapezoidal wave rises along the ramp.

Command Syntax:

[SOURce:]ARB:CURRent:TRAPezoid:RTIMe <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:TRAPezoid:RTIMe? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:CURR:TRAP:RTIM 1,(@1) //Set the current rise time of the trapezoidal waveform of Channel 1 to 1s

ARB:CURR:TRAP:RTIM? (@1) //Query the current rise time of the

trapezoidal waveform of Channel 1.

[SOURce:]ARB:VOLTage:TRAPezoid:RTIMe <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:TRAPezoid:RTIMe? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the trapezoidal wave rises along the ramp.

Command Syntax:

[SOURce:]ARB:VOLTage:TRAPezoid:RTIMe <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:TRAPezoid:RTIMe?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:VOLT:TRAP:RTIM 1,(@1) //Set the voltage rise time of the trapezoidal waveform of Channel 1 to 1s

ARB:VOLT:TRAP:RTIM? (@1) //Query the voltage rise time of the trapezoidal waveform of Channel 1.

[SOURce:]ARB:POWer:TRAPezoid:RTIMe <value>, (@<chanlist>)

[SOURce:]ARB:POWer:TRAPezoid:RTIMe? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the trapezoidal wave rises along the ramp. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:TRAPezoid:RTIMe <value>, (@<chanlist>)

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Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:TRAPezoid:RTIMe? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:POW:TRAP:RTIM 1,(@1) //Set the power rise time of the trapezoidal waveform of Channel 1 to 1s

ARB:POW:TRAP:RTIM? (@1) //Query the power rise time of the trapezoidal waveform of Channel 1

[SOURce:]ARB:RESistance:TRAPezoid:RTIMe <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:TRAPezoid:RTIMe? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the time at which the trapezoidal wave rises along the ramp. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:TRAPezoid:RTIMe <value>, (@<chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:TRAPezoid:RTIMe?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<time>

Example:

ARB:RES:TRAP:RTIM 1,(@1) //Set the resistance rise time of the trapezoidal waveform of Channel 1 to 1s

ARB:RES:TRAP:RTIM? (@1) //Query the resistance rise time of the trapezoidal waveform of Channel 1.

[SOURce:]ARB:CURRent:TRAPezoid:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:TRAPezoid:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the ladder waveform start level.

Command Syntax:

[SOURce:]ARB:CURRent:TRAPezoid:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:TRAPezoid:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<current value>

Example:

ARB:CURR:TRAP:STAR 1,(@1) //Set the starting current of the trapezoidal waveform of Channel 1 to 1A

ARB:CURR:TRAP:STAR? (@1) //Queries the starting current of the trapezoidal waveform of Channel 1.

[SOURce:]ARB:VOLTage:TRAPezoid:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:TRAPezoid:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the ladder waveform start level.

Command Syntax:

[SOURce:]ARB:VOLTage:TRAPezoid:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

Minimum-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:TRAPezoid:STARt[:LEVel]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:VOLT:TRAP:STAR 1,(@1) //Set the starting voltage of the trapezoidal waveform of Channel 1 to 1V

ARB:VOLT:TRAP:STAR? (@1) //Queries the starting voltage of the trapezoidal waveform of Channel 1.

[SOURce:]ARB:POWer:TRAPezoid:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:POWer:TRAPezoid:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the ladder waveform start level. only load module supports.

Command Syntax:

[SOURce:]ARB:POWer:TRAPezoid:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:TRAPezoid:STARt[:LEVel]?
[MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<power value>

Example:

ARB:POW:TRAP:STAR 1,(@1) //Set the starting power of the trapezoidal waveform of Channel 1 to 1W

ARB:POW:TRAP:STAR? (@1) //Query the starting power of the trapezoidal waveform of Channel 1

[SOURce:]ARB:RESistance:TRAPezoid:STARt[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:TRAPezoid:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Sets and queries the ladder waveform start level. only load module supports.

Command Syntax:

[SOURce:]ARB:RESistance:TRAPezoid:STARt[:LEVel] <value>, (@<chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:TRAPezoid:STARt[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:RES:TRAP:STAR 1,(@1) //Set the starting resistance of the trapezoidal waveform of Channel 1 to 1Ω.

ARB:RES:TRAP:STAR? (@1) //Queries the starting resistance of the trapezoidal waveform of Channel 1.

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[SOURce:]ARB:CURRent:TRAPezoid:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:TRAPezoid:STARt:TIME? [MINimum|MAXimum,] (@<chanlist>)

This command is used to set and query the delay period between receiving the trigger signal and the start of the trapezoidal wave.

Command Syntax:

[SOURce:]ARB:CURRent:TRAPezoid:STARt:TIME <value>, (@<chanlist>)

Arguments

0.000000-10000.0|MINimum|MAXimum (unit: s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:TRAPezoid:STARt:TIME? [MINimum|MAXimum,] (@<chanlist>)

Returns

<time>

Example

ARB:CURRent:TRAPezoid:STARt:TIME 1,(@1) // Set the start time of the trapezoidal waveform current for Channel 1 to 1 second.

ARB:CURRent:TRAPezoid:STARt:TIME? (@1) // Set the start time of the trapezoidal waveform current for Channel 1 to 1 second.

[SOURce:]ARB:VOLTage:TRAPezoid:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:TRAPezoid:STARt:TIME? [MINimum|MAXimum,] (@<chanlist>)

This command is used to set and query the delay period between receiving the trigger signal and the start of the trapezoidal wave.

Command Syntax:

[SOURce:]ARB:VOLTage:TRAPezoid:STARt:TIME <value>, (@<chanlist>)

Arguments

0.000000-10000.0|MINimum|MAXimum (unit:s)

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<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:TRAPezoid:STARt:TIME? [MINimum|MAXimum,] (@<chanlist>)

Returns

<time>

Example

ARB:VOLTage:TRAPezoid:STARt:TIME 1,(@1) // Set the start time of the trapezoidal waveform current for Channel 1 to 1 second.

ARB:VOLTage:TRAPezoid:STARt:TIME? (@1) // Query the start time of the current in the trapezoidal waveform of Channel 1

[SOURce:]ARB:POWer:TRAPezoid:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:POWer:TRAPezoid:STARt:TIME? [MINimum|MAXimum,] (@<chanlist>)

Set and query the delay time between receiving the trigger signal and the occurrence of the step. Supported only by load modules.

Command Syntax:

[SOURce:]ARB:POWer:TRAPezoid:STARt:TIME <value>, (@<chanlist>)

Arguments

0.000000-10000.0|MINimum|MAXimum (unit:s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:TRAPezoid:STARt:TIME? [MINimum|MAXimum,] (@<chanlist>)

Returns

<time>

Example

ARB:POWer:TRAPezoid:STARt:TIME 1,(@1) // Set the power start time of the

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trapezoidal waveform for Channel 1 to 1 second.

ARB:POWer:TRAPezoid:STARt:TIME? (@1) // Query the power start time of the trapezoidal waveform on Channel 1

[SOURce:]ARB:RESistance:TRAPezoid:STARt:TIME <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:TRAPezoid:STARt:TIME? [MINimum|MAXimum,] (@<chanlist>)

Set and query the delay time between receiving the trigger signal and the occurrence of the step. Supported only by load modules.

Command Syntax:

[SOURce:]ARB:RESistance:TRAPezoid:STARt:TIME <value>, (@<chanlist>)

Arguments

0.000000-10000.0|MINimum|MAXimum (Unit:s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:TRAPezoid:STARt:TIME? [MINimum|MAXimum,] (@<chanlist>)

Returns

<time>

Example

ARB:RESistance:TRAPezoid:STARt:TIME 1,(@1) // Set the start time of the resistor for the trapezoidal waveform on Channel 1 to 1 second.

ARB:RESistance:TRAPezoid:STARt:TIME? (@1) // Resistance start time of the trapezoidal waveform on channel 1

[SOURce:]ARB:CURRent:TRAPezoid:TOP[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:TRAPezoid:TOP[:LEVel]? [MINimum|MAXimum,] (@<chanlist>)

This command is used to set and query the peak value of a trapezoidal wave.

Command Syntax:

[SOURce:]ARB:CURRent:TRAPezoid:TOP[:LEVel] <value>, (@<chanlist>)

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Arguments

MINimum -MAXimum|MINimum|MAXimum (unit:V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:TRAPezoid:TOP[:LEVel]? [MINimum|MAXimum,] (@<chanlist>)

Returns

<I>

Example

ARB:CURRent:TRAPezoid:TOP 1,(@1) // Set the peak current of the trapezoidal waveform on Channel 1 to 1A.

ARB:CURRent:TRAPezoid:TOP? (@1) // Peak current of the trapezoidal waveform on channel 1

[SOURce:]ARB:VOLTage:TRAPezoid:TOP[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:TRAPezoid:TOP[:LEVel]? [MINimum|MAXimum,] (@<chanlist>)

This command is used to set and query the peak value of a trapezoidal wave.

Command Syntax:

[SOURce:]ARB:VOLTage:TRAPezoid:TOP[:LEVel] <value>, (@<chanlist>)

Arguments

MINimum -MAXimum|MINimum|MAXimum (unit:V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:TRAPezoid:TOP[:LEVel]? [MINimum|MAXimum,] (@<chanlist>)

Returns

<V>

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Example

ARB:VOLTage:TRAPezoid:TOP 10,(@1) // Set the peak voltage of the trapezoidal waveform on Channel 1 to 1A.

ARB:VOLTage:TRAPezoid:TOP? (@1) // Peak voltage of the trapezoidal waveform on channel 1

[SOURce:]ARB:POWer:TRAPezoid:TOP[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:POWer:TRAPezoid:TOP[:LEVel]? [MINimum|MAXimum,] (@<chanlist>)

This command is used to set and query the peak value of the trapezoidal wave. Supported only by load modules.

Command Syntax:

[SOURce:]ARB:POWer:TRAPezoid:TOP[:LEVel] <value>, (@<chanlist>)

Arguments

MINimum -MAXimum|MINimum|MAXimum (unit:V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:TRAPezoid:TOP[:LEVel]? [MINimum|MAXimum,] (@<chanlist>)

Returns

<P>

Example

ARB:POWer:TRAPezoid:TOP 100,(@1) // Set the peak power of the trapezoidal waveform on Channel 1 to 1A.

ARB:POWer:TRAPezoid:TOP? (@1) // Peak power of the trapezoidal waveform on channel 1

[SOURce:]ARB:RESistance:TRAPezoid:TOP[:LEVel] <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:TRAPezoid:TOP[:LEVel]? [MINimum|MAXimum,] (@<chanlist>)

This command is used to set and query the peak value of the trapezoidal wave. Supported only by load modules.

Command Syntax:

[SOURce:]ARB:RESistance:TRAPezoid:TOP[:LEVel] <value>, (@<chanlist>)

Arguments

MINimum -MAXimum|MINimum|MAXimum (unit: V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:TRAPezoid:TOP[:LEVel]? [MINimum|MAXimum,] (@<chanlist>)

Returns

<P>

Example

ARB:RESistance:TRAPezoid:TOP 1000,(@1) // Set the peak resistance of the trapezoidal waveform for Channel 1 to 1A.

ARB:RESistance:TRAPezoid:TOP? (@1) // Peak resistance of the trapezoidal waveform on channel 1

[SOURce:]ARB:CURRent:TRAPezoid:TOP:TIME <value>, (@<chanlist>)

[SOURce:]ARB:CURRent:TRAPezoid:TOP:TIME? [MINimum|MAXimum,] (@<chanlist>)

This command is used to set and query the peak width of a trapezoidal wave.

Command Syntax:

[SOURce:]ARB:CURRent:TRAPezoid:TOP:TIME <value>, (@<chanlist>)

Arguments

0.000005-10000.0|MINimum|MAXimum (Unit: s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:TRAPezoid:TOP:TIME? [MINimum|MAXimum,] (@<chanlist>)

Returns

<time>

Example

ARB:CURRent:TRAPezoid:TOP:TIME 1,(@1) // Set the peak current duration of the trapezoidal waveform for Channel 1 to 1 second.

ARB:CURRent:TRAPezoid:TOP:TIME? (@1) // Peak current duration of the trapezoidal waveform on Channel 1

[SOURce:]ARB:VOLTage:TRAPezoid:TOP:TIME <value>, (@<chanlist>)

[SOURce:]ARB:VOLTage:TRAPezoid:TOP:TIME? [MINimum|MAXimum,] (@<chanlist>)

This command is used to set and query the delay period between receiving the trigger signal and the start of the trapezoidal wave.

Command Syntax:

[SOURce:]ARB:VOLTage:TRAPezoid:TOP:TIME <value>, (@<chanlist>)

Arguments

0.000005-10000.0|MINimum|MAXimum (unit: s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:TRAPezoid:TOP:TIME? [MINimum|MAXimum,] (@<chanlist>)

Returns

<time>

Example

ARB:VOLTage:TRAPezoid:TOP:TIME 1,(@1) // Set the peak voltage duration of the trapezoidal waveform for Channel 1 to 1 second.

ARB:VOLTage:TRAPezoid:TOP:TIME? (@1) // Peak voltage time of the trapezoidal waveform on channel 1

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[SOURce:]ARB:POWer:TRAPezoid:TOP:TIME <value>, (@<chanlist>)

[SOURce:]ARB:POWer:TRAPezoid:TOP:TIME? [MINimum|MAXimum,] (@<chanlist>)

Set and query the delay time between receiving the trigger signal and the occurrence of the step. Supported only by load modules.

Command Syntax:

[SOURce:]ARB:POWer:TRAPezoid:TOP:TIME <value>, (@<chanlist>)

Arguments

0.000005-10000.0|MINimum|MAXimum (Unit:s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:TRAPezoid:TOP:TIME? [MINimum|MAXimum,] (@<chanlist>)

Returns

<time>

Example

ARB:POWer:TRAPezoid:TOP:TIME 1,(@1) //Set the peak power duration of the trapezoidal waveform for Channel 1 to 1 second.

ARB:POWer:TRAPezoid:TOP:TIME? (@1) //Peak power duration of the trapezoidal waveform on Channel 1

[SOURce:]ARB:RESistance:TRAPezoid:TOP:TIME <value>, (@<chanlist>)

[SOURce:]ARB:RESistance:TRAPezoid:TOP:TIME? [MINimum|MAXimum,] (@<chanlist>)

Set and query the delay time between receiving the trigger signal and the occurrence of the step. Supported only by load modules.

Command Syntax:

[SOURce:]ARB:RESistance:TRAPezoid:TOP:TIME <value>, (@<chanlist>)

Arguments

0.000005-10000.0|MINimum|MAXimum (unit:s)

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<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:TRAPezoid:TOP:TIME? [MINimum|MAXimum,] (@<chanlist>)

Returns

<time>

Example

ARB:RESistance:TRAPezoid:TOP:TIME 1,(@1) //Set the peak resistance time of the trapezoidal waveform for Channel 1 to 1 second.

ARB:RESistance:TRAPezoid:TOP:TIME? (@1) //Peak resistance time of the trapezoidal waveform on channel 1

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Chapter18 Sequence Subsystem

[SOURce:]ARB:SEQuence:COUNt <value>, (@<chanlist>)

[SOURce:]ARB:SEQuence:COUNt? [MINimum|MAXimum,] (@<chanlist>)

Sets and queries the number of repetitions for a sequence waveform. When the repetition count is 0, it indicates infinite repetition.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:COUNt <value>, <step#>, (@<chanlist>)

Parameters:

0-65535|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:COUNt? [|MINimum|MAXimum,] <step#>, (@<chanlist>)

Return Parameters:

<repeat count>

Example:

ARB:SEQuence:STEP:COUN 10,(@1) //Set the number of running repetitions of the sequence waveform for CH1 to 10

ARB:SEQuence:STEP:COUN? (@1) //Query the number of repetitions for the sequence waveform on Channel 1

[SOURce:]ARB:SEQuence:STEP:COUNt <value>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:COUNt? [MINimum|MAXimum,](@chanlist>)

Sets and queries the repeat count of the sequence waveform. When the repeat count is 0, it means None limit repeat.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:COUNt <value>,(@chanlist>)

Parameters:

0-65535|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:COUNt? [MINimum|MAXimum,](@chanlist>)

Return Parameters:

<repeat count>

Example:

ARB:SEQuence:STEP:COUN 10,(@1) //Set the number of running repetitions of the sequence waveform for Channel 1 to 10

ARB:SEQuence:STEP:COUN? (@1) //Query Channel 1 for the number of repetitions of the sequence waveform run

[SOURce:]ARB:SEQuence:COUNt:RUN? (@chanlist>)

Queries the number of times the current ARB has been run.

Command Syntax:

[SOURce:]ARB:SEQuence:COUNt:RUN?

Return Parameters:

<number of runs>

Example:

ARB:SEQuence:COUN:RUN? (@1) //Query Channel 1 for the number of runs in the current sequence

[SOURce:]ARB:SEQuence:LENGth <value>,(@<chanlist>)

[SOURce:]ARB:SEQuence:LENGth? (@<chanlist>)

Sets and queries the maximum total number of steps for the sequence.

Command Syntax:

[SOURce:]ARB:SEQuence:LENGth <value>,(@<chanlist>)

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Parameters:

1-500|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:LENGth? (@<chanlist>)

Return Parameters:

<total number>

Example:

ARB:SEQuence:LENGth 20,(@1) // Set the number of steps for the sequence waveform on CH1 to 20 steps.

ARB:SEQuence:LENGth? (@1) // Total number of steps in the sequence waveform for CH1

[SOURce:]ARB:SEQuence:STEP:CURRent:CONVert <step#>,(@chanlist>)

Converts a specified sequence of waveforms to specify the points of a sequence of steps shaped as current custom waveforms.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:CONVert <step#>,(@chanlist>)

Parameters:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

ARB:SEQ:CURR:STEP:CONV 2,(@1) //Converting the second step waveform of a Channel 1 sequence to a custom waveform

[SOURce:]ARB:SEQuence:STEP:VOLTage:CONVert <step#>,(@chanlist>)

Converts a specified sequence of waveforms to a specified number of points in a sequence shaped as a voltage-customized waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:CONVert <step#>,(@chanlist>)

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Parameters:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

ARB:SEQ:STEP:VOLT:CONV 2,(@1) //Convert the second step waveform of Channel 1's current sequence into a customized waveform

[SOURce:]ARB:SEQuence:STEP:POWer:CONVert <step#>,(@chanlist>)

Converts the specified sequence waveform to the point of the sequence where the specified step is shaped as a power custom waveform. Supported only in load mode.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:CONVert <step#>,(@chanlist>)

Parameters:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example:

ARB:SEQ:STEP:POW:CONV 2,(@1) //Convert the second step waveform of Channel 1's current sequence into a customized waveform

[SOURce:]ARB:SEQuence:RESistance:CONVert <step#>,(@chanlist>)

Converts a specified sequence of waveforms by specifying the points of the sequence shaped in steps as resistor custom waveforms. Supported only in load mode.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:CONVert <step#>,(@chanlist>)

Parameters:

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

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Example:

ARB:SEQ:STEP:RES:CONV 2,(@1) //Convert the waveform of the current sequence of CH1 to a customized waveform

[SOURce:]ARB:SEQuence:FUNCtion:SHAPe <function>,<step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:FUNCtion:SHAPe? <step#>,(@chanlist>)

Sets and queries the type of step specified by the sequence waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:FUNCtion:SHAPe <function>,<step#>,(@chanlist>)

Parameters:

<STEP|RAMP|STAircase|SINusoid|PULSe|TRAPezoid|EXPonential|UDEFined>
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:FUNCtion:SHAPe? <step#>,(@chanlist>)

Return Parameters:

STEP,RAMP,STA,SIN,PULS,TRAP,EXPorUDEF

Example:

ARB:SEQuence:STEP:FUNC:SHAP STEP,2,(@1) //Set the waveform of the second step of Channel 1 to STEP

ARB:SEQuence:STEP:FUNC:SHAP? 2,(@1) //Query the waveform type of the second step of Channel 1

[SOURce:]ARB:SEQuence:FUNCtion:TYPE <type>,(@chanlist>)

[SOURce:]ARB:SEQuence:FUNCtion:TYPE? (@chanlist>)

Sets and queries the priority mode of the sequence waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:FUNCtion:TYPE <type>, (@chanlist>)

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Parameters:

CURRent|VOLTage|POWer|RESistance
POWer and RESistance modes are only available in load mode
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:FUNCtion:TYPE? (@chanlist>)

Return Parameters:

CURR,VOLT,POWorRES

Example:

ARB:SEQuence:FUNCtion:TYPE VOLTage,(@1) //Sets Channel 1 sequence operating mode to Voltage Priority

ARB:SEQuence:FUNCtion:TYPE? (@1) //Query Channel 1 sequence operating mode

[SOURce:]ARB:SEQuence:TERMinate:MODE <mode>,(@chanlist>)

[SOURce:]ARB:SEQuence:TERMinate:MODE? (@chanlist>)

Sets and queries how the sequence ends.

Command Syntax:

ARB:SEQuence:TERMinate:MODE (@chanlist>)

Parameters:

NORMal|LAST|OFF
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:FUNCtion:TYPE? (@chanlist>)

Return Parameters:

NORM,LASTorOFF

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Example:

ARB:SEQuence:TERMinate:MODE LAST,(@1) //Set the end-of-sequence mode of Channel 1 to Last

ARB:SEQuence:TERMinate:MODE? (@1) //Query the end-of-sequence mode of Channel 1

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:END[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the end value of the specified step of the sequence current index waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<current value>

Example:

ARB:SEQ:STEP:CURR:EXP:END 1,2,(@1) //Set the ending current of the exponential waveform of the second step of the sequence of Channel 1 to 1A

ARB:SEQ:STEP:CURR:EXP:END? 2,(@1) //Query the end current of the exponential waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:END[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the end value of the sequence voltage index waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum -MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:SEQ:VOLT:EXP:END 10,2,(@1) //Set the ending voltage of the exponential waveform of the second step of the sequence of Channel 1 to 10V

ARB:SEQ:VOLT:EXP:END? 2,(@1) //Query the end voltage of the exponential waveform of the second step of the sequence for Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:END[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the end value of the Sequential Power Index waveform. Supported in load mode only.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum -MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:END[:LEVel]?
[MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<power value>

Example:

ARB:SEQ:POW:EXP:END 100,2,(@1) //Set the ending power of the exponential waveform of the second step of the sequence of Channel 1 to 100W

ARB:SEQ:POW:EXP:END? 2,(@1) // Query the power of the exponential waveform at the end of the second step of the sequence for Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:END[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the end value of the sequence resistance index waveform. Supported in load mode only.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:END[:LEVel]?
[MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:SEQ:RES:EXP:END 1000,2,(@1) //Set the ending resistance of the exponential waveform of the second step of the sequence of Channel 1 to 1000Ω

ARB:SEQ:RES:EXP:END? 2,(@1) //Query Channel 1 for the exponential waveform end resistor of the second step of the sequence

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Specifies the initial value of the current for the specified step index waveform of the sequence waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<current value>

Example:

ARB:SEQ:CURR:EXP:STAR 1,2,(@1) //Set the starting current of the exponential waveform of the second step of the sequence of Channel 1 to 1A

ARB:SEQ:CURR:EXP:STAR? 2,(@1) //Query the starting current of the exponential waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Specifies the initial value of the voltage for the specified step index waveform of the sequence waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum -MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:SEQ:VOLT:EXP:STAR 10,(@1) //Set the starting voltage of the exponential waveform of the second step of the sequence of Channel 1 to 10V

ARB:SEQ:VOLT:EXP:STAR? (@1) //Query the start voltage of the exponential waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Specifies the initial value of the power for the specified step index waveform of the sequence waveform. Supported in load mode only.

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Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum -MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<power value>

Example:

ARB:SEQ:POW:EXP:STAR 100,2,(@1) //Set the starting power of the exponential waveform of the second step of the sequence of Channel 1 to 100W

ARB:SEQ:POW:EXP:STAR? 2,(@1) //Query the starting power of the exponential waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Specifies the initial value of the resistor for the sequence waveform specified step index waveform. Supported in load mode only.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

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Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:SEQ:RES:EXP:STAR 1000,2,(@1) //Set the starting resistance of the exponential waveform in the second step of the sequence for Channel 1 to 1000Ω.

ARB:SEQ:RES:EXP:STAR? 2,(@1) //Query the starting resistance of the exponential waveform in the second step of the sequence for Channel 1.

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Specifies the initial value of the current for the specified step index waveform of the sequence waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<current value>

Example:

ARB:SEQ:CURR:EXP:STAR 1,2.(@1) //Set the starting current of the exponential waveform of the second step of the sequence of Channel 1 to 1A

ARB:SEQ:CURR:EXP:STAR? 1,(@1) //Query the starting current of the exponential waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Specifies the initial value of the voltage for the specified step index waveform of the sequence waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum -MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:SEQ:VOLT:EXP:STAR 10,2,(@1) //Set the starting voltage of the exponential waveform of the second step of the sequence of Channel 1 to 10V

ARB:SEQ:VOLT:EXP:STAR? 2,(@1) //Query the start voltage of the exponential waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Specifies the initial value of the power for the specified step index waveform of the sequence waveform. Supported in load mode only.

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Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum -MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<power value>

Example:

ARB:SEQ:POW:EXP:STAR 100,2,(@1) //Set the starting power of the exponential waveform of the second step of the sequence of Channel 1 to 100W

ARB:SEQ:POW:EXP:STAR? 2,(@1) //Query the starting power of the exponential waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Specifies the initial value of the resistor for the sequence waveform specified step index waveform. Supported in load mode only.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

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Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:SEQ:RES:EXP:STAR 1000,2,(@1) //Set the starting resistance of the exponential waveform in the second step of the sequence for Channel 1 to 1000Ω.

ARB:SEQ:RES:EXP:STAR? 2,(@1) //Query the starting resistance of the exponential waveform in the second step of the sequence for Channel 1.

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Specifies the delay time after the current sequence waveform specified step index waveform is triggered.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:STARt:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (unit:s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:EXP:STAR:TIME 1,2,(@1) //Set the start time of the exponential waveform current in the second step of the sequence of Channel 1 to 1s

ARB:SEQ:CURR:EXP:STAR:TIME? 2,(@1) //Query the start time of the

exponential waveform current of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:STARt:TIME<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Specifies the delay time after the specified step index waveform of the voltage sequence waveform is triggered.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:STARt:TIME<value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:VOLT:EXP:STAR:TIME 1,2,(@1) //Set the exponential waveform voltage start time of the second step of the sequence for Channel 1 to 1s

ARB:SEQ:VOLT:EXP:STAR:TIME? 2,(@1) //Set the exponential waveform voltage start time of the second step of the sequence for Channel 1 to 1s

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:STARt:TIME<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Specifies the delay time after the Power Sequence Waveform Specifies the step index waveform trigger. Supported only in load mode.

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Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:STARt:TIME<value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:POW:EXP:STAR:TIME 1,2,(@1) //Set the exponential waveform power onset time of the second step of the sequence for Channel 1 to 1s

ARB:SEQ:POW:EXP:STAR:TIME? 2,(@1) //Query the exponential waveform power onset time of the second step of the sequence for Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:STARt:TIME<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:STARt:TIME? [|MINimum|MAXimum,]<step#>,(@chanlist>)

Specifies the delay time after the Resistor Sequence Waveform Specifies the step index waveform trigger. Supported only in load mode.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:STARt:TIME<value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

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Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:RES:EXP:STAR:TIME 1,2,(@1) //Set the exponential waveform resistor start time of the second step of the sequence for Channel 1 to 1s

ARB:SEQ:RES:EXP:STAR:TIME? 2,(@1) //Query the start time of the exponential waveform resistance of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential: TCONstant <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential: TCONstant? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the time constant of the current sequence index waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:TCONstant <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:TCONstant? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:EXP:TCON 1,2,(@1) //Set the current time constant of the exponential waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:CURR:EXP:TCON? 2,(@1) //Query the current time constant of the exponential waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:TCONstant <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:TCONstant? [|MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the time constant of the voltage sequence index waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:TCONstant <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:TCONstant? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:VOLT:EXP:TCON 1,2,(@1) //Set the voltage time constant of the exponential waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:VOLT:EXP:TCON? 2,(@1) //Query the voltage time constant of the exponential waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:TCONstant <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:TCONstant? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the time constant of the Power Sequence Index waveform. Supported only in load mode.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:TCONstant <value>, <step#>,(@chanlist>)

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Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:TCONstant? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:POW:EXP:TCON 1,2,(@1) //Set the power time constant of the exponential waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:POW:EXP:TCON? 2,(@1) //Query the power time constant of the exponential waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:TCONstant <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:TCONstant? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the time constant of the exponential waveform of a resistor sequence. Supported only in load mode.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:TCONstant <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:TCONstant? [MINimum|MAXimum,]<step#>,(@chanlist>)

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Return Parameters:

<time>

Example:

ARB:SEQ:RES:EXP:TCON 1,2,(@1) //Set the resistance time constant of the exponential waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:RES:EXP:TCON? 2,(@1) //Query the resistance time constant of the exponential waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the current sequence index change start time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:EXPonential:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:EXP:TIME 1,2,(@1) //Set the current start time of the exponential waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:CURR:EXP:TIME? 2,(@1) //Query the current start time of the exponential waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Setting and querying the voltage sequence index change start time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:EXPonential:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:VOLT:EXP:TIME 1,2,(@1) //Set the voltage start time of the exponential waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:VOLT:EXP:TIME? 2,(@1) //Query the voltage start time of the exponential waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the power series index change start time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)

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<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:EXPonential:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:POW:EXP:TIME 1,2,(@1) //Set the power onset time of the exponential waveform in the second step of the sequence of Channel 1 to 1s

ARB:SEQ:POW:EXP:TIME? 2,(@1) //Query the power onset time of the exponential waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the resistance sequence index change start time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:EXPonential:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:RES:EXP:TIME 1,2,(@1) //Set the power onset time of the

exponential waveform in the second step of the sequence of Channel 1 to 1s

ARB:SEQ:RES:EXP:TIME? 2,(@1) //Query the power onset time of the exponential waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the time at which the output hold ends after the current sequence pulse is completed.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:END:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:PULS:END:TIME 1,2,(@1) //Set the current end time of the pulse waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:CURR:PULS:END:TIME? 2,(@1) //Query the current end time of the pulse waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:END:TIME<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Setting and querying the time for the output hold to end after the completion of a voltage sequence pulse

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Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:END:TIME<value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:VOLT:PULS:END:TIME 1,2,(@1) //Set the voltage end time of the pulse waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:VOLT:PULS:END:TIME? 2,(@1) //Query the voltage end time of the pulse waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:END:TIME<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Setting and querying the time for the output hold to end after the power sequence pulse is completed.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:END:TIME<value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:END:TIME?

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[MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:POW:PULS:END:TIME 1,2,(@1) //Set the power end time of the pulse waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:POW:PULS:END:TIME? 2,(@1) //The power end time of the pulse waveform of the second step of the sequence of the query Channel 1 is 1s

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:END:TIME<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Setting and querying the time at which the output hold ends after the resistor sequence pulse is completed.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:END:TIME<value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:RES:PULS:END:TIME 1,2,(@1) //Set the resistive end time of the pulse waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:RES:PULS:END:TIME? 2,(@1) //Query the resistance end time of the pulse waveform of the second step of the sequence for Channel 1.

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[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:FREQuency<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:FREQuency? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the current frequency of the sequence pulse.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:FREQuency<value>, <step#>,(@chanlist>)

Parameters:

0.01-500|MINimum|MAXimum (in Hz)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:FREQuency? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<frequency>

Example:

ARB:SEQ:CURR:PULS:FREQ 1,2,(@1) //Set the current frequency of the pulse waveform in the second step of the sequence of Channel 1 to 1Hz.

ARB:SEQ:CURR:PULS:FREQ? 2,(@1) //Query the current frequency of the pulse waveform in the second step of the sequence of Channel 1.

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:FREQuency<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:FREQuency? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the voltage frequency of the sequence pulse.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:FREQuency<value>, <step#>,(@chanlist>)

Parameters:

0.01-500|MINimum|MAXimum (in Hz)

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<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:FREQuency? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<frequency>

Example:

ARB:SEQ:VOLT:PULS:FREQ 1,2,(@1) //Set the voltage frequency of the pulse waveform in the second step of the sequence of Channel 1 to 1Hz.

ARB:SEQ:VOLT:PULS:FREQ? 2,(@1) //Query the voltage frequency of the pulse waveform in the second step of the sequence of Channel 1.

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:FREQuency<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:FREQuency? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the power frequency of the serial power pulse.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:FREQuency<value>, <step#>,(@chanlist>)

Parameters:

0.01-500|MINimum|MAXimum (in Hz)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:FREQuency? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<frequency>

Example:

ARB:SEQ:POW:PULS:FREQ 1,2,(@1) //Set the power frequency of the pulse

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waveform in the second step of the sequence of Channel 1 to 1Hz.

ARB:SEQ:POW:PULS:FREQ? 2,(@1) //Query the power frequency of the pulse waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:FREQuency<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:FREQuency? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the resistor frequency of the sequence power pulse.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:FREQuency<value>, <step#>,(@chanlist>)

Parameters:

0.01-500|MINimum|MAXimum (in Hz)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:FREQuency? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<frequency>

Example:

ARB:SEQ:RES:PULS:FREQ 1,2,(@1) //Set the resistor frequency of the pulse waveform in the second step of the sequence of Channel 1 to 1Hz.

ARB:SEQ:RES:PULS:FREQ? 2,(@1) //Query the resistance frequency of the pulse waveform in the second step of the sequence of Channel 1.

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the starting current before the sequence pulse.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:STARt[:LEVel] <value>,

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<step#>,(@chanlist>)

Parameters:

0.01-500|MINimum|MAXimum (in Hz)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<current value>

Example:

ARB:SEQ:CURR:PULS:STAR 1,2,(@1) //Set the starting current of the pulse waveform of the second step of the sequence of Channel 1 to 1A

ARB:SEQ:CURR:PULS:STAR? 2,(@1) //Query the start current of the pulse waveform in the second step of the sequence of Channel 1.

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the starting voltage before the sequence pulse.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

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Return Parameters:

<voltage value>

Example:

ARB:SEQ:VOLT:PULS:STAR 1,2,(@1) //Set the start voltage of the pulse waveform of the second step of the sequence of Channel 1 to 1V.

ARB:SEQ:VOLT:PULS:STAR? 2,(@1) //Query the start voltage of the pulse waveform in the second step of the sequence for Channel 1.

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the starting power before the sequence pulse.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<power value>

Example:

ARB:SEQ:POW:PULS:STAR 1,2,(@1) //Set the starting voltage of the pulse waveform of the second step of the sequence of Channel 1 to 1W

ARB:SEQ:POW:PULS:STAR? 2,(@1) //Query the start voltage of the pulse waveform in the second step of the sequence for Channel 1.

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[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the starting resistance before the sequence pulse.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:SEQ:RES:PULS:STAR 1,2,(@1) //Set the starting resistance of the pulse waveform of the second step of the sequence of Channel 1 to 1Ω.

ARB:SEQ:RES:PULS:STAR? 2,(@1) //Query the start resistance of the pulse waveform in the second step of the sequence of Channel 1.

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the current start time of the sequence pulse.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:STARt:TIME <value>, <step#>,(@chanlist>)

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Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:PULS:STAR:TIME 1,2,(@1) //Set the current start time of the pulse waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:CURR:PULS:STAR:TIME? 2,(@1) //Query the current start time of the pulse waveform in the second step of the sequence for Channel 1.

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the voltage start time of the sequence pulse.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:STARt:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

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Example:

ARB:SEQ:VOLT:PULS:STAR:TIME 1,2,(@1) //Set the voltage start time of the pulse waveform in the second step of the sequence of Channel 1 to 1s

ARB:SEQ:VOLT:PULS:STAR:TIME? 2,(@1) //Query the voltage start time of the pulse waveform in the second step of the sequence for Channel 1.

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the power onset time of the sequence pulse.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:STARt:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:POW:PULS:STAR:TIME 1,2,(@1) //Set the power onset time of the pulse waveform in the second step of the sequence of Channel 1 to 1s

ARB:SEQ:POW:PULS:STAR:TIME? 2,(@1) //Query the power onset time of the pulse waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the resistor start time for sequence pulses.

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Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:STARt:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:RES:PULS:STAR:TIME 1,2,(@1) //Set the resistive start time of the pulse waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:RES:PULS:STAR:TIME? 2,(@1) //Query the resistance start time of the pulse waveform of the second step of the sequence of Channel 1.

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:TOP[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:TOP[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the current peak value of the sequence pulse.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:TOP[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:TOP[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

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Return Parameters:

<current value>

Example:

ARB:SEQ:CURR:PULS:TOP 1,2,(@1) //Set the peak current of the pulse waveform of the second step of the sequence of Channel 1 to 1A

ARB:SEQ:CURR:PULS:TOP? 2,(@1) //Query the peak current of the pulse waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:TOP[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:TOP[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Setting and querying the voltage peaks of sequence pulses.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:TOP[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:TOP[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:SEQ:VOLT:PULS:TOP 1,2,(@1) //Set the peak voltage of the pulse waveform of the second step of the sequence of Channel 1 to 1V

ARB:SEQ:VOLT:PULS:TOP? 2,(@1) //Query the peak voltage of the pulse waveform in the second step of the sequence for Channel 1.

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[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:TOP[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:TOP[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the power peak of the sequence pulse.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:TOP[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:TOP[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<power value>

Example:

ARB:SEQ:POW:PULS:TOP 1,2,(@1) //Set the peak power of the pulse waveform of the second step of the sequence of Channel 1 to 1W

ARB:SEQ:POW:PULS:TOP? 2,(@1) //Query the peak power of the pulse waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:TOP[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:TOP[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the resistor peak value of the sequence pulse.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:TOP[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)

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<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:TOP[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:SEQ:RES:PULS:TOP 1,2,(@1) //Set the peak resistance of the pulse waveform of the second step of the sequence of Channel 1 to 1Ω.

ARB:SEQ:RES:PULS:TOP? 2,(@1) //Query the peak resistance of the pulse waveform in the second step of the sequence for Channel 1.

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:TOP:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:TOP:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the serial current pulse peak time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:TOP:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:PULSe:TOP:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:PULS:TOP:TIME 1,2,(@1) //Set the current peak time of the

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pulse waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:CURR:PULS:TOP:TIME? 2,(@1) //Query the current peak time of the pulse waveform in the second step of the sequence of Channel 1.

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:TOP:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:TOP:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the serial pulse voltage peak time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:TOP:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:PULSe:TOP:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:VOLT:PULS:TOP:TIME 1,2,(@1) //Set the voltage peak time of the pulse waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:VOLT:PULS:TOP:TIME? 2,(@1) //Query the voltage peak time of the pulse waveform in the second step of the sequence of Channel 1.

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:TOP:TIME<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:TOP:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the serial power pulse peak time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:TOP:TIME<value>,

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<step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:PULSe:TOP:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:POW:PULS:TOP:TIME 1,2,(@1) //Set the power peak time of the pulse waveform in the second step of the sequence of Channel 1 to 1s

ARB:SEQ:POW:PULS:TOP:TIME? 2,(@1) //Query the power peak time of the pulse waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:TOP:TIME<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:TOP:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Setting and querying the serial pulse resistance peak time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:TOP:TIME<value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:PULSe:TOP:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

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Return Parameters:

<Resistance>

Example:

ARB:SEQ:RES:PULS:TOP:TIME 1,2,(@1) //Set the power peak time of the pulse waveform in the second step of the sequence of Channel 1 to 1s

ARB:SEQ:RES:PULS:TOP:TIME? 2,(@1) //Query the power peak time of the pulse waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:END[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the ending current after the sequence ramp wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<current value>

Example:

ARB:SEQ:CURR:RAMP:END 1,2,(@1) //Set the end current of the ramp waveform of the second step of the sequence of Channel 1 to 1A

ARB:SEQ:CURR:RAMP:END? 2,(@1) //Query the end current of the ramp waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:END[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the ending voltage after the sequence ramp wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:SEQ:VOLT:RAMP:END 1,2,(@1) //Set the end voltage of the ramp waveform of the second step of the sequence of Channel 1 to 1V

ARB:SEQ:VOLT:RAMP:END? 2,(@1) //Query the end voltage of the ramp waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:END[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the end power after the sequence ramp wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in W)

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<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<power value>

Example:

ARB:SEQ:POW:RAMP:END 1,2,(@1) //Set the end power of the ramp waveform of the second step of the sequence of Channel 1 to 1W

ARB:SEQ:POW:RAMP:END? 2,(@1) //Query the end power of the ramp waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:END[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the ending resistor after the sequence ramp wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:SEQ:RES:RAMP:END 1,2,(@1) //Set the end resistor of the ramp

waveform of the second step of the sequence of Channel 1 to 1Ω.

ARB:SEQ:RES:RAMP:END? 2,(@1) //Query the end resistance of the ramp waveform in the second step of the sequence for Channel 1.

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the time at which the current held by the output ends after the sequence ramp wave completes.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:RAMP:END:TIME 1,2,(@1) //Set the current end time of the ramp waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:CURR:RAMP:END:TIME? 2,(@1) //Query the current end time of the ramp waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the time at which the voltage held by the output ends after the sequence ramp is completed.

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Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:VOLT:RAMP:END:TIME 1,2,(@1) //Set the voltage end time of the ramp waveform in the second step of the sequence of Channel 1 to 1s

ARB:SEQ:VOLT:RAMP:END:TIME? 2,(@1) //Query the voltage end time of the ramp waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the time at which the power held by the output ends after the sequence ramp is complete.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:END:TIME?

[MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:POW:RAMP:END:TIME 1,2,(@1) //Set the power end time of the ramp waveform in the second step of the sequence of Channel 1 to 1s

ARB:SEQ:POW:RAMP:END:TIME? 2,(@1) //Query the power end time of the ramp waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the time at which the resistor held by the output ends after the sequence ramp wave completes.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:RES:RAMP:END:TIME 1,2,(@1) //Set the current end time of the ramp waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:RES:RAMP:END:TIME? 2,(@1) //Query the current end time of the ramp waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:RTIM e <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:RTIM e? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the current rise time of the sequential ramp waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:RTIMe <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:RTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:RAMP:RTIM 1,2,(@1) //Set the current rise time of the ramp waveform in the second step of the sequence of Channel 1 to 1s

ARB:SEQ:CURR:RAMP:RTIM? 2,(@1) //Query the current rise time of the ramp waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:RTIM e <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:RTIM e? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the voltage rise time of the sequential ramp waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:RTIMe <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)

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<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:RTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:VOLT:RAMP:RTIM 1,2,(@1) //Set the voltage rise time of the ramp waveform in the second step of the sequence of Channel 1 to 1s

ARB:SEQ:VOLT:RAMP:RTIM? 2,(@1) //Query the voltage rise time of the ramp waveform in the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:RTIMe <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:RTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the power rise time of the sequential ramp waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:RTIMe <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:RTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:POW:RAMP:RTIM 1,2,(@1) //Set the power rise time of the ramp waveform in the second step of the sequence for Channel 1 to 1s

ARB:SEQ:POW:RAMP:RTIM? 2,(@1) //The power rise time of the ramp waveform in the second step of the sequence of query Channel 1 is 1s

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:RTIMe <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:RTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the resistor rise time of the sequential ramp waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:RTIMe <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:RTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:RES:RAMP:RTIM 1,2,(@1) //Set the resistor rise time of the ramp waveform in the second step of the sequence for Channel 1 to 1s

ARB:SEQ:RES:RAMP:RTIM? 2,(@1) //Query the resistance rise time of the ramp waveform in the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the starting current of the serial ramp waveform.

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Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<current value>

Example:

ARB:SEQ:CURR:RAMP:STAR 1,2,(@1) //Set the starting current of the ramp waveform of the second step of the sequence of Channel 1 to 1AARB:SEQ:CURR:RAMP:STAR? 2,(@1) //Query the start current of the second step of the sequence ramp waveform of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the starting voltage of the serial ramp waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

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Return Parameters:

<voltage value>

Example:

ARB:SEQ:VOLT:RAMP:STAR 1,2,(@1) //Set the start voltage of the ramp waveform of the second step of the sequence of Channel 1 to 1V

ARB:SEQ:VOLt:RAMP:STAR? 2,(@1) //Query the start current of the second step of the sequence ramp waveform of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the starting power of the sequential ramp waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<power value>

Example:

ARB:SEQ:POW:RAMP:STAR 1,2,(@1) //Set the starting power of the ramp waveform of the second step of the sequence of Channel 1 to 1W

ARB:SEQ:POW:RAMP:STAR? 2,(@1) //Query the starting power of the ramp waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the starting resistance of the sequential ramp waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:SEQ:RES:RAMP:STAR 1,2,(@1) //Set the starting resistance of the ramp waveform of the second step of the sequence of Channel 1 to 1Ω

ARB:SEQ:RES:RAMP:STAR? 2,(@1) //Query the starting resistance of the ramp waveform in the second step of the sequence for Channel 1.

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the current start time of the sequential ramp waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:STARt:TIME <value>, <step#>,(@chanlist>)

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Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:RAMP:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:RAMP:STAR:TIME 1,2,(@1) //Set the current start time of the ramp waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:CURR:RAMP:STAR:TIME? 2,(@1) //Query the current start time of the ramp waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the voltage start time of the sequential ramp waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:STARt:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:RAMP:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:VOLT:RAMP:STAR:TIME 1,2,(@1) //Set the voltage start time of the ramp waveform in the second step of the sequence of Channel 1 to 1s

ARB:SEQ:VOLT:RAMP:STAR:TIME? 2,(@1) //Query the voltage start time of the ramp waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the power onset time of the sequential ramp waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:STARt:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:RAMP:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:POW:RAMP:STAR:TIME 1,2,(@1) //Set the power onset time of the ramp waveform in the second step of the sequence for Channel 1 to 1s

ARB:SEQ:POW:RAMP:STAR:TIME? 2,(@1) //Query the power onset time of the ramp waveform in the second step of the sequence for Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the resistor start time of the sequential ramp waveform.

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Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:STARt:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:RAMP:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:RES:RAMP:STAR:TIME 1,2,(@1) //Set the resistor start time of the ramp waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:RES:RAMP:STAR:TIME? 2,(@1) //Query the resistance start time of the ramp waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:CURRent:SINusoid:AMPLitude<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:SINusoid:AMPLitude? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the sine wave amplitude of the sequence current.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:SINusoid:AMPLitude<value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:SINusoid:AMPLitude? [MINimum|MAXimum,]<step#>,(@chanlist>)

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Return Parameters:

<current value>

Example:

ARB:SEQ:CURR:SIN:AMPL 1,2,(@1) //Set the current amplitude of the sinusoidal waveform in the second step of the sequence for Channel 1 to 1A

ARB:SEQ:CURR:SIN:AMPL? 2,(@1) //Query the current amplitude of the sinusoidal waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:SINusoid:AMPLitude<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:SINusoid:AMPLitude? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the sine wave amplitude of the sequence voltage.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:SINusoid:AMPLitude<value>, <step#>,(@chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:SINusoid:AMPLitude? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:SEQ:VOLT:SIN:AMPL 1,2,(@1) //Set the voltage amplitude of the sinusoidal waveform in the second step of the sequence for Channel 1 to 1V

ARB:SEQ:VOLT:SIN:AMPL? 2,(@1) //Query the voltage amplitude of the sinusoidal waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:POWer:SINusoid:AMPLitude<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:SINusoid:AMPLitude? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the sine wave amplitude of the sequence power.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:SINusoid:AMPLitude<value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:SINusoid:AMPLitude? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<power value>

Example:

ARB:SEQ:POW:SIN:AMPL 1,2,(@1) //Set the amplitude power of the sine waveform in the second step of the sequence for Channel 1 to 1A

ARB:SEQ:POW:SIN:AMPL? 2,(@1) //Query the amplitude power of the sine waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:SINusoid:AMPLitude<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:SINusoid:AMPLitude? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the sine wave amplitude of the sequence resistor.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:SINusoid:AMPLitude<value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)

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<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:SINusoid:AMPLitude? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:SEQ:RES:SIN:AMPL 1,2,(@1) //Set the amplitude resistance of the sine waveform of the second step of the sequence of Channel 1 to 1Ω

ARB:SEQ:RES:SIN:AMPL? 2,(@1) //Query the amplitude resistance of the sinusoidal waveform in the second step of the sequence for Channel 1.

[SOURce:]ARB:SEQuence:STEP:CURRent:SINusoid:FREQuency<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:SINusoid:FREQuency? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the current frequency of the serial sine wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:SINusoid:FREQuency<value>, <step#>,(@chanlist>)

Parameters:

0.01-0.01-100000||MINimum|MAXimum (in Hz)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:SINusoid:FREQuency? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<frequency>

Example:

ARB:SEQ:CURR:SIN:FREQ 1,2,(@1) //Set the current frequency of the

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sinusoidal waveform in the second step of the sequence for Channel 1 to 1 Hz

ARB:SEQ:CURR:SIN:FREQ? 2,(@1) //Query the current frequency of the sinusoidal waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:SINusoid:FREQuency<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:SINusoid:FREQuency? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the voltage frequency of the serial sine wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:SINusoid:FREQuency<value>, <step#>,(@chanlist>)

Parameters:

0.01-500|MINimum|MAXimum (in Hz)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:SINusoid:FREQuency? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<frequency>

Example:

ARB:SEQ:VOLT:SIN:FREQ 1,2,(@1) //Set the voltage frequency of the sinusoidal waveform in the second step of the sequence for Channel 1 to 1 Hz

ARB:SEQ:VOLT:SIN:FREQ? 2,(@1) //Query the voltage frequency of the sinusoidal waveform in the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:SINusoid:FREQuency<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:SINusoid:FREQuency? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the power frequency of the serial sine wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:SINusoid:FREQuency<value>,

<step#>,(@chanlist>)

Parameters:

0.01-500|MINimum|MAXimum (in Hz)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:SINusoid:FREQuency? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<frequency>

Example:

ARB:SEQ:POW:SIN:FREQ 1,2,(@1) //Set the power frequency of the sinusoidal waveform in the second step of the sequence for Channel 1 to 1 Hz

ARB:SEQ:POW:SIN:FREQ? 2,(@1) //Query the power frequency of the sinusoidal waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:SINusoid:FREQuency<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:SINusoid:FREQuency? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the resistor frequency of a sequential sine wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:SINusoid:FREQuency<value>, <step#>,(@chanlist>)

Parameters:

0.01-500|MINimum|MAXimum (in Hz)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:SINusoid:FREQuency? [MINimum|MAXimum,]<step#>,(@chanlist>)

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Return Parameters:

<frequency>

Example:

ARB:SEQ:RES:SIN:FREQ 1,2,(@1) //Set the current frequency of the sinusoidal waveform in the second step of the sequence for Channel 1 to 1 Hz

ARB:SEQ:RES:SIN:FREQ? 2,(@1) //Query the current frequency of the sinusoidal waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:CURRent:SINusoid:OFFSet<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:SINusoid:OFFSet? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the current bias of a sequential sine wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:SINusoid:OFFSet<value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:SINusoid:OFFSet? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<current value>

Example:

ARB:SEQ:CURR:SIN:OFFS 1,2,(@1) //Set the bias current of the sine wave of the second step of the sequence of Channel 1 to 1A

ARB:SEQ:CURR:SIN:OFFS? 2,(@1) //Query the bias current of the sine wave of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:VOLTage:SINusoid:OFFSet<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:SINusoid:OFFSet? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the voltage bias of a sequential sine wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:SINusoid:OFFSet<value>, <step#>,(@chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:SINusoid:OFFSet? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:SEQ:VOLT:SIN:OFFS 1,2,(@1) //Set the bias voltage of the sine wave of the second step of the sequence of Channel 1 to 1V

ARB:SEQ:VOLT:SIN:OFFS? 2,(@1) //Query the bias voltage of the sine wave of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:SINusoid:OFFSet<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:SINusoid:OFFSet? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the power bias of the sequential sine wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:SINusoid:OFFSet<value>, <step#>,(@chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in W)

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<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:SINusoid:OFFSet? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<power value>

Example:

ARB:SEQ:POW:SIN:OFFS 1,2,(@1) //Set the bias power of the sine wave of the second step of the sequence of Channel 1 to 1W

ARB:SEQ:POW:SIN:OFFS? 2,(@1) //Query the bias power of the sine wave of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:SINusoid:OFFSet<value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:SINusoid:OFFSet? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the resistor bias of a sequential sine wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:SINusoid:OFFSet<value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:SINusoid:OFFSet? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:SEQ:RES:SIN:OFFS 1,2,(@1) //Set the bias resistor of the sine wave of

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the second step of the sequence of Channel 1 to 1Ω

ARB:SEQ:RES:SIN:OFFS? 2,(@1) //Query the bias resistor of the sine wave of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:END[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the end current of the sequence step wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<current value>

Example:

ARB:SEQ:CURR:STA:END 1,2,(@1) //Set the ending current of the step waveform of the second step of the sequence of Channel 1 to 1A

ARB:SEQ:CURR:STA:END? 2,(@1) //Query the end current of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:END[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the ending voltage of the sequence step wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:END[:LEVel] <value>,

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<step#>,(@chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:SEQ:VOLT:STA:END 1,2,(@1) //Set the ending voltage of the step waveform of the second step of the sequence of Channel 1 to 1V

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:END[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the end power of the sequence step wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<power value>

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Example:

ARB:SEQ:POW:STA:END 1,2,(@1) //Set the ending power of the step waveform of the second step of the sequence of Channel 1 to 1W

ARB:SEQ:POW:STA:END? 2,(@1) //Query the end power of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:END[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the ending resistance of the sequence step wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:SEQ:RES:STA:END 1,2,(@1) //Set the ending resistor of the step waveform of the second step of the sequence of Channel 1 to 1Ω

ARB:SEQ:RES:STA:END? 2,(@1) //Query the ending resistance of the step waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the time at which the current held by the sequential step waveform ends.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:END:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:STAircase:END:TIME 1,2,(@1) //Set the current end time of the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:CURR:STAircase:END:TIME? 2,(@1) //Query the current end time of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the time at which the voltage held by the sequence step waveform ends.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:END:TIME <value>, <step#>,(@chanlist>)

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Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:VOLT:STAircase:END:TIME 1,2,(@1) //Set the voltage end time of the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:VOLT:STAircase:END:TIME? 2,(@1) //Query the voltage end time of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the time at which the power end of the sequence step waveform is held.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:END:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

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Example:

ARB:SEQ:POW:STAircase:END:TIME 1,2,(@1) //Set the power end time of the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:POW:STAircase:END:TIME? 2,(@1) //Query the power end time of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the time at which the resistor for the sequence step waveform hold ends.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:END:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:RES:STAircase:END:TIME 1,2,(@1) //Set the resistive end time of the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:RES:STAircase:END:TIME? 2,(@1) //Query the resistance end time of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:NSTeps <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:NSTeps? [MINimum|MAXimum,]<step#>,(@chanlist>)

Setting and querying the number of sequence step current steps

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Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:NSTeps <value>, <step#>,(@chanlist>)

Parameters:

0-500|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:NSTeps? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<steps>

Example:

ARB:SEQ:CURR:STA:NST 1,2,(@1) //Set the number of current steps in the second step of the sequence of Channel 1's step waveform to 1

ARB:SEQ:CURR:STA:NST? 2,(@1) //Query the number of current steps of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:NSTeps <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:NSTeps? [MINimum|MAXimum,]<step#>,(@chanlist>)

Setting and querying the number of sequence step voltage steps

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:NSTeps <value>, <step#>,(@chanlist>)

Parameters:

0-500|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:NSTeps? [MINimum|MAXimum,]<step#>,(@chanlist>)

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Return Parameters:

<steps>

Example:

ARB:SEQ:VOLT:STA:NST 1,2,(@1) //Set the number of voltage steps in the second step of the sequence of Channel 1's step waveform to 1

ARB:SEQ:VOLT:STA:NST? 2,(@1) //Query the number of voltage steps of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:NSTeps <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:NSTeps? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the number of sequence step power steps.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:NSTeps <value>, <step#>,(@chanlist>)

Parameters:

0-500|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:NSTeps? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<steps>

Example:

ARB:SEQ:POW:STA:NST 1,2,(@1) //Set the power step of the second step of the sequence of Channel 1 to 1 for the step waveform

ARB:SEQ:POW:STA:NST? 2,(@1) //Query the power step of the step waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:NSTeps <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:NSTeps? [MINimum|MAXimum,]<step#>,(@chanlist>)

Setting and querying the number of sequence step resistor steps

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:NSTeps <value>, <step#>,(@chanlist>)

Parameters:

0-500|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:NSTeps? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<steps>

Example:

ARB:SEQ:RES:STA:NST 1,2,(@1) //Set the number of resistor steps of the second step of the sequence of Channel 1 to 1 for the step waveform

ARB:SEQ:RES:STA:NST? 2,(@1) //Query the number of resistor steps of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the starting current of the sequence ladder.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum(in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:STARt[:LEVel]?
[MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<current value>

Example:

ARB:SEQ:CURR:STA:STAR 1,2,(@1) //Set the starting current of the step waveform of the second step of the sequence of Channel 1 to 1A

ARB:SEQ:CURR:STA:STAR? 2,(@1) //Query the starting current of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the starting voltage of the sequence ladder.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum(in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:STARt[:LEVel]?
[MINimum|MAXimum,]<step#>,(@chanlist>)

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Return Parameters:

<voltage value>

Example:

ARB:SEQ:VOLT:STA:STAR 1,2,(@1) //Set the starting voltage of the step waveform of the second step of the sequence of Channel 1 to 1V

ARB:SEQ:VOLT:STA:STAR? 2,(@1) //Query the start voltage of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Setting and querying the starting power of a sequence ladder

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum(in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<power value>

Example:

ARB:SEQ:POW:STA:STAR 1,2,(@1) //Set the starting power of the step waveform of the second step of the sequence of Channel 1 to 1W

ARB:SEQ:POW:STA:STAR? 2,(@1) //Query the starting power of the step waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Setting and querying the starting resistance of the sequence ladder

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum(in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:SEQ:RES:STA:STAR 1,2,(@1) //Set the starting resistance of the step waveform of the second step of the sequence of Channel 1 to 1Ω

ARB:SEQ:RES:STA:STAR? 2,(@1) //Query the starting resistance of the step waveform in the second step of the sequence for Channel 1.

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the current start time of the sequence step waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:STARt:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:STA:STAR:TIME 1,2,(@1) //Set the current start time of the second step waveform of the sequence of Channel 1 to 1s

ARB:SEQ:CURR:STA:STAR:TIME? 2,(@1) //Query the current start time of the second step waveform of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the sequence step waveform voltage start time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:STARt:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

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Example:

ARB:SEQ:VOLT:STA:STAR:TIME 1,2,(@1) //Set the voltage start time of the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:VOLT:STA:STAR:TIME? 2,(@1) //Query the voltage start time of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the sequence step waveform power start time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:STARt:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:POW:STA:STAR:TIME 1,2,(@1) //Set the power onset time of the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:POW:STA:STAR:TIME? 2,(@1) //Query the power onset time of the step waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the sequence step waveform resistor start time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:STARt:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:RES:STA:STAR:TIME 1,2,(@1) //Set the resistor start time for the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:RES:STA:STAR:TIME? 2,(@1) //Query the resistance start time of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query completion of all current sequence step end times.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:TIME <value>, <step#>,(@chanlist>)

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Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STAircase:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:STA:TIME 1,2,(@1) //Set the current end time of the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:CURR:STA:TIME? 2,(@1) //Query the current end time of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query completion of all voltage sequence step end times.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STAircase:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:VOLT:STA:TIME 1,2,(@1) //Set the voltage end time of the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:VOLT:STA:TIME? 2,(@1) //Query the voltage end time of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the completion of all power sequence ladder step end times.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STAircase:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:POW:STA:TIME 1,2,(@1) //Set the power end time of the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:POW:STA:TIME? 2,(@1) //Query the power end time of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the completion of all resistor sequence ladder end times.

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Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STAircase:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:RES:STA:TIME 1,2,(@1) //Set the resistive end time of the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:RES:STA:TIME? 2,(@1) //Query the resistance end time of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:CURRent:STEP:END[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:STEP:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the end current after the sequence current step.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STEP:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STEP:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

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Return Parameters:

<current value>

Example:

ARB:SEQ:CURR:STEP:END 1,2,(@1) //Set the ending current of the step waveform of the second step of the sequence of Channel 1 to 1A

ARB:SEQ:CURR:STEP:END? 2,(@1) //Query the end current of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:STEP:END[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:STEP:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the end voltage after the step of the voltage sequence.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STEP:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STEP:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:SEQ:VOLT:STEP:END 1,2,(@1) //Set the ending voltage of the step waveform of the second step of the sequence of Channel 1 to 1V

ARB:SEQ:VOLT:STEP:END? 2,(@1) //Query the end voltage of the step waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:POWer:STEP:END[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:STEP:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the end power after the power sequence step.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STEP:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STEP:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<power value>

Example:

ARB:SEQ:POW:STEP:END 1,2,(@1) //Set the ending power of the step waveform of the second step of the sequence of Channel 1 to 1W

ARB:SEQ:POW:STEP:END? 2,(@1) //Query the end power of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:STEP:END[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:STEP:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the ending resistance after a resistor sequence step.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STEP:END[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)

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<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STEP:END[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:SEQ:RES:STEP:END 1,2,(@1) //Set the ending resistance of the step waveform of the second step of the sequence of Channel 1 to 1Ω

ARB:SEQ:RES:STEP:END? 2,(@1) //Query the end resistance of the step waveform of the second step of the sequence of Channel 1.

[SOURce:]ARB:SEQuence:STEP:CURRent:STEP:STARt [:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:STEP:STARt [:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the starting current before the current sequence step.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STEP:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STEP:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<current value>

Example:

ARB:SEQ:CURR:STEP:STAR 1,2,(@1) //Set the starting current of the step

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waveform of the second step of the sequence of Channel 1 to 1A

ARB:SEQ:CURR:STEP:STAR? 2,(@1) //Query the starting current of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:STEP:STARt [:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:STEP:STARt [:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the starting voltage before the voltage sequence step.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STEP:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STEP:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<voltage value>

Example:

ARB:SEQ:VOLT:STEP:STAR 1,2,(@1) //Set the start voltage of the step waveform of the second step of the sequence of Channel 1 to 1V

ARB:SEQ:VOLT:STEP:STAR? 2,(@1) //Query the start voltage of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:STEP:STARt[: LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:STEP:STARt[: LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the starting power before the power sequence step.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STEP:STARt[:LEVel] <value>,

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<step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STEP:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<power value>

Example:

ARB:SEQ:POW:STEP:STAR 1,2,(@1) //Set the starting power of the step waveform of the second step of the sequence of Channel 1 to 1W

ARB:SEQ:POW:STEP:STAR? 2,(@1) //Query the starting power of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:STEP:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:STEP:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the starting resistance before the resistor sequence step.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STEP:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STEP:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

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Return Parameters:

<resistance value>

Example:

ARB:SEQ:RES:STEP:STAR 1,2,(@1) //Set the starting resistance of the step waveform of the second step of the sequence of Channel 1 to 1Ω.

ARB:SEQ:RES:STEP:STAR? 2,(@1) //Query the starting resistance of the step waveform of the second step of the sequence of Channel 1.

[SOURce:]ARB:SEQuence:STEP:CURRent:STEP:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:STEP:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the sequence current step start time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STEP:STARt:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STEP:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:STEP:STAR:TIME 1,2,(@1) //Set the current start time of the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:CURR:STEP:STAR:TIME? 2,(@1) //Query the current start time of the step waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:VOLTage:STEP:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:STEP:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the sequence voltage step start time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STEP:STARt:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STEP:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:VOLT:STEP:STAR:TIME 1,2,(@1) //Set the voltage start time of the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:VOLT:STEP:STAR:TIME? 2,(@1) //Query the voltage start time of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:STEP:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:STEP:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the sequence power step start time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STEP:STARt:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)

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<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STEP:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:POW:STEP:STAR:TIME 1,2,(@1) //Set the power onset time of the step waveform of the second step of the sequence of Channel 1 to 1sARB:SEQ:POW:STEP:STAR:TIME? 2,(@1) //Query the power onset time of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:STEP:STARt:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:STEP:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the sequence resistance step start time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STEP:STARt:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000000-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STEP:STARt:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:RES:STEP:STAR:TIME 1,2,(@1) //Set the resistive start time of the

step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:RES:STEP:STAR:TIME? 2,(@1) //Query the resistive start time of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:CURRent:STEP:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:STEP:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the sequence current step end time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STEP:END:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:STEP:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:STEP:END:TIME 1,2,(@1) //Set the current end time of the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:CURR:STEP:END:TIME? 2,(@1) //Query the current end time of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:STEP:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:STEP:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the sequence voltage step end time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STEP:END:TIME <value>,

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<step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:STEP:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:VOLT:STEP:END:TIME 1,2,(@1) //Set the voltage end time of the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:VOLT:STEP:END:TIME? 2,(@1) //Query the voltage end time of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:STEP:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:STEP:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the sequence power step end time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STEP:END:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:STEP:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

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Return Parameters:

<time>

Example:

ARB:SEQ:POW:STEP:END:TIME 1,2,(@1) //Set the current end time of the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:POW:STEP:END:TIME? 2,(@1) //Query the current end time of the step waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:STEP:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:STEP:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the sequence resistance step end time.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STEP:END:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:STEP:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:RES:STEP:END:TIME 1,2,(@1) //Set the resistive end time of the step waveform of the second step of the sequence of Channel 1 to 1s

ARB:SEQ:RES:STEP:END:TIME? 2,(@1) //Query the resistance end time of the step waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:CURRent:TRAPezoid:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:TRAPezoid:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the time at which the current held by the output ends after the completion of the sequence ladder wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:TRAPezoid:END:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:TRAPezoid:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:TRAP:END:TIME 1,2,(@1) //Set the current end time of the trapezoidal waveform in the second step of the sequence of Channel 1 to 1s

ARB:SEQ:CURR:TRAP:END:TIME? 2,(@1) //Query the current end time of the trapezoidal waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:TRAPezoid:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:TRAPezoid:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the time at which the voltage held by the output ends after the completion of the sequence ladder waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:TRAPezoid:END:TIME <value>, <step#>,(@chanlist>)

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Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:TRAPezoid:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:VOLT:TRAP:END:TIME 1,2,(@1) //Set the voltage end time of the trapezoidal waveform in the second step of the sequence for Channel 1 to 1s

ARB:SEQ:VOLT:TRAP:END:TIME? 2,(@1) //Query the voltage end time of the trapezoidal waveform in the second step of the sequence for Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:TRAPezoid:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:TRAPezoid:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the time at which the power held by the output ends after the completion of the sequence ladder wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:TRAPezoid:END:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:TRAPezoid:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:POW:TRAP:END:TIME 1,2,(@1) //Set the power end time of the trapezoidal waveform in the second step of the sequence for Channel 1 to 1s

ARB:SEQ:POW:TRAP:END:TIME? 2,(@1) //Query the power end time of the trapezoidal waveform of the second step of the sequence for Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:TRAPezoid:END:TIME <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:TRAPezoid:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the time at which the resistor held by the output ends after the completion of the sequence ladder waveform.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:TRAPezoid:END:TIME <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:TRAPezoid:END:TIME? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:RES:TRAP:END:TIME 1,2,(@1) //Set the resistive end time of the trapezoidal waveform in the second step of the sequence for Channel 1 to 1s

ARB:SEQ:RES:TRAP:END:TIME? 2,(@1) //Query the resistance end time of the trapezoidal waveform of the second step of the sequence of Channel 1

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[SOURce:]ARB:SEQuence:STEP:CURRent:TRAPezoid:FTIMe <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:TRAPezoid:FTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the time at which the sequence ladder wave falls along the ramp current.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:TRAPezoid:FTIMe <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:TRAPezoid:FTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:TRAP:FTIM 1,2,(@1) //Set the current fall time of the trapezoidal waveform in the second step of the sequence of Channel 1 to 1s

ARB:SEQ:CURR:TRAP:FTIM? (@1) //Query the current fall time of the second step of the trapezoidal waveform of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:TRAPezoid:FTIMe <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:TRAPezoid:FTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the time at which the serial ladder wave falls along the ramp voltage.

Command Syntax:

[SOURce:]ARB:SEQuence:VOLTage:TRAPezoid:FTIMe <value>, <step#>,(@chanlist>)

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Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:TRAPezoid:FTIMe?
[MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:VOLT:TRAP:FTIM 1,2,(@1) //Set the voltage drop time of the trapezoidal waveform in the second step of the sequence for Channel 1 to 1s

ARB:SEQ:VOLT:TRAP:FTIM? 2,(@1) //Query the voltage drop time of the trapezoidal waveform in the second step of the sequence for Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:TRAPezoid:FTIMe <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:TRAPezoid:FTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the time at which the power of the sequential ladder wave falls along the ramp wave.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:TRAPezoid:FTIMe <value>,
<step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:TRAPezoid:FTIMe?
[MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:POW:TRAP:FTIM 1,2,(@1) //Set the power drop time of the trapezoidal waveform in the second step of the sequence for Channel 1 to 1s

ARB:SEQ:POW:TRAP:FTIM? 2,(@1) //Query the power drop time of the trapezoidal waveform in the second step of the sequence for Channel 1

[SOURce:]ARB:SEQuence:STEP:RESistance:TRAPezoid:FTIMe <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:TRAPezoid:FTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the time at which the sequential ladder wave descends along the ramp resistor.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:TRAPezoid:FTIMe <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:TRAPezoid:FTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:RES:TRAP:FTIM 1,2,(@1) //Set the resistance fall time of the trapezoidal waveform in the second step of the sequence for Channel 1 to 1s

ARB:SEQ:RES:TRAP:FTIM? 2,(@1) //Query the resistance fall time of the trapezoidal waveform in the second step of the sequence for Channel 1

[SOURce:]ARB:SEQuence:STEP:CURRent:TRAPezoid:RTIMe <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:TRAPezoid:RTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the time at which the sequential ladder wave rises along the ramp

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current.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:TRAPezoid:RTIMe <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:TRAPezoid:RTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:CURR:TRAP:RTIM 1,2,(@1) //Set the current rise time of the trapezoidal waveform in the second step of the sequence for Channel 1 to 1s
ARB:SEQ:CURR:TRAP:RTIM? 2,(@1) //Query the current rise time of the trapezoidal waveform in the second step of the sequence for Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:TRAPezoid:RTIMe <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:TRAPezoid:RTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the time at which the serial ladder wave rises along the ramp voltage.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:TRAPezoid:RTIMe <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

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Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:TRAPezoid:RTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:VOLT:TRAP:RTIM 1,2,(@1) //Set the voltage rise time of the trapezoidal waveform in the second step of the sequence for Channel 1 to 1s

ARB:SEQ:VOLT:TRAP:RTIM? 2,(@1) //Query the voltage rise time of the trapezoidal waveform in the second step of the sequence for Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:TRAPezoid:RTIMe <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:TRAPezoid:RTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Set and query the time for the sequence ladder wave to rise along the ramp power.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:TRAPezoid:RTIMe <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:TRAPezoid:RTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:POW:TRAP:RTIM 1,2,(@1) //Set the power rise time of the trapezoidal waveform in the second step of the sequence for Channel 1 to 1s

ARB:SEQ:POW:TRAP:RTIM? 2,(@1) //Query the power rise time of the trapezoidal waveform in the second step of the sequence for Channel 1

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[SOURce:]ARB:SEQuence:STEP:RESistance:TRAPezoid:RTIMe <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:TRAPezoid:RTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the time at which the sequential ladder wave rises along the ramp resistor.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:TRAPezoid:RTIMe <value>, <step#>,(@chanlist>)

Parameters:

0.000005-10000.00|MINimum|MAXimum (in s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:TRAPezoid:RTIMe? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<time>

Example:

ARB:SEQ:RES:TRAP:RTIM 1,2,(@1) //Set the resistance rise time of the trapezoidal waveform in the second step of the sequence for Channel 1 to 1s

ARB:SEQ:RES:TRAP:RTIM? 2,(@1) //Query the resistance rise time of the trapezoidal waveform in the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:CURRent:TRAPezoid:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:CURRent:TRAPezoid:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the sequence ladder wave start current.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:TRAPezoid:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:CURRent:TRAPezoid:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<current value>

Example:

ARB:SEQ:CURR:TRAP:STAR 1,2,(@1) //Set the starting current of the trapezoidal waveform in the second step of the sequence for Channel 1 to 1A

ARB:SEQ:CURR:TRAP:STAR? 2,(@1) //Query the starting current of the trapezoidal waveform of the second step of the sequence of Channel 1

[SOURce:]ARB:SEQuence:STEP:VOLTage:TRAPezoid:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:VOLTage:TRAPezoid:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the sequence ladder wave start voltage.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:TRAPezoid:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

0-MAXimum|MINimum|MAXimum (in V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:VOLTage:TRAPezoid:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

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Return Parameters:

<voltage value>

Example:

ARB:SEQ:VOLT:TRAP:STAR 1,2,(@1) //Set the starting voltage of the trapezoidal waveform in the second step of the sequence for Channel 1 to 1V

ARB:SEQ:VOLT:TRAP:STAR? 2,(@1) //Query the starting voltage of the trapezoidal waveform of the second step of the sequence for Channel 1

[SOURce:]ARB:SEQuence:STEP:POWer:TRAPezoid:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:POWer:TRAPezoid:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the Sequential Ladder Wave starting power.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:TRAPezoid:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum -MAXimum|MINimum|MAXimum (in A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:POWer:TRAPezoid:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<NRf+>

Example:

ARB:SEQ:POW:TRAP:STAR 1,2,(@1) //Set the starting power of the trapezoidal waveform in the second step of the sequence for Channel 1 to 1W

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[SOURce:]ARB:SEQuence:STEP:RESistance:TRAPezoid:STARt[:LEVel] <value>, <step#>,(@chanlist>)

[SOURce:]ARB:SEQuence:STEP:RESistance:TRAPezoid:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Sets and queries the sequence ladder wave start resistor.

Command Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:TRAPezoid:STARt[:LEVel] <value>, <step#>,(@chanlist>)

Parameters:

MINimum-MAXimum|MINimum|MAXimum (in Ω)

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:SEQuence:STEP:RESistance:TRAPezoid:STARt[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Return Parameters:

<resistance value>

Example:

ARB:SEQ:RES:TRAP:STAR 1,2,(@1) //Set the starting resistance of the trapezoidal waveform of the second step of the sequence of Channel 1 to 1Ω

ARB:SEQ:RES:TRAP:STAR? 2,(@1) //Query the starting resistance of the trapezoidal waveform in the second step of the sequence for Channel 1.

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Chapter19 CDarb Subsystem

The CDARB command can program a constant dwell arbitrary waveform. A maximum of 8000 points can be assigned to the constant dwell waveform, with each point having the same dwell time.

[SOURce:]ARB:CURRent:CDWell:STEP[:LEVel] <value>,<step#>,(@<chanlist>)

[SOURce:]ARB:CURRent:CDWell:STEP[:LEVel]? [MINimum|MAXimum,]<step#>,(@<chanlist>)

Set and query the specified current values for each step of the constant residence waveform.

Command Syntax:

[SOURce:]ARB:CURRent:CDWell:STEP[:LEVel] <value>,<step#>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum (unit:A)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:CDWell:STEP[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Returns

<I>

Example

ARB:CURR:CDW:STEP 1,2,(@1) // Set the current value at the second point of the constant dwell waveform for Channel 1 to 1A.

ARB:CURR:CDW:STEP? 2,(@1) // Query the current value at the second point of the constant-state waveform for Channel 1.

[SOURce:]ARB:VOLTage:CDWell:STEP[:LEVel] <value>,<step#>,(@<chanlist>)

[SOURce:]ARB:VOLTage:CDWell:STEP[:LEVel]? [MINimum|MAXimum,]<step#>,(@<chanlist>)

Set and query the voltage values at specified steps for the constant dwell

waveform

Command Syntax:

[SOURce:]ARB:VOLTage:CDWell:STEP[:LEVel] <value>,<step#>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum (unit:V)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:CDWell:STEP[:LEVel]?
[MINimum|MAXimum,]<step#>,(@chanlist>)

Returns

<Voltage>

Example

ARB:VOLT:CDW:STEP 1,2,(@1) // Set the voltage value of the first point of the constant dwell waveform for Channel 1 to 2V.

ARB:VOLT:CDW:STEP? 2,(@1) // Query the voltage value at the second point of the constant-state waveform on Channel 1.

[SOURce:]ARB:POWer:CDWell:STEP[:LEVel] <value>,<step#>,(@<chanlist>)

[SOURce:]ARB:POWer:CDWell:STEP[:LEVel]? [MINimum|MAXimum,]<step#>,(@<chanlist>)

Set and query the specified step power values for constant dwell waveforms. Supported only by load modules.

Command Syntax:

[SOURce:]ARB:CURRent:CDWell:STEP[:LEVel] <value>,<step#>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum (Unit:W)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:CDWell:STEP[:LEVel]?

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[MINimum|MAXimum,]<step#>,(@chanlist>)

Returns

<Power>

Example

ARB:POW:CDW:STEP 1,2,(@1) // Set the power value of the first point of the constant dwell waveform for Channel 1 to 2W.

ARB:POW:CDW:STEP? 2,(@1) // Query the power value of the second point on the constant-state waveform for Channel 1

[SOURce:]ARB:RESistance:CDWell:STEP[:LEVel] <value>,<step#>,(@<chanlist>)

[SOURce:]ARB: RESistance:CDWell:STEP[:LEVel]? [MINimum|MAXimum,]<step#>,(@<chanlist>)

Set and query the specified current values for a constant dwell waveform at designated steps. Supported only by load modules.

Command Syntax:

[SOURce:]ARB: Resitance:CDWell:STEP[:LEVel] <value>,<step#>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum (Unit:Ω)

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:Resitance:CDWell:STEP[:LEVel]? [MINimum|MAXimum,]<step#>,(@chanlist>)

Returns

<Resistance>

Example

ARB:RES:CDW:STEP 1,2,(@1) // Set the resistance value of the first point of the constant dwell waveform for Channel 1 to 2Ω.

ARB:RES:CDW:STEP? 2,(@1) // Resistance value of the second point on the constant dwell waveform of Channel 1

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[SOURce:]ARB:CURRent:CDWell:DWELl[:LEVel] <value>,(@<chanlist>)

[SOURce:]ARB:CURRent:CDWell:DWELl[:LEVel]? [MINimum|MAXimum,] (@<chanlist>)

Set and query the constant dwell waveform current hold time

Command Syntax:

[SOURce:]ARB:CURRent:CDWell:DWELl <value>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum (Unit:s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:CDWell:DWELl? [MINimum|MAXimum,](@chanlist>)

Returns

<Time>

Example

ARB:CURR:CDW:DWEL 1,(@1) // Set the constant dwell waveform current hold time for Channel 1 to 1 second.

ARB:CURR:CDW:DWEL? (@1) // Query the constant dwell waveform current hold time for Channel 1

[SOURce:]ARB:VOLTage:CDWell:DWELl[:LEVel] <value>,(@<chanlist>)

[SOURce:]ARB:VOLTage:DWELl[:LEVel]? [MINimum|MAXimum,] (@<chanlist>)

Set and query the voltage hold time for constant-residual waveforms

Command Syntax:

[SOURce:]ARB:VOLTage:CDWell:DWELl <value>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum (Unit:s)
<chanlist>:Channel number.

(@1) – Channel 1

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(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:CDWell:DWELl? [MINimum|MAXimum,](@chanlist>)

Returns

<time>

Example

ARB:VOLT:CDW:DWEL 1,(@1) //Set the constant dwell waveform voltage hold time for Channel 1 to 1 second.

ARB:VOLT:CDW:DWEL? (@1) // Query the constant dwell waveform voltage hold time for Channel 1

[SOURce:]ARB:POWer:CDWell:DWELl[:LEVel] <value>,(@<chanlist>)

[SOURce:]ARB:POWer:CDWell:DWELl[:LEVel]? [MINimum|MAXimum,] (@<chanlist>)

Set and query the constant dwell waveform power hold time; supported only by load modules.

Command Syntax:

[SOURce:]ARB:POWer:CDWell:DWELl <value>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum (unit: s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:CDWell:DWELl? [MINimum|MAXimum,](@chanlist>)

Returns

<time>

Example

ARB:POWer:CDW:DWEL 1,(@1) // Set the constant dwell waveform power hold time for Channel 1 to 1 second.

ARB:POWer:CDW:DWEL? (@1) // Query the constant dwell waveform power hold time for Channel 1

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[SOURce:]ARB:RESistance:CDWell:DWELl[:LEVel] <value>,(@<chanlist>)

[SOURce:]ARB:RESistance:CDWell:DWELl[:LEVel]? [MINimum|MAXimum,] (@<chanlist>)

Set and query the constant dwell waveform current hold time. Supported only by load modules.

Command Syntax:

[SOURce:]ARB:RESistance:CDWell:DWELl <value>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum (Unit: s)
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:CDWell:DWELl? [MINimum|MAXimum,](@chanlist>)

Returns

<time>

Example

ARB:RES:CDW:DWEL 1,(@1) //Set the constant dwell waveform resistance hold time for Channel 1 to 1 second.

ARB:RES:CDW:DWEL? (@1) //Constant dwell waveform resistance hold time for channel 1

[SOURce:]ARB:CURRent:CDWell:POINts <value>,(@<chanlist>)

[SOURce:]ARB:CURRent:CDWell:POINts? [MINimum|MAXimum,] (@<chanlist>)

Set and query the number of constant-residence waveform current points.

Command Syntax:

[SOURce:]ARB:CURRent:CDWell:POINts<value>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum
<chanlist>:Channel number.

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(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CURRent:CDWell:POINts? [MINimum|MAXimum,](@chanlist>)

Returns

<point >

Example

ARB:CURR:CDW: POIN 1,(@1) // Set the number of constant dwell waveform current points for Channel 1 to 1.

ARB:CURR:CDW: POIN? (@1) // Number of constant-residence waveform current points in channel 1

[SOURce:]ARB:VOLTage:CDWell:POINts <value>,(@<chanlist>)

[SOURce:]ARB:VOLTage:CDWell:POINts? [MINimum|MAXimum,] (@<chanlist>)

Set and query the voltage points of the constant-residence waveform.

Command Syntax:

[SOURce:]ARB:VOLTage:CDWell:POINts<value>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:VOLTage:CDWell:POINts? [MINimum|MAXimum,](@chanlist>)

Returns

<point>

Example

ARB:VOLT:CDW:POIN 1,(@1) //Set the constant dwell waveform voltage points for Channel 1 to 1.

ARB:VOLT:CDW:POIN? (@1) //Query the number of constant-residence waveform voltage points for Channel 1

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[SOURce:]ARB:POWer:CDWell:POINts <value>,(@<chanlist>)

[SOURce:]ARB:POWer:CDWell:POINts? [MINimum|MAXimum,] (@<chanlist>)

Set and query constant residence waveform power points; supported only by load modules.

Command Syntax:

[SOURce:]ARB:POWer:CDWell:POINts<value>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:CDWell:POINts? [MINimum|MAXimum,](@chanlist>)

Returns

<point>

Example

ARB:POW:CDW: POIN 1,(@1) // Set the constant dwell waveform power point for Channel 1 to 1.

ARB:POW:CDW: POIN? (@1) // Query the constant-residence waveform power points for Channel 1

[SOURce:]ARB:RESistance:CDWell:POINts <value>,(@<chanlist>)

[SOURce:]ARB:RESistance:CDWell:POINts? [MINimum|MAXimum,] (@<chanlist>)

Set and query the number of constant residence waveform resistance points. Supported only by load modules.

Command Syntax:

[SOURce:]ARB:RESistance:CDWell:POINts<value>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum

<chanlist>:Channel number.

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(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:RESistance:CDWell:POINts? [MINimum|MAXimum,](@chanlist>)

Returns

<point>

Example

ARB:RES:CDW:POIN 1,(@1) //Set the constant dwell waveform resistance value for Channel 1 to 1.

ARB:RES:CDW:POIN? (@1) //Number of constant-resistance points for the stationary waveform in Channel 1

[SOURce:]ARB:CDWell:FUNCtion:TYPE <type>, (@<chanlist>)

[SOURce:]ARB:CDWell:FUNCtion:TYPE? (@<chanlist>)

Set and query the constant residence waveform priority mode.

Command Syntax:

[SOURce:]ARB:CDWell:FUNCtion:TYPE <type>, (@<chanlist>)

Arguments

CURRent|VOLTage|POWer|RESistance

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CDWell:FUNCtion:TYPE? (@<chanlist>)

Returns

CURR,VOLT,POW or RES

Example

ARB:CDW: FUNC:TYPE VOLTage,(@1) //Set Channel 1's constant dwell waveform priority mode to voltage.

ARB:CDW: FUNC:TYPE? (@1) //Query Channel 1's Constant Residual Waveform Priority Mode

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[SOURce:]ARB:CDWell:TERMinate:MODE <NORMal|LAST|OFF>,(@<chanlist>)
[SOURce:]ARB:CDWell:TERMinate:MODE? (@<chanlist>)

Set and Query Constant-Residence Waveform Ending Mode.

Command Syntax:

[SOURce:]ARB:CDWell:TERMinate:MODE <NORMal|LAST|OFF>, (@<chanlist>)

Arguments

NORMal|LAST|OFF

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CDWell:TERMinate:MODE? (@<chanlist>)

Returns

NORM, LAST or OFF

Example

ARB:CDW:TERM:MODE NORM,(@1) // Set the end mode of the constant dwell waveform for Channel 1 to NORMal.

ARB:CDW:TERM:MODE? (@1) // Query the termination method for the constant-state waveform on Channel 1

[SOURce:]ARB:POWer:CDWell:COUNt <count>,(@<chanlist>)

[SOURce:]ARB:POWer:CDWell:COUNt? [MINimum|MAXimum,] (@<chanlist>)

Set and query the number of repetitions for constant dwell waveforms. Supported only by load modules.

Command Syntax:

[SOURce:]ARB:POWer:CDWell:COUNt<value>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum

<chanlist>:Channel number.

(@1) – Channel 1

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(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:POWer:CDWell:COUNt? [MINimum|MAXimum,](@chanlist>)

Returns

<count>

Example

ARB:POW:CDW:COUN 1,(@1) // Set the number of repetitions for the constant dwell waveform on Channel 1 to 1.

ARB:POW:CDW:COUN? (@1) // Number of repetitions of the constant dwell waveform for Channel 1

[SOURce:]ARB:CDWell:EXPort <file_name>,(@<chanlist>)

Export the waveform files permanently stored on the USB drive.

Command Syntax:

[SOURce:]ARB:CDWell:EXPort <file_name>,(@<chanlist>)

Arguments

<file_name> File name, string type
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example

ARB:CDW:EXP “test.csv”,(@1) // Export the constant-state waveform file named test.csv to the USB drive.

[SOURce:]ARB:CDWell:IMPort <file_name>,(@<chanlist>)

Import waveform files into the USB drive for permanent storage.

Command Syntax:

[SOURce:]ARB:CDWell:IMPort <file_name>,(@<chanlist>)

Arguments

<file_name> File name, string type
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

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(@1:4) – Channels 1 through 4, including channel 1234.

Example

ARB:CDW:IMP “test.csv”,(@1) // Import the constant-state waveform file named test.csv from the USB drive.

[SOURce:]ARB:CDWell:SAVE <file_name>,(@<chanlist>)

Save CDarb waveform files to USB (default import/export only supported to USB).

Command Syntax:

[SOURce:]ARB:CDWell:SAVE <file_name>,(@<chanlist>)

Arguments

<file_name> File name, string type
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example

ARB:CDWell:SAVE "cdarb.csv",(@1) // Save CDarb waveform files to USB (default import/export only to USB)

[SOURce:]ARB:CDWell:OPEN <file_name>,(@<chanlist>)

Open the cdarb waveform file to USB (default import/export is only supported via USB).

Command Syntax:

[SOURce:]ARB:CDWell:OPEN <file_name>,(@<chanlist>)

Arguments

<file_name>
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Example

ARB:CDWell:OPEN "cdarb.csv",(@1) // Open the CDarb waveform file for Channel 1

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[SOURce:]ARB:CDWell:INTerpolation[:STATe] <0|OFF|1|ON>,(@<chanlist>)

[SOURce:]ARB:CDWell:INTerpolation[:STATe]?(@<chanlist>)

This command is used to configure and query the cdarb linear interpolation function.

Command Syntax:

[SOURce:]ARB:CDWell:INTerpolation[:STATe] <0|OFF|1|ON>,(@<chanlist>)

Arguments

0|OFF|1|ON

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]ARB:CDWell:INTerpolation[:STATe]?(@<chanlist>)

Returns

0|1

Example

ARB:CDWell:INTerpolation 1,(@1) // configure the cdarb linear interpolation function for CH1

ARB:CDWell:INTerpolation?(@1)

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Chapter20 SSWeep Subsystem

[SOURce:]SSWeep:VOLTage[:AMPLitude]:STARt <value>,(@<chanlist>)

[SOURce:]SSWeep:VOLTage[:AMPLitude]:STARt? [MINimum|MAXimum,] (@<chanlist>)

Set and query the initial voltage amplitude value for the sine wave scan. (Unit: V)

Command Syntax:

[SOURce:]SSWeep:VOLTage[:AMPLitude]:STARt <value>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]SSWeep:VOLTage[:AMPLitude]:STARt? [MINimum|MAXimum,] (@<chanlist>)

Returns

<Voltage>

Example

SSW:VOLT:STAR 12,(@1) // Set the initial amplitude voltage for the sine wave scan on Channel 1 to 12V.

SSW:VOLT:STAR? (@1) // Query Channel 1 Sine Wave Scan Initial Amplitude Voltage

[SOURce:]SSWeep:VOLTage[:AMPLitude]:END <value>,(@<chanlist>)

[SOURce:]SSWeep:VOLTage[:AMPLitude]:END? [MINimum|MAXimum,] (@<chanlist>)

Set and query the end value of the voltage amplitude for the sine wave scan. (Unit: V)

Command Syntax:

[SOURce:]SSWeep:VOLTage[:AMPLitude]:END <value>,(@<chanlist>)

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Arguments

MINimum-MAXimum|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]SSWeep:VOLTage[:AMPLitude]:END? [MINimum|MAXimum,] (@<chanlist>)

Returns

<Voltage>

Example

SSW:VOLT:END 12,(@1) // Set the amplitude voltage at the end of the sine wave sweep for Channel 1 to 12V.

SSW:VOLT:END? (@1) // Query Channel 1 sine wave scan end amplitude voltage

[SOURce:]SSWeep:VOLTage:OFFSet:STARt <value>,(@<chanlist>)

[SOURce:]SSWeep:VOLTage:OFFSet:STARt? [MINimum|MAXimum,] (@<chanlist>)

Set and query the initial voltage offset value for the sine wave scan. (Unit: V)

Command Syntax:

[SOURce:]SSWeep:VOLTage:OFFSet:STARt <value>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]SSWeep:VOLTage:OFFSet:STARt? [MINimum|MAXimum,] (@<chanlist>)

Returns

<Voltage>

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Example

SSW:VOLT:OFFS:STAR 10,(@1) // Set the starting offset voltage for the sine wave scan on Channel 1 to 10V.

SSW:VOLT:OFFS:STAR? (@1) // Query Channel 1 Sine Wave Scan Start Offset Voltage

[SOURce:]SSWeep:VOLTage:OFFSet:END <value>,(@<chanlist>)

[SOURce:]SSWeep:VOLTage:OFFSet:END? [MINimum|MAXimum,] (@<chanlist>)

Set and query the voltage offset value and end value for the sine wave scan. (Unit: V)

Command Syntax:

[SOURce:]SSWeep:VOLTage:OFFSet:END <value>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]SSWeep:VOLTage:OFFSet:END? [MINimum|MAXimum,] (@<chanlist>)

Returns

<Voltage>

Example

SSW:VOLT:OFFS:END 10,(@1) // Set the bias voltage at the end of the sine wave scan for Channel 1 to 10V.

SSW:VOLT:OFFS:END? (@1) //Query Channel 1 Sine Wave Scan End Bias Voltage

[SOURce:]SSWeep:CURRent[:AMPLitude]:STARt <value>,(@<chanlist>)

[SOURce:]SSWeep:CURRent[:AMPLitude]:STARt? [MINimum|MAXimum,] (@<chanlist>)

Set and query the initial value of the current amplitude for the sine wave scan. (Unit: A)

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Command Syntax:

[SOURce:]SSWeep:CURRent[:AMPLitude]:STARt <value>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]SSWeep:CURRent[:AMPLitude]:STARt? [MINimum|MAXimum,] (@<chanlist>)

Returns

<Current>

Example

SSW:CURR:STAR 12,(@1) //Set the initial amplitude current for the sine wave scan on Channel 1 to 12A.
SSW:CURR:STAR? (@1) //Query Channel 1 Sine Wave Scan Initial Amplitude Current

[SOURce:]SSWeep:CURRent[:AMPLitude]:END <value>,(@<chanlist>)

[SOURce:]SSWeep:CURRent[:AMPLitude]:END? [MINimum|MAXimum,] (@<chanlist>)

Set and query the end value of the current amplitude for the sine wave scan. (Unit: A)

Command Syntax:

[SOURce:]SSWeep:CURRent[:AMPLitude]:END <value>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]SSWeep:CURRent[:AMPLitude]:END? [MINimum|MAXimum,] (@<chanlist>)

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Returns

<Current>

Example

SSW:CURR:END 12,(@1) // Set the amplitude current for the sine wave sweep end of Channel 1 to 12A.

SSW:CURR:END? (@1) // Query Channel 1 sine wave scan end amplitude current

[SOURce:]SSWeep:CURRent:OFFSet:STARt <value>,(@<chanlist>)

[SOURce:]SSWeep:CURRent:OFFSet:STARt? [MINimum|MAXimum,] (@<chanlist>)

Set and query the initial value of the current bias for the sine wave scan. (Unit: A)

Command Syntax:

[SOURce:]SSWeep:CURRent:OFFSet:STARt <value>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]SSWeep:CURRent:OFFSet:STARt? [MINimum|MAXimum,] (@<chanlist>)

Returns

<Current>

Example

SSW:CURR:OFFS:STAR 10,(@1) // Set the starting bias current for the sine wave scan on Channel 1 to 10A.

SSW:CURR:OFFS:STAR? (@1) // Query Channel 1 Sine Wave Scan Start Bias Current

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[SOURce:]SSWeep:CURRent:OFFSet:END <value>,(@<chanlist>)

[SOURce:]SSWeep:CURRent:OFFSet:END? [MINimum|MAXimum,] (@<chanlist>)

Set and query the start and end values for the current bias during sine wave scanning. (Unit: A)

Command Syntax:

[SOURce:]SSWeep:CURRent:OFFSet:END <value>,(@<chanlist>)

Arguments

MINimum-MAXimum|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]SSWeep:CURRent:OFFSet:END? [MINimum|MAXimum,] (@<chanlist>)

Returns

<Current>

Example

SSW:CURR:OFFS:END 10,(@1) // Set the bias current for the end of the sine wave scan on Channel 1 to 10A.

SSW:CURR:OFFS:END? (@1) // Query Channel 1 Sine Wave Scan End Bias Current

[SOURce:]SSWeep:FREQuency:STARt <value>,(@<chanlist>)

[SOURce:]SSWeep:FREQuency:STARt? [MINimum|MAXimum,] (@<chanlist>)

Set and query the initial frequency value for the sine wave scan. (Unit: Hz)

Command Syntax:

[SOURce:]SSWeep:FREQuency:STARt <value>,(@<chanlist>)

Arguments

0.01-10000|MINimum|MAXimum
<chanlist>:Channel number.

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(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]SSWeep:FREQuency:STARt? [MINimum|MAXimum,] (@<chanlist>)

Returns

<frequency>

Example

SSW:FREQ:STAR 10,(@1) //Set the initial frequency of the sine wave scan for Channel 1 to 10 Hz.

SSW:FREQ:STAR? (@1) //Initial Frequency for Sine Wave Scan on Query Channel 1

[SOURce:]SSWeep:FREQuency:END <value>,(@<chanlist>)

[SOURce:]SSWeep:FREQuency:END? [MINimum|MAXimum,] (@<chanlist>)

Set and query the frequency end value for the sine wave scan. (Unit: Hz)

Command Syntax:

[SOURce:]SSWeep:FREQuency:END <value>,(@<chanlist>)

Arguments

0.01-10000|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]SSWeep:FREQuency:END? [MINimum|MAXimum,] (@<chanlist>)

Returns

<frequency>

Example

SSW:FREQ:END 10,(@1) // Set the end frequency of the sine wave scan for Channel 1 to 10 Hz.

SSW:FREQ:END? (@1) // Query Channel 1 Sine Wave Scan End Frequency

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[SOURce:]SSWeep:PHASe:STARt <value>,(@<chanlist>)

[SOURce:]SSWeep:PHASe:STARt? [MINimum|MAXimum,] (@<chanlist>)

Set and query the starting phase angle for the sine wave scan. (Unit:°)

Command Syntax:

[SOURce:]SSWeep:PHASe:STARt <value>,(@<chanlist>)

Arguments

0-365|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]SSWeep:PHASe:STARt? [MINimum|MAXimum,] (@<chanlist>)

Returns

<angle>

Example

SSW:PHAS:STAR 10,(@1) // Set the initial angle for the sine wave scan on Channel 1 to 10°.

SSW:PHAS:STAR? (@1) // Initial angle for sine wave scan on query channel 1

[SOURce:]SSWeep:TIME <value>,(@<chanlist>)

[SOURce:]SSWeep:TIME? [MINimum|MAXimum,] (@<chanlist>)

Set and query the duration of the sine wave scan. (Unit: seconds)

Command Syntax:

[SOURce:]SSWeep:TIME <value>,(@<chanlist>)

Arguments

0.1-3600|MINimum|MAXimum
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

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Query Syntax:

[SOURce:]SSWeep:TIME? [MINimum|MAXimum,] (@<chanlist>)

Returns

<time>

Example

SSW:TIME 10,(@1) // Set the sine wave scan time for Channel 1 to 10 seconds.

SSW:TIME? (@1) // Query Channel 1 Sine Wave Scan Time

[SOURce:]SSWeep:FUNCtion:TYPE <CURRent|VOLTage>,(@<chanlist>)

[SOURce:]SSWeep:FUNCtion:TYPE? (@<chanlist>)

Set and query the operating mode for sine wave scanning. (Unit: s)

Command Syntax:

[SOURce:]SSWeep:FUNCtion:TYPE <CURRent|VOLTage>,(@<chanlist>)

Arguments

VOLTage:voltage source mode
CURRent:current source mode
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]SSWeep:FUNCtion:TYPE? (@<chanlist>)

Returns

VOLT|CURR

Example

SSW:FUNC:TYPE CURR,(@1) //Set Channel 1 sine wave scan operating mode to current mode.

SSW:FUNC:TYPE? (@1) //Query Channel 1 Sine Wave Scan Operating Mode

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[SOURce:]SSWeep:COUNt <value>,(@<chanlist>)

[SOURce:]SSWeep:COUNt? [MINimum|MAXimum,] (@<chanlist>)

Set and query the number of repetitions for the sine wave scan.

Command Syntax:

[SOURce:]SSWeep:COUNt <value>,(@<chanlist>)

Arguments

<0-65535|MINimum|MAXimum>
When Arguments is 0, it indicates an infinite loop.
<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]SSWeep:COUNt? [MINimum|MAXimum,] (@<chanlist>)

Returns

<count>

Example

SSW:COUN 1,(@1) // Set the number of repetitions for the sine wave scan on Channel 1 to 1.

SSW:COUN? (@1) // Number of repetitions for sine wave scanning on query channel 1

[SOURce:]SSWeep:COUNt:RUN? (@<chanlist>)

Query the number of repetitions the sine wave scan has run.

Query Syntax:

[SOURce:]SSWeep:COUNt:RUN? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Returns

<count>

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Example

SSW:COUN:RUN? (@1) // Number of repetitions completed for sine wave scan on Query Channel 1

[SOURce:]SSWeep:TERMinate:MODE <NORMal|LAST|OFF>,(@<chanlist>)

[SOURce:]SSWeep:TERMinate:MODE? (@<chanlist>)

Set and query the end mode for sine wave scanning.

Command Syntax:

[SOURce:]SSWeep:TERMinate:MODE <NORMal|LAST|OFF>,(@<chanlist>)

Arguments

NORMal: Output the settings in Fix mode after the operation completes.

LAST: After execution acceptance, the output remains at the scan value of the last step.

OFF: Close the module's output after operation completes.

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Query Syntax:

[SOURce:]SSWeep:TERMinate:MODE? (@<chanlist>)

Returns

NORM|LAST|OFF

Example

SSW:TERM:MODE LAST,(@1) // Set the sine wave scan termination mode for Channel 1 to LAST.

[SOURce:]SSWeep:FREQuency:RUN? (@<chanlist>)

Query the current operating frequency of the sine wave scan. (Unit: Hz)

Query Syntax:

[SOURce:]SSWeep:FREQuency:RUN? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

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Example

SSW:FREQ:RUN? (@1) // Query Channel 1 sine wave scan currently operating frequency

[SOURce:]SSWeep:RSTate? (@<chanlist>)

Check the operational status of the sine wave scan.

Query Syntax:

[SOURce:]SSWeep:RSTate? (@<chanlist>)

Arguments

<chanlist>:Channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and channel 4

(@1:4) – Channels 1 through 4, including channel 1234.

Returns

IDEL

WTG

RUN

END

Example

SSW:RST? (@1) // Query Channel 1 Sine Wave Scan Current Operating Status

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Chapter21 BATTery CHARge Subsystem (Source)

BATTery:CHARge:MODE <CURRent|VOLTage>,(@<chanlist>)

This commands set the battery charging mode.

Syntax

BATTery:CHARge:MODE <CURRent|VOLTage>,(@<chanlist>)

Arguments

CURRent|VOLTage

Example

BATT:CHAR:MODE CURR,(@1)

Query Syntax

BATTery:CHARge:MODE? (@<chanlist>)

Returns

CURRent|VOLTage

BATTery:CHARge:VOLTage <value>,(@<chanlist>)

Set the voltage for battery charging.

Syntax

BATTery:CHARge:VOLTage <value>,(@<chanlist>)

Arguments

<MINimum-MAXimum|MINimum|MAXimum>

Example

BATT:CHAR:VOLT 5.0,(@1)

Query Syntax

BATTery:CHARge:VOLTage? [MINimum|MAXimum],(@<chanlist>)

Returns

NR3

BATTery:CHARge:CURRent <value>,(@<chanlist>)

Set the current limit value for battery charging.

Syntax

BATTery:CHARge:CURRent <value>,(@<chanlist>)

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Arguments

<MINimum-MAXimum|MINimum|MAXimum>

Example

BATT:CHAR:CURR 3.0,(@1)

Query Syntax

BATTery:CHARge:CURRent? [MINimum|MAXimum],(@<chanlist>)

Returns

NR3

BATTery:CHARge:STOP:VOLTage <value>,(@<chanlist>)

This command sets the stop charging voltage level for battery charging test.

Syntax

BATTery:CHARge:STOP:VOLTage <value>,(@<chanlist>)

Arguments

0-MAXimum|MINimum|MAXimum (Unit is V)

Example

BATT:CHAR:STOP:VOLT 4.0,(@1)

Query Syntax

BATTery:CHARge:STOP:VOLTage? [MINimum|MAXimum,] (@<chanlist>)

Returns

NR3

BATTery:CHARge:STOP:VOLTage:STATe <0|OFF|1|ON>,(@<chanlist>)

This command sets the stop charging voltage status for battery charging test.

Syntax

BATTery:CHARge:STOP:VOLTage:STATe <0|OFF|1|ON>,(@<chanlist>)

Arguments

<0|OFF|1|ON>

Example

BATTery:CHAR:STOP:VOLT:STAT 1,(@1)

Query Syntax

BATTery:CHARge:STOP:VOLTage? [MINimum|MAXimum,] (@<chanlist>)

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Returns

0|1

BATTery:CHARge:STOP:CURRent <value>,(@<chanlist>)

This command sets and queries the stop charging current value for bettery charging test.

Syntax

BATTery:CHARge:STOP:CURRent <value>,(@<chanlist>)

Arguments

0-MAXimum|MINimum|MAXimum (Unit is A)

Example

BATT:CHAR:STOP:CURR 2.0, (@1)

Query Syntax

BATTery:CHARge:STOP:CURRent? <value>,(@<chanlist>)

Returns

NR3

BATTery:CHARge:STOP:CURRent:STATe <0|OFF|1|ON>,(@<chanlist>)

This command sets the stop charging current status for battery charging test.

Syntax

BATTery:CHARge:STOP:CURRent:STATe <0|OFF|1|ON>,(@<chanlist>)

Arguments

<0|OFF|1|ON>

Example

BATT:CHAR:STOP:CURR:STAT 1,(@1)

Query Syntax

BATTery:CHARge:STOP:CURRent:STATe? (@<chanlist>)

Returns

0|1

BATTery:CHARge:STOP:AHOur <value>,(@<chanlist>)

This command sets and queries the stop Ahour for battery charging.

Syntax

BATTery:CHARge:STOP:AHOur <value>,(@<chanlist>)

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Arguments

0.01-999.99|MINimum|MAXimum (unit is AH)

Example

BATT:CHAR:STOP:AHO 0.1,(@1)

Query Syntax

BATTery:CHARge:STOP:AHOur? [MINimum|MAXimum,] (@<chanlist>)

Returns

MINimum|MAXimum

BATTery:CHARge:STOP:AHOur:STATe <0|OFF|1|ON>,(@<chanlist>)

This command sets the stop charging Ahour status for battery charging test.

Syntax

BATTery:CHARge:STOP:AHOur:STATe <0|OFF|1|ON>,(@<chanlist>)

Arguments

<0|OFF|1|ON>

Example

BATT:CHAR:STOP:AHO:STAT 1,(@1)

Query Syntax

BATTery:CHARge:STOP:AHOur:STATe? (@<chanlist>)

Returns

0|1

BATTery:CHARge:STOP:WHOur <value>,(@<chanlist>)

This command sets and queries the stop Whour for battery charging.

Syntax

BATTery:CHARge:STOP:AHOur <value>,(@<chanlist>)

Arguments

0.01-999.99|MINimum|MAXimum (Unit is WH)

Example

BATT:CHAR:STOP:WHO 0.1,(@1)

Query Syntax

BATTery:CHARge:STOP:WHOur? [MINimum|MAXimum,] (@<chanlist>)

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Returns

MINimum|MAXimum

BATTery:CHARge:STOP:WHOur:STATe <0|OFF|1|ON>,(@<chanlist>)

This command sets the stop charging Ahour status for battery charging test.

Syntax

BATTery:CHARge:STOP:WHOur:STATe <0|OFF|1|ON>,(@<chanlist>)

Arguments

<0|OFF|1|ON>

Example

BATT:CHAR:STOP:WHO:STAT 1,(@1)

Query Syntax

BATTery:CHARge:STOP:WHOur:STATe? (@<chanlist>)

Returns

0|1

BATTery:CHARge:STOP:TIME <value>,(@<chanlist>)

This command sets and queries the stop time for battery charging.

Syntax

BATTery:CHARge:STOP:TIME <value>,(@<chanlist>)

Arguments

0-86400|MINimum|MAXimum

Example

BATT:CHAR:STOP:TIME 12.0,(@1)

Query Syntax

BATTery:CHARge:STOP:TIME? [MINimum|MAXimum,] (@<chanlist>)

Returns

MINimum|MAXimum

BATTery:CHARge:STOP:TIME:STATe <0|OFF|1|ON>,(@<chanlist>)

Set and query the status of the battery charging time stop.

Syntax

BATTery:CHARge:STOP:TIME:STATe <0|OFF|1|ON>,(@<chanlist>)

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Arguments

<0|OFF|1|ON>

Example

BATT:CHAR:STOP:TIME:STAT 0,(@1)

Query Syntax

BATTery:CHARge:STOP:TIME:STATe? (@<chanlist>)

Returns

0|1

BATTERY:CHARge:SAVe <"filename">,(@<chanlist>)

This command saves the edit test item as a file, overwriting the current file if the file name already exists, otherwise, saving as a new file.

Syntax

BATTERY:CHARge:SAVe <"filename">,(@<chanlist>)

Arguments

<string>

Example

BATT:CHAR:SAVe (@1),"test.csv"

BATTery:CHARge:RECall <"filename">,(@<chanlist>)

This command recalls the battery charging test file stored in instrument.

Syntax

BATTery:CHARge:RECall <"filename">,(@<chanlist>)

Arguments

<string>

Example

BATT:CHAR:RECall "test.csv",(@1)

BATTery:CHARge:DELete <"filename">,(@<chanlist>)

This command deletes the battery charging test file stored in instrument.

Syntax

BATTery:CHARge:DELete <"filename">,(@<chanlist>)

Arguments

<string>

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Example

BATT:CHAR:DEL "test.csv",(@1)

BATTery:CHARge:TIME? (@<chanlist>)

This command queries how long the battery charge has been running.

Syntax

BATTery:CHARge:TIME? (@<chanlist>)

Returns

<NRF>

Example

BATT:CHAR:TIME? (@1)

BATTery:CHARge:RSTate? (@<chanlist>)

This command queries the battery charge status.

Syntax

BATTery:CHARge:RSTate? (@<chanlist>)

Returns

"Idle","Charging","Voltage Up","Current Up","Energy Up","Time up","Abort"

Example

BATT:CHAR:RST? (@1)

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Chapter22 BATTery DISCharge Subsystem (Load)

BATTery:DISCharge:MODE <mode>,(@<chanlist>)

BATTery:DISCharge:MODE? (@<chanlist>)

The commands set and query the discharge mode of the battery.

Command Syntax:

BATTery:DISCharge:MODE <CURRent|VOLTage>,(@<chanlist>)

Arguments:

CURRent|POWer|RESistance
<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:DISCharge:MODE? (@<chanlist>)

Returns

CURR,POW or RES

Example:

BATT:DISC:MODE CURR,(@1) // Setting the mode of discharging the channel 1 battery to constant current mode
BATT:DISC:MODE? (@1) // Query the mode of discharging the battery of channel 1

BATTery:DISCharge:VOLTage <value>,(@<chanlist>)

BATTery:DISCharge:VOLTage? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the voltage value of battery discharge.

Command Syntax:

BATTery:DISCharge:VOLTage <value>,(@<chanlist>)

Arguments:

0-MAXimum|MINimum|MAXimum

<chanlist>: channel number.

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(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:DISCharge:VOLTage? [MINimum|MAXimum],(@<chanlist>)

Returns

< Discharge voltage value >

Example:

BATT:DISC:VOLT 5.0,(@1) // Set the voltage of channel 1 discharge to 5V
BATT:DISC:VOLT? (@1) // Query the voltage discharged from channel 1

BATTery:DISCharge:CURRent <value>,(@<chanlist>)

BATTery:DISCharge:CURRent? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the current value of battery discharge.

Command Syntax:

BATTery:DISCharge:CURRent <value>,(@<chanlist>)

Arguments:

0-MAXimum|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:DISCharge:CURRent? [MINimum|MAXimum,](@<chanlist>)

Returns

<Discharge current value >

Example:

BATT:DISC:CURR 5.0,(@1) // Set the current of channel 1 discharge to 5A
BATT:DISC:CURR? (@1) // Query the current discharged by channel 1

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BATTery:DISCharge:POWer <value>,(@<chanlist>)

BATTery:DISCharge:POWer? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the power value of battery discharge.

Command Syntax:

BATTery:DISCharge:POWer <value>,(@<chanlist>)

Arguments:

0-MAXimum|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:DISCharge:POWer? [MINimum|MAXimum,] (@<chanlist>)

Returns

< Discharge power value >

Example:

```
BATT:DISC:POW 3.0,(@1)      // Set the discharge power to 3W
BATT:DISC:POW? (@1)         // Inquire about the power of the discharge
```

BATTery:DISCharge:RESistance <value>,(@<chanlist>)

BATTery:DISCharge:RESistance? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the resistance value of battery discharge.

Command Syntax:

BATTery:DISCharge:RESistance <value>,(@<chanlist>)

Arguments:

0-MAXimum|MINimum|MAXimum
<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

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Query Syntax

BATTery:DISCharge:RESistance? [MINimum|MAXimum,] (@<chanlist>)

Returns

<Discharge resistance>

Example:

BATT:DISC:RESistance 1000,(@1) // Set the discharge resistance of channel 1 to 1000Ω
BATT:DISC:RESistance? (@1) //Query the discharge resistance of channel 1

BATTery:DISCharge:STOP:VOLTage <value>,(@<chanlist>)

BATTery:DISCharge:STOP:VOLTage? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the stop voltage of battery discharge.

Command Syntax:

BATTery:DISCharge:STOP:VOLTage <value>,(@<chanlist>)

Arguments:

0-MAXimum|MINimum|MAXimum (Unit is V)
<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:DISCharge:STOP:VOLTage? [MINimum|MAXimum,] (@<chanlist>)

Returns

<stoped voltage >

Example:

BATT:DISC:STOP:VOLT 4.0,(@1) // Set the stopping voltage for discharging the channel 1 battery to 4V
BATT:DISC:STOP:VOLT? (@1) // Query channel 1 battery discharge stop voltage

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BATTery:DISCharge:STOP:VOLTage:STATe <0|OFF|1|ON>,(@<chanlist>)

BATTery:DISCharge:STOP:VOLTage:STATe? (@<chanlist>)

This command queries the enable state of voltage stop for battery discharge.

Command Syntax:

BATTery:DISCharge:STOP:VOLTage:STATe <0|OFF|1|ON>,(@<chanlist>)

Arguments:

<0|OFF|1|ON>
<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:DISCharge:STOP:VOLTage:STATe? (@<chanlist>)

Returns

0 or 1

Example:

BATTery:DISC:STOP:VOLT:STAT 1,(@1) // Enable the voltage stop function for channel 1 battery discharge
BATTery:DISC:STOP:VOLT:STAT? (@1) // Query the voltage stop function enable state of channel 1 battery discharge.

BATTery:DISCharge:STOP:CURRent <value>,(@<chanlist>)

BATTery:DISCharge:STOP:CURRent? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the stop current of battery discharge.

Command Syntax:

BATTery:DISCharge:STOP:CURRent <value>,(@<chanlist>)

Arguments:

0-MAXimum|MINimum|MAXimum (Unit is A)
<chanlist>: channel number.

(@1) – Channel 1

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(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:DISCharge:STOP:CURRent? [MINimum|MAXimum,] (@<chanlist>)

Returns

<Stop current>

Example:

BATT:DISC:STOP:CURR 2.0,(@1) // Set the stopping current for discharging the channel 1 battery to 2A
BATT:DISC:STOP:CURR? (@1) // Query the stopping current

BATTery:DISCharge:STOP:CURRent:STATe <0|OFF|1|ON>,(@<chanlist>)

BATTery:DISCharge:STOP:CURRent:STATe? (@<chanlist>)

The commands set and query the current stop state of battery discharge.

Command Syntax:

BATTery:DISCharge:STOP:CURRent:STATe <0|OFF|1|ON>,(@<chanlist>)

Arguments:

<0|OFF|1|ON>
<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:DISCharge:STOP:CURRent:STATe? (@<chanlist>)

Returns

0 or 1

Example:

BATT:DISC:STOP:CURR:STAT 1,(@1) // Enable current stop function
BATT:DISC:STOP:CURR:STAT? (@1) // Query current stop function enable state

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BATTery:DISCharge:STOP:AHOur <value>,(@<chanlist>)

BATTery:DISCharge:STOP:AHOur [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the stop ampere time of battery charging.

Command Syntax:

BATTery:DISCharge:STOP:AHOur <value>,(@<chanlist>)

Arguments:

0.01-999.99|MINimum|MAXimum (Unit is AH)
<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:DISCharge:STOP:AHOur? [MINimum|MAXimum,] (@<chanlist>)

Returns

<stop AH>

Example:

BATT:DISC:STOP:AHO 0.1,(@1) // Set the stopping ampere time to 0.1AH
BATT:DISC:STOP:AHO? (@1) // Query the stopping ampere time of discharging a channel 1 battery

BATTery:DISCharge:STOP:AHOur:STATe <0|OFF|1|ON>,(@<chanlist>)

BATTery:DISCharge:STOP:AHOur:STATe? (@<chanlist>)

This command is used to set and query the status of the ampere-time stop of battery discharge.

Command Syntax:

BATTery:DISCharge:STOP:AHOur:STATe <0|OFF|1|ON>,(@<chanlist>)

Arguments:

<0|OFF|1|ON>
<chanlist>: channel number.

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(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:DISCharge:STOP:AHOur:STATe? (@<chanlist>)

Returns

0 or 1

Example:

BATT:DISC:STOP:AHO:STAT 1,(@1) //Enable Channel 1 Battery Discharge Amp-Hour Stop Function
BATT:DISC:STOP:AHO:STAT? (@1) //Query Channel 1 Battery Discharge Ampere-Hour Stop Function Enable Status

BATTery:DISCharge:STOP:WHOur <value>,(@<chanlist>)

BATTery:DISCharge:STOP:WHOur? [MINimum|MAXimum,] (@<chanlist>)

This command is used to set and query the stop watt-hours for battery discharge.

Command Syntax:

BATTery:DISCharge:STOP:WHOur <value>,(@<chanlist>)

Arguments:

0.01-999.99|MINimum|MAXimum (unit: WH)
<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:DISCharge:STOP:WHOur? [MINimum|MAXimum,] (@<chanlist>)

Returns

<wh>

Example:

BATT:DISC:STOP:WHO 0.1,(@1) //Set the stop watt-hour for Channel 1 battery discharge to 0.1 Wh.
BATT:DISC:STOP:WHO? (@1) //Query Channel 1 Battery Discharge Stop Watt-Hours

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BATTery:DISCharge:STOP:WHOur:STATe <0|OFF|1|ON>,(@<chanlist>)

BATTery:DISCharge:STOP:WHOur:STATe? (@<chanlist>)

This command is used to set and query the status of the watt-hour cutoff for battery discharge.

Command Syntax:

BATTery:DISCharge:STOP:WHOur:STATe <0|OFF|1|ON>,(@<chanlist>)

Arguments:

<0|OFF|1|ON>
<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:DISCharge:STOP:WHOur:STATe? (@<chanlist>)

Returns

0 or 1

Example:

BATT:DISC:STOP:WHO:STAT 1,(@1) //Watt-hour cutoff function for enabling channel 1 battery discharge
BATT:DISC:STOP:WHO:STAT? (@1) // Query the enable status of the watt-hour cutoff function for battery discharge on Channel 1

BATTery:DISCharge:STOP:TIME <value>,(@<chanlist>)

BATTery:DISCharge:STOP:TIME? (@<chanlist>)

This command is used to set and query the stop time for battery discharge.

Command Syntax:

BATTery:DISCharge:STOP:TIME <value>,(@<chanlist>)

Arguments:

0-86400|MINimum|MAXimum
<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

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Query Syntax

BATTery:DISCharge:STOP:TIME? [MINimum|MAXimum,] (@<chanlist>)

Returns

<time>

Example:

BATT:DISC:STOP:TIME 12.0,(@1) //Set the stop time for Channel 1 battery discharge to 12 seconds.
BATT:DISC:STOP:TIME? (@1) //Query Channel 1 Battery Discharge Stop Time

BATTery:DISCharge:STOP:TIME:STATe <0|OFF|1|ON>,(@<chanlist>)

BATTery:DISCharge:STOP:TIME:STATe? (@<chanlist>)

This command is used to set and query the status of the battery discharge timeout.

Command Syntax:

BATTery:DISCharge:STOP:TIME:STATe <0|OFF|1|ON>,(@<chanlist>)

Arguments:

<0|OFF|1|ON>
<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:DISCharge:STOP:TIME:STATe? (@<chanlist>)

Returns

0 or 1

Example:

BATT:DISC:STOP:TIME:STAT 0,(@1) //Faulty Channel 1 Battery Discharge Timeout Function
BATT:DISC:STOP:TIME:STAT? (@1) //Query the enable status of the time-out function for battery discharge on Channel 1

BATTery:DISCharge:SAVe <"filename">,(@<chanlist>)

Save the currently edited test item as a file. If the file name already exists, the current file will be overwritten. If it does not exist, it will be saved as a new file.

Command Syntax:

BATTERY:DISCharge:SAVe <"filename">,(@<chanlist>)

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Arguments:

<string>: File name must end with the .csv.
<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Example:

BATT:DISC:SAVe "test.csv",(@1) //Save the current editing item under Channel 1 with the filename “test.csv”.

BATTery:DISCharge:RECall <"filename">,(@<chanlist>)

Access files stored using the battery discharge function.

Command Syntax:

BATTery:DISCharge:RECall <"filename">,(@<chanlist>)

Arguments:

<string>: File name must end with the .csv.
<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Example:

BATT:DISC:RECall "test.csv",(@1) // Call the battery discharge storage file under Channel 1, file name “test.csv”

BATTery:DISCharge:DELete <"filename">,(@<chanlist>)

Delete files stored in the battery discharge function.

Command Syntax:

BATTery:DISCharge:DELete <"filename">,(@<chanlist>)

Arguments:

<string>: File name must end with the .csv.
<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Example:

BATT:DISC:DEL "test.csv",(@1) //Delete the battery discharge storage file for Channel 1, named “test.csv”.

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BATTery:DISCharge:TIME? (@<chanlist>)

Check the elapsed time since the battery began discharging.

Query Syntax

BATTery:DISC:TIME? (@<chanlist>)

Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Returns

<time>

Example:

BATT:DISC:TIME? (@1) //Query Channel 1 Battery Discharge Duration

BATTery:DISCharge:RSTate? (@<chanlist>)

Check the status of battery discharge operation.

Query Syntax

BATTery:DISCharge:RSTate? (@<chanlist>)

Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Returns

"Idle","Discharging ","Voltage Up","Current Up","Energy Up","Time up","Abort"

Example:

BATT:DISCharge:RST? (@1) //Query the discharge status of Channel 1 battery

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Chapter23 BATTery EMULator Subsystem

BATTery:EMULator:VOC:FULL <value>,(@<chanlist>)

BATTery:EMULator:VOC:FULL? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the full voltage (in V) of the battery emulation single cell.

Command Syntax:

BATTery:EMULator:VOC:FULL <value>,(@<chanlist>)

Arguments:

MINimum-MAXimum|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:VOC:FULL? [MINimum|MAXimum,](@<chanlist>)

Returns

<Full voltage>

Example:

BATTery:EMULator:VOC:FULL 15,(@1) // Setting the full voltage of a single cell for channel 1 battery emulation 15.0V

BATTery:EMULator:VOC:FULL? (@1) // Query the full voltage of a single cell for channel 1 cell simulation

BATTery:EMULator:VOC:EMPTy <value>|MINimum|MAXimum,(@<chanlist>)

BATTery:EMULator:VOC:EMPTy? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the empty voltage (in V) of the battery emulation single cell.

Command Syntax:

BATTery:EMULator:VOC:EMPT <value>|MINimum|MAXimum,(@<chanlist>)

Arguments:

MINimum-MAXimum|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:VOC:EMPT? [MINimum|MAXimum,](@<chanlist>)

Returns

< empty voltage>

Example:

BATTery:EMULator:VOC: EMPT 10,(@1) // Setting the empty voltage of a single cell for channel 1 battery emulation 10.0V

BATTery:EMULator:VOC: EMPT? (@1) // Query channel 1 battery simulation single cell empty voltage

BATTery:EMULator:VOC:FULL:SUMMary? (@<chanlist>)

The command queries the full voltage (in V) of the battery after emulating the series-parallel connection.

Query Syntax

BATTery:EMULator:VOC:FULL:SUMMary? (@<chanlist>)

Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Returns

< Full voltage after series-parallel aggregation >

Example:

BATTery:EMULator:VOC:FULL:SUMMary? (@1) // Query the full voltage of channel 1 battery after simulating series-parallel connection.

BATTery:EMULator:VOC:EMPTy:SUMMary? (@<chanlist>)

The command queries the null voltage (in V) of the battery after emulating the series-parallel connection.

Query Syntax

BATTery:EMULator:VOC:EMPT:SUMMary? (@<chanlist>)

Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Returns

< Null voltage after series-parallel aggregation >

Example:

BATTery:EMULator:VOC: EMPT:SUMMary? (@1) // Query the empty voltage of channel 1 battery after simulating series-parallel connection.

BATTery:EMULator:CURRent:LIMit[:POSitive] <value>|MINimum|MAXimum,(@<chanlist>)

BATTery:EMULator:CURRent:LIMit[:POSitive]? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the positive current limit (in A) for battery emulation.

Command Syntax:

BATTery:EMULator:CURRent:LIMit[:POSitive] <value>|MINimum|MAXimum,(@<chanlist>)

Arguments:

MINimum-MAXimum|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:CURRent:LIMit[:POSitive]? [MINimum|MAXimum,](@<chanlist>)

Returns

< Positive current limit value >

Example:

BATTery:EMULator:CURRent:LIMit 2,(@1)// Set channel 1 battery emulation positive current limit 2.0A

BATTery:EMULator:CURRent:LIMit? (@1) // Query positive current limit for channel 1 battery emulation

BATTery:EMULator:CURRent:LIMit:NEGative <value>|MINimum|MAXimum,(@<chanlist>)

BATTery:EMULator:CURRent:LIMit:NEGative? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the negative current limit (in A) of the battery emulation.

Command Syntax:

BATTery:EMULator:CURRent:LIMit:NEGative <value>|MINimum|MAXimum,(@<chanlist>)

Arguments:

MINimum-MAXimum|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:CURRent:LIMit:NEGative? [MINimum|MAXimum,](@<chanlist>)

Returns

< Negative current limit value >

Example:

BATTery:EMULator:CURRent:LIMit:NEGative -2.0,(@1)// Set channel 1 battery emulation negative current limit -2.0A

BATTery:EMULator:CURRent:LIMit: NEGative? (@1)// Query Channel 1 Battery Emulation Negative Current Limit

BATTery:EMULator:RESistance <value>|MINimum|MAXimum,(@<chanlist>)

BATTery:EMULator:RESistance? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the internal resistance (in mΩ) of a single cell of the battery emulation.

Command Syntax:

BATTery:EMULator:CURRent:LIMit:RESistance <value>|MINimum|MAXimum,(@<chanlist>)

Arguments:

MINimum-MAXimum|MINimum|MAXimum

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<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:CURRent:LIMit:RESistance? [MINimum|MAXimum,](@<chanlist>)

Returns

<internal resistance >

Example:

BATTery:EMULator:RESistance 10,(@1) // Set the internal resistance of a single cell for channel 1 battery emulation to 10mΩ
BATTery:EMULator:RESistance? (@1)// Querying the internal resistance of a single cell for channel 1 battery simulation

BATTery:EMULator:RESistance:SUMMary? (@<chanlist>)

This command queries the internal resistance (in Ω) of the battery after emulating series-parallel connection.

Query Syntax

BATTery:EMULator:RESistance:SUMMary? (@<chanlist>)

Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Returns

< Full voltage after series-parallel aggregation >

Example:

BATTery:EMULator:RESistance:SUMMary? (@1) // Query the internal resistance of channel 1 battery after simulating series-parallel connection.

BATTery:EMULator:CAPacity:LIMit <value>|MINimum|MAXimum,(@<chanlist>)

BATTery:EMULator:CAPacity:LIMit? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the capacity (in Ah) of a single cell of the battery

emulation.

Command Syntax:

BATTery:EMULator:CAPacity:LIMit <value>|MINimum|MAXimum,(@<chanlist>)

Arguments:

0-999.999|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:CAPacity:LIMit? [MINimum|MAXimum,](@<chanlist>)

Returns

< Capacity of a single battery >

Example:

BATTery:EMULator:CAPacity:LIMit 50,(@1) // Set the capacity of a single battery for 1-channel battery simulation to 50Ah.
BATTery:EMULator:CAPacity:LIMit? (@1) // Queries the capacity of a single cell for channel 1 battery emulation.

BATTery:EMULator:CAPacity:LIMit:SUMMary? (@<chanlist>)

This command queries the capacity (in Ah) of the battery after emulating series-parallel connection.

Query Syntax

BATTery:EMULator:CAPacity:LIMit:SUMMary? (@<chanlist>)

Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Returns

< capacity after series-parallel connection >

Example:

BATTery:EMULator:CAPacity:LIMit:SUMMary? (@1) // Query the capacity of channel 1 battery after simulating series-parallel connection.

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BATTery:EMULator:SOC:LIMit[:UPPer] <value>|MINimum|MAXimum,(@<chanlist>)

BATTery:EMULator:SOC:LIMit[:UPPer]? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the upper limit of battery emulation capacity.

Command Syntax:

BATTery:EMULator:SOC:LIMit[:UPPer] <value>|MINimum|MAXimum,(@<chanlist>)

Arguments:

100-110|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:SOC:LIMit[:UPPer]? [MINimum|MAXimum,](@<chanlist>)

Returns

< Upper limit of capacity >

Example:

BATTery:EMULator:SOC:LIMit 105,(@1) // Set the upper limit of the channel battery emulation capacity to 105 percent

BATTery:EMULator:SOC:LIMit? (@1) // Query the upper limit of 1-channel current simulation capacity

BATTery:EMULator:SOC:LIMit:LOWer <value>|MINimum|MAXimum,(@<chanlist>)

BATTery:EMULator:SOC:LIMit:LOWer? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the lower limit of battery emulation capacity.

Command Syntax:

BATTery:EMULator:SOC:LIMit:LOWer <value>|MINimum|MAXimum,(@<chanlist>)

Arguments:

(-10)-0|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:SOC:LIMit:LOWer? [MINimum|MAXimum,](@<chanlist>)

Returns

< Lower limit of capacity >

Example:

BATTery:EMULator:SOC:LIMit:LOWer -5,(@1) // Set the lower limit of the simulated capacity of channel 1 battery to -5.0%

BATTery:EMULator:SOC:LIMit:LOWer? (@1) // Query the lower limit of ch1 current simulation capacity

BATTery:EMULator:INITial:SOC <value>|MINimum|MAXimum,(@<chanlist>)

BATTery:EMULator:INITial:SOC? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the initial charge value of the battery emulation.

Command Syntax:

BATTery:EMULator:INITial:SOC <value>|MINimum|MAXimum,(@<chanlist>)

Arguments:

0-100|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:INITial:SOC? [MINimum|MAXimum,](@<chanlist>)

Returns

< initial charge >

Example:

BATTery:EMULator:INITial:SOC 10,@1) // Setting the initial charge value for channel 1 battery emulation 10%

BATTery:EMULator:INITial:SOC? @1) // Query the initial charge value of channel 1 battery simulation

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BATTery:EMULator:INITial:VOC <value>|MINimum|MAXimum,(@<chanlist>)

BATTery:EMULator:INITial:VOC? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the initial open-circuit voltage (in V) of the battery emulation.

Command Syntax:

BATTery:EMULator:INITial:VOC <value>|MINimum|MAXimum,(@<chanlist>)

Arguments:

MINimum-MAXimum|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:INITial:VOC? [MINimum|MAXimum,](@<chanlist>)

Returns

< Initial open circuit voltage >

Example:

BATTery:EMULator:INITial:VOC 12,@1) // Set initial open-circuit voltage for channel 1 battery emulation to 12.0V

BATTery:EMULator:INITial:VOC? @1) //Query the initial open-circuit voltage value of channel 1 battery emulation

BATTery:EMULator:INITial:CAPacity <value>|MINimum|MAXimum,(@<chanlist>)

BATTery:EMULator:INITial:CAPacity? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the initial capacity (in Ah) of the battery emulation.

Command Syntax:

BATTery:EMULator:INITial:CAPacity <value>|MINimum|MAXimum,(@<chanlist>)

Arguments:

0-999.999|MINimum|MAXimum

<chanlist>: channel number.

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(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:INITial:CAPacity? [MINimum|MAXimum,](@<chanlist>)

Returns

< Initial open circuit voltage >

Example:

BATTery:EMULator:INITial:CAPacity 12,@1) // Set the initial capacity of channel 1 battery emulation to 12Ah
BATTery:EMULator:INITial:CAPacity? @1) // Query the initial capacity value of channel 1 battery emulation

BATTery:EMULator:PARallel <value>|MINimum|MAXimum,(@<chanlist>)

BATTery:EMULator:PARallel? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the number of parallel connections for battery emulation.

Command Syntax:

BATTery:EMULator:PARallel <value>|MINimum|MAXimum,(@<chanlist>)

Arguments:

1-99|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:PARallel? [MINimum|MAXimum,] (@<chanlist>)

Returns

< Number of parallel connections >

Example:

BATTery:EMULator:PARallel 2,@1) // Set the number of cells in parallel for channel 1 battery emulation to 2
BATTery:EMULator:PARallel? @1) // Query the number of parallel connections for channel 1 battery simulation

BATTery:EMULator:SERies <value>|MINimum|MAXimum,(@<chanlist>)

BATTery:EMULator:SERies? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the number of series connections for battery emulation.

Command Syntax:

BATTery:EMULator:SERies <value>|MINimum|MAXimum,(@<chanlist>)

Arguments:

1-99|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:SERies? [MINimum|MAXimum,] (@<chanlist>)

Returns

< Number of tandem connections >

Example:

BATTery:EMULator:SERies 2,@1) // Set the number of cells in series for channel 1 battery emulation to 2

BATTery:EMULator:SERies? @1) // Query the number of series connections for channel 1 battery emulation

BATTery:EMULator:CURRent:CAPacity? (@<chanlist>)

This command queries the present capacity (in Ah) of the battery emulation.

Query Syntax

BATTery:EMULator:CURRent:CAPacity? (@<chanlist>)

Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Returns

< present capacity >

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Example:

BATTery:EMULator:CURRent:CAPacity? (@1) //Query the present capacity of channel 1 battery emulation

BATTery:EMULator:CURRent:SOC? (@<chanlist>)

This command queries the SOC of battery emulation.

Query Syntax

BATTery:EMULator:CURRent:SOC? (@<chanlist>)

Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Returns

<SOC>

Example:

BATTery:EMULator:CURRent:SOC? (@1) // Query the SOC of the channel 1 battery emulation

BATTery:EMULator:CURRent:VOC? (@<chanlist>)

This command queries the voltage open-circuit (in V) of the battery emulation.

Query Syntax

BATTery:EMULator:CURRent:VOC? (@<chanlist>)

Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Returns

<VOC>

Example:

BATTery:EMULator:CURRent:VOC? (@1) //Query the VOC of channel 1 battery emulation

BATTery:EMULator:CURRent: AH? (@<chanlist>)

The command queries the current current current integral of battery emulation.

Query Syntax

BATTery:EMULator:CURRent:AH? (@<chanlist>)

Returns

< Current Current Integral >

Example:

BATTery:EMULator:CURRent:AH? (@1) //Query the current current integral of channel 1 battery simulation

BATTery:EMULator:CURRent:RESistance? (@<chanlist>)

This command queries the current internal resistance of battery emulation.

Query Syntax

BATTery:EMULator:CURRent:RESistance? (@<chanlist>)

Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Returns

<resistance>

Example:

BATTery:EMULator:CURRent:RESistance? (@1) //Query the internal resistance value of channel 1 battery emulation

BATTery:EMULator:SAVe <"filename">,(@<chanlist>)

This command is used to save the file of the current battery simulation.

Command Syntax:

BATTery:EMULator:Save <"filename">,(@<chanlist>)

Arguments:

<string>
<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

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Example:

BATTery:EMULator:SAVe 1.csv,(@1) //Save the file for the current battery simulation for channel 1 as 1.csv

BATTery:EMULator:RECall<"filename">,(@<chanlist>)

This instruction is used to call open the battery simulation file.

Command Syntax:

BATTery:EMULator:RECall <"filename">,(@<chanlist>)

Arguments:

<string>
<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Example:

BATTery:EMULator:RECall 1.csv,(@1) //Open the 1.csv file for the current battery simulation for channel 1

BATTery:EMULator:RSTatus? (@<chanlist>)

This command queries the running status of the battery emulation.

Query Syntax

BATTery:EMULator:RStatus? (@<chanlist>)

Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Returns

< Running status >

- 0 is the stop state
- 1 is the start state
- 2 is the running state
- 3 is the stop state
- 4 is the active state
- 5 is the end state

Example:

BATTery:EMULator:RStatus? (@1) //Query the operation status of the battery simulation for channel 1

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BATTery:EMULator:CURVe:STEP <step>,<NRf>,<NRf>,<NRf>,(@<chanlist>)

BATTery:EMULator:CURVe:STEP? < step >,(@<chanlist>)

The commands set and query the content information of the specified step of the battery simulation curve.

Command Syntax:

BATTery:EMULator:CURVe:STEP <step>,<NRf1>,<NRf2>,<NRf3>,(@<chanlist>)

Arguments:

<step>: Step value in battery analog curve mode, range 1-200

< NRf1 >: Charge value for a step in the battery analog curve mode, range 0-100

< NRf2 >: Open circuit voltage value in battery analog curve mode. The range is <MINimum-MAXimum>.

< NRf3 >: Internal resistance value in battery analog curve mode. The range is <MINimum-MAXimum>.

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:CURVe:STEP? < step >,(@<chanlist>)

Returns

<SOC, VOC, Resistance>

Example:

BATTery:EMULator:CURVe:STEP 1,10,20,50,(@1) // Set the content information of the specified step, insert step 1 with soc value of 10%, voc value of 20V, and internal resistance value of 50mΩ.
BATTery:EMULator:CURVe:STEP? 1,(@1) // Queries the content information of step 1.

BATTery:EMULator:CURVe:COUNt? (@<chanlist>)

This command queries the current total number of steps of the battery simulation curve.

Query Syntax

BATTery:EMULator:CURVe:COUNt? (@<chanlist>)

Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

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(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Returns

< Total steps in curve mode >

Example:

BATTery:EMULator:CURRent: COUNt? (@1) // Queries the current total number of steps for the ch1 battery simulation

BATTery:EMULator:MODE BASic|CURVe,(@<chanlist>)

BATTery:EMULator:MODE? (@<chanlist>)

The commands set and query the operating mode of battery emulation.

Command Syntax:

BATTery:EMULator:MODE BASic|CURVe,(@<chanlist>)

Arguments:

BASic | CURVe

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:MODE? (@<chanlist>)

Returns

BAS or CURV

Example:

BATTery:EMULator:MODE CURve,(@1) // Set the operating mode of channel 1 battery emulation to curve mode

BATTery:EMULator:MODE? (@1) // Query the operating mode of channel 1 battery emulation

BATTery:EMULator:END:MODE HOLD|OFF,(@<chanlist>)

BATTery:EMULator:END:MODE? (@<chanlist>)

The commands set and query the end mode of battery emulation.

Command Syntax:

BATTery:EMULator:END:MODE HOLD|OFF,(@<chanlist>)

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Arguments:

HOLD | OFF

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:END:MODE? (@<chanlist>)

Returns

HOLD or OFF

Example:

BATTery:EMULator:END:MODE HOLD,(@1) // Set the end mode of channel 1 battery emulation to HOLD mode

BATTery:EMULator:END:MODE? (@1) // Query the end mode of channel 1 battery simulation

BATTery:EMULator:INITial:PARameter SOC|VOC|CAPacity,(@<chanlist>)

BATTery:EMULator:INITial:PARameter? (@<chanlist>)

The commands set and query the initial parameters of battery emulation.

Command Syntax:

BATTery:EMULator:INITial:PARameter SOC|VOC|CAPacity,(@<chanlist>)

Arguments:

SOC |VOC|CAPacity

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

BATTery:EMULator:INITial:PARameter? (@<chanlist>)

Returns

SOC, VOC, CAP

Example:

BATTery:EMULator:INITial:PARameter SOC,(@1) // Set the initial parameters for channel 1 battery simulation to SOC

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BATTery:EMULator:INITial:PARameter? (@1) // Query the initial parameters of channel 1 battery simulation

Chapter24 TRANsient Subsystem (Load module supports)

[SOURce:]TRANsient:COUNt <value>|MINimum|MAXimum,(@<chanlist>)

[SOURce:]TRANsient:COUNt? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the number of repetitions of consecutive transients. 0 for infinite loop.

Note: Continuous mode only.

Command Syntax:

[SOURce:]TRANsient:COUNt <value>|MINimum|MAXimum,(@<chanlist>)

Arguments:

0-65535|MINimum|MAXimum, 0 represents an infinite number of times

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

[SOURce:]TRANsient:COUNt? [MINimum|MAXimum,](@<chanlist>)

Returns

<repeat count>

Example:

TRAN:coun 10,(@1) // Set the number of repetitions for the continuous mode of the channel 1 transient function to 10

TRAN:coun? (@1) // Query the number of repetitions of the continuous mode of the transient function of channel 1

[SOURce:]TRANsient:COUNt:RUN? (@<chanlist>)

This command queries the number of times the continuous mode inside the transient has been run.

Query Syntax

[SOURce:]TRANsient:COUNt:RUN?

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Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Returns

<repeat count>

Example:

TRAN:COUN:RUN? (@1) // Query the current number of times channel 1 has been run

[SOURce:]TRANsient:DCYCle <value>|MINimum|MAXimum,(@<chanlist>)

[SOURce:]TRANsient:DCYCle? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the duty cycle of continuous dynamic.

Command Syntax:

[SOURce:]TRANsient:DCYCle <value>|MINimum|MAXimum,(@<chanlist>)

Arguments:

1- 99|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

[SOURce:]TRANsient:DCYCle? [MINimum|MAXimum,](@<chanlist>)

Returns

< Continuous Mode Duty Cycle >

Example:

TRAN::DCYCle 10,(@1) // Set the duty cycle of the continuous mode of the transient function of channel 1 to 10 percent.

TRAN:DCYC? (@1) // Queries the duty cycle of the continuous mode of the transient function of channel 1

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[SOURce:]TRANsient:FREQuency <value>|MINimum|MAXimum,(@<chanlist>)

[SOURce:]TRANsient:FREQuency? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the frequency of continuous dynamics. (Unit is Hz)

Command Syntax:

[SOURce:]TRANsient:FREQuency <value>|MINimum|MAXimum, (@<chanlist>)

Arguments:

0.0025- 1000|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

[SOURce:]TRANsient:FREQuency? [MINimum|MAXimum,](@<chanlist>)

Returns

< Continuous mode frequency >

Example:

TRAN:FREQ 20,(@1) // Setting the frequency of the continuous mode of the channel 1 transient function 20Hz

TRAN:FREQ? (@1) // Queries the frequency of the continuous mode of the transient function of channel 1

[SOURce:]TRANsient:PERiod <value>|MINimum|MAXimum,(@<chanlist>)

[SOURce:]TRANsient:PERiod? [MINimum | MAXimum,](@<chanlist>)

The commands set and query the period of continuous mode transient. (Unit is s)

Command Syntax:

[SOURce:]TRANsient:PERiod <value>|MINimum|MAXimum,(@<chanlist>)

Arguments:

minimum-maximum|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

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(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

[SOURce:]TRANsient:PERiod? [MINimum|MAXimum,](@<chanlist>)

Returns

< Continuous mode period >

Example:

TRANsient:PERiod 20,(@1) // Set the period of channel 1 continuous mode transient to 20s

TRAN:PER? (@1) // Query the period of channel 1 continuous mode transient

[SOURce:]TRANsient:MODE CONTinuous|PULSe|TOGGle,(@<chanlist>)

[SOURce:]TRANsient:MODE? (@<chanlist>)

The commands set and query the mode of transient.

Command Syntax:

[SOURce:]TRANsient:MODE CONTinuous | PULSe | TOGGle,(@<chanlist>)

Arguments:

CONTinuous | PULSe | TOGGle

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

[SOURce:]TRANsient:MODE? (@<chanlist>)

Returns

CONT,PULS or TOGG

Example:

TRAN:MODE CONT (@1) // Set the operation mode of channel 1 transients to continuous

TRAN:MODE? (@1) // Query the operating mode of channel 1 transient

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[SOURce:]TRANsient:TWIDth <value>|MINimum|MAXimum,(@<chanlist>)

[SOURce:]TRANsient:TWIDth? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the width of the transient pulse mode pulse. (Unit is s)

Command Syntax:

[SOURce:]TRANsient:TWIDth <value> | MINimum | MAXimum,(@<chanlist>)

Arguments:

minimum-maximum|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

[SOURce:]TRANsient:TWIDth? [MINimum | MAXimum,](@<chanlist>)

Returns

< pulse width >

Example:

TRANsient:TWIDth 10,(@1) // Set the pulse width of channel 1 to 10

TRANsient:TWIDth? (@1) // Queries the pulse width of 1 channel

[SOURce:]TRANsient:CURRent:TLEVel<transient level>|MINimum|MAXimum,(@<chanlist>)

[SOURce:]TRANsient:CURRent:TLEVel? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the current level during transient. (Unit is A)

This command is the same as [SOURce:]CURRent:TLEVel

Command Syntax:

[SOURce:]TRANsient:CURRent:TLEVel <transient level>|MINimum|MAXimum,(@<chanlist>)

Arguments:

minimum-maximum|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

[SOURce:]TRANsient:CURRent:TLEVel? [MINimum|MAXimum,](@<chanlist>)

Returns

<transient current >

Example:

TRANsient:CURRent:TLEVel 15,(@1) // Set current level at channel 1 transient to 15A

TRAN:CURR:TLEV? (@1) // Query the current level during transient of channel 1

[SOURce:]TRANsient:VOLTage:TLEVel <transient level>|MINimum|MAXimum,(@<chanlist>)

[SOURce:]TRANsient:VOLTage:TLEVel? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the voltage level during transient. (Unit is V)

This command is the same as [SOURce:]VOLTage:TLEVel

Command Syntax:

[SOURce:]TRANsient:VOLTage:TLEVel <transientlevel>|MINimum|MAXimum,(@<chanlist>)

Arguments:

minimum-maximum|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

[SOURce:]TRANsient:VOLTage:TLEVel? [MINimum|MAXimum,](@<chanlist>)

Returns

<transient voltage>

Example:

TRANsient:VOLTage:TLEVel 15,(@1) // Setting the voltage level at channel 1

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transient 15V

TRANsient:VOLTage:TLEV? (@1) // Query the voltage level during transient of channel 1

[SOURce:]TRANsient:POWer:TLEVel <transient level>|MINimum|MAXimum,(@<chanlist>)

[SOURce:]TRANsient:POWer:TLEVel? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the power level at transient. (Unit is W)

This command is the same as [SOURce:]POWer:TLEVel

Command Syntax:

[SOURce:]TRANsient:POWer:TLEVel <transient level>|MINimum|MAXimum,(@<chanlist>)

Arguments:

minimum-maximum|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

[SOURce:]TRANsient:POWer:TLEVel? [MINimum|MAXimum,](@<chanlist>)

Returns

<transient power>

Example:

TRANsient:POWer:TLEVel 15,(@1) // Setting the power level of channel 1 at transient 15W

TRAN:POW:TLEV? (@1) // Query the power level during transient of channel 1

[SOURce:]TRANsient:RESistance:TLEVel <transient level>|MINimum|MAXimum,(@<chanlist>)

[SOURce:]TRANsient:RESistance:TLEVel? [MINimum|MAXimum,](@<chanlist>)

The commands set and query the resistance level during transient. (Unit is Ω)

This command is the same as [SOURce:]RESistance:TLEV

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Command Syntax:

[SOURce:]TRANsient:RESistance:TLEVel

<transient level>|MINimum|MAXimum,(@<chanlist>)

Arguments:

minimum-maximum|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

[SOURce:]TRANsient:RESistance:TLEVel? [MINimum|MAXimum,](@<chanlist>)

Returns

<transient resistance>

Example:

TRANsient:RESistance:TLEVel 2,(@1) // Set channel 1 resistor level to 2 ohms

TRAN:RES:TLEV? (@1) // Query the resistor level during transient of channel 1

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Chapter25 Trigger Subsystem

The Trigger command provides trigger source selection, trigger condition settings, and instant trigger settings for transient systems, acquisition systems, data loggers, and external loggers. The Trigger command also provides settings for trigger output and trigger output mapping.

TRIGger:ACQuire[:IMMediate] (@<chanlist>)

This command is used to generate an immediate trigger signal for the measurement function.

Command Syntax:

TRIGger:ACQuire[:IMMediate] (@<chanlist>)

Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Example:

TRIG:ACQ (@1) //Generate a trigger signal for the channel 1 measurement function

TRIGger:ACQuire:SOURce <source>,(@<chanlist>)

TRIGger:ACQuire:SOURce? (@<chanlist>)

The commands set and query the trigger source of the measurement function. If the trigger source is selected as the voltage and current of a certain channel, the channel should exist and not be a slave, otherwise an error is reported.

Command Syntax:

TRIGger:ACQuire:SOURce <source>,(@<chanlist>)

Arguments:

IMMediate|BUS|PIN<1-7>|CURRent<1-8>|VOLTage<1-8>

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

TRIGger:ACQuire:SOURce? (@<chanlist>)

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Returns

IMM,BUS,,PIN<1-7>,CURR<1-8> or VOLT<1-8>

Example:

TRIG:ACQ:SOUR VOLT8,(@1) // Setting the trigger source bit of the channel 1 measurement function VOLT8

TRIG:ACQ:SOUR? (@1) // Query the trigger source of the channel 1 measurement function

TRIGger:ACQuire:VOLTage[:LEVel] <value>,(@<chanlist>)

TRIGger:ACQuire:VOLTage[:LEVel]? (@<chanlist>)

The commands set and query the level of voltage trigger for the measurement function. (Unit is V)

Command Syntax:

TRIGger:ACQuire:VOLTage[:LEVel] <value>,(@<chanlist>)

Arguments:

MINimum-MAXimum|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

TRIGger:ACQuire:VOLTage[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Returns

< Voltage Trigger Level >

Example:

TRIG:ACQ:VOLT 2.0,(@1) // Setting the measurement function channel 1 voltage trigger level to 2.0V

TRIG:ACQ:VOLT? (@1) // Query Measurement Function Channel 1 Voltage Trigger Level

TRIGger:ACQuire:VOLTage:SLOPe <POSitive|NEGative>,(@<chanlist>)

TRIGger:ACQuire:VOLTage:SLOPe? (@<chanlist>)

The commands set and query the voltage-triggered edge of the measurement function.

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Command Syntax:

TRIGger:ACQuire:VOLTage:SLOPe <POSitive|NEGative>,(@<chanlist>)

Arguments:

POSitive|NEGative

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

TRIGger:ACQuire:VOLTage:SLOPe? (@<chanlist>)

Returns

POS or NEG

Example:

TRIG:ACQ:VOLT:SLOP NEG,(@1) // Setting the voltage trigger edge of measurement function channel 1 to falling edge

TRIG:ACQ:VOLT:SLOP? (@1) // Query Measurement Function Channel 1 Voltage Trigger Edge

TRIGger:ACQuire:VOLTage[:LEVel]:HYSTeresis <value>,(@<chanlist>)

TRIGger:ACQuire:VOLTage[:LEVel]:HYSTeresis? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the level hysteresis triggered by the measurement function voltage. (Unit is V)

Command Syntax:

TRIGger:ACQuire:VOLTage[:LEVel]:HYSTeresis <value>,(@<chanlist>)

Arguments:

MINimum-MAXimum|MINimum|MAXimum

Query Syntax

TRIGger:ACQuire:VOLTage[:LEVel]:HYSTeresis? [MINimum|MAXimum,] (@<chanlist>)

Returns

< voltage trigger level hysteresis >

Example:

TRIG:ACQ:VOLT:HYST 0.1,(@1) // Setting the voltage trigger level hysteresis

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of channel 1 of the measurement function to 0.1V

TRIG:ACQ:VOLT:HYST? (@1) // Voltage trigger level hysteresis for channel 1 of the query measurement function

TRIGger:ACQuire:CURRent[:LEVel] <value>,(@<chanlist>)

TRIGger:ACQuire:CURRent[:LEVel]? (@<chanlist>)

The commands set and query the level of current trigger for measurement function. (Unit is A)

Command Syntax:

TRIGger:ACQuire:CURRent[:LEVel] <value>,(@<chanlist>)

Arguments:

MINimum-MAXimum|MINimum|MAXimum

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

TRIGger:ACQuire:CURRent[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Returns

< Current Trigger Level >

Example:

TRIG:ACQ:CURR 1.0,(@1) // Setting the current trigger level of measurement function channel 1 to 1.0A

TRIG:ACQ:CURR? (@1) // Query Measurement Function Channel 1 Current Trigger Level

TRIGger:ACQuire:CURRent:SLOPe <POSitive|NEGative>,(@<chanlist>)

TRIGger:ACQuire:CURRent:SLOPe? (@<chanlist>)

The commands set and query the current trigger edge of the measurement function.

Command Syntax:

TRIGger:ACQuire:CURRent:SLOPe <POSitive|NEGative>,(@<chanlist>)

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Arguments:

POSitive|NEGative

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax

TRIGger:ACQuire:CURRent:SLOPe? (@<chanlist>)

Returns

POS or NEG

Example:

TRIG:ACQ:CURR:SLOP NEG,(@1) // Setting the current trigger edge of measurement function channel 1 to falling edge

TRIG:ACQ:CURR:SLOP? (@1) // Query Measurement Function Channel 1 Current Trigger Edge

TRIGger:ACQuire:VOLTage[:LEVel]:HYSTeresis <value>,(@<chanlist>)

TRIGger:ACQuire:CURRent[:LEVel]:HYSTeresis <value>,(@<chanlist>)

TRIGger:ACQuire:CURRent[:LEVel]:HYSTeresis? [MINimum|MAXimum,] (@<chanlist>)

This command is used to set and query the hysteresis of the current trigger level in the acquisition system.

Command Syntax:

TRIGger:ACQuire:CURRent[:LEVel]:HYSTeresis <value>,(@<chanlist>)

Arguments

<value>

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Query Syntax:

TRIGger:ACQuire:CURRent[:LEVel]:HYSTeresis? [MINimum|MAXimum,] (@<chanlist>)

Returns

<value>

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Example

TRIG:ACQ:CURR:HYST 0.5,(@1) //Set the hysteresis of the system current trigger level for Channel 1 acquisition to 0.5A.

TRIGger:ELOG[:IMMediate] (@<chanlist>)

This command is used to generate an immediate trigger signal for an external data logger.

Command Syntax:

TRIGger:ELOG[:IMMediate] (@<chanlist>)

Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Example:

TRIG:ACQ (@1)

TRIGger:ELOG:SOURce <source>,(@<chanlist>)

TRIGger:ELOG:SOURce? (@<chanlist>)

The commands set and query the trigger source of external data records.

Command Syntax:

TRIGger:ELOG:SOURce? (@<chanlist>)

Arguments:

IMMediate|BUS|PIN<1-7>

Query Syntax

TRIGger:ELOG:SOURce? (@<chanlist>)

Returns

IMM,BUS or PIN<1-7>

Example:

TRIG:ELOG:SOUR BUS,(@1) // Setting the trigger source bit of channel 1 external recorder BUS

TRIG:ELOG:SOUR? (@1) //Query the trigger source of channel 1 external logger

TRIGger:DLOG[:IMMediate] (@<chanlist>)

This command is used to generate an immediate trigger signal for the data logger.

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Command Syntax:

TRIGger:DLOG[:IMMediate] (@<chanlist>)

Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Example:

TRIG:DLOG (@1) //Generate an instant trigger signal for the data logger

TRIGger:DLOG:SOURce <source>

TRIGger:DLOG:SOURce?

The commands set and query the trigger source of data records.

Command Syntax:

TRIGger:DLOG:SOURce <source>

Arguments:

IMMediate|BUS|MANual|PIN<1-7>|CURRent<1-8>|VOLTage<1-8>

Query Syntax

TRIGger:DLOG:SOURce?

Returns

IMM,BUS,MAN,PIN<1-7>,CURR<1-8> or VOLT<1-8>

Example:

TRIG:DLOG:SOUR BUS // Set the trigger source of the data logger to BUS

TRIG:DLOG:SOUR? // Query the trigger source of the data logger

TRIGger:DLOG:VOLTage[:LEVel] <value>,(@<chanlist>)

TRIGger:DLOG:VOLTage[:LEVel]? (@<chanlist>)

The commands set and query the voltage trigger level of the data logger. (Unit is V)

Command Syntax:

TRIGger:DLOG:VOLTage[:LEVel] <value>,(@<chanlist>)

Arguments:

MINimum-MAXimum|MINimum|MAXimum

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Query Syntax

TRIGger:DLOG:VOLTage[:LEVel]? [MINimum|MAXimum,] (@<chanlist>)

Returns

< Voltage Trigger Level >

Example:

TRIG:DLOG:VOLT 1.0,(@1) // Setting the voltage trigger level of channel 1 of the data logger to 1.0V

TRIG:DLOG:VOLT? (@1) // Query the voltage trigger level of channel 1 of the data logger

TRIGger:DLOG:VOLTage:SLOPe <POSitive|NEGative>,(@<chanlist>)

TRIGger:DLOG:VOLTage:SLOPe? (@<chanlist>)

The commands set and query the edge of data logger voltage trigger.

Command Syntax:

TRIGger:DLOG:VOLTage:SLOPe POSitive|NEGative,(@<chanlist>)

Arguments:

POSitive|NEGative

Query Syntax

TRIGger:DLOG:VOLTage:SLOPe? (@<chanlist>)

Returns

POS or NEG

Example:

TRIG:ACQ:VOLT:SLOP NEG,(@1) // Setting the voltage trigger edge of channel 1 of the data logger to falling edge

TRIG:ACQ:VOLT:SLOP?(@1) // Query the voltage trigger edge of channel 1 of the data logger

TRIGger:DLOG:VOLTage[:LEVel]:HYSTeresis <value>,(@<chanlist>)

TRIGger:DLOG:VOLTage[:LEVel]:HYSTeresis? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the level hysteresis triggered by the data logger voltage. (Unit is V)

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Command Syntax:

TRIGger:DLOG:VOLTage[:LEVel]:HYSTeresis <value>,(@<chanlist>)

Arguments:

MINimum-MAXimum|MINimum|MAXimum

Query Syntax

TRIGger:DLOG:VOLTage[:LEVel]:HYSTeresis? [MINimum|MAXimum,](@<chanlist>)

Returns

< Voltage trigger level hysteresis >

Example:

TRIG:DLOG:VOLT:HYST 0.1,(@1) // Setting the voltage trigger level hysteresis of channel 1 of the data logger to 0.1V

TRIG:DLOG:VOLT:HYST? (@1) // Voltage trigger level hysteresis on channel 1 of the query data logger

TRIGger:DLOG:CURRent[:LEVel] <value>,(@<chanlist>)

TRIGger:DLOG:CURRent[:LEVel]? (@<chanlist>)

The commands set and query the level of data logger current trigger. (Unit is A)

Command Syntax:

TRIGger:DLOG:CURRent[:LEVel] <value>,(@<chanlist>)

Arguments:

MINimum-MAXimum|MINimum|MAXimum

Query Syntax

TRIGger:DLOG:CURRent[:LEVel]? [MINimum|MAXimum,](@<chanlist>)

Returns

< Current Trigger Level >

Example:

TRIG:DLOG:CURR 1.0,(@1) // Setting the current trigger level of channel 1 of the data logger to 1.0A

TRIG:DLOG:CURR? (@1) // Query the current trigger level of channel 1 of the data logger

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TRIGger:DLOG:CURRent:SLOPe <POSitive|NEGative>,(@<chanlist>)

TRIGger:DLOG:CURRent:SLOPe? (@<chanlist>)

The commands set and query the edge of data logger current trigger.

Command Syntax:

TRIGger:DLOG:CURRent:SLOPe POSitive|NEGative, (@<chanlist>)

Arguments:

POSitive|NEGative

Query Syntax

TRIGger:DLOG:CURRent:SLOPe? (@<chanlist>)

Returns

POS or NEG

Example:

TRIG:ACQ:CURR:SLOP NEG,(@1) // Setting the current trigger edge of channel 1 of the data logger to falling edge

TRIG:ACQ:CURR:SLOP?(@1) // Query the current trigger edge of channel 1 of the data logger

TRIGger:DLOG:CURRent[:LEVel]:HYSTeresis <value>,(@<chanlist>)

TRIGger:DLOG:CURRent[:LEVel]:HYSTeresis? [MINimum|MAXimum,] (@<chanlist>)

The commands set and query the current-triggered level hysteresis of the data logger. (Unit is A)

Command Syntax:

TRIGger:DLOG:CURRent[:LEVel]:HYSTeresis <value>,(@<chanlist>)

Arguments:

MINimum-MAXimum|MINimum|MAXimum

Query Syntax

TRIGger:DLOG:CURRent[:LEVel]:HYSTeresis? [MINimum|MAXimum,] (@<chanlist>)

Returns

< Current trigger level hysteresis >

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Example:

TRIG:DLOG:CURR:HYST 0.1,(@1) //Setting the current trigger level hysteresis of channel 1 of the data logger to 0.1A

TRIG:DLOG:CURR:HYST? (@1) //Query the current trigger level hysteresis of channel 1 of the data logger

TRIGger:TRANsient[:IMMediate] (@<chanlist>)

This command is used to generate an immediate trigger signal for a transient system.

Include ARB,sequence,cdarb,sine sweep,load transient,list.

Command Syntax:

TRIGger:TRANsient[:IMMediate] (@<chanlist>)

Arguments:

<chanlist>: channel number.

(@1) – Channel 1

(@1,4) – Channel 1 and Channel 4

(@1:4) – Channels 1 through 4. Include ch1,ch2,ch3,ch4

Example:

TRIG:TRAN (@1) // Trigger transient system

TRIGger:MAP:DC[:STATe] <0|OFF|1|ON>,(@<chanlist>)

TRIGger:MAP:DC[:STATe]? (@<chanlist>)

Sets and queries the mapping state of the external output trigger signal when the DC value is changed.

Command Syntax:

TRIGger:MAP:DC[:STATe] 0|OFF|1|ON, (@<chanlist>)

Arguments:

0|OFF|1|ON

Query Syntax

TRIGger:MAP:DC[:STATe]? (@<chanlist>)

Returns

0 or 1

Example:

TRIG:MAP:DC ON,(@1) // Setting the trigger signal generated externally when the DC value changes can be mapped.

TRIG:MAP:DC? (@1) // Queries whether the external trigger signal generated

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when the DC value changes is mapped.

TRIGger:MAP:DC:PIN PIN<1-7>,(@chanlist)

TRIGger:MAP:DC:PIN? (@chanlist)

Set and query the mapping pin for the external trigger signal output when the DC value is changed.

Command Syntax:

TRIGger:MAP:DC:PIN PIN<1-7>, (@chanlist)

Arguments:

PIN<1-7>

Query Syntax

TRIGger:MAP:DC:PIN? (@chanlist)

Returns

PIN<1-7>

Example:

TRIG:MAP:DC:PIN PIN1,(@1) // Setting the DC value change to generate a trigger signal mapped to PIN1

TRIG:MAP:DC:PIN?(@1) // Queries to which PIN the trigger signal generated by a change in the DC value is sent.

TRIGger:MAP:TRANsient[:STATe] <0|OFF|1|ON>,(@chanlist)

TRIGger:MAP:TRANsient[:STATe]? (@chanlist)

Sets and queries the mapping status of the transient system to the external output trigger signal.

Note: Both the beginning and the end of a transient system can generate external trigger signals that can be pointed out by the commands TRIGger:TRANsient:TOUTput:BEGinning and TRIGger:TRANsient:TOUTput:END.

The pre-step and post-step of the List function can also be configured as to whether or not an external trigger signal is generated as well as each step of the Sequence function can be configured as to whether or not an external trigger signal is generated.

Include ARB,sequence,cdarb,sine sweep,load transient,list.

Command Syntax:

TRIGger:MAP:TRANsient[:STATe] <0|OFF|1|ON>, (@chanlist)

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Arguments:

0|OFF|1|ON

Query Syntax

TRIGger:MAP:TRANsient[:STATe]? (@chanlist)

Returns

0 or 1

Example:

TRIG:MAP:TRAN ON,(@1) // Trigger signals that enable the external output of the transient system can be mapped

TRIG:MAP:TRAN? (@1) // Queries whether the trigger signal output from the transient system to the outside world is mapped or not.

TRIGger:MAP:TRANsient:PIN PIN<1-7>,(@chanlist)

TRIGger:MAP:TRANsient:PIN? (@chanlist)

Set and query the pins mapped to the external output trigger signals of the transient system.

Command Syntax:

TRIGger:MAP:TRANsient:PIN PIN<1-7>, (@chanlist)

Arguments:

PIN<1-7>

Query Syntax

TRIGger:MAP:TRANsient:PIN? (@chanlist)

Returns

PIN<1-7>

Example:

TRIG:MAP:TRAN:PIN PIN1,(@1) // Setting up the transient system to generate a trigger signal mapped to PIN1

TRIG:MAP:TRAN:PIN?(@1) // Query the pin number of the output trigger signal that generated by the output transient system

TRIGger:TRANsient:TOUTput:BEGinning <0|OFF|1|ON>,(@chanlist)

TRIGger:TRANsient:TOUTput:BEGinning? (@chanlist)

Sets and queries whether an external trigger signal is generated at the start of the transient system function.

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Command Syntax:

TRIGger:TRANsient:TOUTput:BEGinnig <0|OFF|1|ON>, (@chanlist)

Arguments:

0|OFF|1|ON

Query Syntax

TRIGger:TRANsient:TOUTput:BEGinnig? (@chanlist)

Returns

0 or 1

Example:

TRIG:TRAN:TOUT:BEG ON,(@1) // Setting a trigger signal to be generated at the start of the transient system function on channel 1

TRIG:TRAN:TOUT:BEG? (@1) // Queries whether a trigger signal is generated at the start of the transient system function of channel 1.

TRIGger:TRANsient:TOUTput:END <0|OFF|1|ON>,(@chanlist)

TRIGger:TRANsient:TOUTput:END? (@chanlist)

Sets and queries whether an external trigger signal is generated at the end of the transient system.

Command Syntax:

TRIGger:TRANsient:TOUTput:END <0|OFF|1|ON>, (@chanlist)

Arguments:

0|OFF|1|ON

Query Syntax

TRIGger:TRANsient:TOUTput:END? (@chanlist)

Returns

0 or 1

Example:

TRIG:TRAN:TOUT:END ON,(@1) // Setting a trigger signal to be generated at the end of the transient system function of channel 1

TRIG:TRAN:TOUT:END? (@1) // Queries whether a trigger signal is generated at the end of the transient system function of channel 1.

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Chapter26 Digital Subsystem

Digital commands are used to program settings for the digital control ports on the rear panel of the instrument.

[SOURce:]DIGital:INPut:DATA?

This command is used to read the status of the digital IO inputs.

Command Syntax:

[SOURce:]DIGital:INPut:DATA?

Returns

<IO input state>

Example:

DIG:INP:DATA? //read the status of the digital IO inputs

[SOURce:]DIGital:OUTPut:DATA <value>

[SOURce:]DIGital:OUTPut:DATA?

The commands set and query the status of digital IO output.

Command Syntax:

[SOURce:]DIGital:OUTPut:DATA <value>

Arguments:

<0-127>

Query Syntax

[SOURce:]DIGital:OUTPut:DATA?

Returns

<IO input state >

Example:

DIG:OUTP:DATA 21 //21 corresponds to bit value 0001 0101. Indicates that the output state of IO-1 IO-3 IO-5 is on.

DIG:OUTP:DATA? //Query the output status of IO pins

[SOURce:]DIGital:PIN<1-7>:FUNCtion <function>

[SOURce:]DIGital:PIN<1-7>:FUNCtion?

The commands set and query the functions of digital IOs. DIO|DINPut|TINput|TOUTput are generic functions, and the all of the IO1-IO7

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support the generic functions.

INHibit: IO1 support

FAULt: IO2 support

PSCLear: IO3 support

ONOFfstate: IO4 support

SYNCin,SYNCout: IO5 support

ONCouple: IO6 support

OFFCouple: IO7 support

Command Syntax:

[SOURce:]DIGital:PIN<1-7>:FUNCtion <function>

Arguments:

<DIO|DINPut|FAULt|INHibit|ONCouple|OFFCouple|TINPut|TOUTput|ONOFfstate|SIN|SOUT|PSCLear>

Query Syntax

[SOURce:]DIGital:PIN<1-7>:FUNCtion?

Returns

DIO,DINP,FAUL,INH,ONC,OFFC,TINP,TOUT,ONOF,SIN,SOUT or PSCL

Example:

DIG:PIN1:FUNC DIO //Set the PIN1 to DIO

DIG:PIN2:FUNC? //Query the function of PIN2

[SOURce:]DIGital:PIN<1-7>:POLarity POSitive|NEGative

[SOURce:]DIGital:PIN<1-7>:POLarity?

The commands set and query the polarity of the digital IO pin.

Polarity set to Positive indicate IO Pin set to enable inverse

Polarity set to Negative indicate IO Pin set to disable inverse

Command Syntax:

[SOURce:]DIGital:PIN<1-7>:POLarity POSitive|NEGative

Arguments:

POSitive|NEGative

Query Syntax

[SOURce:]DIGital:PIN<1-7>:POLarity?

Returns

POS or NEG

Example:

DIG:PIN1:POL POS // Set the polarity of the IO1 pin to positive

DIG:PIN2:POL? // Query the polarity of the IO2 pin

[SOURce:]DIGital:ONOFf:STATe:CHANnel <1-8>

[SOURce:]DIGital:ONOFf:STATe:CHANnel?

This command is used to set and query the corresponding channel number of IO4 indicating onoff state.

Command Syntax:

[SOURce:]DIGital:ONOFf:STATe:CHANnel <1-8>

Arguments:

<1-8>

Query Syntax

[SOURce:]DIGital:ONOFf:STATe:CHANnel?

Returns

<1-8>

Example:

DIG:ONOF:STAT:CHAN 2 // Setting the on/off state of the status bit channel 2 indicated by IO4

DIG:ONOF:STAT:CHAN? // Query IO4 to indicate which channel is on/off.

[SOURce:]DIGital:PLUSe:WIDTh <value>

[SOURce:]DIGital:PLUSe:WIDTh? [MAXimum|MINimum|DEFault]

The commands set and query the width of digital IO pulse. Unit is second (s).

Command Syntax:

[SOURce:]DIGital:PLUSe:WIDTh <value>

Arguments:

5-500|MINimum|MAXimum| DEFault

Query Syntax

[SOURce:]DIGital:PLUSe:WIDTh? [MAXimum|MINimum|DEFault]

Returns

< pulse width >

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Example:

DIG:PLUS:WIDT 30 // Set the width of the IO recognition pulse and the width of the output pulse to 30us

DIG:PLUS:WIDT? // Queries the width of the IO recognition pulse and the width of the output pulse.

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Chapter27 MMemory Subsystem

The memory command is used to query files stored in the instrument's internal memory.

MMEMory:ARB:UDEFined:CATalog? (@<chanlist>)

This command reads locally stored ARB custom waveform files catalog. If no relevant files exist, it returns nothing.

Command Syntax:

MMEMory:ARB:UDEFined:CATalog? (@<chanlist>)

Returns

“file1.csv”,file2.csv", …

Example

MMEM:ARB:UDEF:CAT? (@1) //Return the custom wave file for arb in Channel 1

MMEMory:ARB:SEQuence:CATalog? (@<chanlist>)

This command reads locally stored Sequence waveform files catalog. If no relevant files exist, it returns nothing.

Command Syntax:

MMEMory:ARB:SEQuence:CATalog? (@<chanlist>)

Returns

“file1.csv”,file2.csv", …

Example

MMEM:ARB:SEQ:CAT? (@1) // Return the sequence waveform file for Channel 1

MMEMory:LIST:CATalog? (@<chanlist>)

This command reads locally stored list waveform files. If no relevant files exist, it returns nothing.

Command Syntax:

MMEMory:LIST:CATalog? (@<chanlist>)

Returns

“file1.csv”,file2.csv", …

Example

MMEMory:LIST:CATalog? (@1) // Return the list wave file for Channel 1

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MMEMory:BATTERY:EMULator:CATalog? (@<chanlist>)

This command reads a locally stored battery simulation waveform file catalog. If no such file exists, it returns nothing.

Command Syntax:

MMEMory:BATTERY:EMULator:CATalog? (@<chanlist>)

Returns

“file1.csv”,file2.csv", …

Example

MMEM:BATTERY:EMUL:CAT? (@1>) //Return the list wave file for Channel 1

MMEMory:BATTERY:CHARge:CATalog? (@<chanlist>)

This command reads locally stored battery charging waveform files catalog. If no relevant files exist, it returns nothing.

Command Syntax:

MMEMory:BATTERY:CHARge:CATalog? (@<chanlist>)

Returns

“file1.csv”,file2.csv", …

Example

MMEM:BATTERY:CHAR:CAT? (@1) //Return the list wave file for Channel 1

MMEMory:BATTERY:DISCharge:CATalog? (@<chanlist>)

This command reads locally stored battery discharge waveform files catalog. If no relevant files exist, it returns nothing.

Command Syntax:

MMEMory:BATTERY:DISCharge:CATalog? (@<chanlist>)

Returns

“file1.csv”,file2.csv", …

Example

MMEM:BATTERY:DISC:CAT? (@1) // Return the list wave file for Channel 1

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Chapter28 IEEE-488 Common Commands

IEEE-488 Common commands usually control all instrument functions, such as reset, status, and synchronization. All common commands consist of a three-letter mnemonic, and are preceded by an asterisk, E.g:*RST *IDN? *SRE 8.

*CLS

Clear Status Command. Clears the event registers in all register groups. Clears both the status byte and the error queue. If *CLS immediately follows the programming message terminator (<NL>), the output queue and MAV bits are also cleared.

Command Syntax:

*CLS

Arguments:

Null

Example:

*CLS //Clear event memory bits and status queues

*ESE <0-255>

*ESE?

Sets and queries the bits of the Standard Event Status Enable Register.

Command Syntax:

*ESE <NR1>

Arguments:

0～255 //This is a decimal value with a binary weighted sum.

Query Syntax

*ESE?

Returns

<NR1> // This is a decimal value with a binary weighted sum.

Example:

*ESE 128 //Enable Status Register 128

*ESE? //Query Enable Status Register

Standard event enable register bit definitions:

Bit	7	6	5	4	3	2	1	0

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Name	PON	n.u	CME	EXE	DDE	QYE	n.u	OPC
Value Bit Weighted	128		32	16	8	4		1

*ESR?

This command can be used to read the value of the standard event register. Reads and clears the event registers of the standard event status group. The event registers are read-only registers that latch all standard events.

Query Syntax

*ESR?

Returns

<NR1> //This is a decimal value with a binary weighted sum.

Example:

*ESR? // Read event register

*IDN?

Identification query. Returns the instrument identification string, which consists of four comma-separated fields. The first field is the manufacturer's name, the second is the instrument model, the third is the serial number, and the fourth is the CPU version.

Query Syntax

*IDN?

Returns

Manufacture, model, serial number, CPU version

Example:

*IDN? //Returns the instrument's identification string

*OPC

*OPC?

Set the OPC (operation complete) bit in the standard event register. This occurs after the suspension operation is completed.

Command Syntax:

*OPC

Query Syntax

*OPC?

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Returns

1

Example:

* OPC //Generate an SRQ after the operation is complete

* OPC? //Wait for the operation to complete and return to “1”.

*RST

This command is used to reset the instrument to typical or safe predefined values.

Command Syntax:

*RST

Example:

* RST //Reset the device to the factory-set state

*SRE <0-255>

*SRE?

Service Request Enable Command. Used to set the value of the Service Request Enable Register. This determines the bits to be summed from the Status Byte Register to set the Request for Service (RQS) Summary bit. A one in any Service Request Enable Register bit position will enable the corresponding Status Byte Register bit.

Command Syntax:

*SRE <NR1>

Arguments:

0~255

Query Syntax

*SRE?

Returns

<NR1>

Example:

*SRE 24 // Enable bits 3 and 4 in the Enable Register

*SRE? // Query Enable Register

*STB?

Status byte lookup. The “status byte” is a read-only register that is read without clearing bits.

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Query Syntax

*STB?

Returns

<NR1>

Example:

* STB? // Read status byte

Bit definition of the status byte register:

Bit	7	6	5	4	3	2	1	0
Name	OPER	RQS	ESB	MAV	QUES	EAV	n.u	n.u
Value Bit Weighted	128	64	32	16	8	4		

*SAV <1-100>

Save some parameter settings of the instrument into 100 non-volatile memories, the position can be set to 1~100. when shipped from the factory, the position 1 to 100 is empty. The specific parameter information is described in the user's manual.

Command Syntax:

*SAV <1-100>

Arguments:

1~100

Example:

*SAV 1 // Store instrument status to internal memory area 1

*RCL <1-100>

Recalls the parameter settings of the instrument from the specified storage area.

Command Syntax:

*RCL <1-100>

Arguments:

1~100

Example:

*RCL 1 // Recall instrument status from internal memory area 1

*TST?

Make the device self-test, return to the device self-test situation

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Query Syntax

*TST?

Returns

0 or 1

Example:

* TST? // Execution Self-inspection

*WAI

Suspend other command processing until all pending operations are completed.

Command Syntax:

*WAI

Example:

*WAI // Wait for all pending operations to complete

*TRG

When the device trigger source is BUS, generating a trigger.

Command Syntax:

*TRG

Example:

*TRG // Send a trigger signal

*RDT?

Returns a description of each channel power module installed.

Query Syntax

* RDT?

Returns

AARD

Example:

* RDT? //Returns a description of each channel power module installed

Chapter29 Status Tutorial

This section describes individual registers and register groups in detail.

status register

The Operational and Suspicious Status groups will use four different types of registers to track qualifying, tagging, and enabling instrument events. The Standard Events group will only use the Event and Allow registers.

- The condition register monitors the status of the instrument without interruption. The bits in the condition register are updated in real time and the bits are not latched.
- The PTR/NTR register will qualify the signal passed to the event register. With the PTR bit set, signals with positive edge jumps will be passed to the event register. When the NTR bit is set, signals with negative edge jumps are passed to the event register. When both bits are set, all signals are passed. When no bit is set, no signal is passed.
- The event register locks out jumps through the PTR register and the NTR register. Setting the event bit retains the setting until the event register is read. It is cleared when the event register is read.
- The Enable Register defines which bits in the Event Register will be reported to the Status Byte Register. The enable register is readable and writable.

Operational status group

These registers will record signals that occur during normal operation. The group consists of condition, PTR/NTR, event and enable registers. The output of the Operation Status Register group will be logically “ORed” into the OPERation Accumulation bit of the Status Byte register.

Bit	Name	Decimal Value	Description
0	ACQ-WTG	1	Measuring system waiting to be triggered
1	TRAN-WTG	2	Transient System Waiting Trigger
2	DLOG-WTG	4	DLOG system waiting to be triggered
3	ACQ-Active	8	Measurement system in operation
4	TRAN-Active	16	Transient system in operation
5	DLOG-Active	32	DLOG system is operational
6	OFF	64	Output off state
7	CC	128	Output operates in constant current mode
8	CV	256	Output operates in constant voltage mode

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9	CW	512	Output operates in constant power mode
10	CR	1024	Output operates in constant resistance mode
11	CC-	2048	Output operates in constant negative current mode
12	CP-	4096	Output operates in constant negative power mode
13	ON_DELAY	8192	Output open delay state
14	OFF_DELAY	16384	Output off delay state
15	UVI	32768	Undervoltage inhibit output
NU	16-31	0	Reserved bit

suspicious state group (computing)

These registers register signals indicating abnormal operation. The group consists of condition, PTR/NTR, event and enable registers. The output of the Questionable Status group is logically “ORed” into the QUEStionable accumulator bit of the Status Byte register.

Bit	Name	Decimal Value	Description
0	OV	1	Measuring system waiting to be triggered
1	OC	2	Transient System Waiting Trigger
2	OP	4	DLOG system waiting to be triggered
3	UV	8	Measurement system in operation
4	UC	16	Transient system in operation
5	Rvs	32	Output is shut down by reverse connection protection
6	Errsense	64	Outputs are turned off by sense protection
7	OT	128	Output is shut down by over-temperature protection
8	PCP	256	Output is turned off by output coupling protection
9	FOLDBACK	512	Output is turned off by foldback protection
10	WDOG	1024	Output protected by watchdog
11	LIVING	2048	The output is disabled by Living protection, which does not affect the onoff state, but stops the output.

12	LATCH	4096	Outputs shut down by Latch protection
13	PF	8192	Output is shut down by APFC power module protection
14	MF	16384	Output shut down by power module protection
NU	15-31	0	Reserved bit

Standard Event Status Group

These registers are programmed and set by general-purpose commands. The group consists of event registers and enable registers. The standard event event register will latch on to events related to communication status. This register is read-only and will be cleared when read. The Standard Event Enable Register functions similarly to the Enable Register of the Operational and Suspicious Status Group. The following table describes the standard event status register bit assignments.

Bit	Name	Decimal Value	Description
0	OPC Operational Integrity	1	All commands before and including *OPC have been executed.
1	No used	No used	Return 0
2	QYE inquiry error	4	The instrument tries to read the output buffer, but it is empty, a new command line was received before the last query was read, or both the input and output buffers are full.
3	DDE Device-specific errors	8	A device-specific error occurs, including a self-test error, calibration error, or other device-specific error.
4	EXE implementation error	16	An execution error occurred.
5	CME command	32	A command syntax error occurred
6	No used	No used	Return 0
7	PON Power On	128	Power has been turned off and on since the last time the event register was read or cleared.

Status Byte Register

This register, as defined in the IEEE 488.2 Standard Digital Interface for Programmable Instruments, summarizes all other status groups in the information.

The following table describes the status byte register bit assignments.

Bit	Name	Decimal Value	Description
0	No used	No used	Return 0
1	No used	No used	Return 0
2	ERR error queue	4	One or more errors in the error queue. Use SYSTem:ERRor? to read and delete errors.
3	QUES Summary of suspicious status	8	One or more bits are set in the Suspect Data Register. You must enable the bit, see STATus:QUEStionable:ENABle.
4	MAV Available Information	16	Data available in the instrument output buffer.
5	ESB Event Status Summary	32	One or more bits are set in the standard event register. The bits must be enabled, see *ESE.
6	MSS Master Status Summary	64	One or more bits are set in the Status Byte Register and may generate service requests. Bits must be enabled, see *SRE.
7	OPER Operational Status Summary	128	One or more bits are set in the Operational Status Register. The bits must be enabled, see STATus:OPERation:ENABle.

Master Status Summary and Service Bit Request

The MSS is a real-time (unlocked) summary of all status byte register bits enabled by the Service Request Enable Register. The MSS is set whenever the instrument has one or more reasons for requesting service.*STB? reads the MSS in response bit position 6, but does not clear any bits in the Status Byte Register.

The RQS bit is a locked version of the MSS bit. When the instrument requests service, this RQS bit sets the SRQ interrupt line to True.

This RQS bit sets the SRQ interrupt line to True when the instrument requests service and latches RQS into bit 6 of the Status Byte Register. When the controller performs a serial poll, it clears the RQS in the register and returns the RQS to the responding bit position 6. The remaining bits of the Status Byte Register are not disturbed.

Error and output queues

The Error Queue is a first-in-first-out (FIFO) data register used to store numeric and textual descriptions of errors or events. Error messages are stored until they are read using SYSTem:ERRor? If the queue overflows, the last error/event in the queue will be replaced with error -350, “Queue overflow”. The output queue is a first-in, first-out (FIFO) data register used to store an instrument-to-controller message before the controller reads it. The MAV bit of the Status Byte Register is set when the queue contains this message.

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Chapter30 Trigger Tutorial

The Trigger System is a flexible, multi-purpose system for controlling the operation of an instrument to suit a variety of user-defined applications.

The Trigger System provides a variety of trigger inputs, providing methods including commands, panel operations, IO pins, and measurements as trigger sources, as well as trigger output functions that can be realized through IO pins.

DC value change, transient system and acquisition system provide trigger output configurations, and trigger outputs can be mapped to IO pins. trigger input and trigger output functions of IO pins provide users with cascadable trigger systems, and a single trigger can realize the startup of different functions running between different channels or even different machines.

Transient systems include arbitrary wave (arb), sequence (sequence), constant dwell arbitrary wave (CDARB), list (LIST), sine sweep, load transient.

Trigger source

Source	Description
IMMediate	Set the trigger source as IMM, and the INIT instruction can trigger the relevant function to run.
BUS	Set the trigger source as BUS, through the “TRG” can trigger the related function function operation.
MANual	Set the trigger source as MAN, the Trig button on the panel can trigger the relevant function to run.
PIN<1-7>	Set the trigger source as PIN and select the digital port pin configured as the trigger input. <1-7> Specify the pin number, by inputting a pulse signal can trigger the related function to run
VOLT<1-8>	Set the trigger source as VOLT, and you can configure the trigger level and trigger edge to trigger the relevant function to run.
CURR<1-8>	Set the trigger source as VOLT, and you can configure the trigger level and trigger edge to trigger the relevant function to run.
TRIG:TRAN:IMM	Instant trigger transient system
TRIG:ACQ:IMM	Immediate triggering of the acquisition system
TRIG:DLOG:IMM	Immediate triggering of the DLOG system
TRIG:ELOG:IMM	Immediate triggering of the ELOG system

Trigger target

target	Description
transient system	Trigger transient system operation
acquisition system	Trigger acquisition system operation
DLOG system	Trigger a DLOG run
ELOG system	Trigger ELOG to run

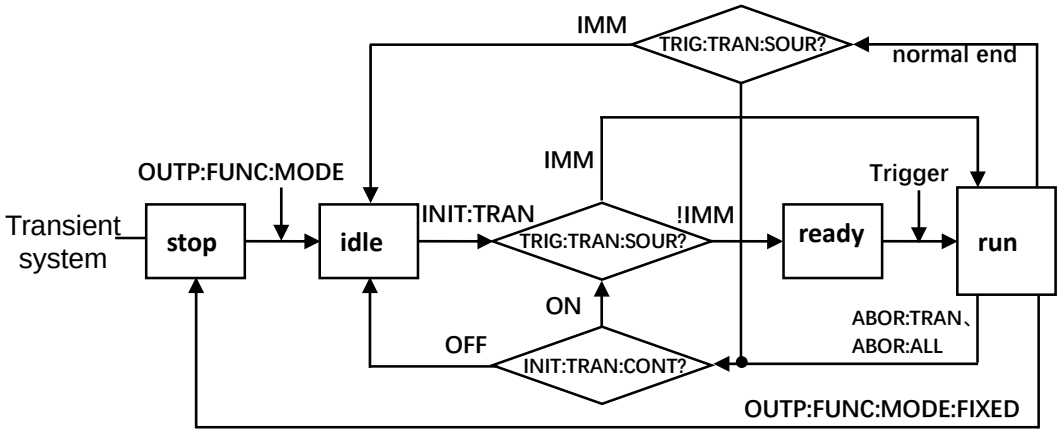
Copyright ©ITECH Electronic Co., Ltd.

IO pin	Transient system functions and acquisition functions can be configured to trigger to IOs

functional status diagram

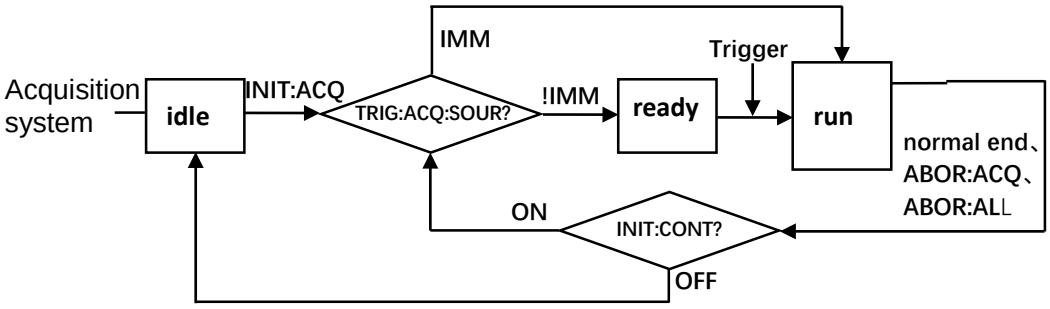
transient system

The transient system can be brought back to the STOP state by OUTP:FUNC:MODE FIXED in any state.



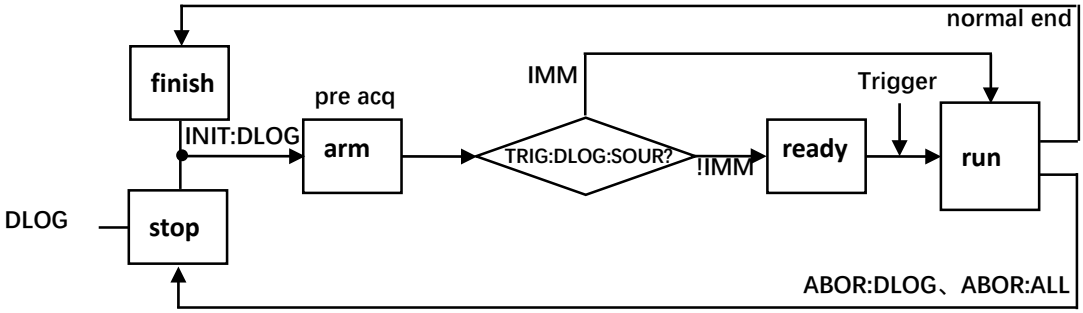
transient system status

Acquisition system



acquisition system status

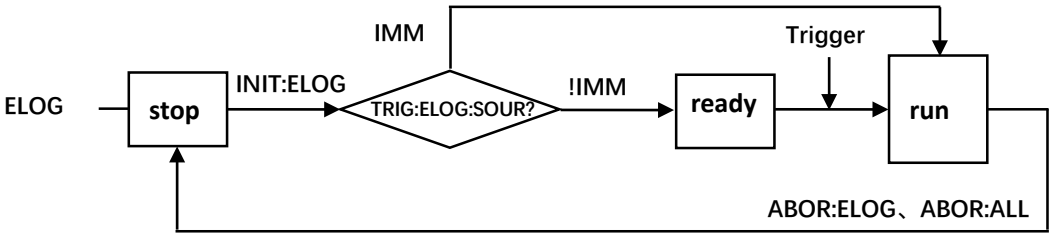
DLOG System:



DLOG System status

ELOG System:

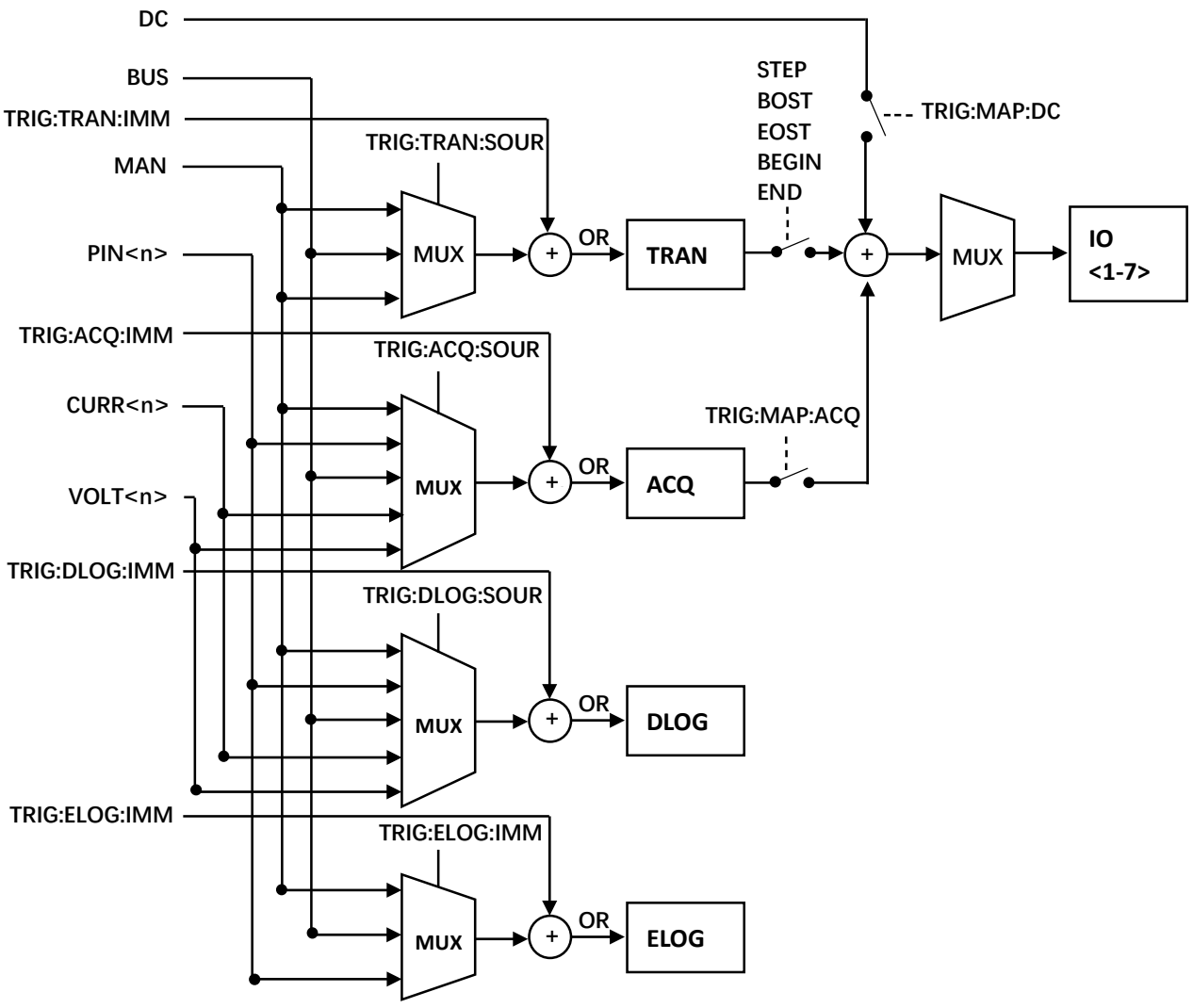
After the ELOG system enters the run state, it will remain in the run state for data logging, and the ELOG system can be terminated to return to the stop state by ABOR:ELOG.



DLOG System status

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Trigger Diagram



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Chapter31 Example of Common Commands

This chapter displays the programming examples to remotely control IT2700 power supply using SCPI commands.

If the user want to change the settings of the instrument, for instance, the output setting value, the command SYST:REM must be sent to the instrument after finishing the connection between the instrument and PC.

Example 1:Identify the power supply

You can verify that you are communicating with the correct IT2700 power supply.

To query the identification of the power supply, enter the following command:

-> *IDN?

To check the error queue of the power supply, enter the following command:

-> SYST:ERR?

Example 2:Common output commands

The IT2700 series instruments can be connected to a PC via a communication interface to realize remote operation of the instrument by sending SCPI commands.

The following common commands to help you quickly implement common operations. For more command information, refer to the corresponding section. You can omit the small write parts in the following commands.

SYST:REM // Set the power supply to remote operation mode.

SYST:CHAN:STAT? (@1:8) \\ Read the online channels

CV Priority:

FUNC VOLT ,(@3,5) // Setting CH3 and CH5 voltage priority mode

VOLT 10,(@3,5) //Set CH3 and CH5 voltage to 10V.

CURR:LIM:COUP OFF,(@3,5) // Setting the tracking state of CH3 and CH5 current positive and negative limit values to not coupling.

CURR:LIM 2,(@3,5) // Setting CH3 and CH5 I+ to 2A

CURR:LIM:NEG -2.5,(@3,5) // Setting CH3 和 CH5 I- to -2.5A ;

CC Priority:

FUNC CURR ,(@3,5) // Setting CH3 and CH5 current priority mode

CURRent 5,(@3,5) // Setting CH3 and CH5 current to 5A

CURRent -5,(@3,5) // Setting CH3 and CH5 current to -5A

VOLT:LIM:HIGH 10,(@3,5) // Setting CH3 and CH5 VH to 10V

VOLT:LIM:LOW 0,(@3,5) // Setting CH3 and CH5 VL to 0V

Open the output

OUTP ON ,(@3,5) // Setting CH3 and CH5 Instrument output state on

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```
MEAS:VOLT? (@3,5)          // Query the CH3 and CH5 output
MEAS:CURR? (@3,5)          // Query the CH3 and CH5 output current
```

Example 3: DLOG commands

Use the data logger to record 1 minute of data on channel 3 and channel 5, connect the outputs of channel 3 and channel 5, configure the trigger source of the data logger to be BUS, and the trigger offset bit of 20%, and measure the data in the specified time period through the data analysis function after the data recording is completed.

Set channel 3 to output a 10V amplitude 1Hz sine wave.

```
ARB:FUNC:TYPE VOLT,(@3)          //Set arb priority to voltage
ARB:COUN 0,(@3)                  //Set arb repeat count to 0 and keep looping
ARB:END NORMal,(@3)              //Set the arb termination mode to normal
ARB:FUNC:SHAPE SINusoid,(@3)     // Set arb waveform type to sine
ARB:VOLT:SIN:AMPL 10,(@3)        // Set sine amplitude to 10V
ARB:VOLT:SIN:OFFS 10,(@3)        //Set sineoffset to 10V
ARB:VOLT:SIN:FREQ 1,(@3)         //Set sine frequency to 1Hz
TRIG:TRAN:SOUR BUS,(@3)          //Set arb trigger source to BUS
OUTP:FUNC:MODE ARB,(@3)          // Set output mode to ARB
OUTP 1,(@3)                      //Enable Output
INIT:TRAN (@3)                   //initialize arb
TRIG:TRAN (@3)                   //Trigger ARB
```

Set the output current of channel 5 to be 1A and the end current to be 2A, and the start time and end time are both 1s step wave.

```
ARB:FUNC:TYPE CURR,(@5)          //Set arb priority to current
ARB:COUN 0,(@5)                  //Set arb repeat count to 0 and keep looping
ARB:END NORMal,(@5)              //Set the arb termination mode to normal
ARB:FUNC:SHAPE STEP,(@5)         //Set arb waveform type to STEP
ARB:VOLT:STEP:STAR 1,(@5)        //Set the STEP starting current to 1A
ARB:VOLT:STEP:END 2,(@5)         // Set the STEP starting current to 2A
ARB:VOLT:STEP:STAR:TIME 1,(@5)   // Set step start time to 1s
ARB:VOLT:STEP:END:TIME 1,(@5)    // Set step end time to 1s
TRIG:TRAN:SOUR BUS,(@5)          // Set arb trigger source to BUS
OUTP:FUNC:MODE ARB,(@5)          // Set output mode to ARB
OUTP 1,(@3)                      //Enable output
INIT:TRAN (@5)                   //initialize arb
TRIG:TRAN (@5)                   //Trigger an ARB run
```

Set the sampling integration time of the recorder to 0.002s to record the voltage and current of channel 3 and channel 5 as well as the maximum value and minimum value of 60s, the trigger source bit BUS, the trigger offset is 20% (record

the 12s data before the trigger as well as the 48s data after the trigger session):

SENS:DLOG:FUNC:VOLT NO,(@3) // Enable dlog to record voltage data for channel 3

SENS:DLOG:FUNC:CURR NO,(@3) // Enable dlog to record current data for channel 3

SENS:DLOG:FUNC:VOLT NO,(@5) // Enable dlog to record voltage data for channel 5

SENS:DLOG:FUNC:CURR NO,(@5) // Enable dlog to log current data for channel 5

SENS:DLOG:FUNC:MINM NO // Turn on the dlog voltage and current maximum and minimum value logging function

SENS:DLOG:PER 0.002 //Setting the dlog sample rate bit 2ms

SENS:DLOG:OFFS 20 //Setting the trigger bias of dlog to 20%

SENS:DLOG:TIME 60 //Set dlog logging time to 30s

TRIG:DLOG:SOUR BUS //Setting the dlog trigger source bit BUS

INIT:DLOG "Dlog_flile1" // Run the dlog function and specify the file name of the exported data as “Dlog_file1”.

TRIG:DLOG // Trigger dlog logging

SENSe:DLOG:MARKer1:POINt -5 // Setting the position bit of measurement marker point 1 to the position 5s before the trigger

SENSe:DLOG:MARKer2:POINt 15 // Set the position of the measurement marker point to the position 15s after the trigger.

Wait for DLOG recording to complete Use the measurement function to measure the relevant data from 5s before to 15s after the trigger:

FETC:DLOG:ALL? (@3) //Getting dlog measurements

FETC:DLOG:AHO? (@3) // Obtaining time-of-day data

FETC:DLOG:WHO? (@3) // Getting watt-hour data

FETC:DLOG:CURR? (@3) //Getting the average value of current

FETC:DLOG:VOLT? (@3) //Getting the average voltage value

FETC:DLOG:CURR:MAX? (@3) // Getting the maximum value of current

FETC:DLOG:VOLT:MAX? (@3) // Getting the maximum value of voltage

FETC:DLOG:CURR:MIN? (@3) // Getting the maximum value of current

FETC:DLOG:VOLT:MIN? (@3) // Getting the maximum value of voltage

FETC:DLOG:CURR:PTP? (@3) // Get current peak-to-peak value

FETC:DLOG:VOLT:PTP? (@3) //Get voltage peak-to-peak

Example 4: ELOG commands

Use the external Logger function of channel 3. Set the external logger integration time to 1ms, the trigger source to BUS, and set channel 3 to output a sine wave

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with an amplitude of 10V, 1Hz. Initialize the external logger and trigger the recording of channel 3 data.

Set channel 3 to output a 10V amplitude 1Hz sine wave.

```
ARB:FUNC:TYPE VOLT,(@3)                 // Set arb priority to voltage
ARB:COUN 0,(@3)                         // Set arb repeat count to 0 and keep looping
ARB:END NORMal,(@3)                     // Set the arb termination mode to normal
ARB:FUNC:SHAPE SINusoid,(@3)            // Set arb waveform type to sine
ARB:VOLT:SIN:AMPL 10,(@3)               // Set sine amplitude to 10V
ARB:VOLT:SIN:OFFS 10,(@3)               // Set sine bias to 10V
ARB:VOLT:SIN:FREQ 1,(@3)                // Set the sine frequency to 1Hz
TRIG:TRAN:SOUR BUS,(@3)                 // Set arb trigger source to BUS
OUTP:FUNC:MODE ARB,(@3)         // Set output mode to ARB
OUTP 1,(@3)                             //Enable output
INIT:TRAN (@3)                  //initialize arb
TRIG:TRAN (@3)                          //Trigger an ARB run
```

Set the enable to trigger the external logger to record data:

```
SENS:ELOG:FUNC:VOLT ON,(@3)             //Setting to enable external recorder voltage recording for channel 3
SENS:ELOG:FUNC:CURR ON,(@3)             //Setting to enable external logger current logging for channel 3
SENS:ELOG:FUNC:VOLT:MINM ON,(@3)        //Setting to enable the external recorder voltage maximum and minimum value recording function for channel 3
SENS:ELOG:FUNC:CURR:MINM OFF,(@3)       //Setting the external recorder voltage max. min. recording function for deactivation channel 3
TRIG:ELOG:SOUR BUS,(@3)                 //Setting the trigger source to BUS
SENS:ELOG:PER 0.001,(@3)                // Setting the integration time of the external recorder to 1ms
INIT:ELOG (@3)                  //Initialize external logger for channel 3
TRIG:ELOG (@3)                  //Trigger external logger operation
FORM ASC                        // Set the query data format to ASC
FETC:ELOG? 1000,(@3)            // When the running time of the external logger of channel 3 is more than 1s, it will be possible to query 1000 pieces of data in the format of “volt-max,volt-min,volt,curr”.
FORM REAL                       // Set the query data format to REAL
REAL                // Returns the data in a finite-length data block as an IEEE single-precision floating-point number.
FORM:BORD NORM
FETC:ELOG? 10,(@3)                      // Query 10 data, the data will be returned in data block MSB's
FORM:BORD SWAP
```

FETC:ELOG? 100,(@3) // Query 100 pieces of data, data will be returned in data block LSB format

ABOR:ELOG (@3) //Stop CH3 External Recorder

FETC:ELOG? 100,(@3) // The external logger stops and no data is returned to the query.

Example 5: BATTery CHARge Commands:

Execute the following commands to run the battery discharge test for channel3.

BATTery:CHARge:MODE CURRent,(@3) // Battery charging mode is current mode

BATTery:CHARge:VOLTage 5,(@3) // Set battery charging voltage to 5V

BATTery:CHARge:CURRent 3,(@3) // Set battery charge flow to 3A

BATTery:CHARge:STOP:VOLTage:STATe 1,(@3) // Enable voltage stop function

BATTery:CHARge:STOP:VOLTage 8,(@3) // Setting the battery charging stop voltage to 8V

BATTery:CHARge:STOP:CURRent:STATe 1,(@3) // Enable current stop function

BATTery:CHARge:STOP:CURRent 2,(@3) // Set battery charging stop current to 2A

BATTery:CHARge:STOP:AHOur:STATe 1,(@3) // Enable the time-stop function

BATTery:CHARge:STOP:AHOur 1,(@3) // Setting the battery charging stop ampere time to 1AH

BATTery:CHARge:STOP:WHOur:STATe 1,(@3) // Enable watt-hour stop function

BATTery:CHARge:STOP:WHOur 1,(@3) // Setting the battery charging stop watt-hour to 1WH

BATTery:CHARge:STOP:TIME:STATe 1,(@3) // Enable time stop function

BATTery:CHARge:STOP:TIME 60,(@3) // Setting the battery charging stop time to 60s

OUTP ON,(@3) //Enable output

OUTPut:FUNCtion:MODE BCHarge,(@3) // Enable battery charge test

Example 6: BATTery DISCharge Commands:

BATTery:DISCharge:MODE CURRent,(@4) // Battery discharge mode is current mode

BATTery:DISCharge:VOLTage 5,(@4) // Set the battery discharge voltage to 5V

BATTery:DISCharge:CURRent 2,(@4) // Set the battery discharge current to 3A

BATTery:DISCharge:STOP:VOLTage:STATe 1,(@4) // Enabled voltage

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stop function

BATTery:DISCharge:STOP:VOLTage 4,(@4) // Battery discharge stop voltage set to 8V

BATTery:DISCharge:STOP:CURRent:STATe 1,(@4) // Enabled current stop function

BATTery:DISCharge:STOP:CURRent 2,(@4) // Battery discharge stop current set to 2A

BATTery:DISCharge:STOP:AHOur:STATe 1,(@4) // Enabled Anchor Stop Function

BATTery:DISCharge:STOP:AHOur 1,(@4) // The set battery discharge stopping ampere time is 1AH

BATTery:DISCharge:STOP:WHOur:STATe 1,(@4) // Enabled watt-hour stop function

BATTery:DISCharge:STOP:WHOur 1,(@4) // The set battery discharge stop watt-hour is 1WH

BATTery:DISCharge:STOP:TIME:STATe 1,(@4) // Enabled time stop function

BATTery:DISCharge:STOP:TIME 60,(@4) // The set battery discharge stopping time is 60s

OUTP ON,(@ 4) //Enable output

OUTPut:FUNCtion:MODE BDIScharge,(@4) // Enable battery charge test

Example 7: BATTery EMULator Commands:

BATTery:EMULator:VOC:FULL 5,(@3) // Set the battery analog full charge voltage to 5V

BATTery:EMULator:VOC:EMPT 3,(@3) // Set the battery analog empty voltage to 3V

BATTery:EMULator:CURRent:LIMit 2,(@3) // Set the positive current limit for battery emulation to 2A

BATTery:EMULator:CURRent:LIMit:NEG -2,(@3) // Set the negative current limit for battery emulation to -2.0A

BATTery:EMULator:SOC:LIMit 105,(@3) // Setting the upper limit of battery simulation capacity to 105%

BATTery:EMULator:SOC:LIMit:LOWer -5,(@3) // Setting the lower limit of battery simulation capacity to -5.0%

BATTery:EMULator:INITial:SOC 50,@3) // Setting the initial state of charge of the battery simulation by 50%

BATTery:EMULator:PARallel 1,(@3) // Set the number of parallel connections for battery emulation to 1

BATTery:EMULator:SERies 1,(@3) // Set the number of series connections for battery emulation to 1

BATTery:EMULator:INITial:CAPacity 5,(@3) // Set the initial capacity of the battery emulation to 5Ah

BATTery:EMULator:END:MODE HOLD,(@3) // Setting the end mode of battery emulation to HOLD mode

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OUTP ON,(@3) //Enable output

OUTPut:FUNCtion:MODE BEMulation,(@3) //Enable the battery discharge test

Example 8: SINE SWEEp Commands:

SSWEEP SSW:FUNC:TYPE CURR,(@3) // Set the scan type of SSWEEP to current
SSW:CURR:STAR 2,(@3) // Set SSWEEP's sine start amplitude to 2A
SSW:CURR:END 3,(@3) // Setting the end-of-sine amplitude of SSWEEP to 3A
SSW:CURR:OFFS:STAR 3,(@3) // Set SSWEEP's sine start bias to 2A
SSWCURR:OFFS:END 3,(@3) // Set the end-of-sine bias of SSWEEP to 3A
SSW:FREQ:STAR 200,(@3) // Setting SSWEEP's sine start frequency to 200Hz
SSW:FREQ:END 500,(@3) // Setting SSWEEP's sine end frequency to 500Hz
SSW:TIME 5,(@3) // Set the scan time of SSWEEP to 5S
SSW:COUN 1,(@3) // Setting the number of SSWEEP scans to 1
OUTP:FUNC:MODE SSW,(@3) // Set the output mode to SSWEEP
OUTP ON,(@3) // Turn on module outputs
INIT:TRAN (@3) // Initializing the Dynamic Trigger System
TRIG:TRAN:SOUR BUS,(@3) // Set the trigger dynamic trigger source to BUS
*TRG // Trigger, SSWEEP running

Example 9: List Commands:

LIST:FUNC:TYPE VOLTage,(@2) // Set list output type to voltage

LIST:COUN 1,(@2) // Set the list repeat count to 1

LIST:END NORMal,(@2) // Set the list termination mode to normal

LIST:VOLT:POIN 2,(@2) // Set the total number of list points to 2

LIST:VOLT 5,1,(@2) // Set the first step of the list to 5V

LIST:VOLT:SLEW 1.5,1,(@2) // Setting the slope of the first step of list to 1.5V/ms

LIST:DWEL 1,1,(@2) // Set list first step time to 1s

LIST:TOUT:BOST 0,1,(@2) // Trigger output off before setting list step 1

LIST:TOUT:EOST 1,1,(@2) // Trigger the output to turn on after the first step of setting up the list.

LIST:VOLT 10,2,(@2) // Set list second step voltage to 10V

LIST:VOLT:SLEW 1.5,2,(@2) // Set list second step slope to 1.5V/ms

LIST:DWEL 1,2,(@2) // Set list second step time to 1s

LIST:TOUT:BOST 0,2,(@2) // Trigger the output off before setting the second step of the list

LIST:TOUT:EOST 1,2,(@2) // Trigger the output to turn on after the second step of setting up the list.

LIST:STEP AUTO,(@2) // Set list stepping mode to auto

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```
TRIG:TRAN:SOUR BUS,(@2)          // Set list trigger source to BUS
OUTP:FUNC:MODE LIST,(@2)         // Set the output mode to list
INIT:TRAN (@2)                   // Initialize list
*TRG                             // Trigger list to run
```

Example 10: CDarb Commands:

```
ARB:CDWell:FUNCtion:TYPE VOLTage,(@2)        // Set the output type of cdarb to voltage
ARB:CDW:COUN 1,(@2)              // Set the number of repetitions of cdarb to 1
ARB:CDW:TERM NORMal,(@2)         // Set the end of cdarb to normal
ARB:VOLT:CDW:POIN 3,(@2)         // Set the total number of points in the cdarb to 3
ARB:VOLT:CDW:DWEL 1,(@2)         // Set the step time at each point of cdarb to 1s
ARB:VOLT:CDW:LEV 1,1,(@2)        // Set the amplitude of the first point of cdarb to 1V
ARB:VOLT:CDW:LEV 2,2,(@2)        // Set the amplitude of the second point of cdarb to 2V
ARB:VOLT:CDW:LEV 3,3,(@2)        // Set the amplitude of the third point of cdarb to 3V
TRIG:TRAN:SOUR BUS,(@2)          // Setting the cdarb trigger source to bus
OUTP:FUNC:MODE CDARB,(@2)        // Set output mode to cdarb
INIT:TRAN (@2)                   // Initialize cdarb
*TRG                             //Run cdarb
```

ARB Commands:

Sine Waveform

```
ARB:FUNC:TYPE VOLTage,(@2)       // Set arb priority to voltage
ARB:COUN 1,(@2)                  // Set arb repeat count to 1
ARB:END NORMal,(@2)              // Set the arb termination mode to normal
ARB:FUNC:SHAPE SINusoid,(@2)     // Set arb waveform type to sine
ARB:VOLT:SIN:AMPL 10,(@2)        // Set sine amplitude to 10V
ARB:VOLT:SIN:OFFS 10,(@2)        // Set sine bias to 10V
ARB:VOLT:SIN:FREQ 1,(@2)         // Set the sine frequency to 1Hz
TRIG:TRAN:SOUR BUS,(@2)          // Set arb trigger source to BUS
OUTP:FUNC:MODE ARB,(@2)          // Set output mode to ARB
INIT:TRAN (@2)                   // Initialization arb
*TRG                             //Running arb
```

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User defined waveform

```
ARB:FUNC:TYPE VOLTage,(@2)              // Set arb priority to voltage
ARB:COUN 1,(@2)                 // Set arb repeat count to 1
ARB:END NORMal,(@2)                 // Set the arb termination mode to normal
ARB:FUNC:SHAPE UDEFined,(@2)          // Set arb waveform to udefined
ARB:VOLT:UDEF:LEV:POIN 2,(@2)              // Set udefined total points to 2
ARB:VOLT:UDEF:LEV 5,1,(@2)            // Set udefined first step voltage to 5V
ARB:VOLT:UDEF:DWEL 1,1,(@2)           // Set udefined first step time to 1s
ARB:VOLT:UDEF:LEV 10,2,(@2)           // Set udefined second step voltage to
10V
ARB:VOLT:UDEF:DWEL 2,2,(@2)           // Set udefined second step time to 2s
TRIG:TRAN:SOUR BUS,(@2)               // Set output mode to BUS
OUTP:FUNC:MODE ARB,(@2)               // Set output mode to ARB
INIT:TRAN (@2)              // Initialization arb
*TRG                    // Running arb
```

SEQUence command:

```
ARB:SEQ:FUNC:TYPE VOLTage,(@2)          // Set the sequence output type to
voltage
ARB:SEQ:COUN 1,(@2)               // Set the number of times the first step of the
sequence is repeated to 1
ARB:SEQ:TERM:MODE NORMal,(@2)          // Set the sequence end mode to
normal
ARB:SEQ:LENG 9,(@2)               // Set the total number of waveforms in the
sequence to 9
ARB:SEQ:STEP:FUNC:SHAP DC,1,(@2)              // Set the first waveform of the
sequence to DC
ARB:SEQ:STEP:COUN 1,1,(@2)            // Set the number of repetitions of the
first waveform of the sequence to 1
ARB:SEQ:STEP:PAC DWELl,1,(@2)              // Setting the first waveform of a
sequence to run in time mode
ARB:SEQ:STEP:TOUT 0,1,(@2)            // Set the first waveform trigger output of
the sequence to Off
ARB:SEQ:STEP:VOLT:DC:END 1,1,(@2)              // Set the end voltage of the
first waveform DC of the sequence to 1V
ARB:SEQ:STEP:VOLT:DC:END:TIME 1,1,(@2)              // Set the end time of the
first waveform DC of the sequence to 1s
ARB:SEQ:STEP:VOLT:DC:SLEW 1,1,(@2)              // Set the slope time of the first
waveform DC of the sequence to 1s
ARB:SEQ:STEP:FUNC:SHAP SINusoid,2,(@2)              // Set the second
waveform of the sequence to SINE
ARB:SEQ:STEP:COUN 1,2,(@2)              // Set the number of times the
second step of the sequence is repeated to 1
```

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ARB:SEQ:STEP:PAC DWELl,2,(@2) // Setting the second wave sine of the sequence to run in time mode

ARB:SEQ:STEP:TOUT 0,2,(@2) // Set the second waveform trigger output of the sequence to Off

ARB:SEQ:STEP:VOLT:SIN:AMPL 1,2,(@2) // Set the second waveform ac voltage of the sequence to 1V

ARB:SEQ:STEP:VOLT:SIN:OFFS 1,2,(@2) // Set the second waveform bias voltage of the sequence to 1V

ARB:SEQ:STEP:VOLT:SIN:FREQ 1,2,(@2) // Set the second waveform frequency of the sequence to 1 Hz

ARB:SEQ:STEP:FUNC:SHAP STEP,3,(@2) // Set the third waveform of the sequence to STEP

ARB:SEQ:STEP:COUN 1,3,(@2) // Set the number of repetitions of the third waveform of the sequence to 1

ARB:SEQ:STEP:PAC DWELl,3,(@2) // Setting the third waveform of a sequence to run in time mode

ARB:SEQ:STEP:TOUT 0,3,(@2) // Set the third waveform trigger output of the sequence to Off

ARB:SEQ:STEP:VOLT:STEP:STAR 0,3,(@2) // Set the start voltage of the third waveform of the sequence to 0V

ARB:SEQ:STEP:VOLT:STEP:END 1,3,(@2) // Set the end voltage of the third waveform of the sequence to 1V

ARB:SEQ:STEP:VOLT:STEP:STAR:TIME 1,3,(@2) // Set the start time of the third waveform of the sequence to 1s

ARB:SEQ:STEP:VOLT:STEP:END:TIME 1,3,(@2) // Set the end time of the third waveform of the sequence to 1s

ARB:SEQ:STEP:FUNC:SHAP RAMP,4,(@2) // Set the fourth waveform of the sequence to RAMP

ARB:SEQ:STEP:COUN 1,4,(@2) // Set the number of repetitions of the fourth waveform of the sequence to 1

ARB:SEQ:STEP:PAC DWELl,4,(@2) // Setting the fourth waveform of the sequence to run in time mode

ARB:SEQ:STEP:TOUT 0,4,(@2) // Set the fourth waveform trigger output of the sequence to Off

ARB:SEQ:STEP:VOLT:RAMP:STAR 0,4,(@2) // Set the start voltage of the fourth waveform of the sequence to 0V

ARB:SEQ:STEP:VOLT:RAMP:END 1,4,(@2) // Set the end voltage of the fourth waveform of the sequence to 1V

ARB:SEQ:STEP:VOLT:RAMP:STAR:TIME 1,4,(@2) // Set the start time of the fourth waveform of the sequence to 1s

ARB:SEQ:STEP:VOLT:RAMP:RTIM 1,4,(@2) // Set the runtime of the fourth waveform of the sequence to 1s

ARB:SEQ:STEP:VOLT:RAMP:END:TIME 1,4,(@2) // Set the end time of the fourth waveform of the sequence to 1s

ARB:SEQ:STEP:FUNC:SHAP PULSe,5,(@2) // Set the fifth waveform of the sequence to PULSE

ARB:SEQ:STEP:COUN 1,5,(@2) // Set the number of repetitions of the fifth waveform of the sequence to 1

ARB:SEQ:STEP:PAC DWELl,5,(@2) // Setting the fifth waveform of a sequence to run in time mode

ARB:SEQ:STEP:TOUT 0,5,(@2) // Set the fifth waveform trigger output of the sequence to Off

ARB:SEQ:STEP:VOLT:PULS:STAR 0,5,(@2) // Set the start voltage of the fifth waveform of the sequence to 0V

ARB:SEQ:STEP:VOLT:PULS:TOP 1,5,(@2) // Set the peak voltage of the fifth waveform of the sequence to 1V

ARB:SEQ:STEP:VOLT:PULS:STAR:TIME 1,5,(@2) // Set the start time of the fifth waveform of the sequence to 1s

ARB:SEQ:STEP:VOLT:PULS:TOP:TIME 2,5,(@2) // Set the peak time of the fifth waveform of the sequence to 2s

ARB:SEQ:STEP:VOLT:PULS:END:TIME 1,5,(@2) // Set the end time of the fifth waveform of the sequence to 1s

ARB:SEQ:STEP:VOLT:PULS:FREQ 0.25,5,(@2) // Set the fifth waveform frequency of the sequence to 0.25 Hz

ARB:SEQ:STEP:FUNC:SHAP TRAPezoid,6,(@2) // Set the sixth waveform of the sequence to TRAPezoid

ARB:SEQ:STEP:COUN 1,6,(@2) // Set the number of repetitions of the sixth waveform of the sequence to 1

ARB:SEQ:STEP:PAC DWELl,6,(@2) // Setting the sixth waveform of a sequence to run in time mode

ARB:SEQ:STEP:TOUT 0,6,(@2) // Set the sixth waveform trigger output of the sequence to Off

ARB:SEQ:STEP:VOLT:TRAP:STAR 0,6,(@2) // Set the start voltage of the sixth waveform of the sequence to 0V

ARB:SEQ:STEP:VOLT:TRAP:TOP 1,6,(@2) // Set the peak voltage of the sixth waveform of the sequence to 1V

ARB:SEQ:STEP:VOLT:TRAP:STAR:TIME 1,6,(@2) // Set the start time of the sixth waveform of the sequence to 1s

ARB:SEQ:STEP:VOLT:TRAP:RTIM 1,6,(@2) // Set the rise time of the sixth waveform of the sequence to 1s

ARB:SEQ:STEP:VOLT:TRAP:TOP:TIME 1,6,(@2) // Set the peak time of the sixth waveform of the sequence to 1s

ARB:SEQ:STEP:VOLT:TRAP:FTIM 1,6,(@2) // Set the rise time of the sixth waveform of the sequence to 1s

ARB:SEQ:STEP:VOLT:TRAP:END:TIME 1,6,(@2) // Set the end time of the sixth waveform of the sequence to 1s

ARB:SEQ:STEP:FUNC:SHAP Exponential,7,(@2) // Set the seventh waveform of the sequence to Exponential

ARB:SEQ:STEP:COUN 1,7,(@2) // Set the number of repetitions of the seventh waveform of the sequence to 1

ARB:SEQ:STEP:PAC DWELl,7,(@2) // Setting the seventh waveform of a sequence to run in time mode

ARB:SEQ:STEP:TOUT 0,7,(@2) // Set the seventh waveform trigger output of the sequence to Off

ARB:SEQ:STEP:VOLT:EXP:STAR 0,7,(@2) // Set the start voltage of the seventh waveform of the sequence to 0V

ARB:SEQ:STEP:VOLT:EXP:END 1,7,(@2) // Set the end voltage of the seventh waveform of the sequence to 1V

ARB:SEQ:STEP:VOLT:EXP:STAR:TIME 1,7,(@2) // Set the start time of the seventh waveform of the sequence to 1s

ARB:SEQ:STEP:VOLT:EXP:TIME 2,7,(@2) // Set the seventh waveform of the sequence to hold the index change hold time to 2s

ARB:SEQ:STEP:VOLT:EXP:TCON 1,7,(@2) // Set the time of change of the seventh waveform index of the sequence to 1s

ARB:SEQ:STEP:FUNC:SHAP STAircase,8,(@2) // Set the eighth waveform of the sequence to STAircase

ARB:SEQ:STEP:COUN 1,8,(@2) // Set the number of repetitions of the eighth waveform of the sequence to 1

ARB:SEQ:STEP:PAC DWELl,8,(@2) // Setting the eighth waveform of a sequence to run in time mode

ARB:SEQ:STEP:TOUT 0,8,(@2) // Set the eighth waveform trigger output of the sequence to Off

ARB:SEQ:STEP:VOLT:STA:STAR 0,8,(@2) // Set the start voltage of the eighth waveform of the sequence to 0V

ARB:SEQ:STEP:VOLT:STA:END 1,8,(@2) // Set the end voltage of the eighth waveform of the sequence to 1V

ARB:SEQ:STEP:VOLT:STA:STAR:TIME 1,8,(@2) // Set the start time of the eighth waveform of the sequence to 1s

ARB:SEQ:STEP:VOLT:STA:TIME 1,8,(@2) // Set the eighth waveform step time of the sequence to 1s

ARB:SEQ:STEP:VOLT:STA:END:TIME 1,8,(@2) // Set the end time of the eighth waveform of the sequence to 1s

ARB:SEQ:STEP:VOLT:STA:NST 5,8,(@2) // Set the eighth waveform step of the sequence to 5 steps

ARB:SEQ:STEP:FUNC:SHAP UDEFined,9,(@2) // Set the ninth waveform of the sequence to UDEFined, customized waveforms

ARB:SEQ:STEP:COUN 1,9,(@2) // Set the number of repetitions of the ninth waveform of the sequence to 1

ARB:SEQ:STEP:PAC DWELl,9,(@2) // Setting the ninth waveform of a sequence to run in time mode

ARB:SEQ:STEP:TOUT 0,9,(@2) // Set the ninth waveform trigger output of the sequence to Off

ARB:SEQ:STEP:VOLT:UDEF:LEV:POIN 2,9,(@2) // Set the ninth waveform step of the sequence to 2 steps

ARB:SEQ:STEP:VOLT:UDEF:STEP:LEV 2,9,1,(@2) // Set the voltage value of the first step of the ninth waveform of the sequence to 2V

ARB:SEQ:STEP:VOLT:UDEF:DWEL 1,9,1,(@2) // Set the value of the first step time of the ninth waveform of the sequence to 1S

ARB:SEQ:STEP:VOLT:UDEF:STEP:LEV 3,9,2,(@2) // Set the voltage value of the second step of the ninth waveform of the sequence to 3V

ARB:SEQ:STEP:VOLT:UDEF:DWEL 1,9,2,(@2) // Set the value of the second step time of the ninth waveform of the sequence to 1S

TRIG:TRAN:SOUR BUS,(@2) // Setting the trigger mode to BUS

OUTP:FUNC:MODE SEQ,(@2) // Set output mode to sequence

INIT:TRAN (@2) // Initialize sequence

*TRG //Runing ARB sequence

Examples of Load Transient Functions

Current Dynamic Continuous Mode Command

FUNC CURR,(@1) // Configure the load mode to CC mode

TRAN:MODE CONT,(@1) // Set dynamic mode to continuous mode

CURR 0.2,(@1) // Current level configured to 0.2A

CURR:RISE 4.0,(@1) // Rising slope of 4A/ms

CURR:FALL 3.0,(@1) // Current drop slope set to 3.0A/ms

CURR:TLEV 2.0,(@1) // Dynamic transient current set to 2.0

TRAN:FREQ 50,(@1) // Frequency set to 50Hz

TRAN:DCYC 20,(@1) // Duty cycle set to 20.0%

TRAN:COUN 20,(@1) // The number of repetitions is set to 20

TRIG:TRAN:SOUR BUS (@1) //trigger source set to BUS

OUTP:FUNC:MODE TRAN,(@1) // Enable transient function

INP ON,(@) //Enable load input

INIT:TRAN (@1) // Initialize transient function

*TRG //trigger to run

Resistor Dynamic Pulse Mode Command

FUNC RES,(@1) // Configure the load mode to CR mode

TRAN:MODE PULS,(@1) // Set dynamic mode to Puls mode

RES 100,(@1) //resistance level configured to 100Ω

RES:TRAN 50.0,(@1) // Dynamic transient resistance 50Ω

TRAN:TWID 0.5,(@1) // The dynamic pulse width of the load is set to 0.5s

TRIG:TRAN:SOUR BUS,(@1) // trigger source set to BUS

OUTP:FUNC:MODE TRAN,(@1) // Enable transient function

INP ON,(@1) // Enable load input

INIT:TRAN,(@1) // Initialize transient function

*TRG // trigger to run

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Power Dynamic Flip Mode Command

```
FUNC POW,(@1)                    // Configure the load mode to CP mode
TRAN:MODE TOGG,(@1)              // Set dynamic mode to toggle mode
POW 10,(@1)                      // Power level configured to 10w
POW:TLEV 20,(@1)                 // Dynamic transient power 20w
TRIG:TRAN:SOUR BUS,(@1)          // Set the trigger source to BUS
OUTP:FUNC:MODE TRAN,(@1)         // Enable dynamic function
INP ON,(@1)                      // Enable load input
INIT:TRAN,(@1)                   // Initialize transient function
*TRG                             // trigger to run
```

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Appendix:Module Parameter List

IT27334					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	30	30.6	0	0	V
Current positive	15	15.3	0	1.5	A
Current negative	-15	-15.3	0	-1.5	A
Resistance(Sink)	\	200	0.02	200	Ω
Power positive	200	204	0	204	W
Power negative	-200	-204	0	-204	W
Voltage Slew	\	6	0.001	0.3	V/ms
Current Slew	\	30	0.001	0.15	A/ms
Time Slew	\	60	0.001	0.03	s
OVP	\	30.6	0	30.6	V
UVP	\	31.5	0	31.5	V
OPP Positive	\	210	0	210	W
OPP Negative	\	0	-210	-210	W
OCP Positive	\	15.75	0	15.75	A
OCP Negative	\	0	-15.75	-15.75	A
UCP Positive	\	15.75	0.15	-15.75	A
UCP Negative	\	-0.15	-15.75	-15.75	A
Inter Resistance	\	1	0	0	Ω

IT27354					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	30	30.6	0	0	V
Current positive	30	30.6	0	3	A
Current negative	-30	-30.6	0	-3	A
Resistance(Sink)	\	100	0.01	100	Ω
Power positive	500	510	0	510	W
Power negative	-500	-510	0	-510	W
Voltage Slew	\	6	0.001	0.3	V/ms
Current Slew		60	0.001	0.3	A/ms
Time Slew	\	60	0.001	0.03	s
OVP	\	30.6	0	30.6	V
UVP	\	31.5	0	31.5	V
OPP Positive	\	525	0	525	W
OPP Negative	\	0	-525	-525	W
OCP Positive	\	31.5	0	31.5	A
OCP Negative	\	0	-31.5	-31.5	A
UCP Positive	\	31.5	0.3	-31.5	A
UCP Negative	\	-0.3	-31.5	-31.5	A
Inter Resistance	\	1	0	0	Ω

IT27335

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Parameter	Rated	Maxmun	Minmun	Default	Unit
Voltage	60	61.2	0	0	V
Current positive	10	10.2	0	1	A
Current negative	-10	-10.2	0	-1	A
Resistance(Sink)	\	600	0.06	600	Ω
Power positive	200	204	0	204	W
Power negative	-200	-204	0	-204	W
Voltage Slew	\	12	0.001	0.6	V/ms
Current Slew	\	20	0.001	0.1	A/ms
Time Slew	\	60	0.001	0.03	s
OVP	\	61.2	0	61.2	V
UVP	\	63	0	63	V
OPP Positive	\	210	0	210	W
OPP Negative	\	0	-210	-210	W
OCP Positive	\	10.5	0	10.5	A
OCP Negative	\	0	-10.5	-10.5	A
UCP Positive	\	10.5	0.1	-10.5	A
UCP Negative	\	-0.1	-10.5	-10.5	A
Inter Resistance	\	1	0	0	Ω

IT27355					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	60	61.2	0	0	V
Current positive	20	20.4	0	2	A
Current negative	-20	-20.4	0	-2	A
Resistance(Sink)	\	300	0.03	300	Ω
Power positive	500	510	0	510	W
Power negative	-500	-510	0	-510	W
Voltage Slew	\	12	0.001	0.6	V/ms
Current Slew	\	40	0.001	0.2	A/ms
Time Slew	\	60	0.001	0.03	s
OVP	\	61.2	0	61.2	V
UVP	\	63	0	63	V
OPP Positive	\	525	0	525	W
OPP Negative	\	0	-525	-525	W
OCP Positive	\	21	0	21	A
OCP Negative	\	0	-21	-21	A
UCP Positive	\	21	0.2	-21	A
UCP Negative	\	-0.2	-21	-21	A
Inter Resistance	\	1	0	0	Ω

IT27337					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	150	153	0	0	V
Current positive	5	5.1	0	0.51	A
Current negative	-5	-5.1	0	-0.51	A

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Resistance(Sink)	\	3000	0.3	3000	Ω
Power positive	200	204	0	204	W
Power negative	-200	-204	0	-204	W
Voltage Slew	\	30	0.001	1.5	V/ms
Current Slew	\	10	0.001	0.05	A/ms
Time Slew	\	60	0.001	0.03	s
OVP	\	153	0	153	V
UVP	\	156	0	156	V
OPP Positive	\	210	0	210	W
OPP Negative	\	0	-210	-210	W
OCP Positive	\	5.25	0	5.25	A
OCP Negative	\	0	-5.25	-5.25	A
UCP Positive	\	5.25	0.05	-5.25	A
UCP Negative	\	-0.05	-5.25	-5.25	A
Inter Resistance	\	1	0	0	Ω

IT27357					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	150	153	0	0	V
Current positive	10	10.2	0	1	A
Current negative	-10	-10.2	0	-1	A
Resistance(Sink)	\	1500	0.15	1500	Ω
Power positive	500	510	0	510	W
Power negative	-500	-510	0	-510	W
Voltage Slew	\	30	0.001	1.5	V/ms
Current Slew	\	20	0.001	0.1	A/ms
Time Slew	\	60	0.001	0.03	s
OVP	\	153	0	153	V
UVP	\	156	0	156	V
OPP Positive	\	525	0	525	W
OPP Negative	\	0	-525	-525	W
OCP Positive	\	10.5	0	10.5	A
OCP Negative	\	0	-10.5	-10.5	A
UCP Positive	\	10.5	0.1	-10.5	A
UCP Negative	\	-0.1	-10.5	-10.5	A
Inter Resistance	\	1	0	0	Ω

IT27134					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	30	30.6	0	0	V
Current	15	15.3	0	1.5	A
Resistance	\	200	0.02	200	Ω
Power	200	204	0	204	W
Voltage Slew	\	6	0.001	0.3	V/ms
Current Slew	\	30	0.001	0.15	A/ms
Time Slew	\	60	0.001	0.03	s

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OVP	\	30.6	0	30.6	V
UVP	\	31.5	0	31.5	V
OPP	\	210	0	210	W
OCP	\	15.75	0	15.75	A
UCP	\	15.75	0.15	15.75	A
Inter Resistance	\	1	0	0	Ω

IT27154					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	30	30.6	0	0	V
Current	30	30.6	0	3	A
Resistance	\	100	0.01	100	Ω
Power	500	510	0	510	W
Voltage Slew	\	6	0.001	0.3	V/ms
Current Slew	\	60	0.001	0.3	A/ms
Time Slew	\	60	0.001	0.03	s
OVP	\	30.6	0	30.6	V
UVP	\	31.5	0	31.5	V
OPP	\	525	0	525	W
OCP	\	31.5	0	31.5	A
UCP	\	31.5	0.3	31.5	A
Inter Resistance	\	1	0	0	Ω

IT27135					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	60	61.2	0	0	V
Current	10	10.2	0	1	A
Resistance	\	600	0.06	600	Ω
Power	200	204	0	204	W
Voltage Slew	\	12	0.001	0.6	V/ms
Current Slew	\	20	0.001	0.1	A/ms
Time Slew	\	60	0.001	0.03	s
OVP	\	61.2	0	61.2	V
UVP	\	63	0	63	V
OPP	\	210	0	210	W
OCP	\	10.5	0	10.5	A
UCP	\	10.5	0.1	10.5	A
Inter Resistance	\	1	0	0	Ω

IT27155					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	60	61.2	0	0	V
Current	20	20.4	0	2	A
Resistance	\	300	0.03	300	Ω
Power	500	510	0	510	W
Voltage Slew	\	12	0.001	0.6	V/ms

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Current Slew	\	40	0.001	0.2	A/ms
Time Slew	\	60	0.001	0.03	s
OVP	\	61.2	0	61.2	V
UVP	\	63	0	63	V
OPP	\	525	0	525	W
OCP	\	21	0	21	A
UCP	\	21	0.2	21	A
Inter Resistance	\	1	0	0	Ω

IT27137					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	150	153	0	0	V
Current	5	5.1	0	0.5	A
Resistance	\	3000	0.3	3000	Ω
Power	200	204	0	204	W
Voltage Slew	\	30	0.001	1.5	V/ms
Current Slew	\	10	0.001	0.05	A/ms
Time Slew	\	60	0.001	0.03	s
OVP	\	153	0	153	V
UVP	\	156	0	156	V
OPP	\	210	0	210	W
OCP	\	5.25	0	5.25	A
UCP	\	5.25	0.05	5.25	A
Inter Resistance	\	1	0	0	Ω

IT27157					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	150	153	0	0	V
Current	10	10.2	0	1	A
Resistance	\	1500	0.15	1500	Ω
Power	500	510	0	510	W
Voltage Slew	\	30	0.001	1.5	V/ms
Current Slew	\	20	0.001	0.1	A/ms
Time Slew	\	60	0.001	0.03	s
OVP	\	153	0	153	V
UVP	\	156	0	156	V
OPP	\	525	0	525	W
OCP	\	10.5	0	10.5	A
UCP	\	10.5	0.1	10.5	A
Inter Resistance	\	1	0	0	Ω

IT27534					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	30	30.6	0	30.6	V
Current	15	15.3	0	0	A
Resistance	\	200	0.02	200	Ω

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Power	200	204	0	0	W
Voltage Slew	\	6	0.001	0.3	V/ms
Current Slew	\	30	0.001	0.15	A/ms
Resistance Slew	\	200	0.001	200	S/ms
Power Slew	\	500	2	500	W/ms
OPP	\	210	0	210	W
OCP	\	15.75	0	15.75	A

IT27554					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	30	30.6	0	30.6	V
Current	30	30.6	0	0	A
Resistance	\	300	0.03	300	Ω
Power	500	510	0	0	W
Voltage Slew	\	6	0.001	0.3	V/ms
Current Slew	\	60	0.001	0.3	A/ms
Resistance Slew	\	200	0.001	200	S/ms
Power Slew	\	500	2	500	W/ms
OPP	\	525	0	525	W
OCP	\	31.5	0	31.5	A

IT27535					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	60	61.2	0	61.2	V
Current	10	10.2	0	0	A
Resistance	\	600	0.06	600	Ω
Power	200	204	0	0	W
Voltage Slew	\	12	0.001	0.6	V/ms
Current Slew	\	20	0.001	0.1	A/ms
Resistance Slew	\	200	0.001	200	S/ms
Power Slew	\	500	2	500	W/ms
OPP	\	210	0	210	W
OCP	\	10.5	0	10.5	A

IT27555					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	60	61.2	0	61.2	V
Current	20	20.4	0	0	A
Resistance	\	600	0.06	600	Ω
Power	500	510	0	0	W
Voltage Slew	\	12	0.001	0.6	V/ms
Current Slew	\	40	0.001	0.2	A/ms
Resistance Slew	\	200	0.001	200	S/ms
Power Slew	\	500	2	500	W/ms
OPP	\	525	0	525	W
OCP	\	21	0	21	A

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IT27537					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	150	153	0	153	V
Current	5	5.1	0	0	A
Resistance	\	3000	0.3	3000	Ω
Power	200	204	0	0	W
Voltage Slew	\	30	0.001	1.5	V/ms
Current Slew	\	10	0.001	0.05	A/ms
Resistance Slew	\	200	0.001	200	S/ms
Power Slew	\	500	2	500	W/ms
OPP	\	210	0	210	W
OCP	\	5.25	0	5.25	A

IT27557					
Parameter	**Rated**	**Maxmun**	**Minmun**	**Default**	**Unit**
Voltage	150	153	0	153	V
Current	10	10.2	0	0	A
Resistance	\	1500	0.15	1500	Ω
Power	500	510	0	0	W
Voltage Slew	\	30	0.001	1.5	V/ms
Current Slew	\	20	0.001	0.1	A/ms
Resistance Slew	\	200	0.001	200	S/ms
Power Slew	\	500	2	500	W/ms
OPP	\	525	0	525	W
OCP	\	10.5	0	10.5	A

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